

8		7		6		5		4		3		2		1	
D	BOM Groups														
	BOM GROUP		BOM OPTIONS												
	X362_COMMON		ALTERNATE, COMMON, X362_COMMON1, X362_COMMON2, X362_COMMON3, X362_PROGPARTS												
	X362_COMMON1		SE:PROD, BOARD_ID:8, T288_PROG:REV5, TBTTHRM_SNS, S3XCLK:INT												
	X362_COMMON2		EDP_ENABLE, XDP:YES, PCH_CLK:GRNCLK, TBT_DBG, SAMCONN, SKIP_S3V3:AUDIBLE, SOC_BOOT:SPI												
	X362_COMMON3		CPUTHRM:ALRT, TBTTHRM:ALRT, LOADRC:NO, CUMULUS_IPD, S3_STATE:YES, VCCPLLLOC:S3												
	X362_PROGPARTS		BOOTROM_PROG, SMC_PROG, AR_LT_PROG, AR_RT_PROG, WIFI_PROG, BTROM_PROG												
	X362_DEVEL:ENG		ALTERNATE, DBGLED, USBC_DBG, XDP_CONN:YES, WIFI_DBG, S3X_DBG, DEBUG_BUTTON, LOADISNS												
	X362_DEVEL:DVT		ALTERNATE, USBC_DBG, XDP_CONN:YES, WIFI_DBG												
	X362_DEVEL:PVT		ALTERNATE												
C	Module Parts														
	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION									
	337S00266	1	CPU, SKLU, SR2JK, PRQ, 2.9, 28W, 1.05, B1356	U0500	CRITICAL	CPU_SKL23:2.9G									
	337S00267	1	CPU, SKLU, SR2JJ, PRQ, 3.1, 28W, 1.1, B1356	U0500	CRITICAL	CPU_SKL23:3.1G									
	337S00268	1	CPU, SKLU, SR2JH, PRQ, 3.3, 28W, 1.1, B1356	U0500	CRITICAL	CPU_SKL23:3.3G									
	998-04195	1	INTERPOSER, VTT ADAPTER, SKL-U, BGA1356	U0500	CRITICAL	CPU_SOCKET									
	338S00254	2	IC, TBT, ALPINE RIDGE, QSTY, PRQ, C1, CSP337	U2800, UB000	CRITICAL										
	353S00961	4	IC, CD3215, ACE, C00, USB PWR SW, BLNK, BGA96	U3100, U3200, UB300, UB400	CRITICAL										
	338S00276	1	IC, CNTRLR, S3X, B1, FCBGA980	U8600	CRITICAL										
	333S00055	1	IC, LPDDR3-1600, 4GBIT, 25NM, A, 276B	POP8600	CRITICAL	POP_4GBIT									
333S00056	1	IC, LPDDR3-1600, 8GBIT, 25NM, A, 276B	POP8600	CRITICAL	POP_8GBIT										
B	343S00147	1	IC, SL64AP41172, PAK3, STQFN20	U3620	CRITICAL										
	338S00221	1	IC, PMU, SM650039, 7X7MM, BGA168	U7800	CRITICAL										
	353S01016	1	IC, ISL9239HIZ, PMU, TUBA, WCSF40, 2.1X3.3MM	U7000	CRITICAL										
	338S00227	1	IC, PMU, PICCOLO, D2231A0, OTP-AK, WLCSP96	U9300	CRITICAL										
	Programmables (All Builds)														
	EFI ROM														
	341S00698	1	IC, EFI ROM (V0193) DVT, X362	U6100	CRITICAL	BOOTROM_PROG									
	SMC ROM														
	341S00700	1	IC, SMC-B1, EXT (V2.37F6) PVT, X362	U5000	CRITICAL	SMC_PROG									
	TBT ROMs														
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION										
341S00717	1	T29, AR1 (VTB0) PVT, X362	U2890	CRITICAL	AR_LT_PROG										
341S00718	1	T29, AR2 (VTB0) PVT, X362	UB090	CRITICAL	AR_RT_PROG										
WIFI/BT ROM															
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION										
341S00715	1	IC, BT ROM (V32) PVT, X362/X363	U3750	CRITICAL	BTROM_PROG										
341S00716	1	WIFI ROM (P108) PVT, WV1, X362/X363	U3710	CRITICAL	WIFI_PROG										
A	Variable BOM Groups Development/Base BOMs														
	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION									
	685-00055	1	COMMON PARTS, MLB, X362	BASE	CRITICAL	BASE_BOM									
	985-00070	1	DEV, MLB, X362	DEVEL	CRITICAL	DEVEL_BOM									
	Main DRAM Parts														
	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION									
	333S00069	4	IC, SDRAM, LPDDR3-2133, 16GBIT, 20NM, BGA178	U2300, U2400, U2500, U2600	CRITICAL	8G_MICRON_2133									
	333S00070	4	IC, SDRAM, LPDDR3-2133, 32GBIT, 20NM, BGA178	U2300, U2400, U2500, U2600	CRITICAL	16G_MICRON_2133									
	333S00068	4	IC, SDRAM, LPDDR3-2133, 16GBIT, 20NM, BGA178	U2300, U2400, U2500, U2600	CRITICAL	8G_SAMSUNG_2133									
	333S00050	4	IC, SDRAM, LPDDR3-2133, 32GBIT, 20NM, BGA178	U2300, U2400, U2500, U2600	CRITICAL	16G_SAMSUNG_2133									
Main DRAM SPD Straps															
BOM GROUP		BOM OPTIONS													
RAM_8G_MICRON_2133		8G_MICRON_2133, RAMCFG4_L, RAMCFG3_L, RAMCFG2_L, RAMCFG1_L													
RAM_16G_MICRON_2133		16G_MICRON_2133, RAMCFG4_L, RAMCFG3_L, RAMCFG1_L													
RAM_8G_SAMSUNG_2133		8G_SAMSUNG_2133, RAMCFG4_L, RAMCFG3_L, RAMCFG2_L, RAMCFG0_L													
RAM_16G_SAMSUNG_2133		16G_SAMSUNG_2133, RAMCFG4_L, RAMCFG3_L, RAMCFG0_L													
NAND Parts															
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION										
335S00124	4	NAND, 1ZNM, 64GB, TOGG, HPN, 128G, LGA60	U9100, U9120, U9200, U9220	CRITICAL	NAND_SDISK_256GB										
335S00125	4	NAND, 1ZNM, 128GB, TOGG, HPN, 128G, LGA60	U9100, U9120, U9200, U9220	CRITICAL	NAND_SDISK_512GB										
335S00126	4	NAND, 1ZNM, 256GB, TOGG, HPN, 128G, LGA60	U9100, U9120, U9200, U9220	CRITICAL	NAND_SDISK_1TB										
335S00261	4	NAND, TGD0R2, 128GX4, 15NM, HP, USHD, T2, LGA60	U9100, U9120, U9200, U9220	CRITICAL	NAND_TSHBA_256GB										
335S00262	4	NAND, TGD0R2, 128GX8, 15NM, HP, USHD, T2, LGA60	U9100, U9120, U9200, U9220	CRITICAL	NAND_TSHBA_512GB										
335S00263	4	NAND, TGD0R2, 128GX16, 15NM, HP, UHD, T2, LGA60	U9100, U9120, U9200, U9220	CRITICAL	NAND_TSHBA_1TB										
NAND Straps															
BOM GROUP		BOM OPTIONS													
SAND_256G		ALTERNATE, NAND_SDISK_256GB, POP_4GBIT, CAPACITY1, CAPACITY0													
SAND_512G		ALTERNATE, NAND_SDISK_512GB, POP_4GBIT, CAPACITY2													
SAND_1T		ALTERNATE, NAND_SDISK_1TB, POP_8GBIT, CAPACITY2, CAPACITY0													
TOSH_256G		ALTERNATE, NAND_TSHBA_256GB, POP_4GBIT, CAPACITY1, CAPACITY0													
TOSH_512G		ALTERNATE, NAND_TSHBA_512GB, POP_4GBIT, CAPACITY2													
TOSH_1T		ALTERNATE, NAND_TSHBA_1TB, POP_8GBIT, CAPACITY2, CAPACITY0													
Strategic Silicon															
PART#	STRATEGIC VALUE	COMMENT	PART#	STRATEGIC VALUE	COMMENT										
337S00266	08	CPU	338S00227	02	PICCOLO										
337S00267	08	CPU	343S00135	10	T288										
337S00268	08	CPU	343S00136	10	T288										
333S00050	07	MEMORY	343S00137	10	T288										
333S00068	07	MEMORY	343S00138	10	T288										
333S00069	07	MEMORY	338S00193	09	BERKELIUM										
333S00070	07	MEMORY	353S39978	02	MOJAVE										
335S00124	02	NAND	338S00147	02	SECURE ELEMENT										
335S00125	02	NAND	338S00254	08	ALPINE RIDGE										
335S00126	02	NAND	353S00961	09	ACE										
335S00261	02	NAND	338S00142	09	CLIFDEN										
335S00262	02	NAND	353S00685	07	AUDIO AMP										
335S00263	02	NAND	353S4316	08	BAYSIDE										
333S00025	02	S3X DRAM	338S00221	08	BANJO										
333S00026	02	S3X DRAM	353S01016	09	TUBA										
333S00055	02	S3X DRAM	339S00056	05	ICEBOCK										
333S00056	02	S3X DRAM	343S00147	08	PAK										
333S00107	02	S3X DRAM	359S00006	08	GREEN CLOCK										
333S00188	02	S3X DRAM	353S00795	09	DEBUG MUX										
998-06736	02	S3X CONTROLLER													
8		7		6		5		4		3		2		1	
CPU DRAM CFG Chart															
DIE REV		CFG 4													
A		0													
B		1													
SPEED		CFG 3													
2133		0													
1866		1													
CAPACITY		CFG 2													
8GB		0													
16GB		1													
VENDOR		CFG 1	CFG 0												
HYNIX		0	0												
MICRON		0	1												
SAMSUNG		1	0												
N/A		1	1												
BOM Configuration															
DRAWING NUMBER		SIZE													
051-00515		D													
REVISION		9.0.0													
BRANCH		dvt-fab09-0													
PAGE		2 OF 145													
SHEET		2 OF 119													
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Variable BOM Groups Development/Base BOMs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-00055	1	COMMON PARTS,MLB,X362	BASE	CRITICAL	BASE_BOM
985-00070	1	DEV,MLB,X362	DEVEL	CRITICAL	DEVEL_BOM

Main DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S00069	4	IC,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	8G_MICRON_2133
333S00070	4	IC,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_MICRON_2133
333S00068	4	IC,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	8G_SAMSUNG_2133
333S00050	4	IC,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_SAMSUNG_2133

Main DRAM SPD Straps

BOM GROUP	BOM OPTIONS
RAM_8G_MICRON_2133	8G_MICRON_2133, RAMCFG4_L, RAMCFG3_L, RAMCFG2_L, RAMCFG1_L
RAM_16G_MICRON_2133	16G_MICRON_2133, RAMCFG4_L, RAMCFG3_L, RAMCFG1_L
RAM_8G_SAMSUNG_2133	8G_SAMSUNG_2133, RAMCFG4_L, RAMCFG3_L, RAMCFG2_L, RAMCFG0_L
RAM_16G_SAMSUNG_2133	16G_SAMSUNG_2133, RAMCFG4_L, RAMCFG3_L, RAMCFG0_L

NAND Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S00124	4	NAND,1ZNM,64GB,T066,HPN,1286,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_256GB
335S00125	4	NAND,1ZNM,128GB,T066,HPN,1286,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_512GB
335S00126	4	NAND,1ZNM,256GB,T066,HPN,1286,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_1TB
335S00261	4	NAND,TGDDR2,128GX4,15NM,HP,USHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_256GB
335S00262	4	NAND,TGDDR2,128GX8,15NM,HP,USHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_512GB
335S00263	4	NAND,TGDDR2,128GX16,15NM,HP,UHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_1TB

NAND Straps

BOM GROUP	BOM OPTIONS
SAND_256G	ALTERNATE,NAND_SDISK_256GB,POP_4GBIT,CAPACITY1,CAPACITY0
SAND_512G	ALTERNATE,NAND_SDISK_512GB,POP_4GBIT,CAPACITY2
SAND_1T	ALTERNATE,NAND_SDISK_1TB,POP_8GBIT,CAPACITY2,CAPACITY0
TOSH_256G	ALTERNATE,NAND_TSHBA_256GB,POP_4GBIT,CAPACITY1,CAPACITY0
TOSH_512G	ALTERNATE,NAND_TSHBA_512GB,POP_4GBIT,CAPACITY2
TOSH_1T	ALTERNATE,NAND_TSHBA_1TB,POP_8GBIT,CAPACITY2,CAPACITY0

Strategic Silicon

PART#	STRATEGIC VALUE	COMMENT
337S00266	08	CPU
337S00267	08	CPU
337S00268	08	CPU
333S00050	07	MEMORY
333S00068	07	MEMORY
333S00069	07	MEMORY
333S00070	07	MEMORY
335S00124	02	NAND
335S00125	02	NAND
335S00126	02	NAND
335S00261	02	NAND
335S00262	02	NAND
335S00263	02	NAND
333S00025	02	S3X DRAM
333S00026	02	S3X DRAM
333S00055	02	S3X DRAM
333S00056	02	S3X DRAM
333S00107	02	S3X DRAM
333S00108	02	S3X DRAM
998-06736	02	S3X CONTROLLER

PART#	STRATEGIC VALUE	COMMENT
338S00227	02	PICCOLO
343S00135	10	T200
343S00136	10	T200
343S00137	10	T200
343S00138	10	T200
338S00193	09	BERKELIUM
353S3978	02	MOJAVE
338S00147	02	SECURE ELEMENT
338S00254	00	ALPINE RIDGE
353S00961	09	ACE
338S00142	09	CLIFDEN
353S00685	07	AUDIO AMP
353S4316	08	BAYSIDE
338S00221	08	BANJO
353S01016	09	TUBA
339S00056	05	ICEBOCK
343S00147	08	PAK
359S00006	08	GREEN CLOCK
353S00795	09	DEBUG MUX

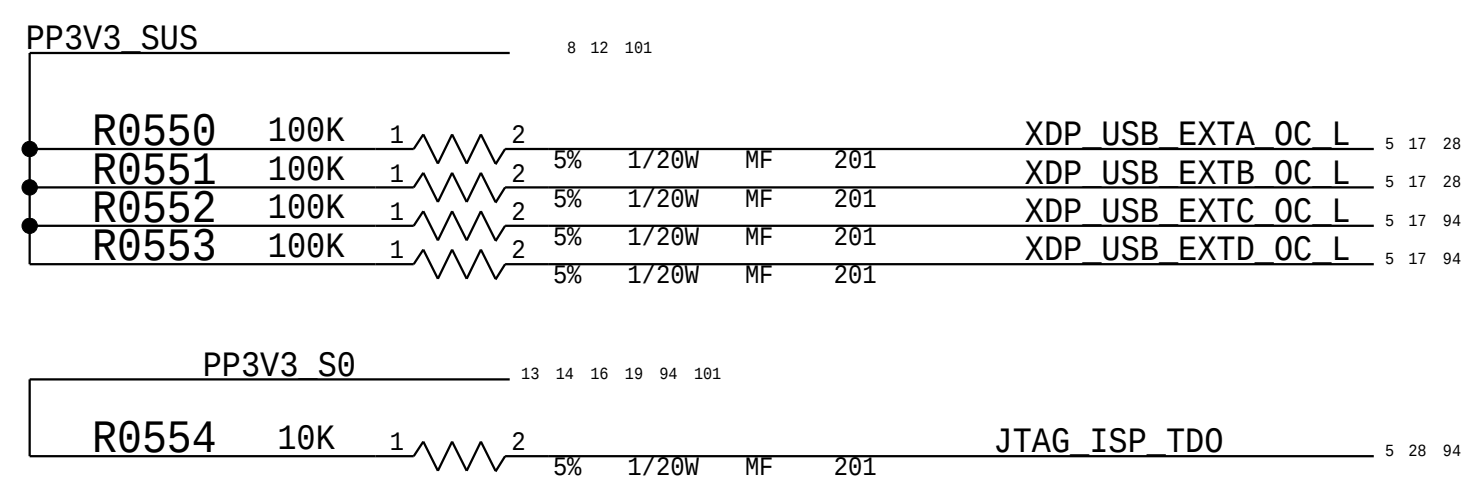
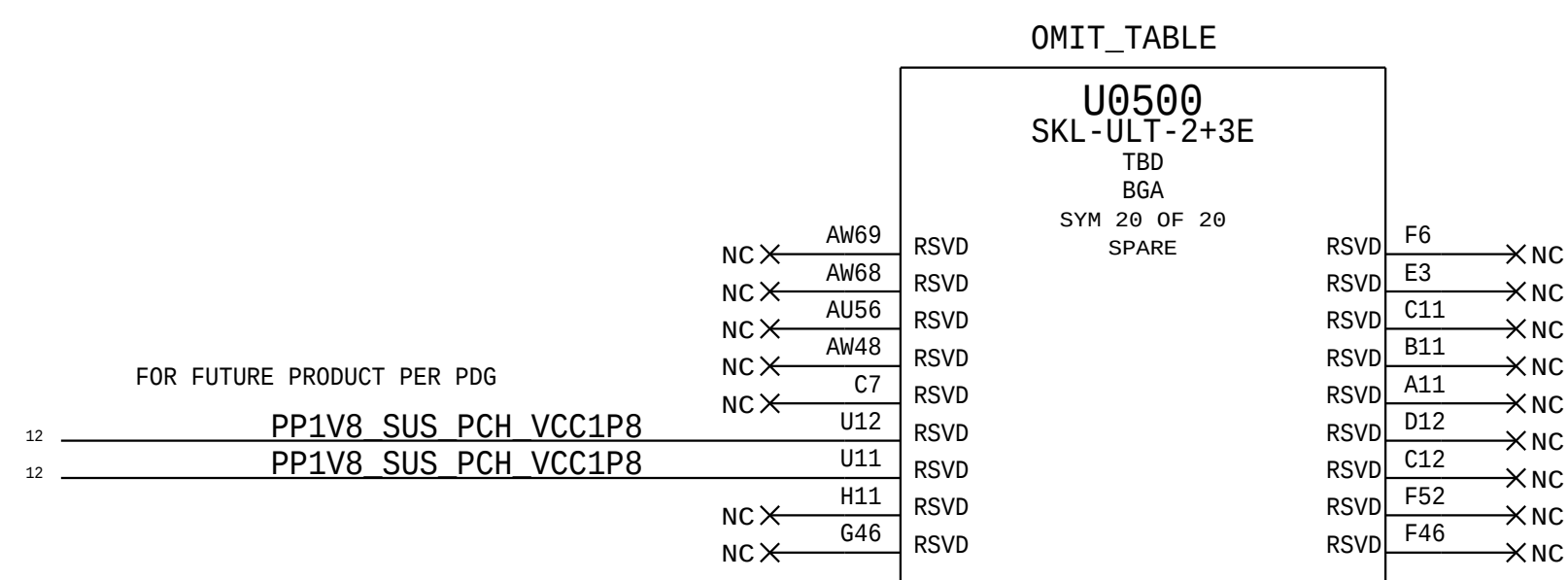
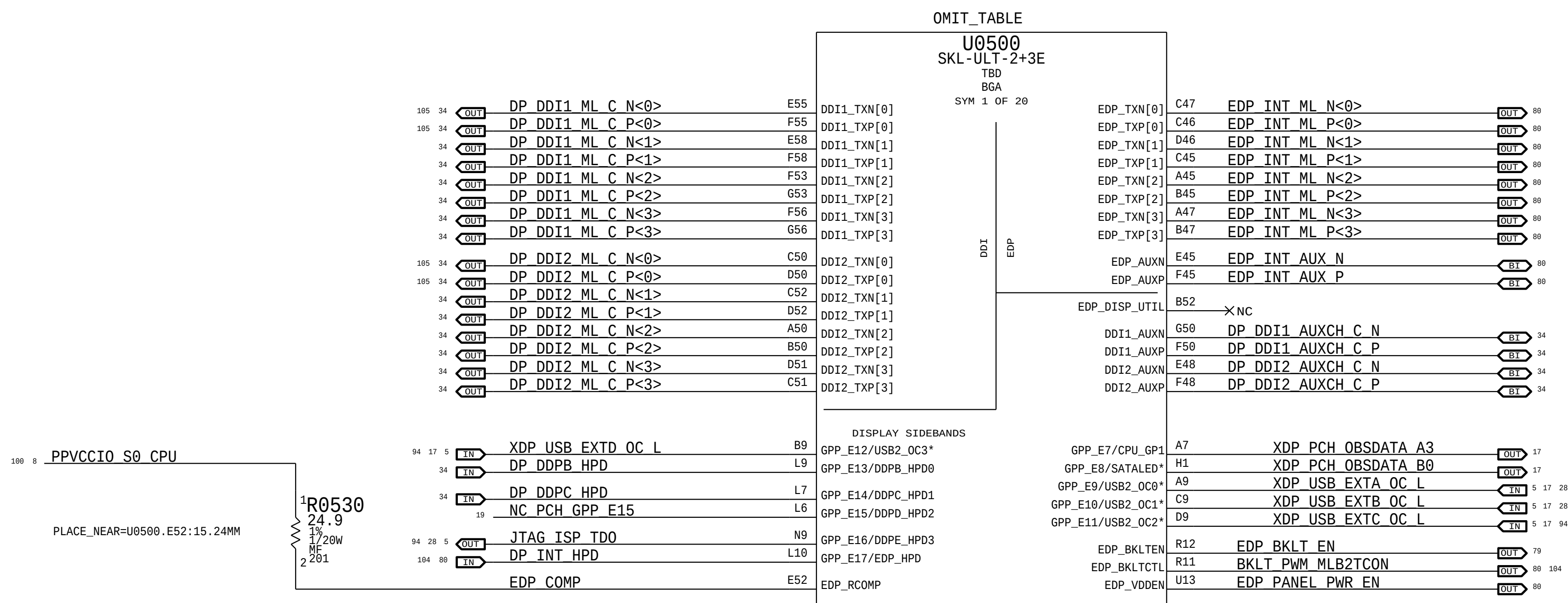
CPU DRAM CFG Chart

DIE REV	CFG 4
A	0
B	1

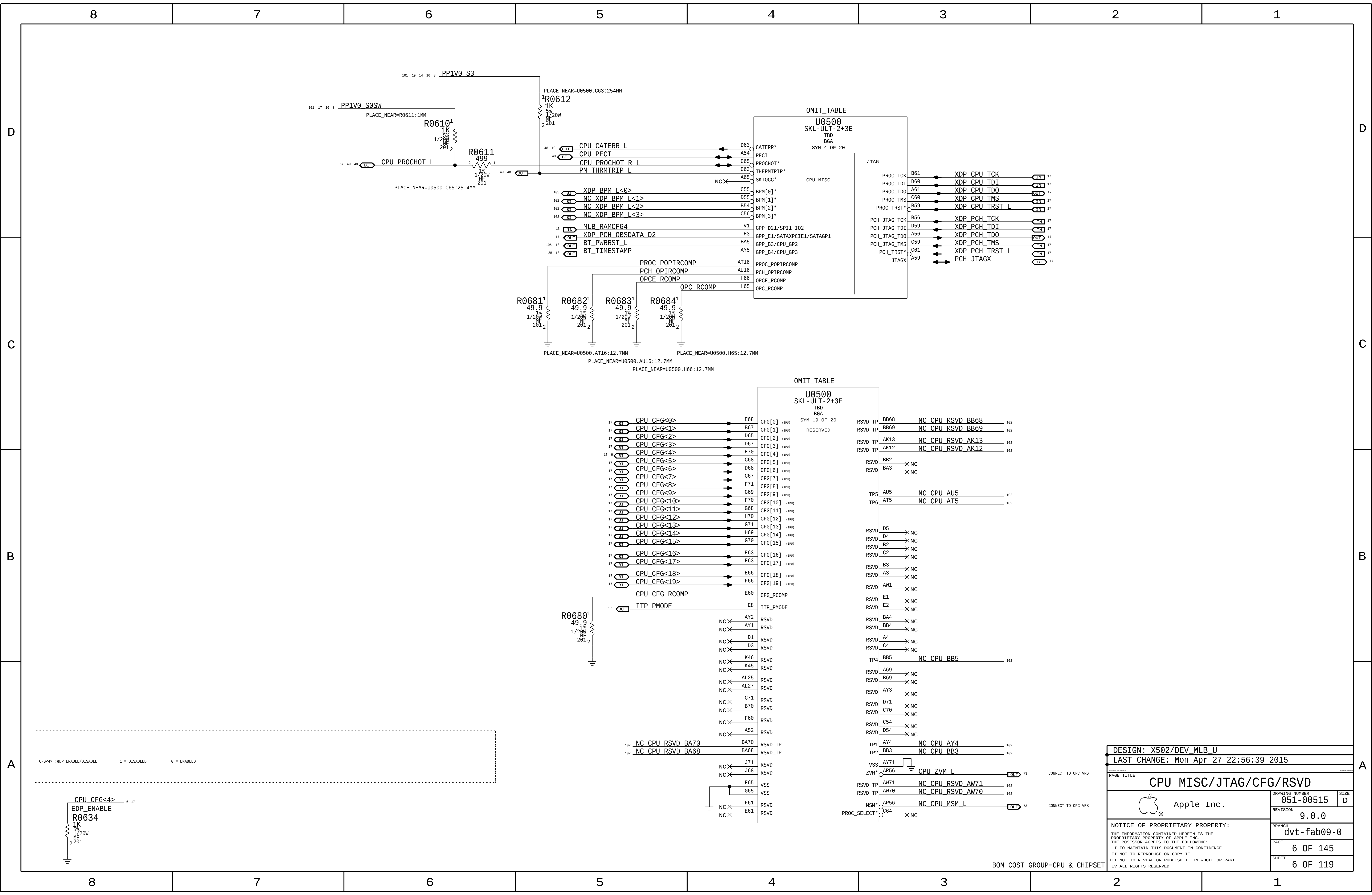
SPEED	CFG 3
2133	0
1866	1

8		7		6		5		4		3		2		1																																																																																																																						
D	BOM Variants														D																																																																																																																					
	<table><tr><th>BOM NUMBER</th><th>BOM NAME</th><th>BOM OPTIONS</th></tr><tr><td>685-00055</td><td>COMMON PARTS, MLB, X362</td><td>X362_COMMON</td></tr><tr><td>985-00070</td><td>DEV, MLB, X362</td><td>X362_DEVEL :DVT</td></tr><tr><td>639-01870</td><td>MLB, 2.9G, SAM-8G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_SAMSUNG_2133, SAND_256G</td></tr><tr><td>639-01871</td><td>MLB, 2.9G, SAM-16G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_SAMSUNG_2133, SAND_256G</td></tr><tr><td>639-01872</td><td>MLB, 2.9G, MIC-8G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_MICRON_2133, SAND_256G</td></tr><tr><td>639-01873</td><td>MLB, 2.9G, MIC-16G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_MICRON_2133, SAND_256G</td></tr><tr><td>639-01984</td><td>MLB, 2.9G, SAM-8G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_SAMSUNG_2133, SAND_512G</td></tr><tr><td>639-01985</td><td>MLB, 2.9G, SAM-16G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_SAMSUNG_2133, SAND_512G</td></tr><tr><td>639-01986</td><td>MLB, 2.9G, MIC-8G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_MICRON_2133, SAND_512G</td></tr><tr><td>639-01987</td><td>MLB, 2.9G, MIC-16G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_MICRON_2133, SAND_512G</td></tr><tr><td>639-02517</td><td>MLB, 2.9G, SAM-8G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_SAMSUNG_2133, SAND_1T</td></tr><tr><td>639-02518</td><td>MLB, 2.9G, SAM-16G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_SAMSUNG_2133, SAND_1T</td></tr><tr><td>639-02519</td><td>MLB, 2.9G, MIC-8G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_MICRON_2133, SAND_1T</td></tr><tr><td>639-02520</td><td>MLB, 2.9G, MIC-16G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_MICRON_2133, SAND_1T</td></tr><tr><td>639-01874</td><td>MLB, 3.1G, SAM-8G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_SAMSUNG_2133, SAND_256G</td></tr><tr><td>639-01875</td><td>MLB, 3.1G, SAM-16G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_SAMSUNG_2133, SAND_256G</td></tr><tr><td>639-01876</td><td>MLB, 3.1G, MIC-8G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_MICRON_2133, SAND_256G</td></tr><tr><td>639-01877</td><td>MLB, 3.1G, MIC-16G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_MICRON_2133, SAND_256G</td></tr><tr><td>639-01883</td><td>MLB, 3.1G, SAM-8G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_SAMSUNG_2133, SAND_512G</td></tr><tr><td>639-01884</td><td>MLB, 3.1G, SAM-16G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_SAMSUNG_2133, SAND_512G</td></tr><tr><td>639-01885</td><td>MLB, 3.1G, MIC-8G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_MICRON_2133, SAND_512G</td></tr><tr><td>639-01886</td><td>MLB, 3.1G, MIC-16G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_MICRON_2133, SAND_512G</td></tr><tr><td>639-01887</td><td>MLB, 3.1G, SAM-8G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_SAMSUNG_2133, SAND_1T</td></tr><tr><td>639-01888</td><td>MLB, 3.1G, SAM-16G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_SAMSUNG_2133, SAND_1T</td></tr><tr><td>639-01889</td><td>MLB, 3.1G, MIC-8G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_MICRON_2133, SAND_1T</td></tr><tr><td>639-01890</td><td>MLB, 3.1G, MIC-16G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_MICRON_2133, SAND_1T</td></tr><tr><td>639-02221</td><td>MLB, 3.3G, SAM-8G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_SAMSUNG_2133, SAND_256G</td></tr><tr><td>639-02222</td><td>MLB, 3.3G, SAM-16G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_SAMSUNG_2133, SAND_256G</td></tr><tr><td>639-02223</td><td>MLB, 3.3G, MIC-8G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_MICRON_2133, SAND_256G</td></tr><tr><td>639-02224</td><td>MLB, 3.3G, MIC-16G, SAND-256G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_MICRON_2133, SAND_256G</td></tr><tr><td>639-01891</td><td>MLB, 3.3G, SAM-8G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_SAMSUNG_2133, SAND_512G</td></tr><tr><td>639-01892</td><td>MLB, 3.3G, SAM-16G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_SAMSUNG_2133, SAND_512G</td></tr><tr><td>639-01893</td><td>MLB, 3.3G, MIC-8G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_MICRON_2133, SAND_512G</td></tr><tr><td>639-01894</td><td>MLB, 3.3G, MIC-16G, SAND-512G, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_MICRON_2133, SAND_512G</td></tr><tr><td>639-01895</td><td>MLB, 3.3G, SAM-8G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_SAMSUNG_2133, SAND_1T</td></tr><tr><td>639-01896</td><td>MLB, 3.3G, SAM-16G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_SAMSUNG_2133, SAND_1T</td></tr><tr><td>639-01897</td><td>MLB, 3.3G, MIC-8G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_MICRON_2133, SAND_1T</td></tr><tr><td>639-01898</td><td>MLB, 3.3G, MIC-16G, SAND-1T, X362</td><td>BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_MICRON_2133, SAND_1T</td></tr></table>															BOM NUMBER	BOM NAME	BOM OPTIONS	685-00055	COMMON PARTS, MLB, X362	X362_COMMON	985-00070	DEV, MLB, X362	X362_DEVEL :DVT	639-01870	MLB, 2.9G, SAM-8G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_SAMSUNG_2133, SAND_256G	639-01871	MLB, 2.9G, SAM-16G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_SAMSUNG_2133, SAND_256G	639-01872	MLB, 2.9G, MIC-8G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_MICRON_2133, SAND_256G	639-01873	MLB, 2.9G, MIC-16G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_MICRON_2133, SAND_256G	639-01984	MLB, 2.9G, SAM-8G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_SAMSUNG_2133, SAND_512G	639-01985	MLB, 2.9G, SAM-16G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_SAMSUNG_2133, SAND_512G	639-01986	MLB, 2.9G, MIC-8G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_MICRON_2133, SAND_512G	639-01987	MLB, 2.9G, MIC-16G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_MICRON_2133, SAND_512G	639-02517	MLB, 2.9G, SAM-8G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_SAMSUNG_2133, SAND_1T	639-02518	MLB, 2.9G, SAM-16G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_SAMSUNG_2133, SAND_1T	639-02519	MLB, 2.9G, MIC-8G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_MICRON_2133, SAND_1T	639-02520	MLB, 2.9G, MIC-16G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_MICRON_2133, SAND_1T	639-01874	MLB, 3.1G, SAM-8G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_SAMSUNG_2133, SAND_256G	639-01875	MLB, 3.1G, SAM-16G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_SAMSUNG_2133, SAND_256G	639-01876	MLB, 3.1G, MIC-8G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_MICRON_2133, SAND_256G	639-01877	MLB, 3.1G, MIC-16G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_MICRON_2133, SAND_256G	639-01883	MLB, 3.1G, SAM-8G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_SAMSUNG_2133, SAND_512G	639-01884	MLB, 3.1G, SAM-16G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_SAMSUNG_2133, SAND_512G	639-01885	MLB, 3.1G, MIC-8G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_MICRON_2133, SAND_512G	639-01886	MLB, 3.1G, MIC-16G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_MICRON_2133, SAND_512G	639-01887	MLB, 3.1G, SAM-8G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_SAMSUNG_2133, SAND_1T	639-01888	MLB, 3.1G, SAM-16G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_SAMSUNG_2133, SAND_1T	639-01889	MLB, 3.1G, MIC-8G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_8G_MICRON_2133, SAND_1T	639-01890	MLB, 3.1G, MIC-16G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_MICRON_2133, SAND_1T	639-02221	MLB, 3.3G, SAM-8G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_SAMSUNG_2133, SAND_256G	639-02222	MLB, 3.3G, SAM-16G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_SAMSUNG_2133, SAND_256G	639-02223	MLB, 3.3G, MIC-8G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_MICRON_2133, SAND_256G	639-02224	MLB, 3.3G, MIC-16G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_MICRON_2133, SAND_256G	639-01891	MLB, 3.3G, SAM-8G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_SAMSUNG_2133, SAND_512G	639-01892	MLB, 3.3G, SAM-16G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_SAMSUNG_2133, SAND_512G	639-01893	MLB, 3.3G, MIC-8G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_MICRON_2133, SAND_512G	639-01894	MLB, 3.3G, MIC-16G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_MICRON_2133, SAND_512G	639-01895	MLB, 3.3G, SAM-8G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_SAMSUNG_2133, SAND_1T	639-01896	MLB, 3.3G, SAM-16G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_SAMSUNG_2133, SAND_1T	639-01897	MLB, 3.3G, MIC-8G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_MICRON_2133, SAND_1T	639-01898	MLB, 3.3G, MIC-16G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_MICRON_2133, SAND_1T
	BOM NUMBER	BOM NAME	BOM OPTIONS																																																																																																																																	
	685-00055	COMMON PARTS, MLB, X362	X362_COMMON																																																																																																																																	
	985-00070	DEV, MLB, X362	X362_DEVEL :DVT																																																																																																																																	
	639-01870	MLB, 2.9G, SAM-8G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_SAMSUNG_2133, SAND_256G																																																																																																																																	
	639-01871	MLB, 2.9G, SAM-16G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_SAMSUNG_2133, SAND_256G																																																																																																																																	
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	639-01873	MLB, 2.9G, MIC-16G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_MICRON_2133, SAND_256G																																																																																																																																	
	639-01984	MLB, 2.9G, SAM-8G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_SAMSUNG_2133, SAND_512G																																																																																																																																	
	639-01985	MLB, 2.9G, SAM-16G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_SAMSUNG_2133, SAND_512G																																																																																																																																	
	639-01986	MLB, 2.9G, MIC-8G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_MICRON_2133, SAND_512G																																																																																																																																	
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	639-02517	MLB, 2.9G, SAM-8G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_SAMSUNG_2133, SAND_1T																																																																																																																																	
	639-02518	MLB, 2.9G, SAM-16G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_SAMSUNG_2133, SAND_1T																																																																																																																																	
	639-02519	MLB, 2.9G, MIC-8G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_8G_MICRON_2133, SAND_1T																																																																																																																																	
	639-02520	MLB, 2.9G, MIC-16G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:2.9G, RAM_16G_MICRON_2133, SAND_1T																																																																																																																																	
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	639-01875	MLB, 3.1G, SAM-16G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_SAMSUNG_2133, SAND_256G																																																																																																																																	
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	639-01884	MLB, 3.1G, SAM-16G, SAND-512G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.1G, RAM_16G_SAMSUNG_2133, SAND_512G																																																																																																																																	
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639-02223	MLB, 3.3G, MIC-8G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_MICRON_2133, SAND_256G																																																																																																																																		
639-02224	MLB, 3.3G, MIC-16G, SAND-256G, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_16G_MICRON_2133, SAND_256G																																																																																																																																		
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639-01895	MLB, 3.3G, SAM-8G, SAND-1T, X362	BASE_BOM, DEVEL_BOM, CPU_SKL23:3.3G, RAM_8G_SAMSUNG_2133, SAND_1T																																																																																																																																		
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	<table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>152S00367</td><td>152S00266</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00368</td><td>152S00269</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00370</td><td>152S00270</td><td></td><td>ALL</td><td></td></tr><tr><td>353S4068</td><td>353S4070</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00772</td><td>353S4070</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00086</td><td>138S0884</td><td></td><td>ALL</td><td></td></tr><tr><td>107S0248</td><td>107S0250</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00434</td><td>152S1829</td><td></td><td>ALL</td><td></td></tr><tr><td>371S00019</td><td>371S0463</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00107</td><td>353S3239</td><td>ANY</td><td>ALL</td><td></td></tr><tr><td>353S00231</td><td>353S3987</td><td></td><td>ALL</td><td></td></tr><tr><td>333S00025</td><td>333S00055</td><td>POP_4GBIT</td><td>ALL</td><td></td></tr><tr><td>333S00026</td><td>333S00056</td><td>POP_8GBIT</td><td>ALL</td><td></td></tr><tr><td>333S00107</td><td>333S00055</td><td>POP_4GBIT</td><td>ALL</td><td></td></tr><tr><td>333S00108</td><td>333S00056</td><td>POP_8GBIT</td><td>ALL</td><td></td></tr></table>															PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	152S00367	152S00266		ALL		152S00368	152S00269		ALL		152S00370	152S00270		ALL		353S4068	353S4070		ALL		353S00772	353S4070		ALL		138S00086	138S0884		ALL		107S0248	107S0250		ALL		152S00434	152S1829		ALL		371S00019	371S0463		ALL		353S00107	353S3239	ANY	ALL		353S00231	353S3987		ALL		333S00025	333S00055	POP_4GBIT	ALL		333S00026	333S00056	POP_8GBIT	ALL		333S00107	333S00055	POP_4GBIT	ALL		333S00108	333S00056	POP_8GBIT	ALL																																						
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8		7		6		5		4		3		2		1	
BOARD MECHANICALS															
Shield Cans - BOTTOM SIDE				Shield Cans - TOP SIDE				POGO PINS				Cowling Bosses - BOTTOM SIDE			
ALPINE RIDGE - LIO (U2800) - 806-06077				T208 (U3900) - 806-06264				LIO and RIO -2X (870-5071)				DFR TOUCH CONN (J4402) - 860-00414			
LPDDR3 (U2300 ~ U2600) - 806-06167				DIPLEXERS - 806-06266				AROUND THE FAN AND CENTER - 8X (870-01518)				USB-C CONN - LIO (J3300) - 860-00392			
NAND - BOTTOM SOUTH (U9120) - 806-05945				NAND - TOP SOUTH (U9100) - 806-06262											
S3X (U8600) - 806-06023				NAND - TOP NORTH (9220) - 806-06258											
NAND - BOTTOM NORTH (U9200) - 806-06265				ALPINE RIDGE - RIO (UB000) - 806-06077				USB-C CONN - RIO (JB500) - 860-00392				KBD CONN (J4500) - 860-00412			
Shield CAN Alignment Slots 14X - 998-04440 (1.2mm X 0.4mm)				Thermal Stage Mounting Holes				AUDIO JACK CONN (J6600) - 860-00399				MESA CONN (J4900) - 860-00399			



DESIGN: X502/DEV_MLB_U				051	
LAST CHANGE: Tue Apr 28 20:32:21 2015					
CPU GFX					
 Apple Inc.			DRAWING NUMBER 051-00515		SIZE D
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			BRANCH dvt-fab09-0		
			PAGE 5 OF 145		
			SHEET 5 OF 119		



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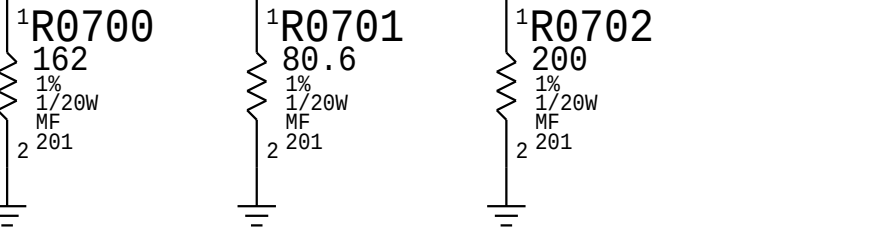
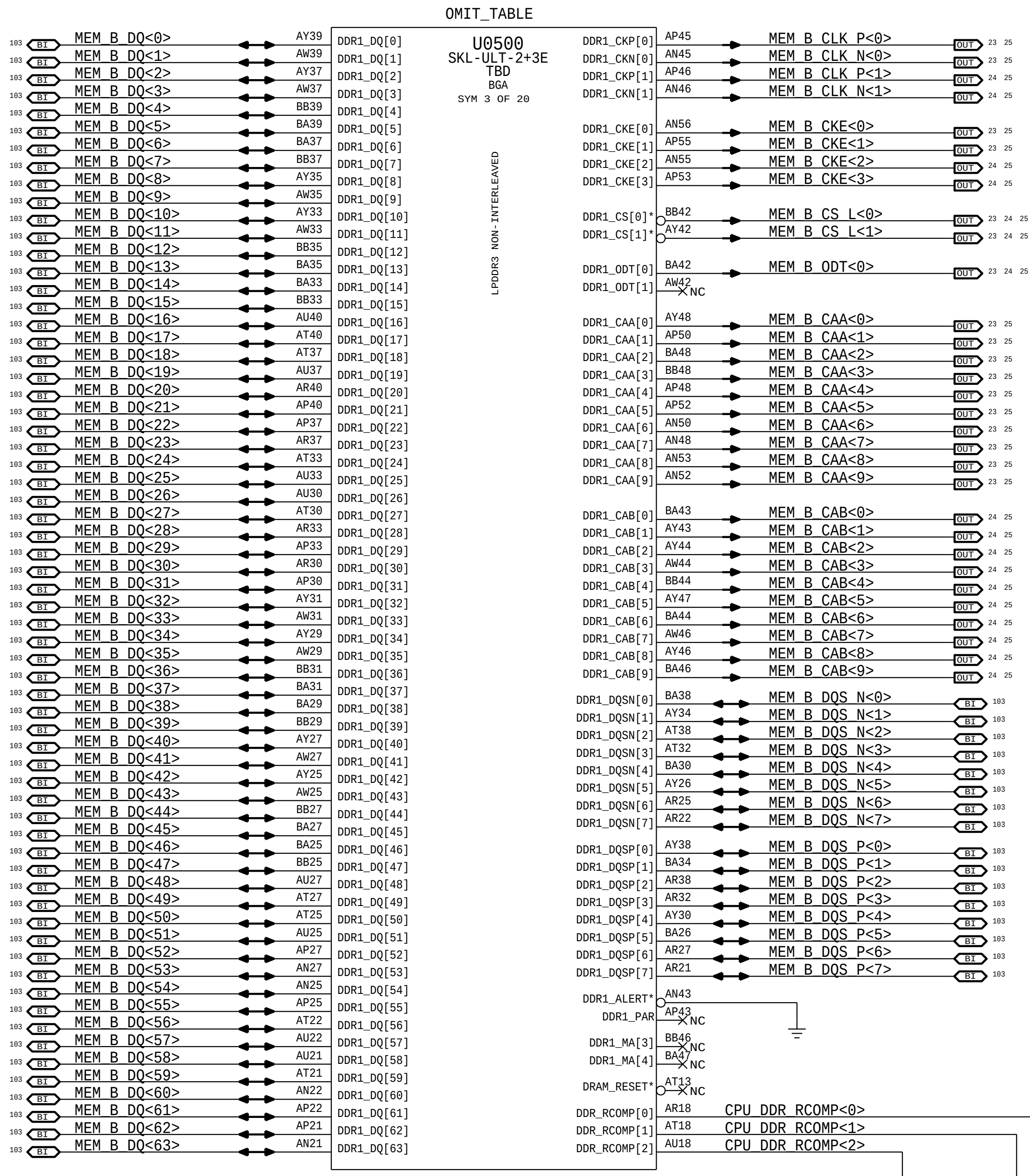
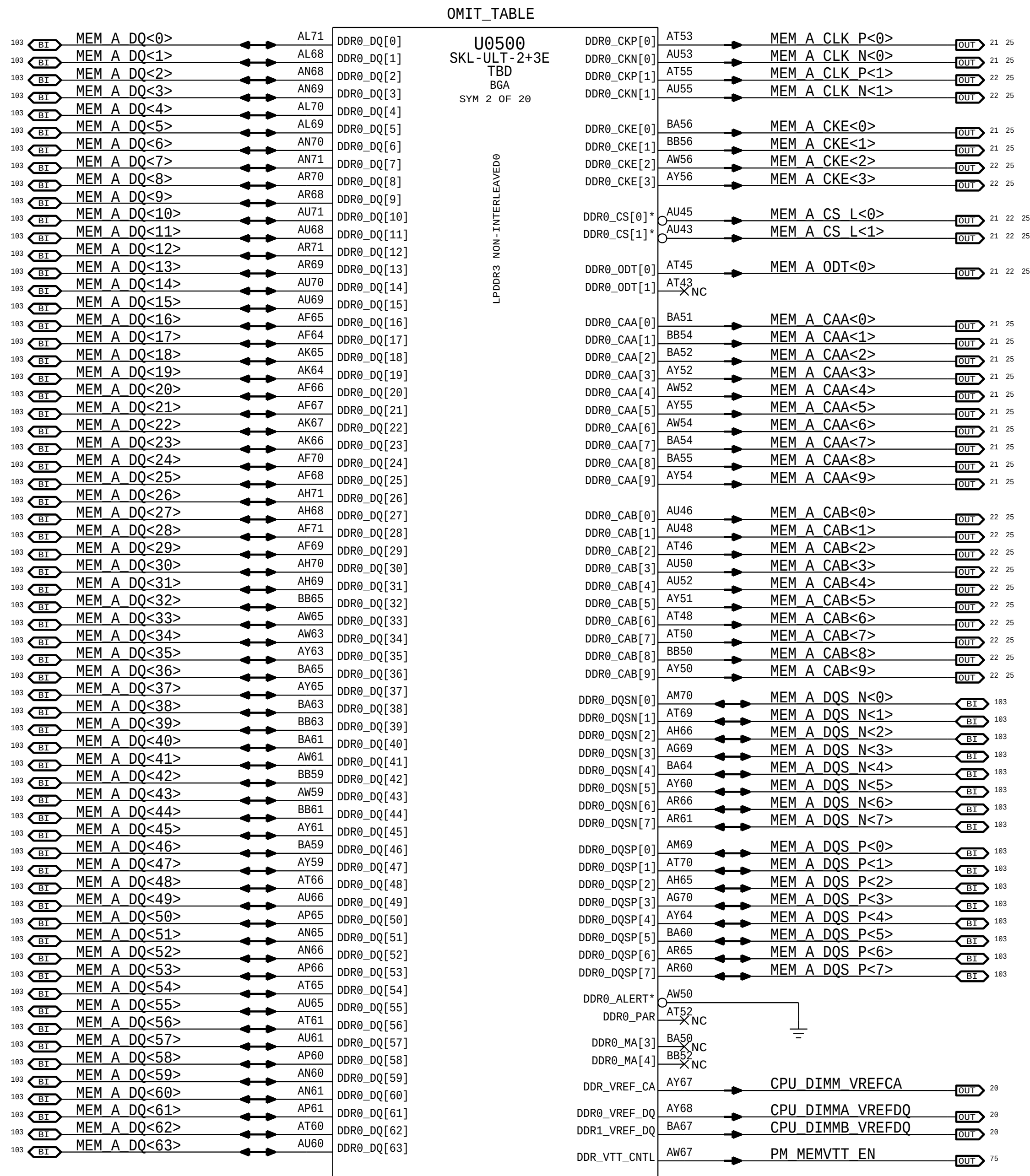
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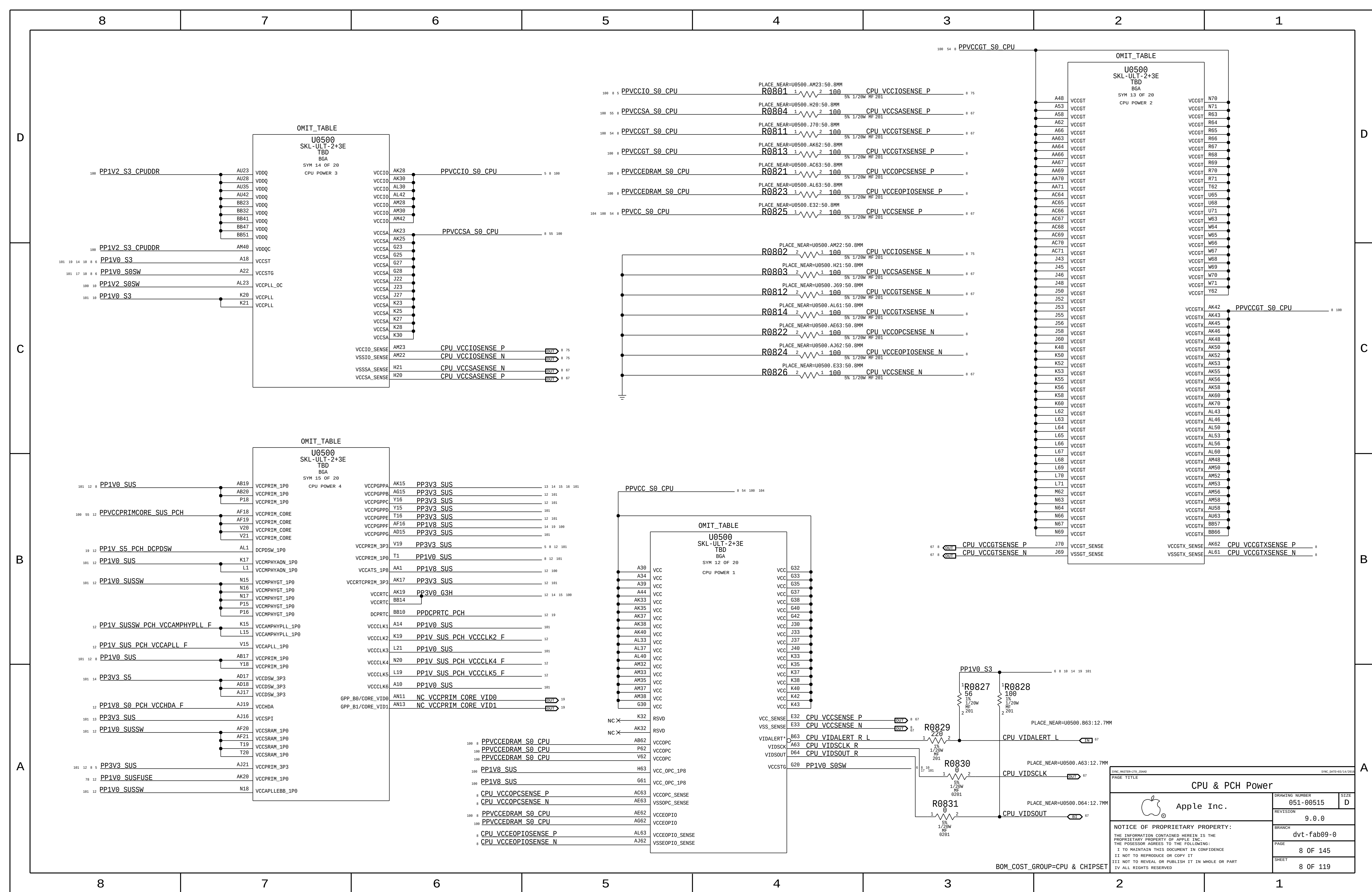
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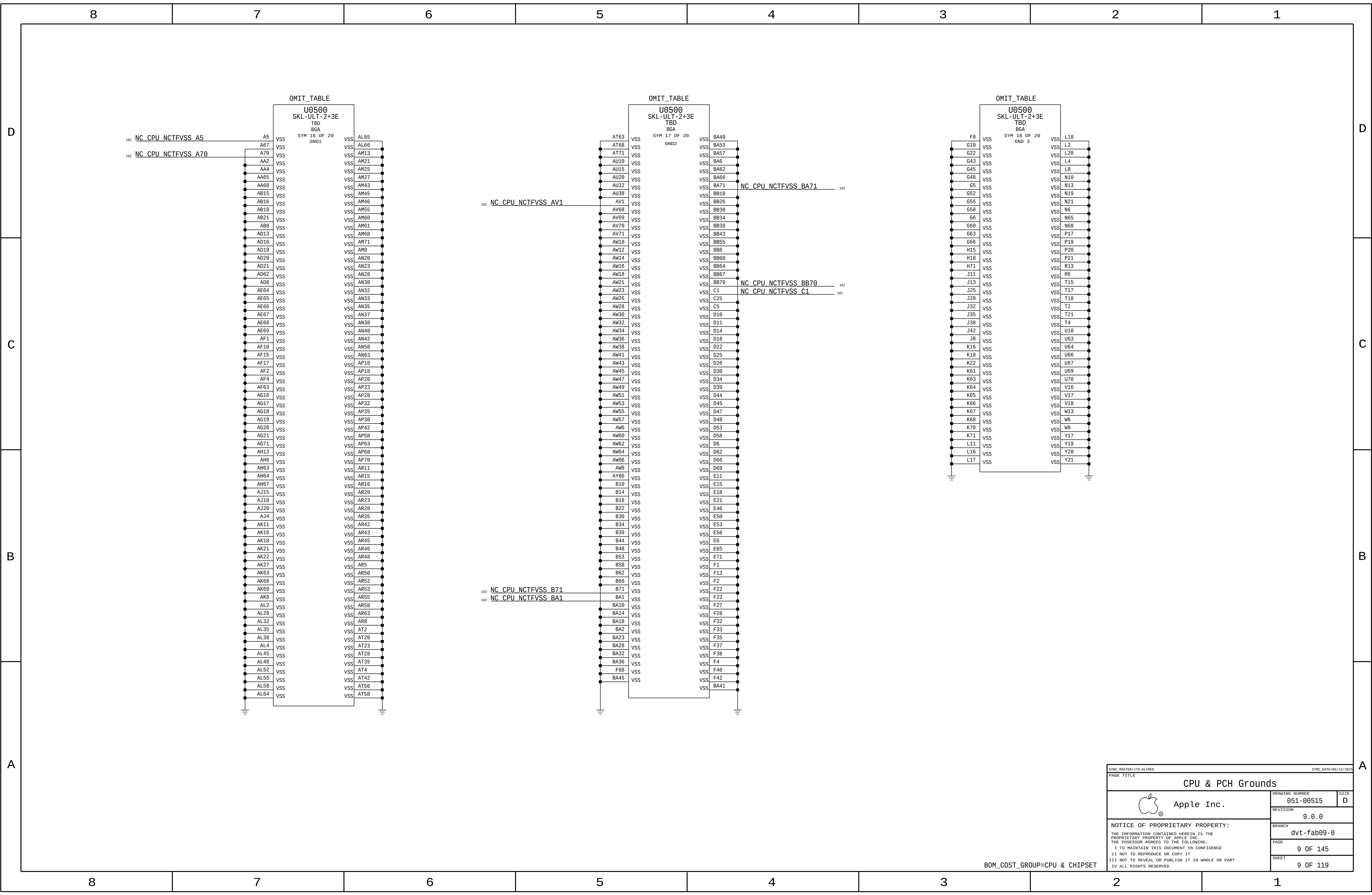
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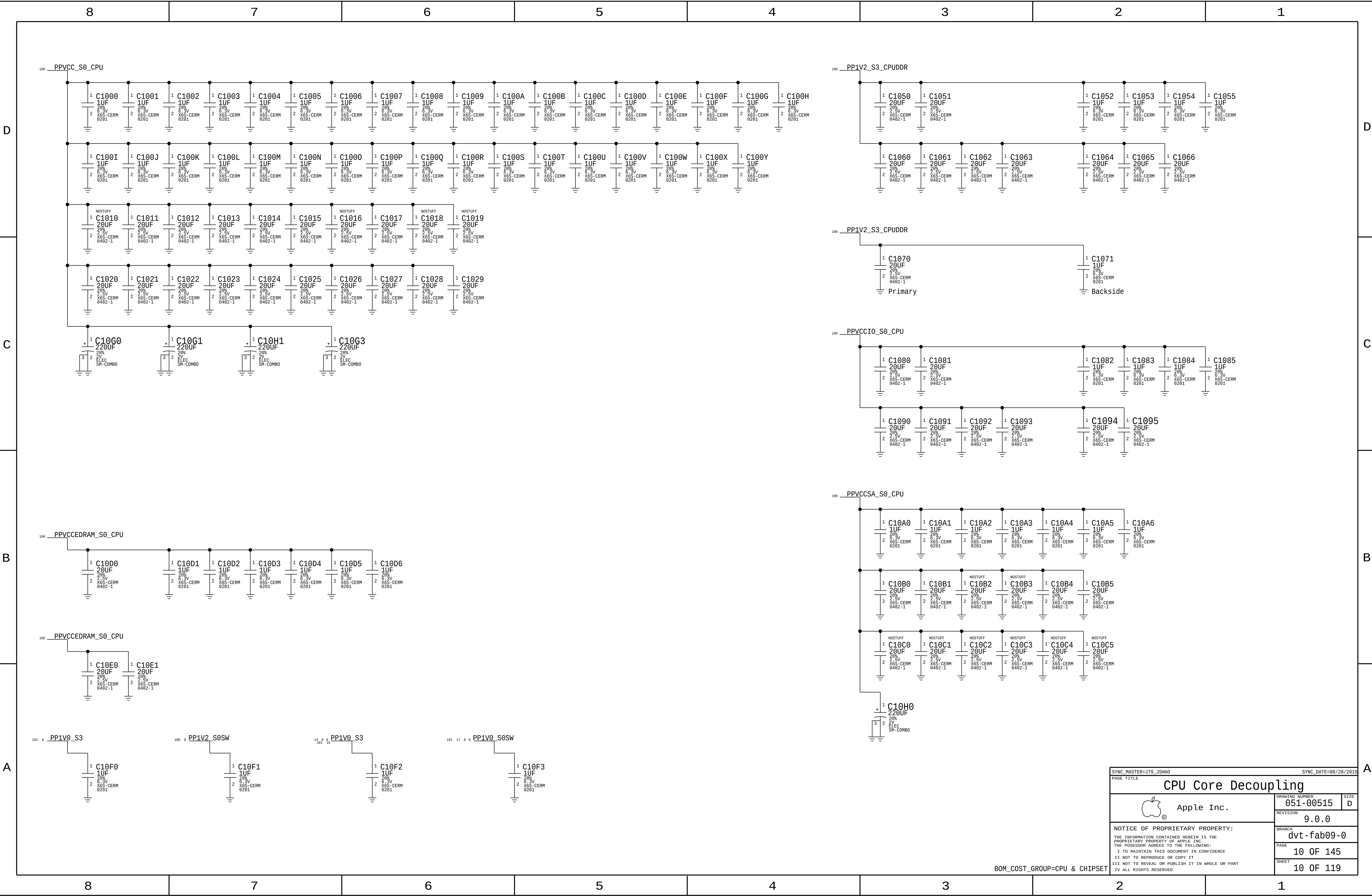


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		SIZE	D
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		BRANCH	dvt-fab09-0
		PAGE	7 OF 145
		SHEET	7 OF 119

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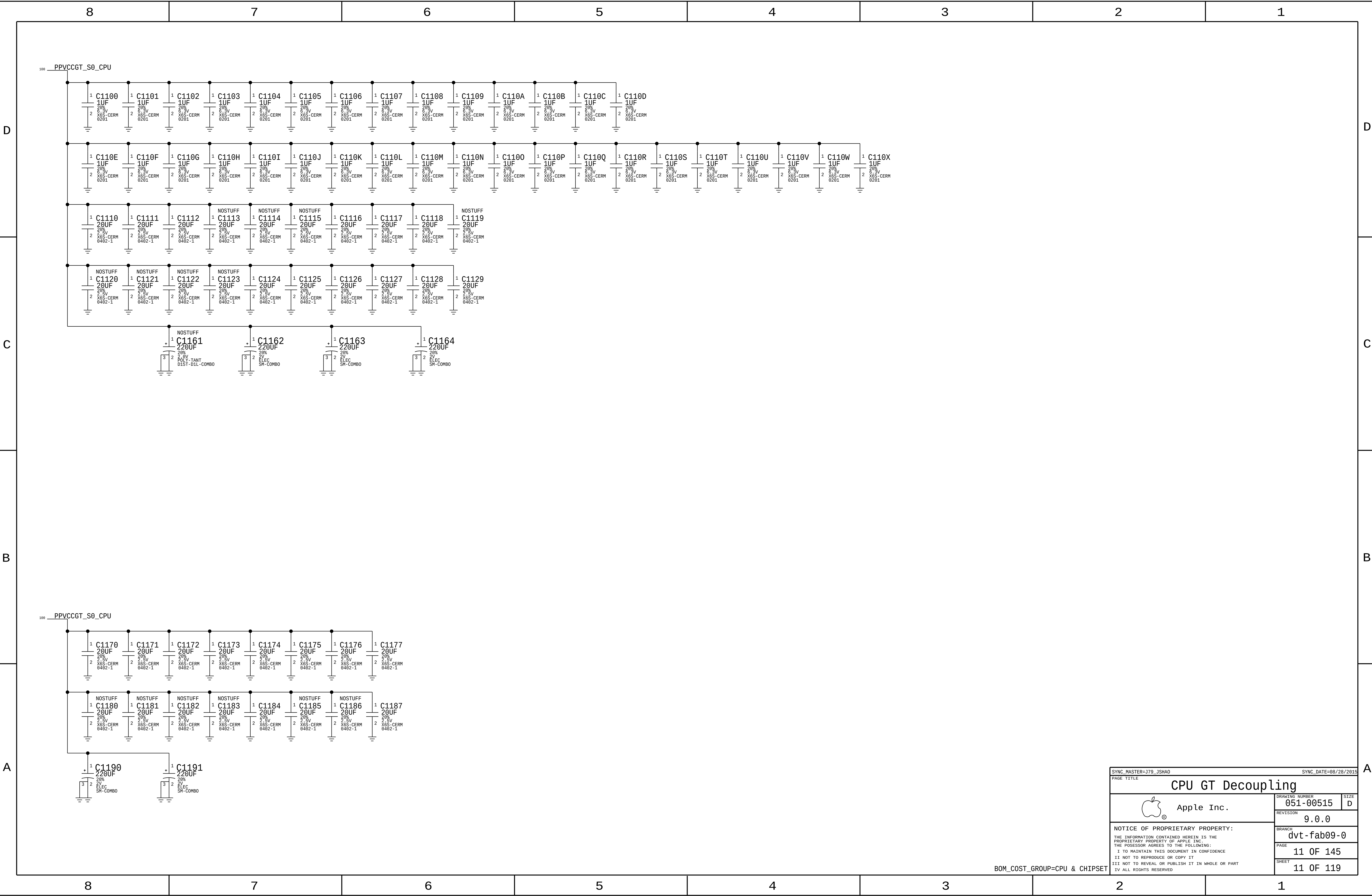






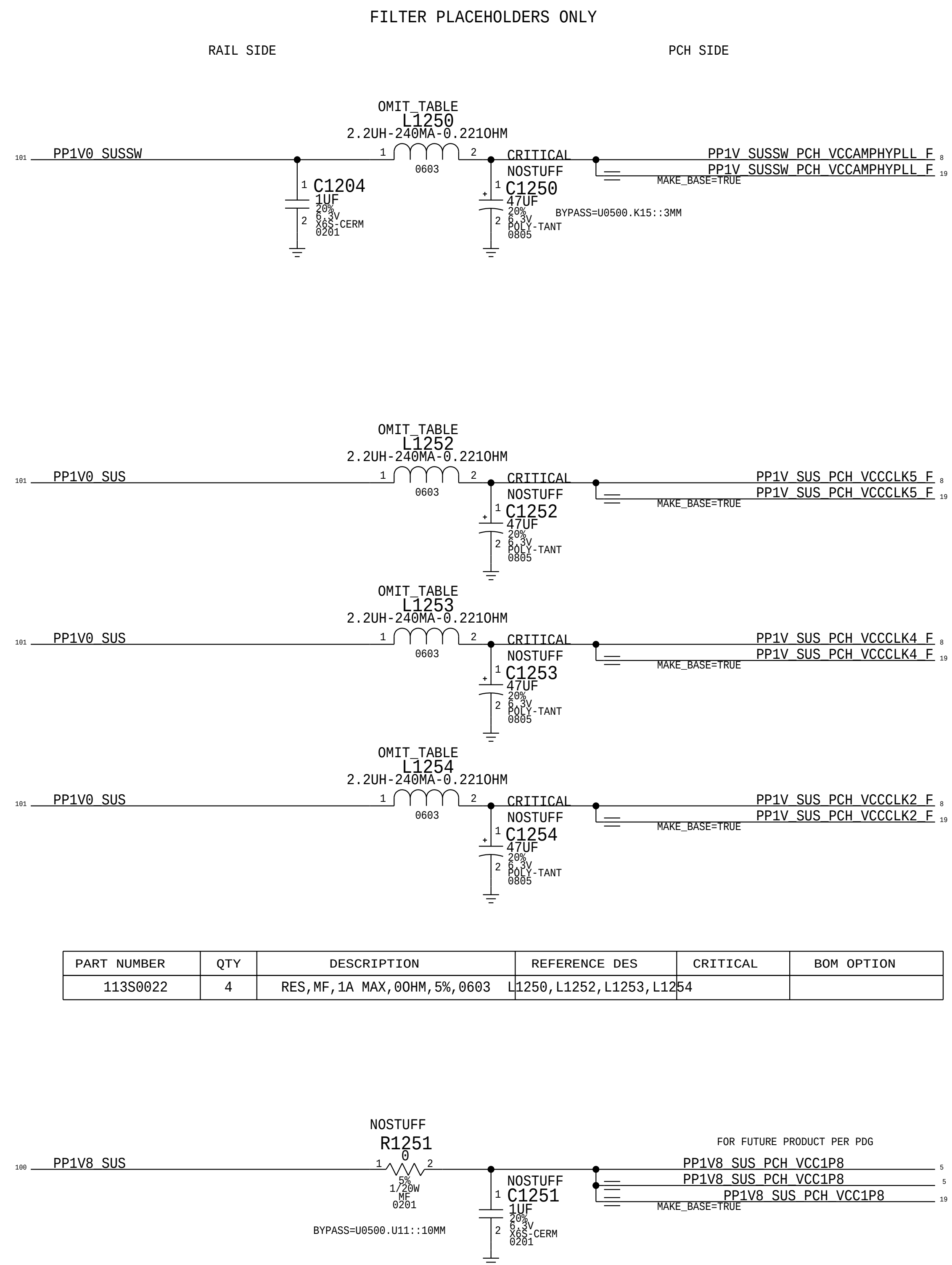
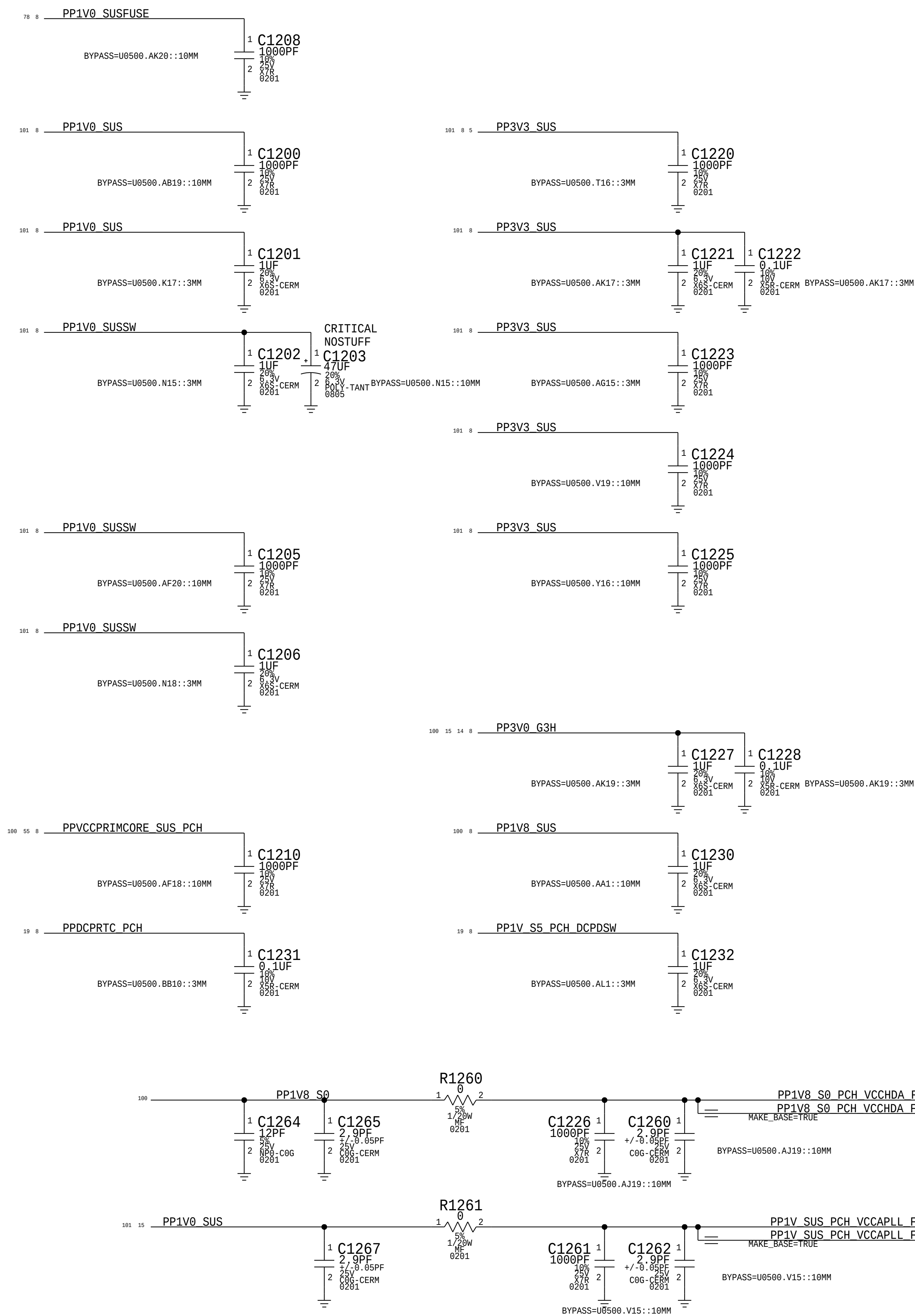
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PAGE TITLE			
CPU Core Decoupling			
 Apple Inc.		DRAWING NUMBER	051-00515
		REVISON	9.0.0
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		PAGE	10 OF 145
		SHEET	10 OF 119

BOM_COST_GROUP=CPU & CHIPSET

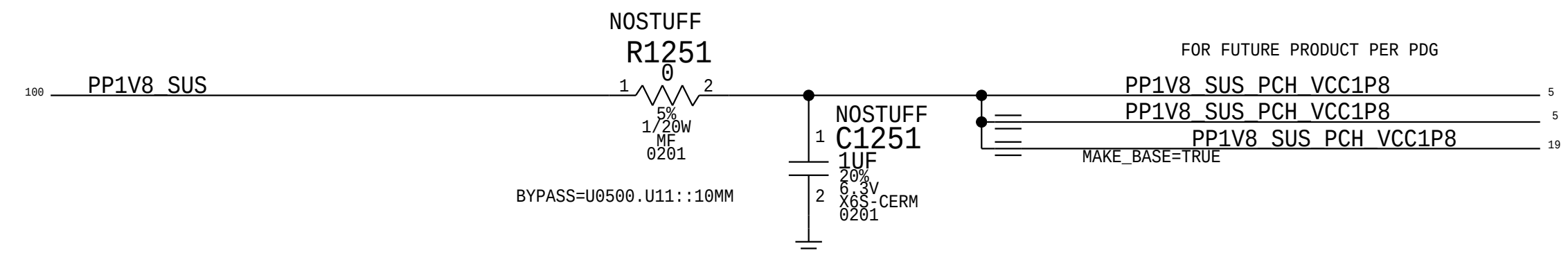


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	dvt-fab09-0		
	PAGE		11 OF 145
	SHEET		11 OF 119

BOM_COST_GROUP=CPU & CHIPSET

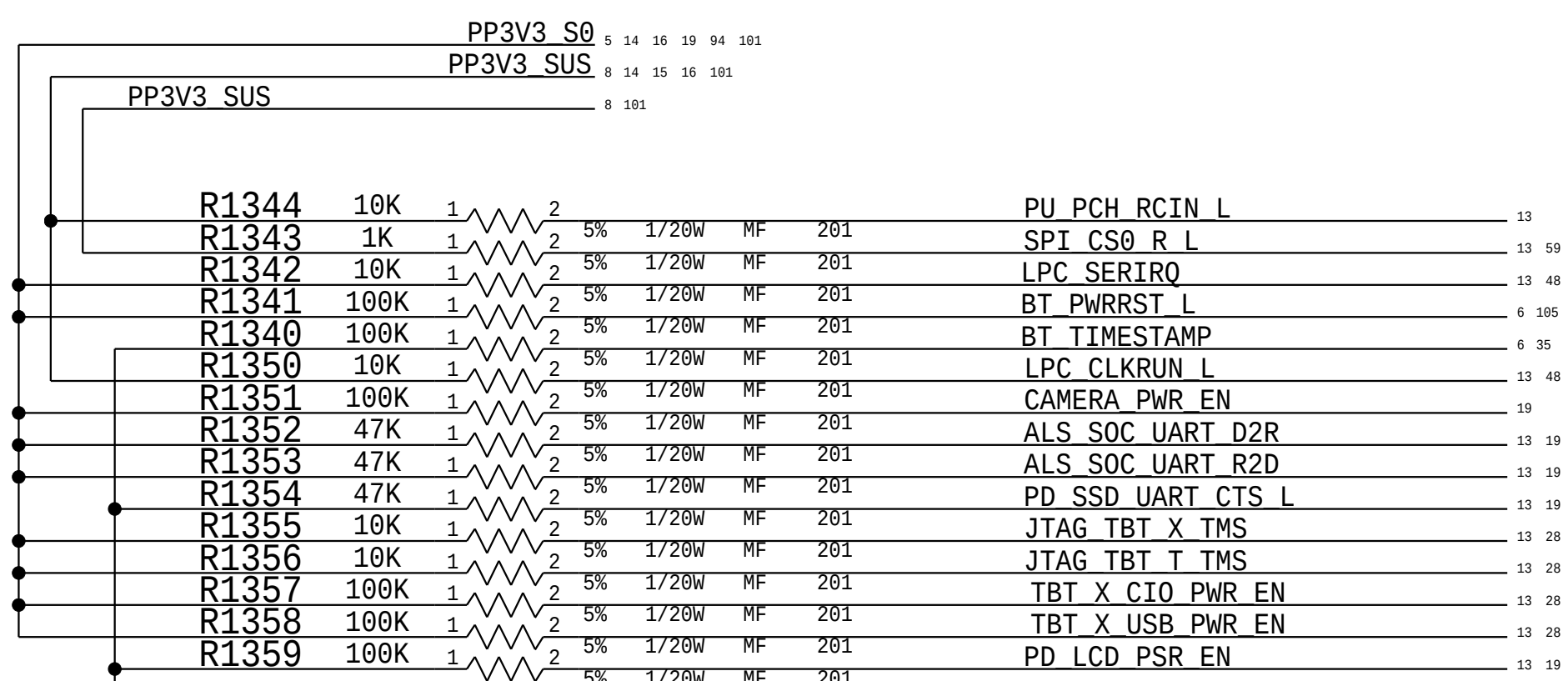
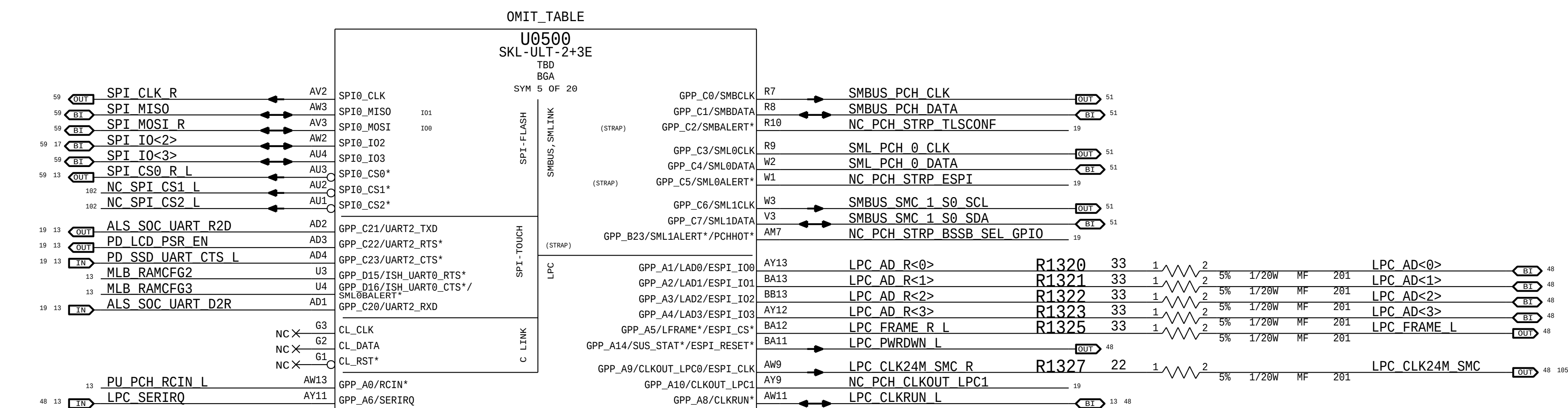
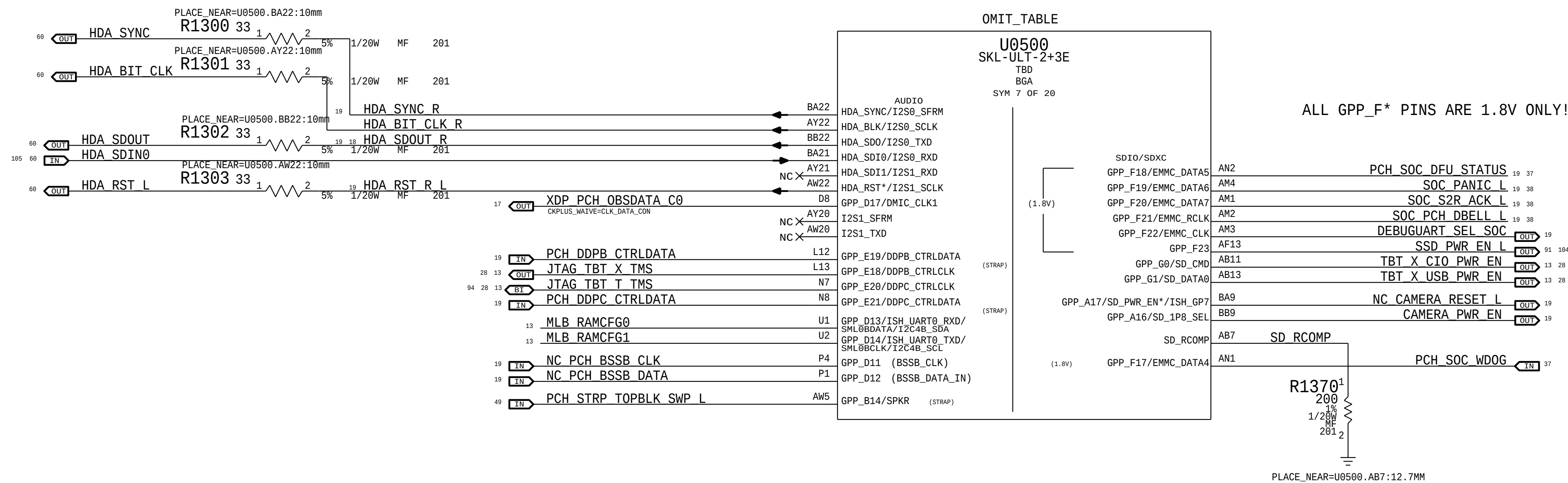


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
113S0022	4	RES,MF,1A MAX,00HM,5%,0603	L1250,L1252,L1253,L1254		



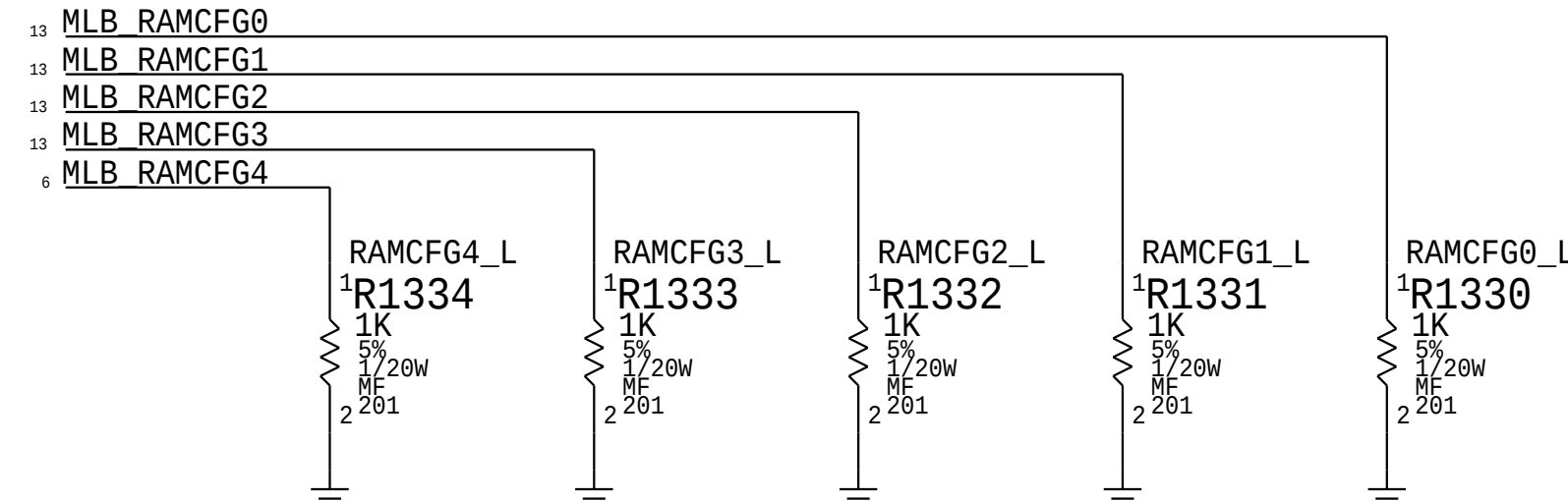
PAGE TITLE		PCH Decoupling	
 Apple Inc.	DRAWING NUMBER		SIZE
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		SHEET	12 OF 119

BOM_COST_GROUP=CPU & CHIPSET



MEMORY CONFIGURATION STRAPS.

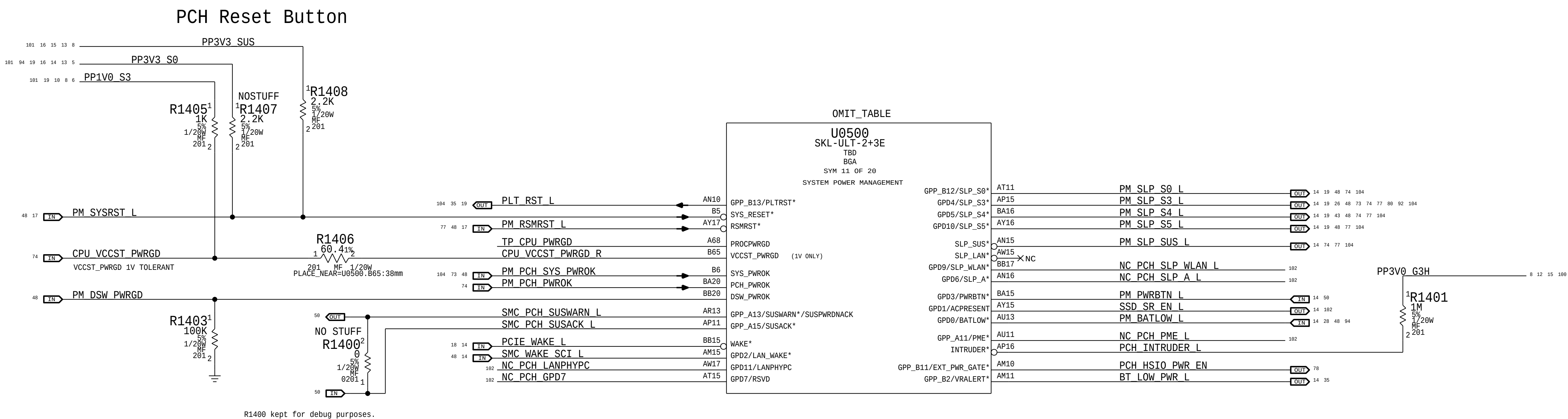
PCH INTERNAL PULL-UPS ARE TO 3.3V.



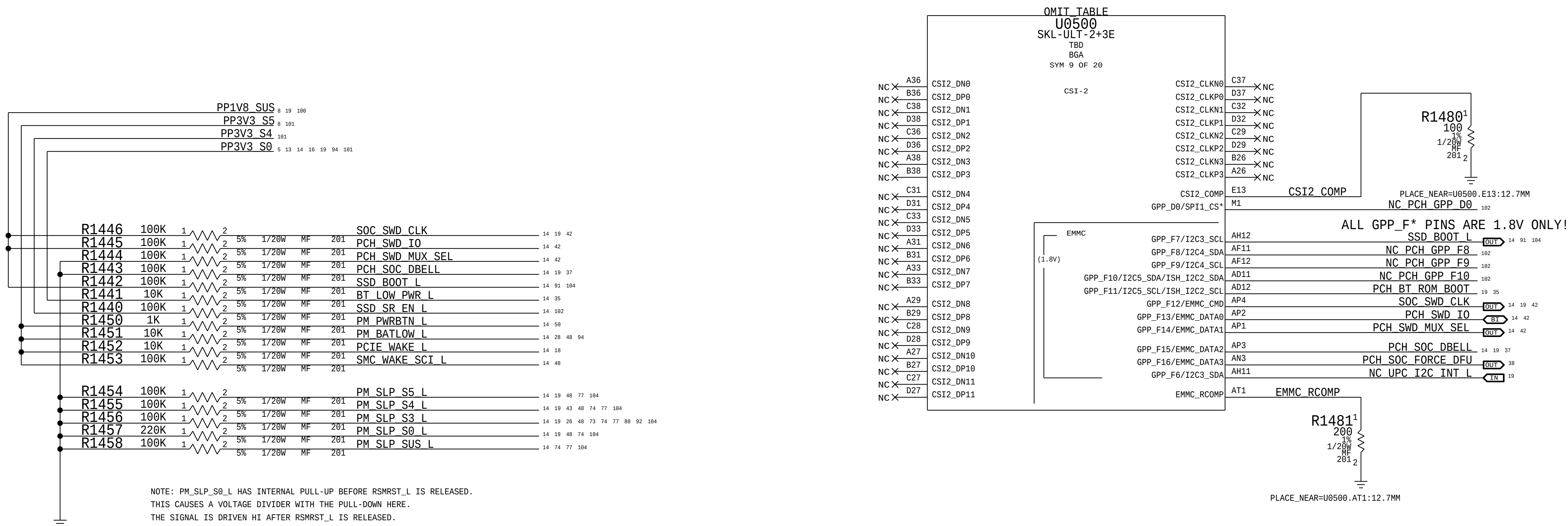
BOM GROUP	BOM OPTIONS
RAMCFG_SL0T	RAMCFG4_L, RAMCFG3_L, RAMCFG2_L, RAMCFG1_L, RAMCFG0_L

BOM_COST_GROUP=CPU & CHIPSET

DESIGN: X502/MLB			
LAST CHANGE: Tue Feb 2 13:18:21 2016			
PAGE TITLE			
PCH Audio/LPC/SPI/SMBus			
 Apple Inc.	DRAWING NUMBER 051-00515		SIZE D
	REVISION 9.0.0		
	BRANCH dvt-fab09-0		
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PAGE		13 OF 145	
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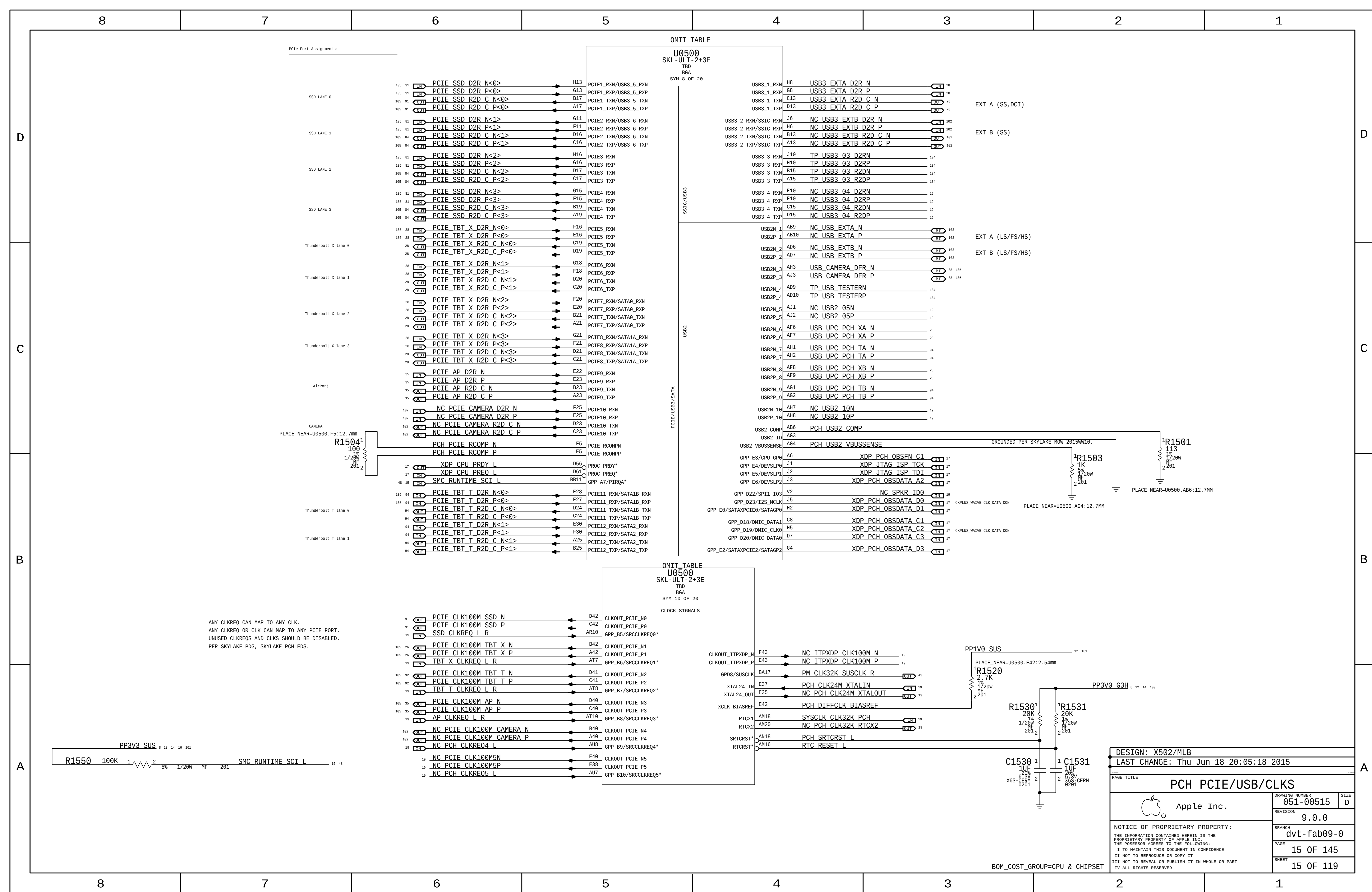
R1400 kept for debug purposes

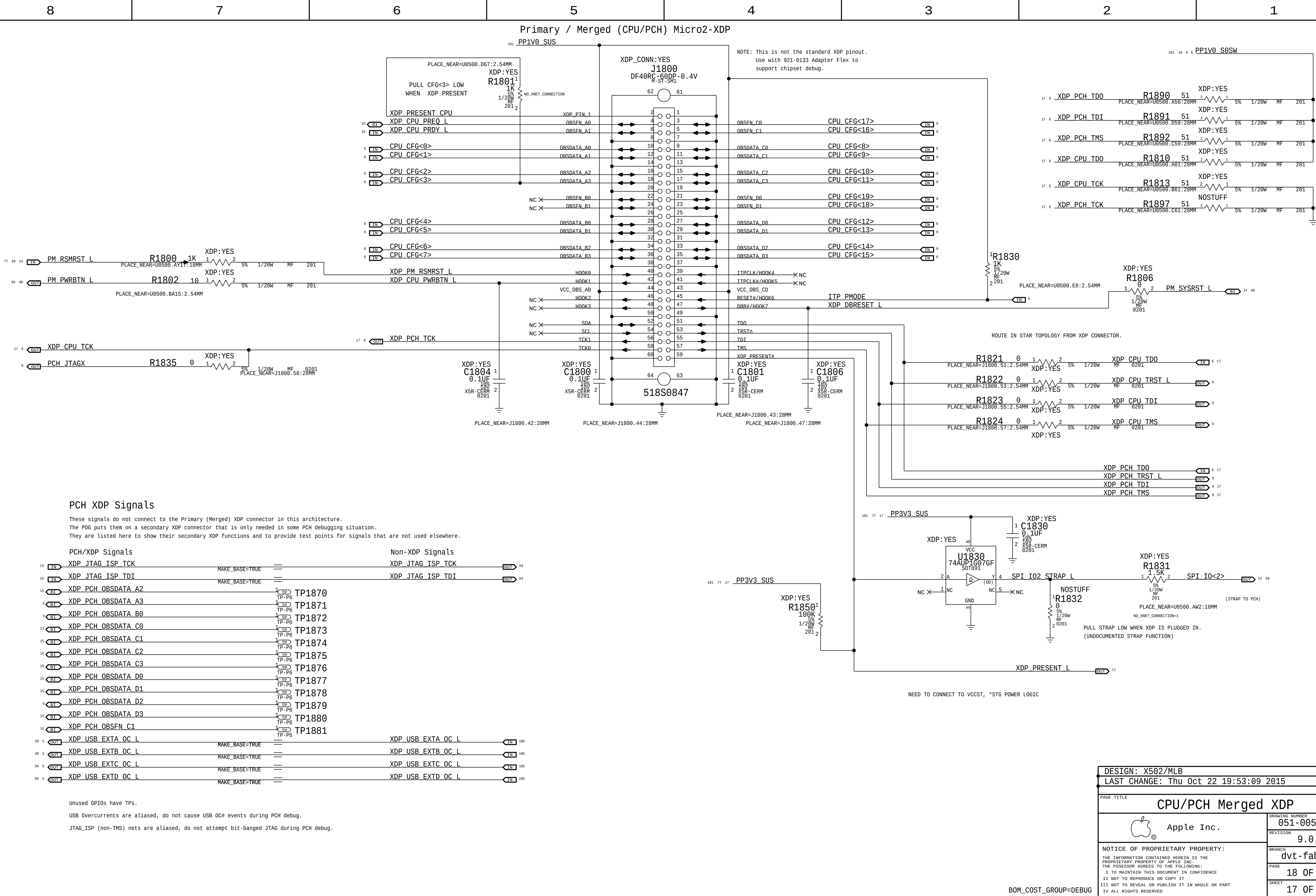


NOTE: PM_SLP_S0_L HAS INTERNAL PULL-UP BEFORE RSMRST_L IS RELEASED.
THIS CAUSES A VOLTAGE DIVIDER WITH THE PULL-DOWN HERE.
THE SIGNAL IS DRIVEN HI AFTER RSMRST_L IS RELEASED.

DESIGN: X502/MLB			
LAST CHANGE: Tue May 3 17:45:28 2016			
PAGE TITLE			
PCH Power Management			
 Apple Inc.		DRAWING NUMBER	051-00515
		REVISION	9.0.0
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		PAGE	14 OF 145
		SHEET	14 OF 119

BOM_COST_GROUP=CPU & CHIPSET





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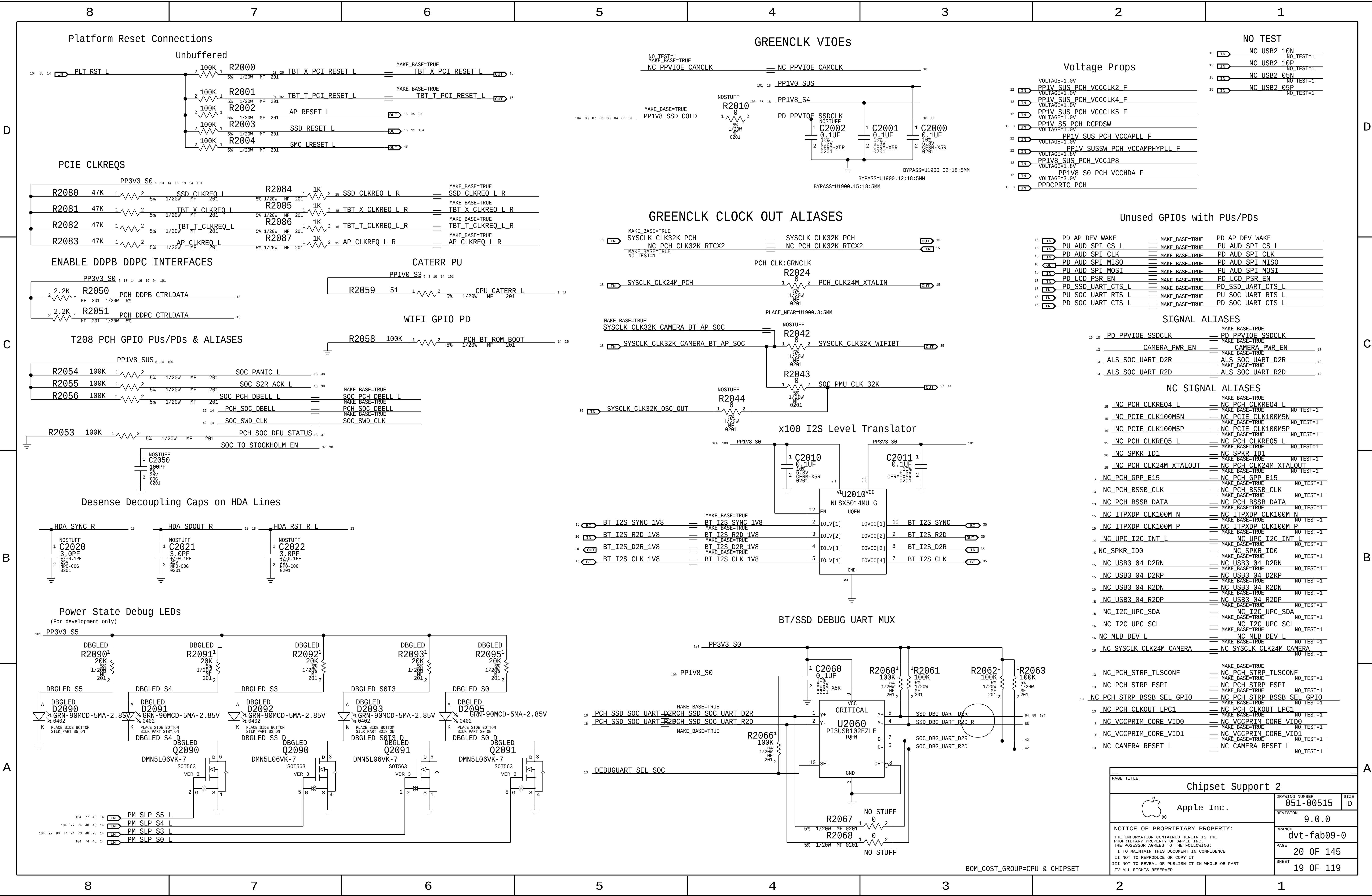


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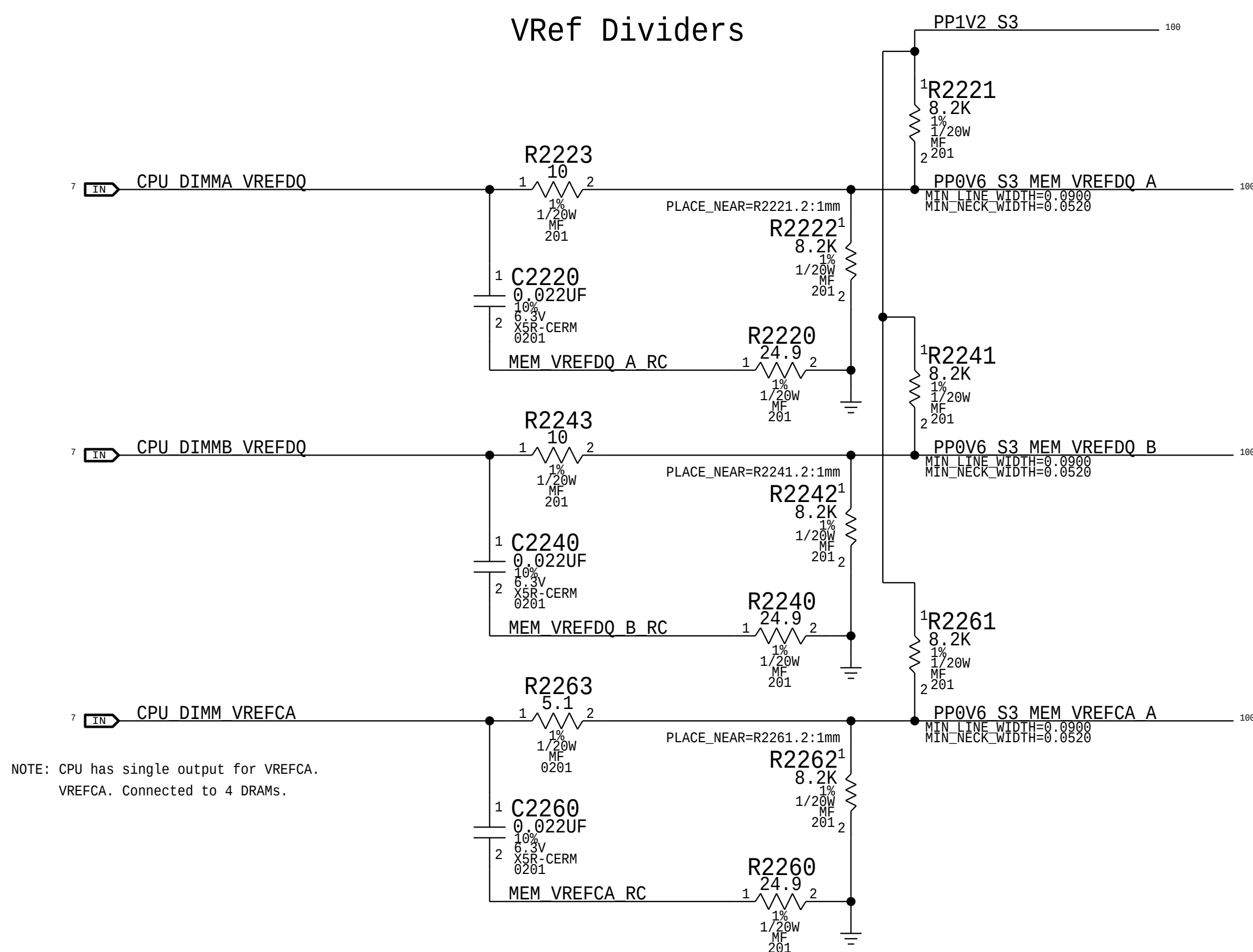


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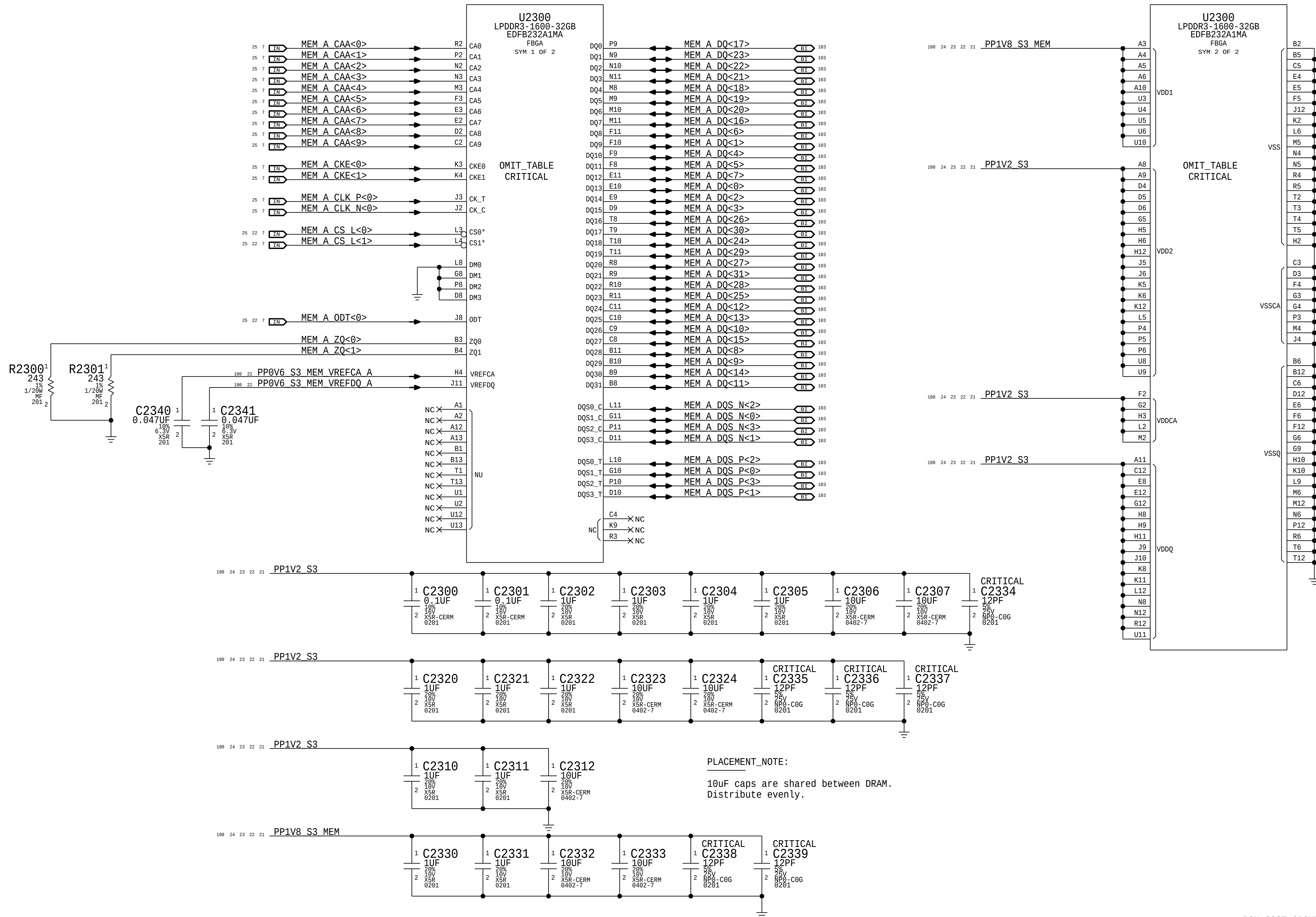
CPU-Based Margining



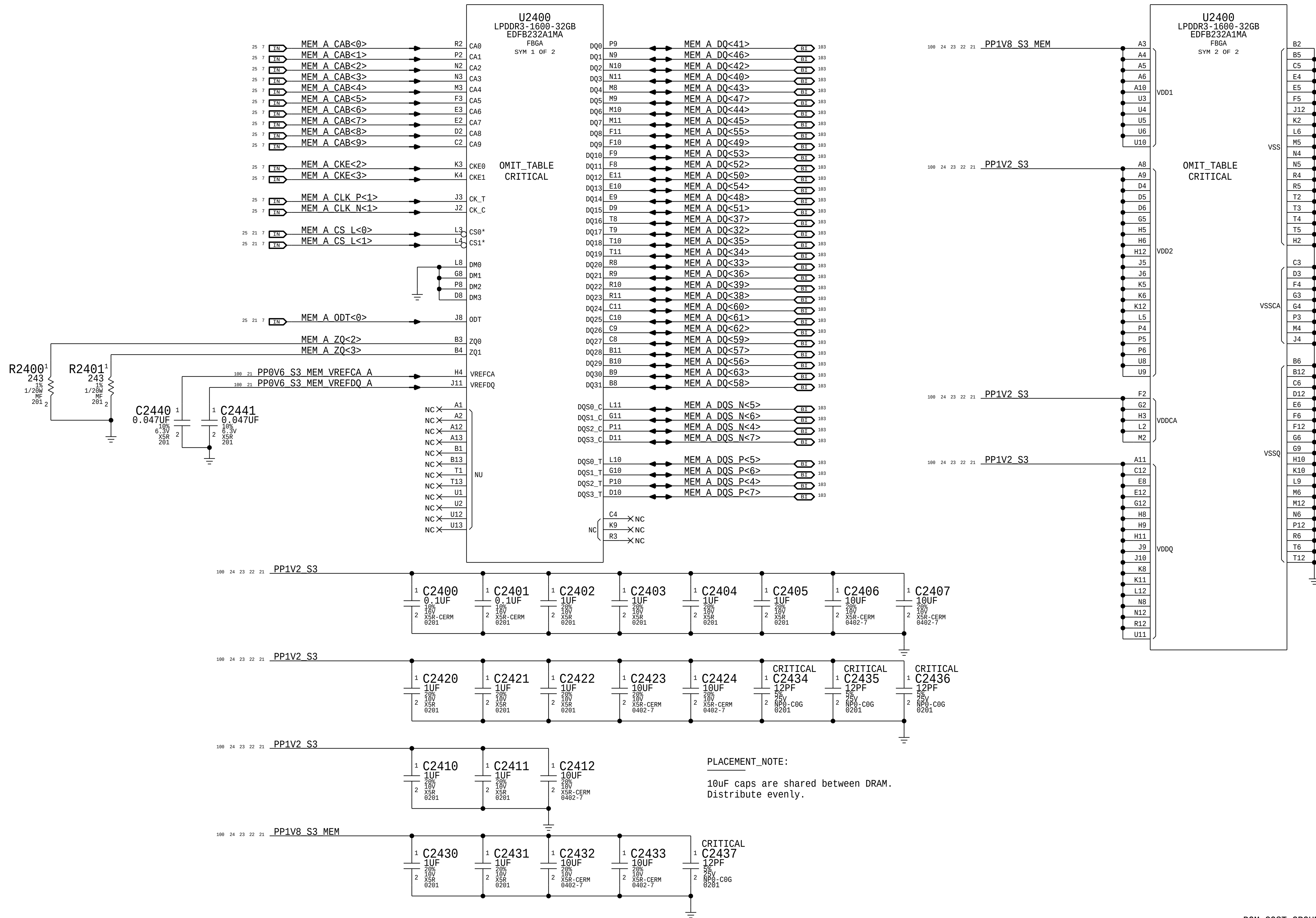
NOTE: CPU has single output for VREFCA
VREFCA. Connected to 4 DRAMs.

SYNC_MASTER=352_MLB		SYNC_DATE=05/12/2015	
PAGE TITLE			
LPDDR3 VREF Margining			
 Apple Inc.		DRAWING NUMBER	
		051-00515	
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LPDDR3 CHANNEL A (0-31)



LPDDR3 CHANNEL A (32-63)



SYNC_MASTER=J52_MLB		SYNC_DATE=05/12/2015	
PAGE TITLE			
LPDDR3 DRAM Channel A (32-63)			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
	REVISION		
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IV ALL RIGHTS RESERVED		22 OF 119	

BOM_COST_GROUP=DRAM

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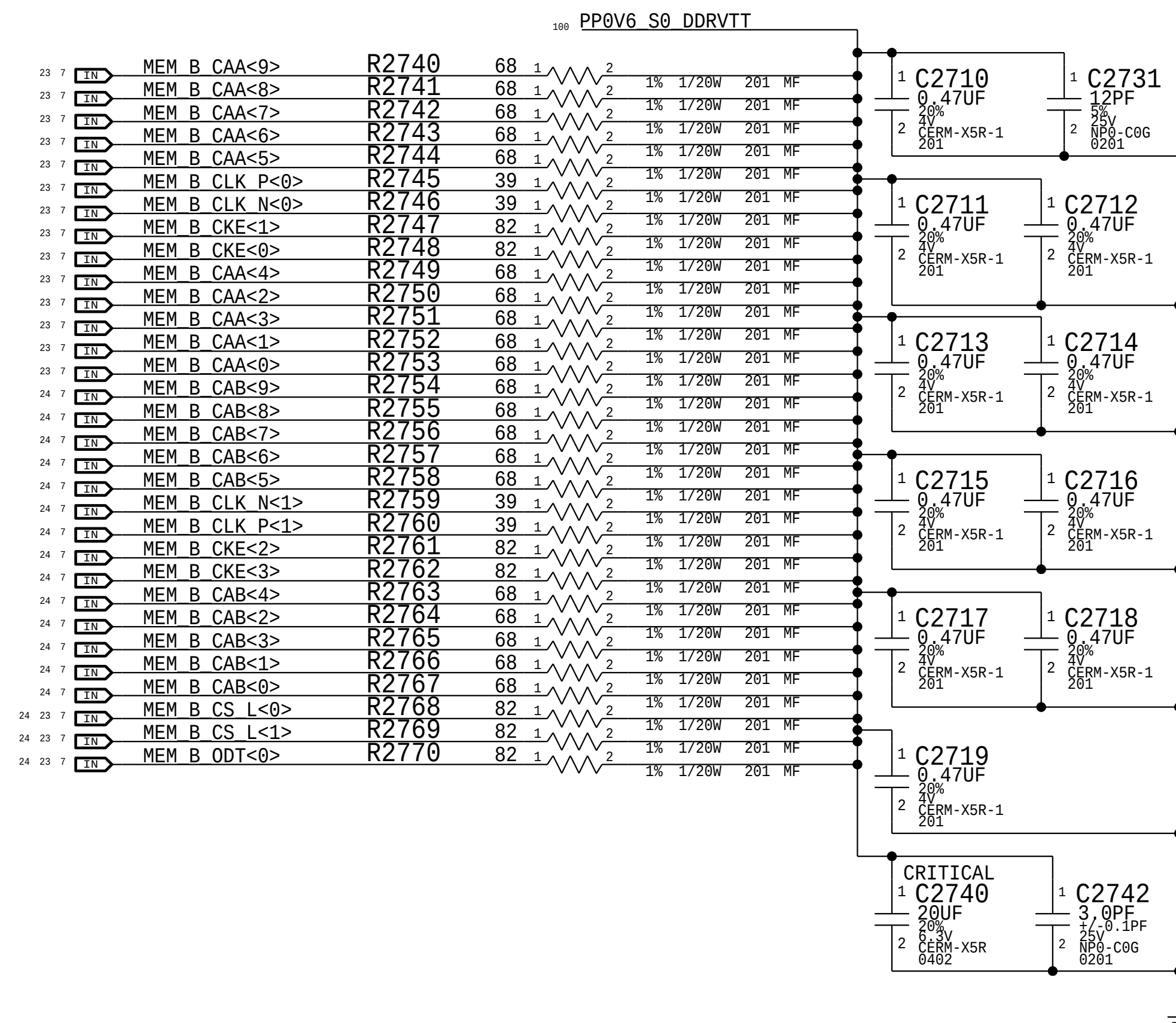
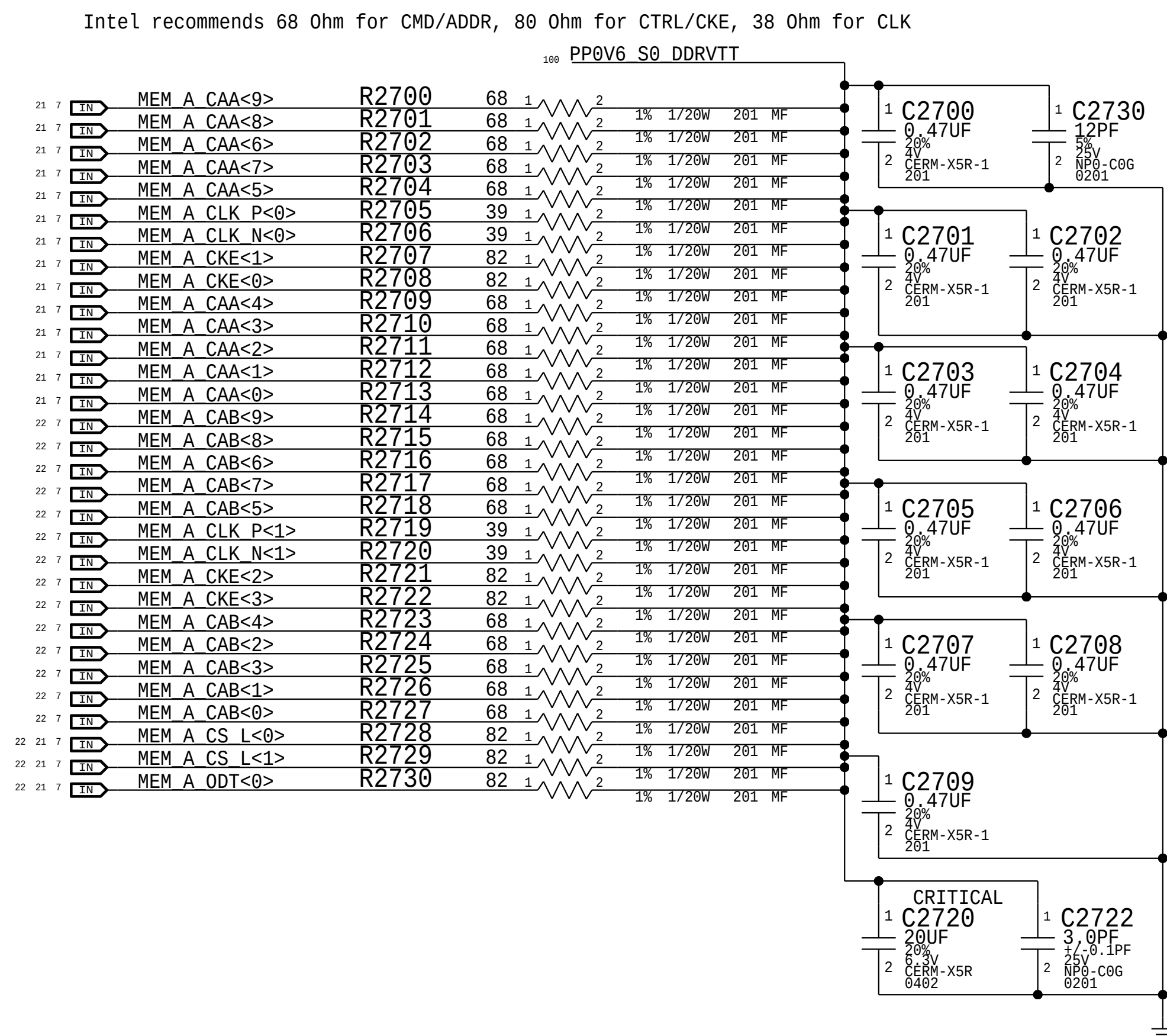
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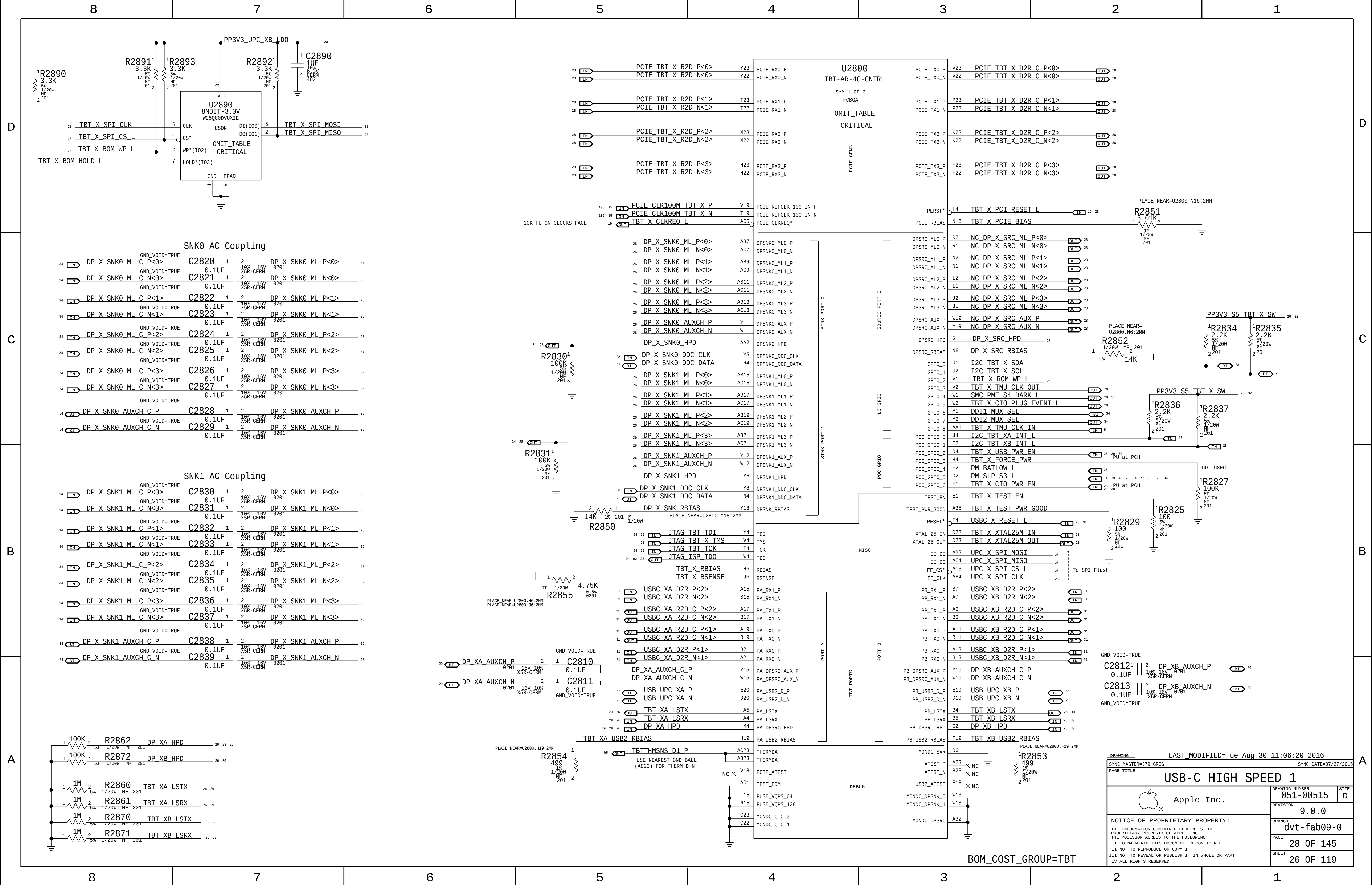


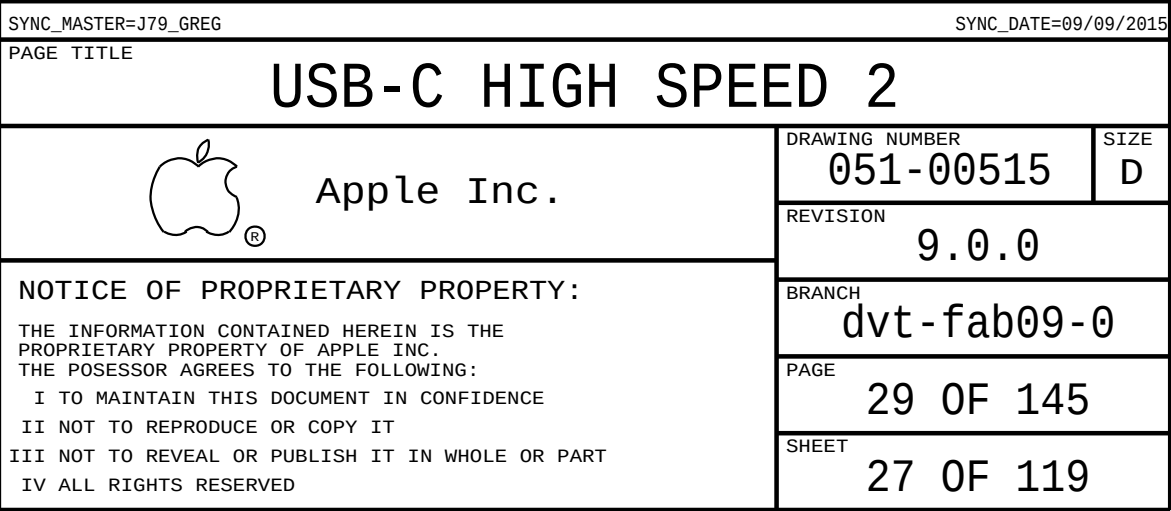
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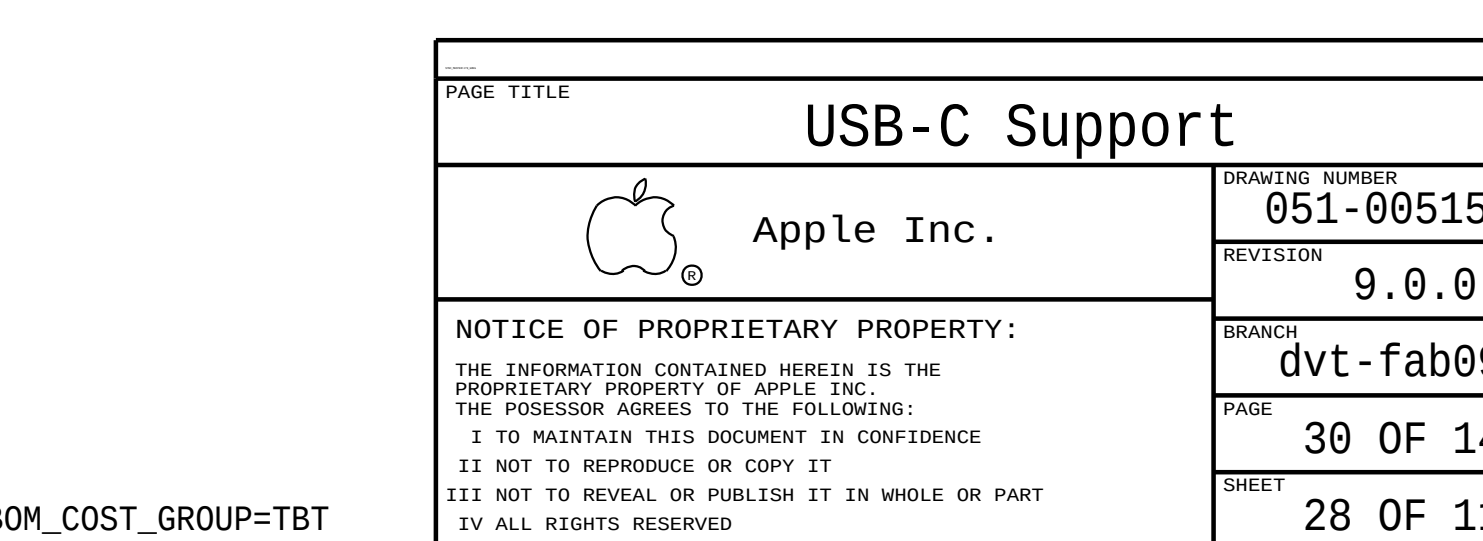
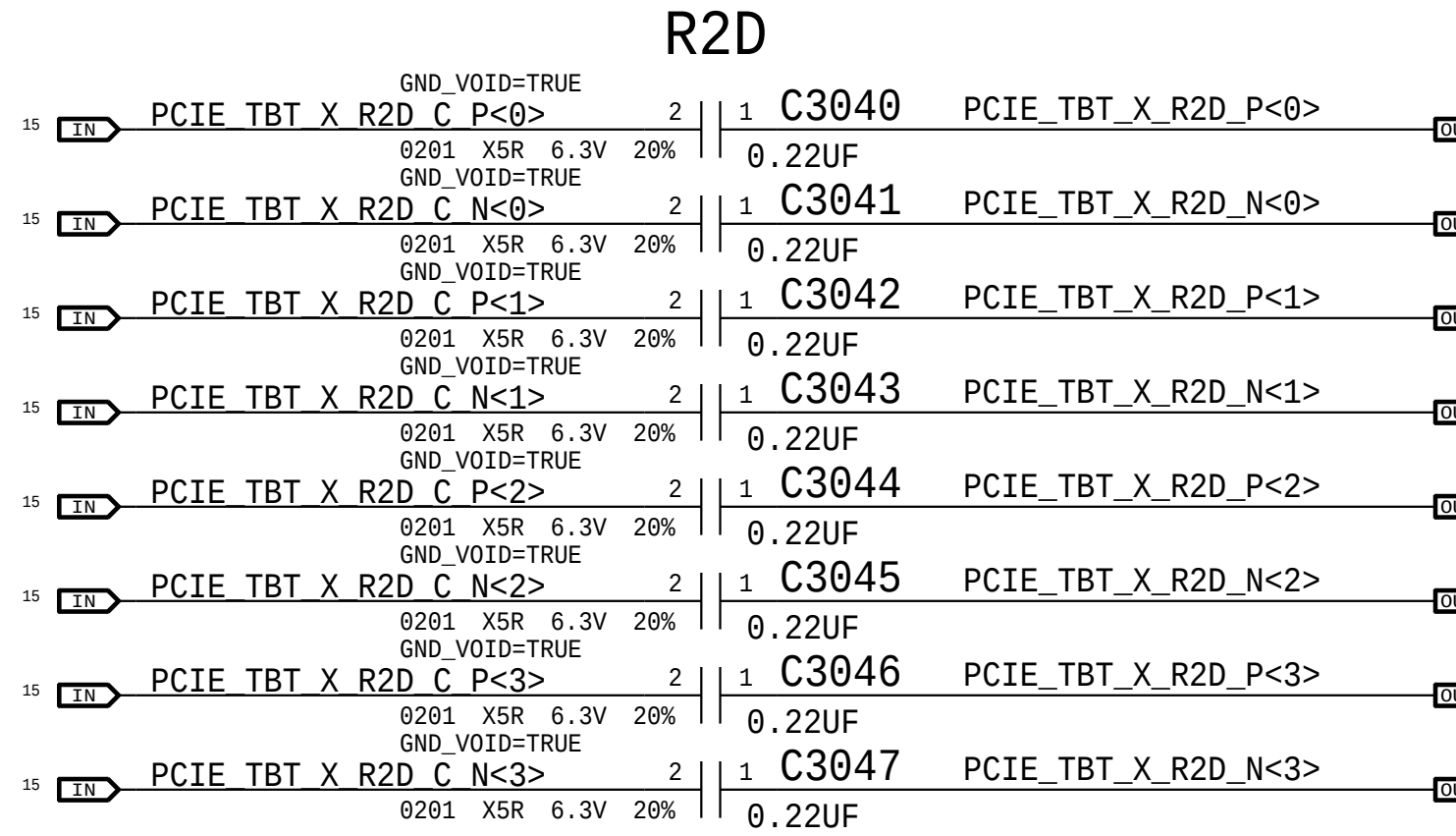
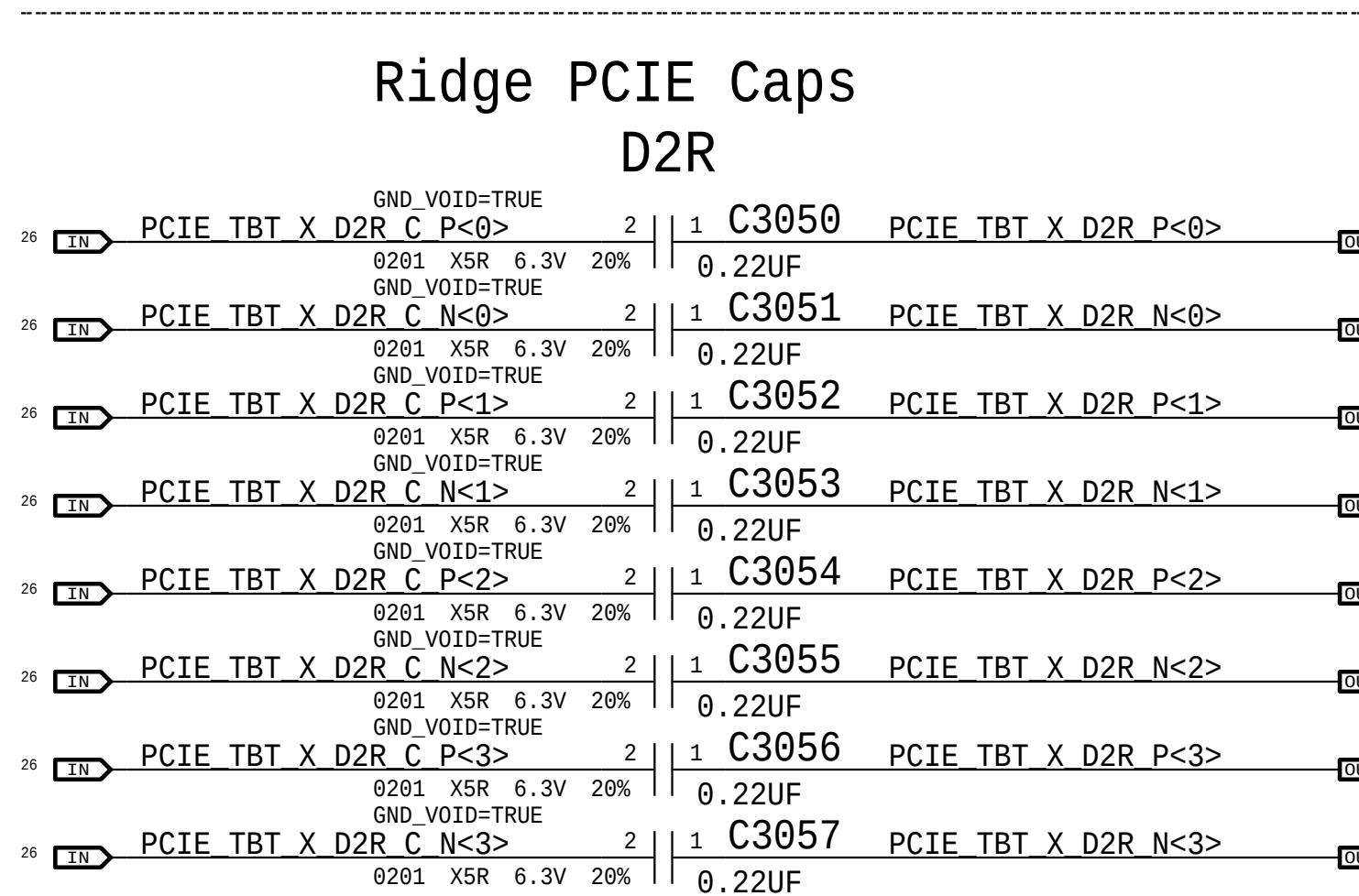
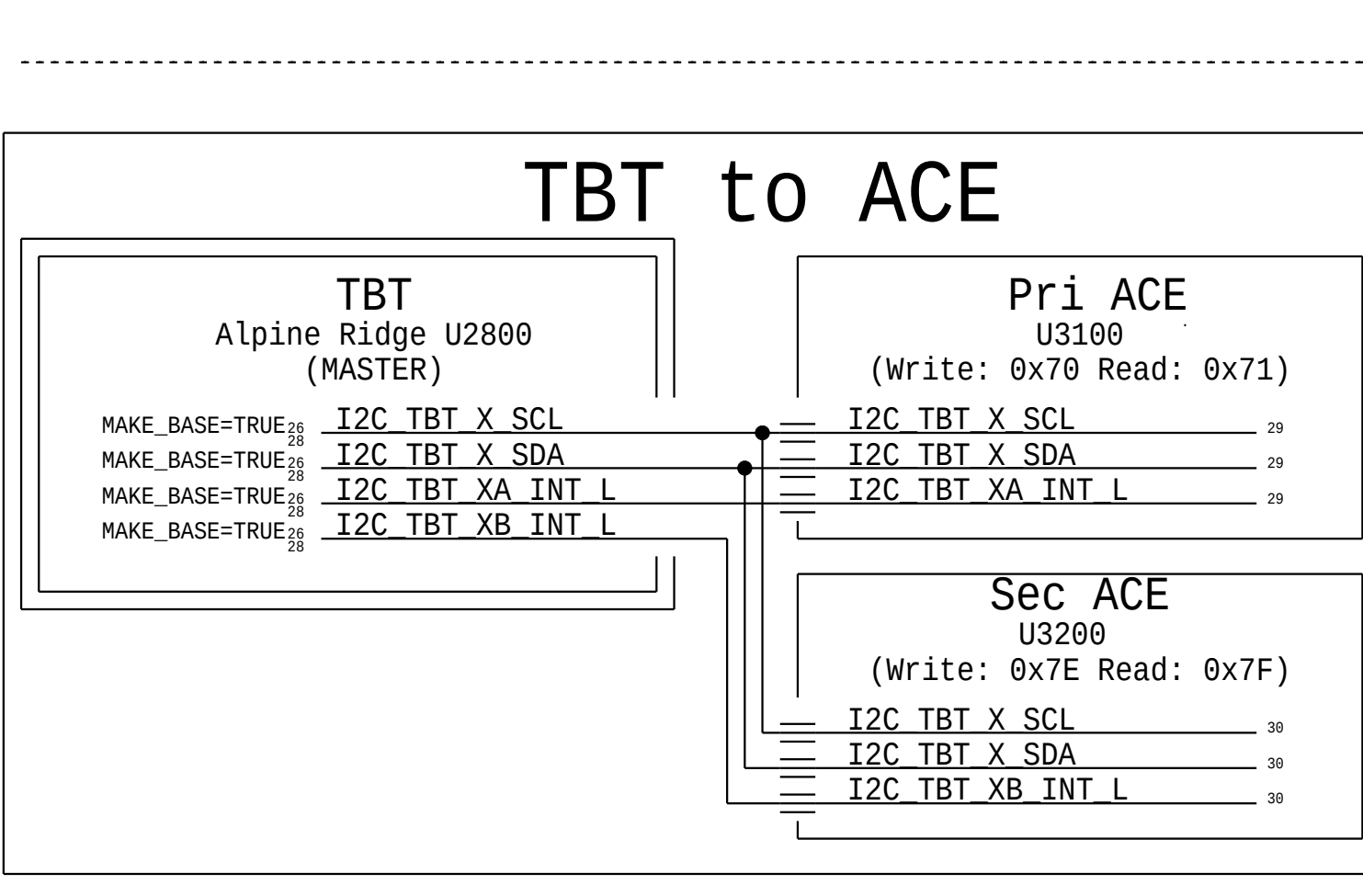
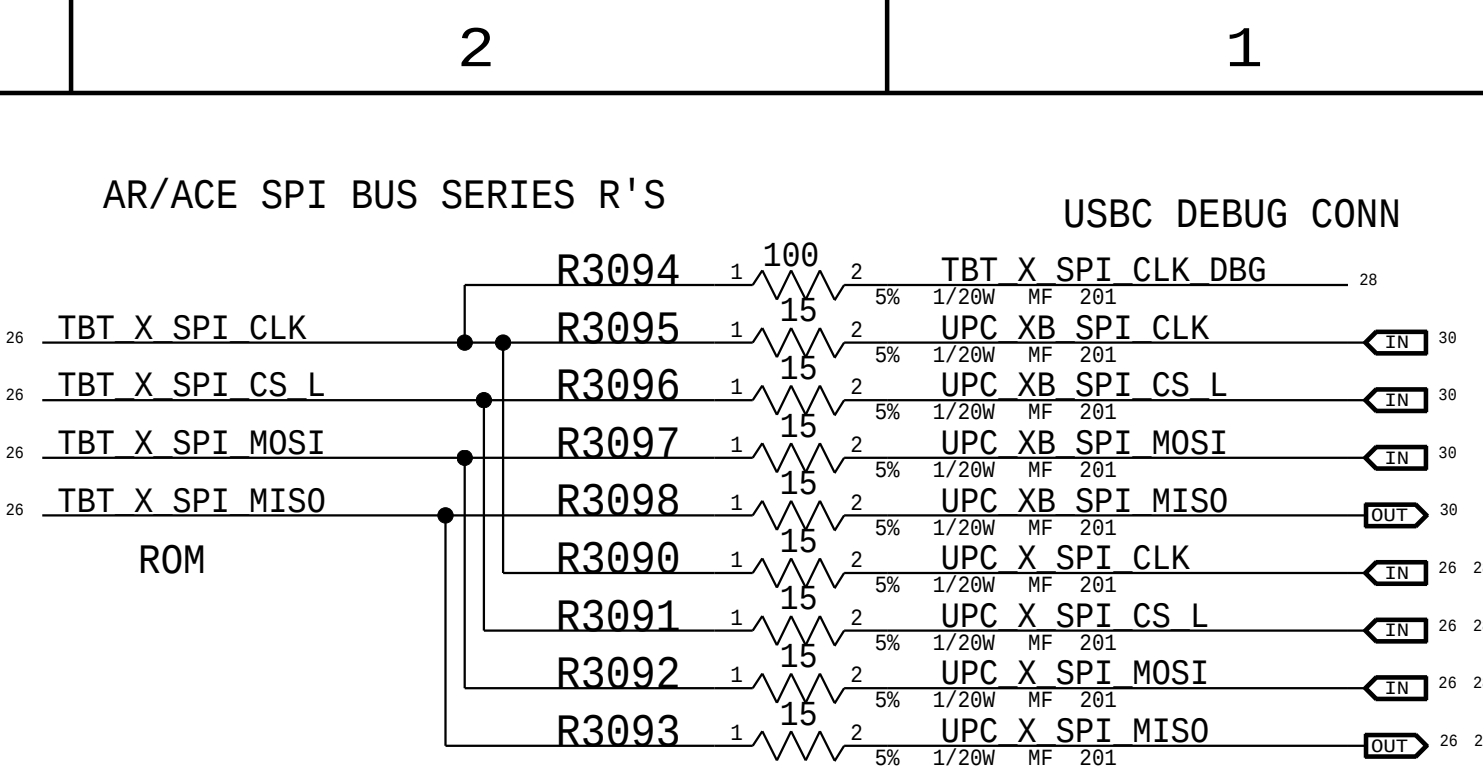
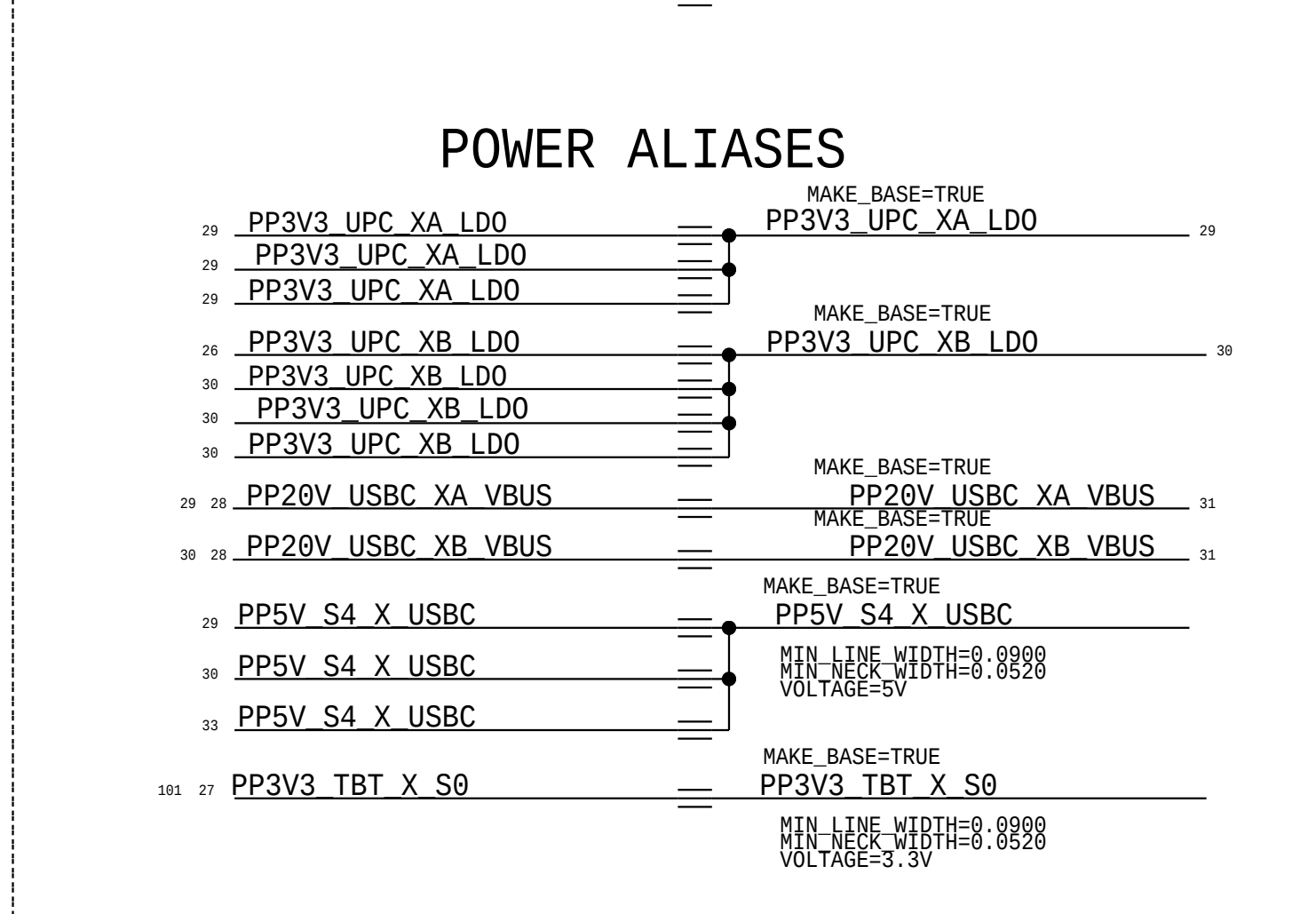
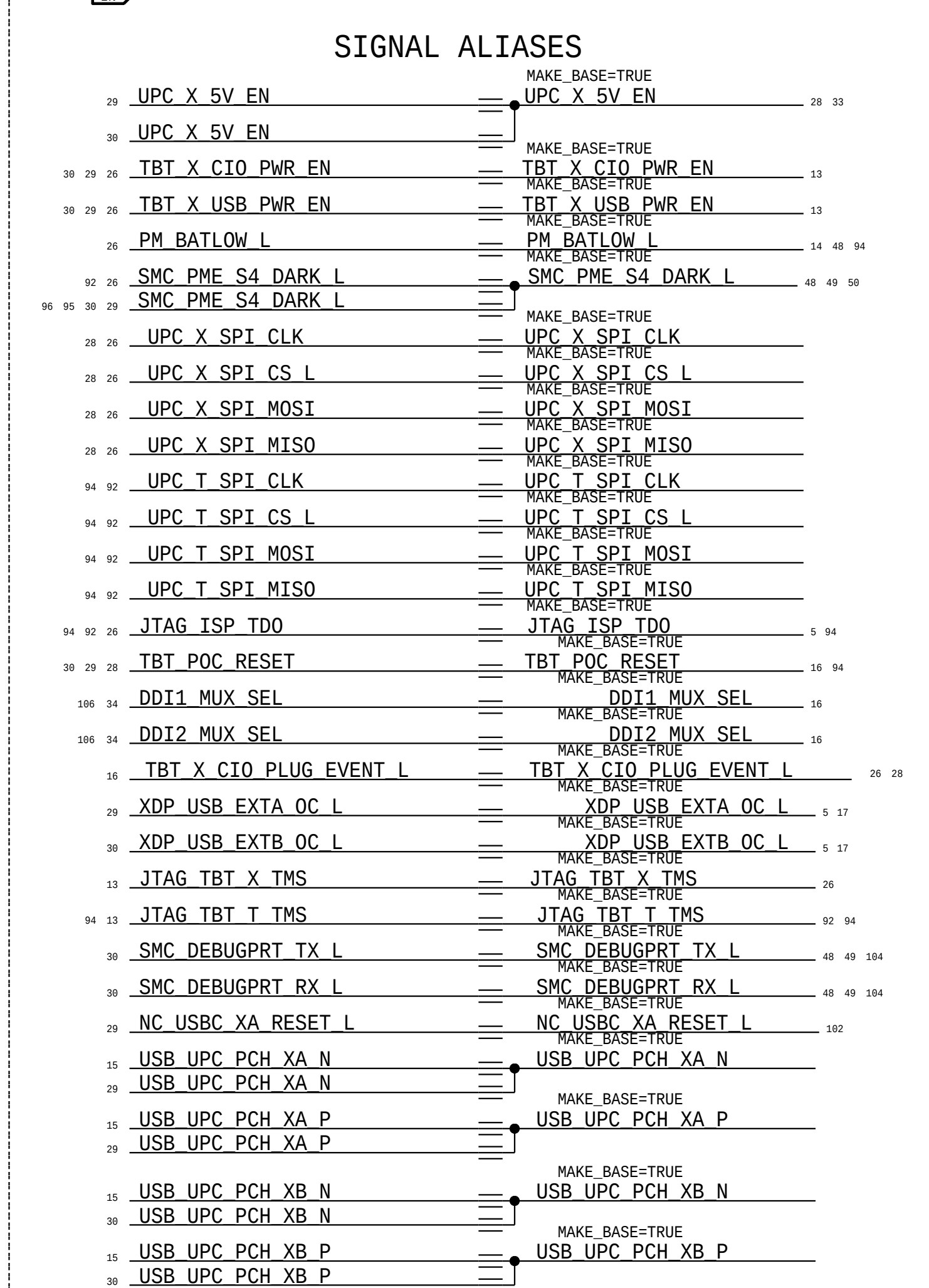
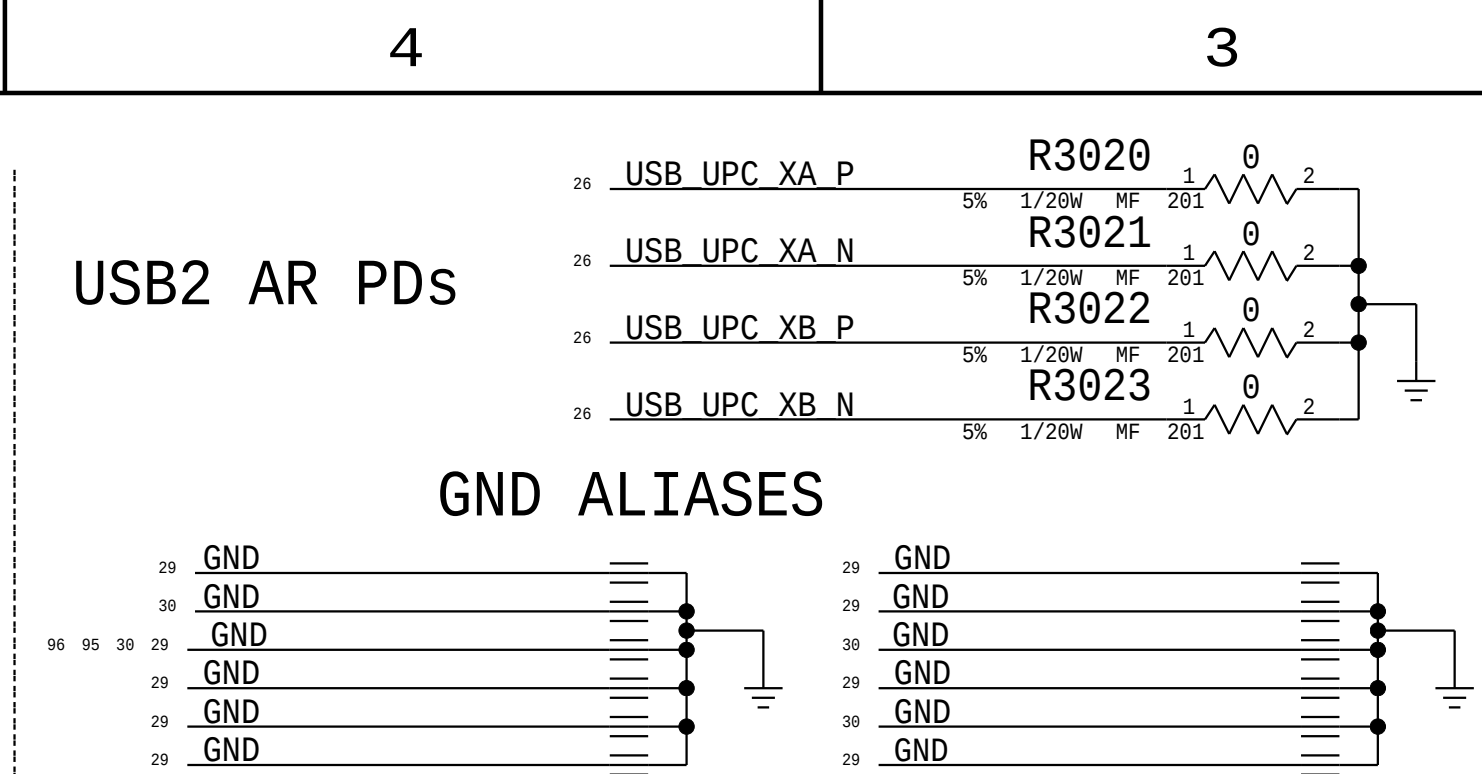
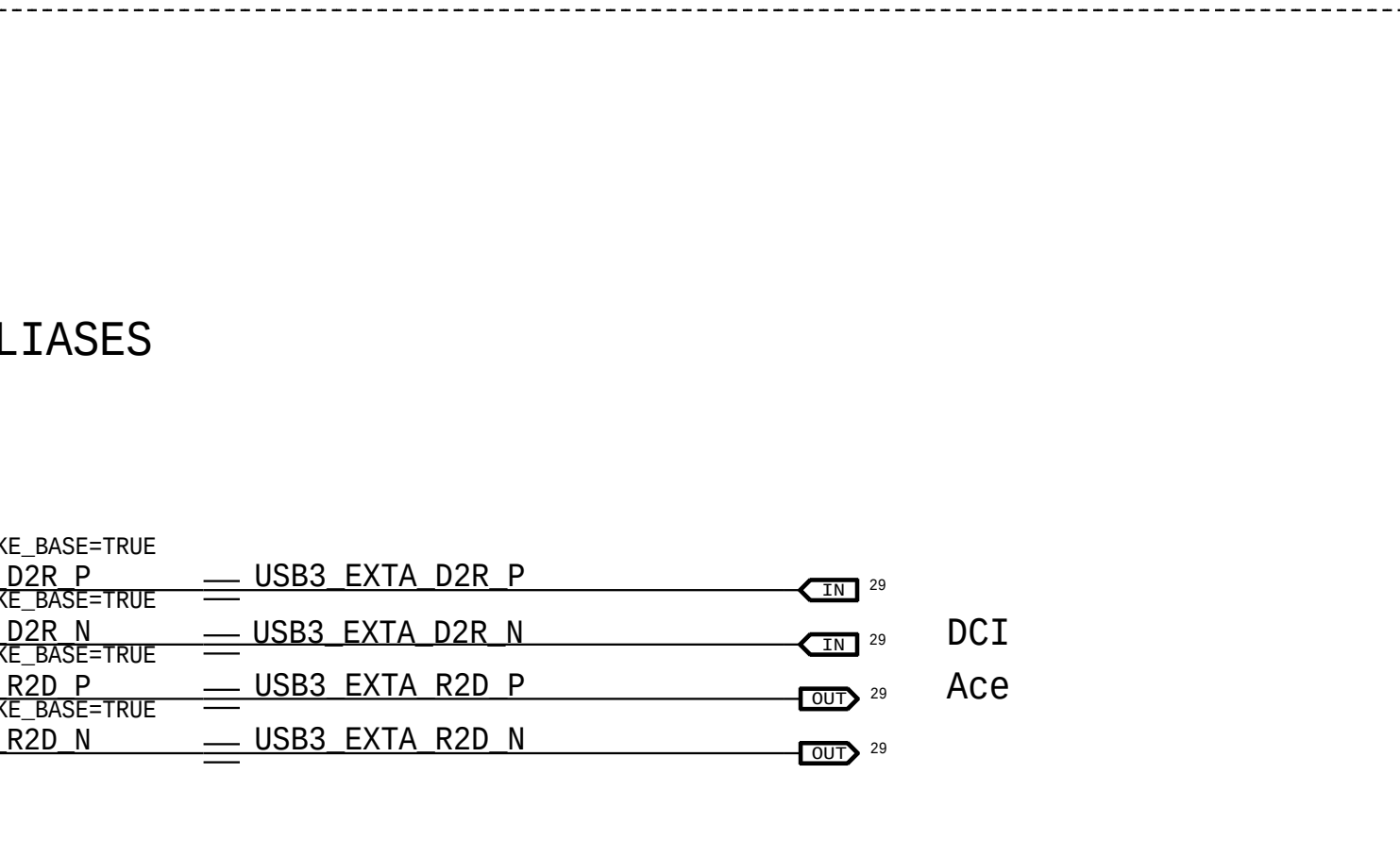
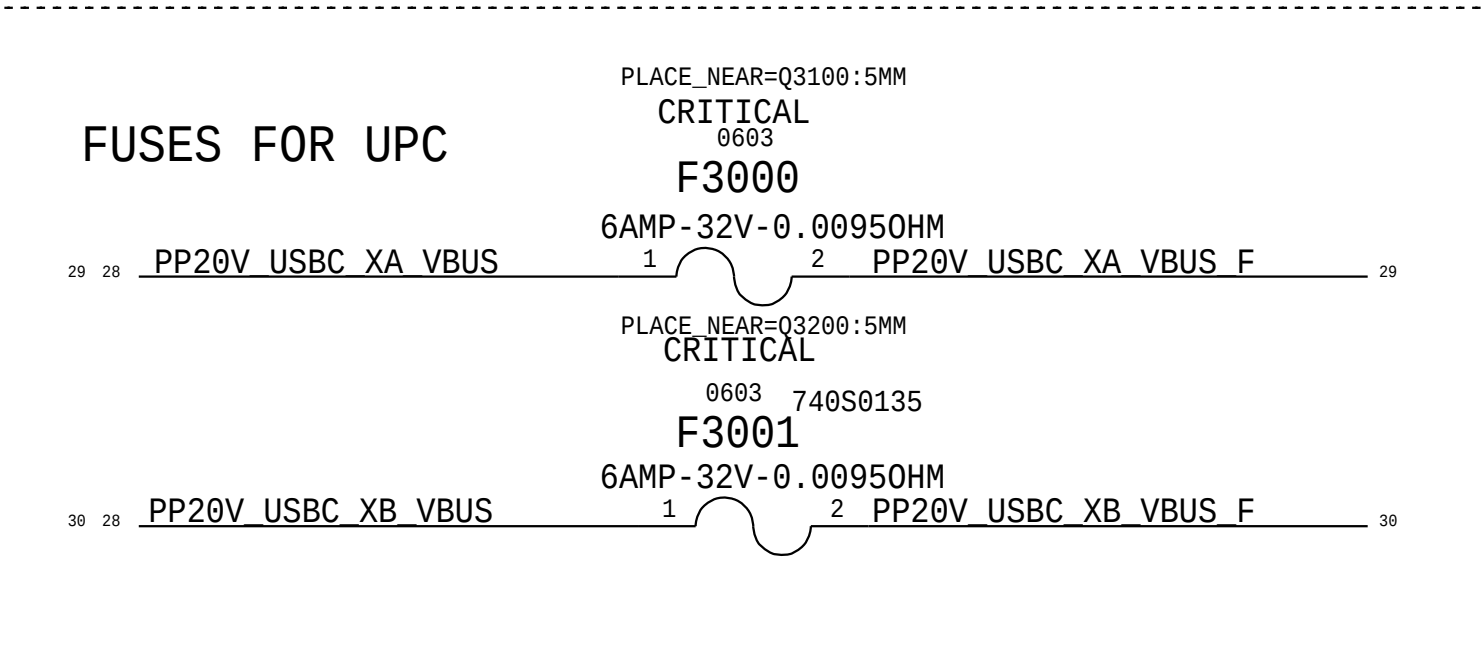
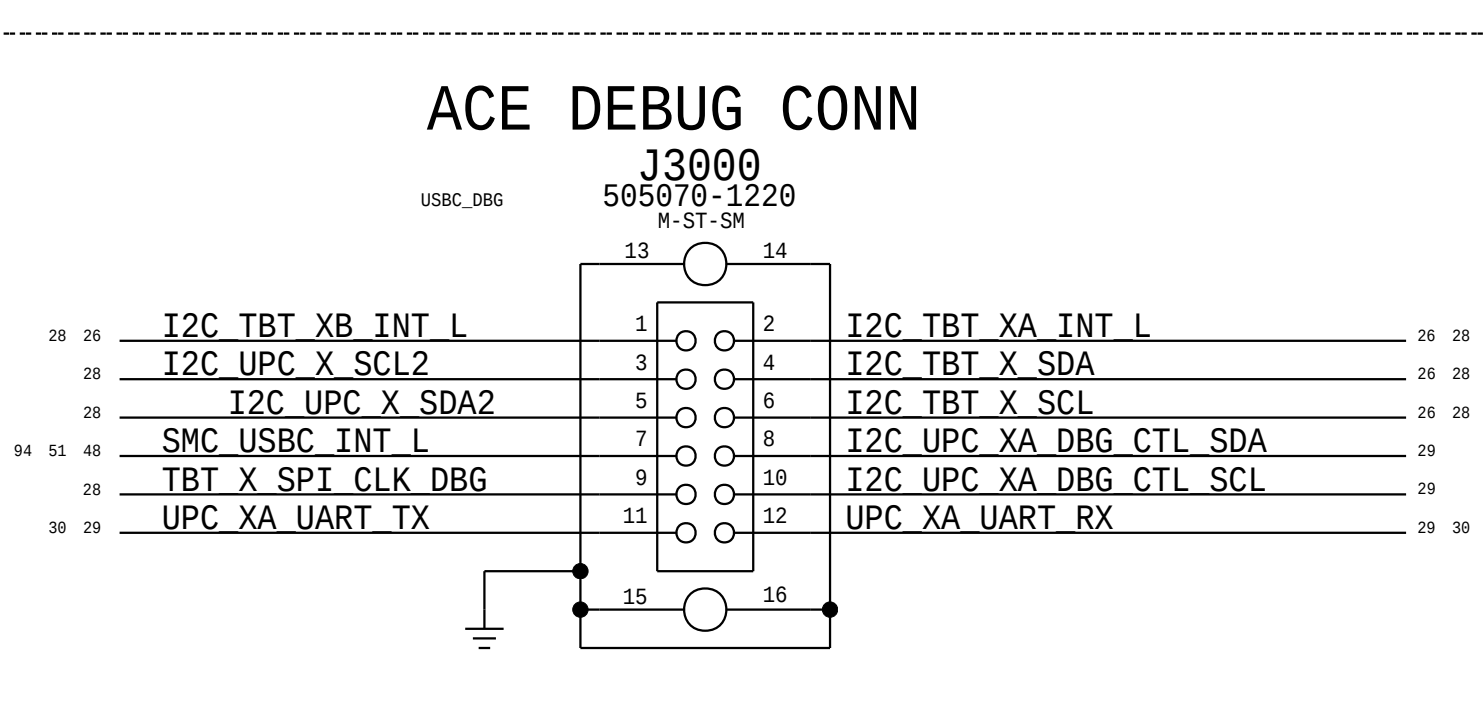
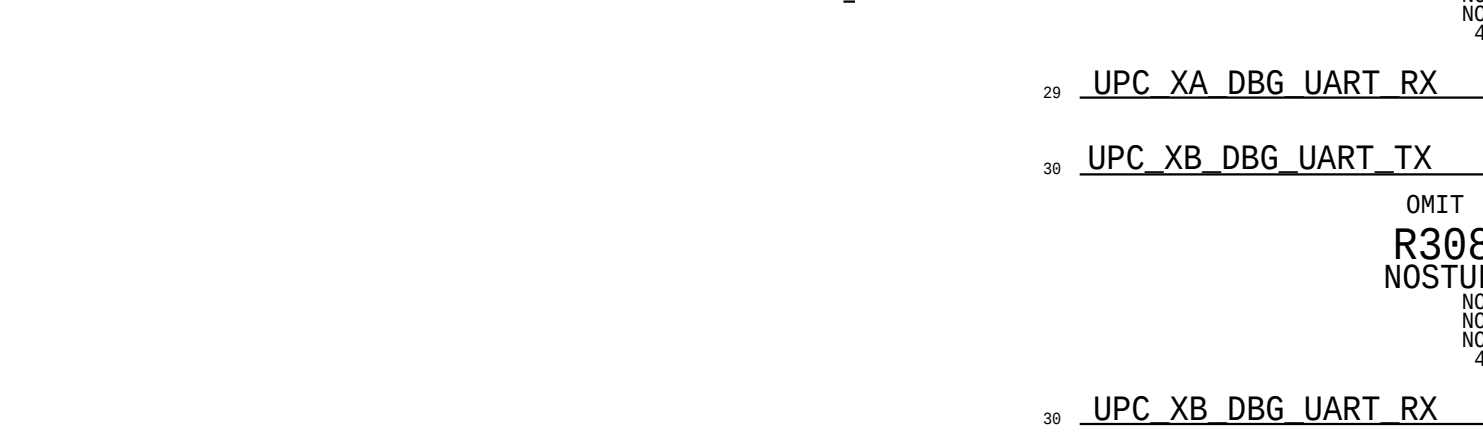
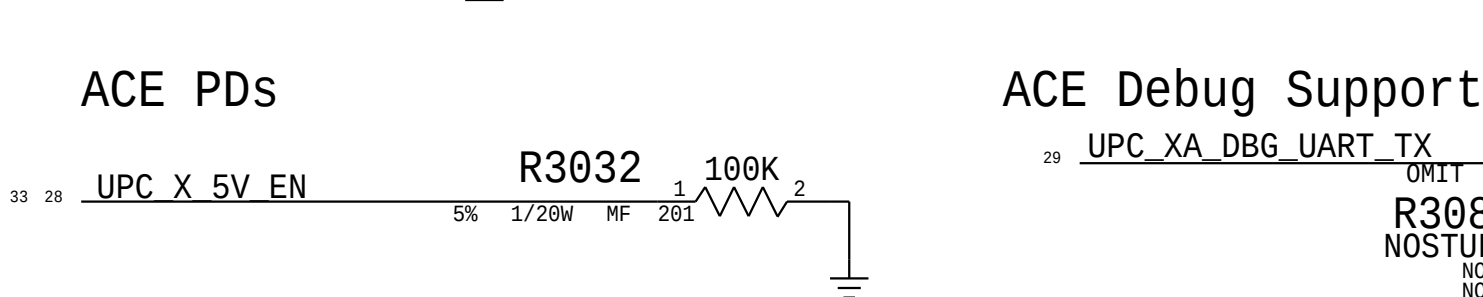
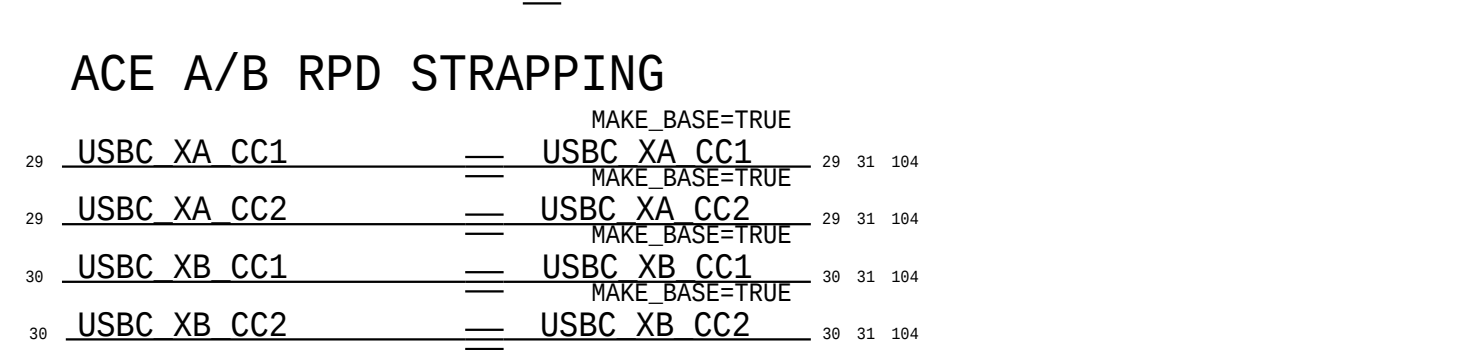
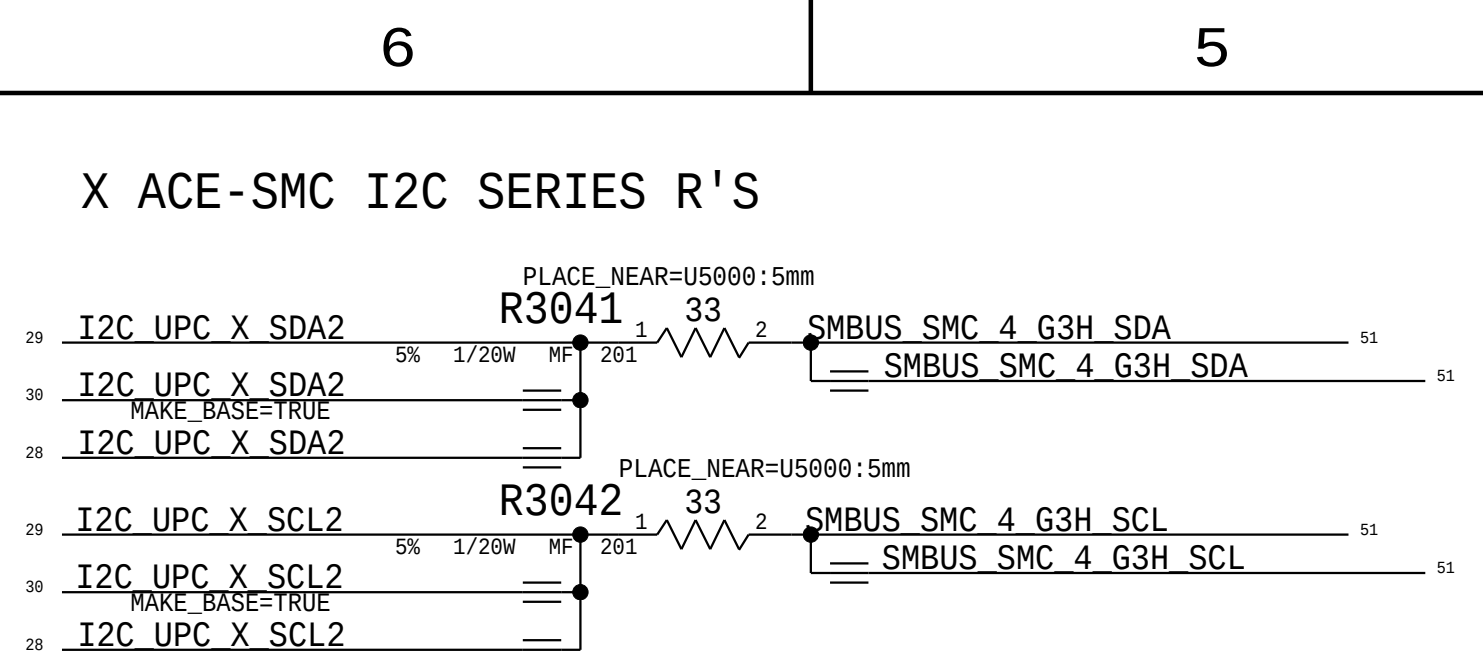
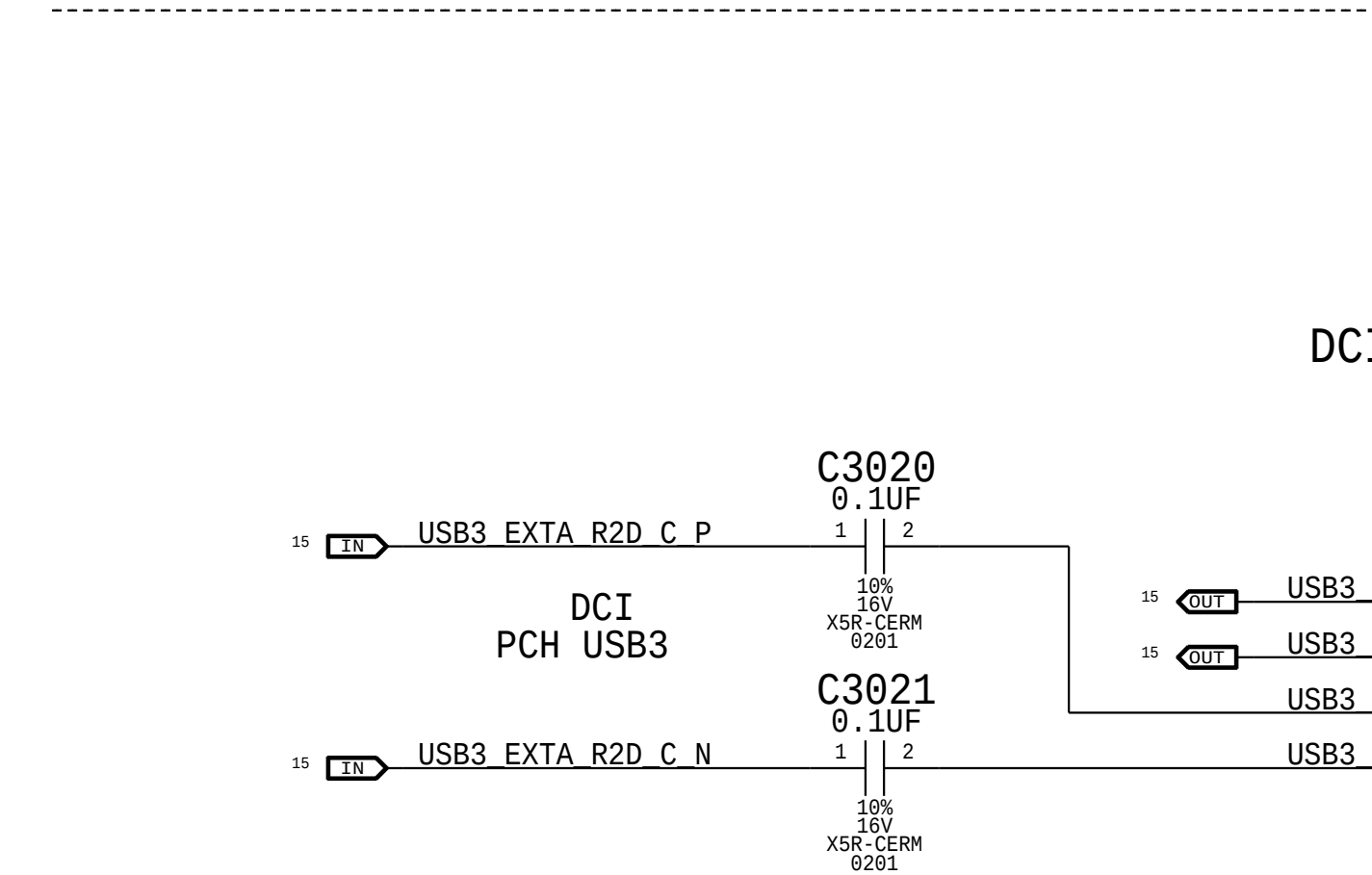
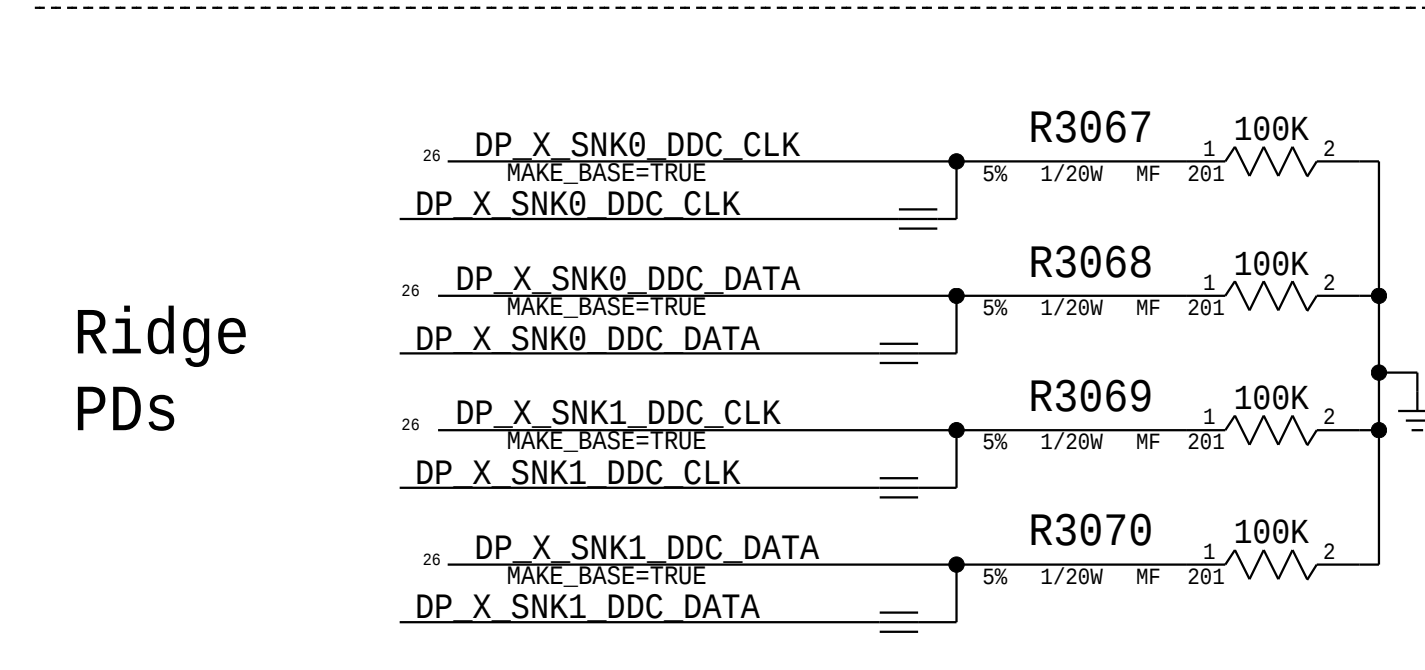
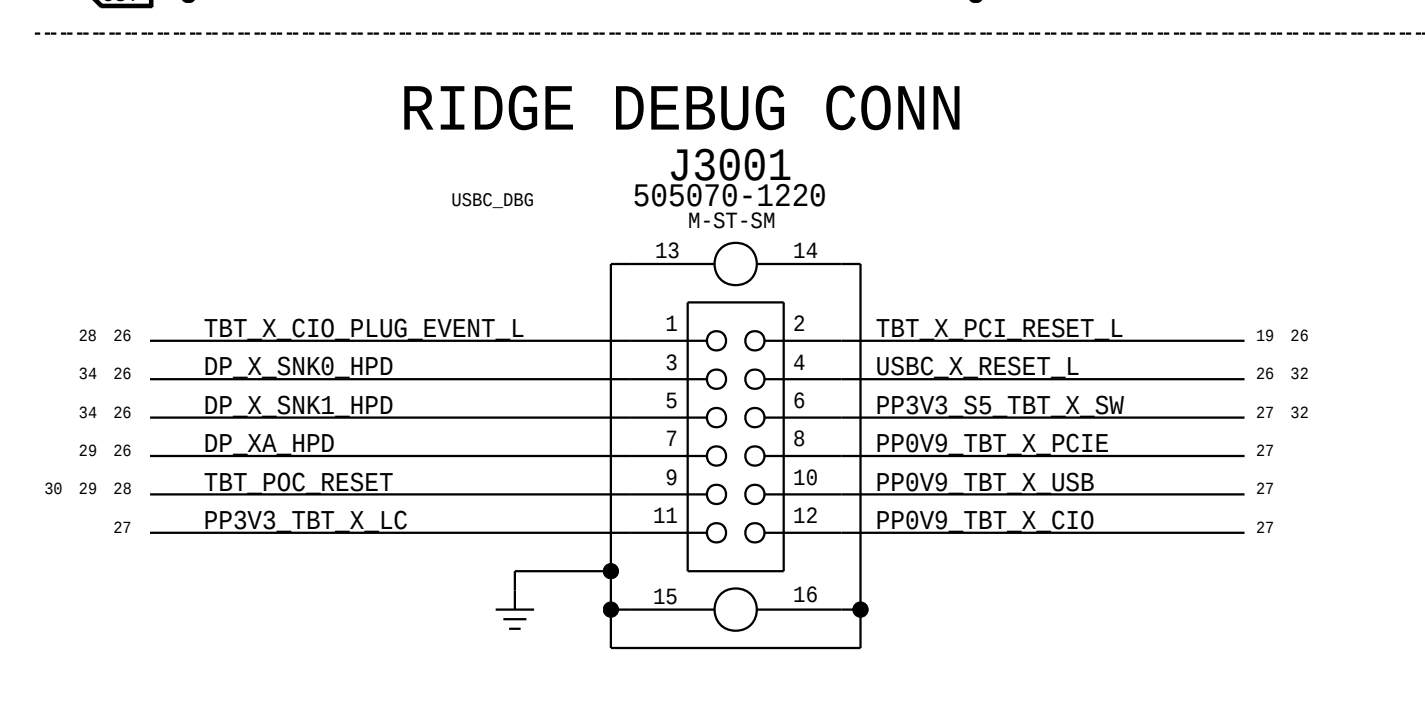
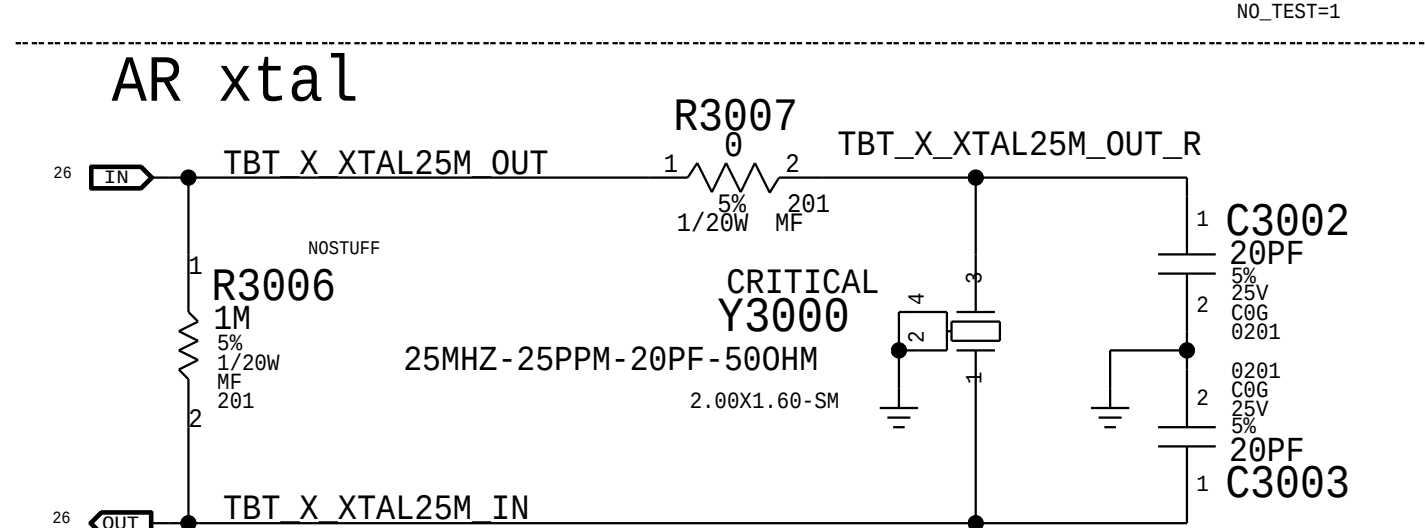
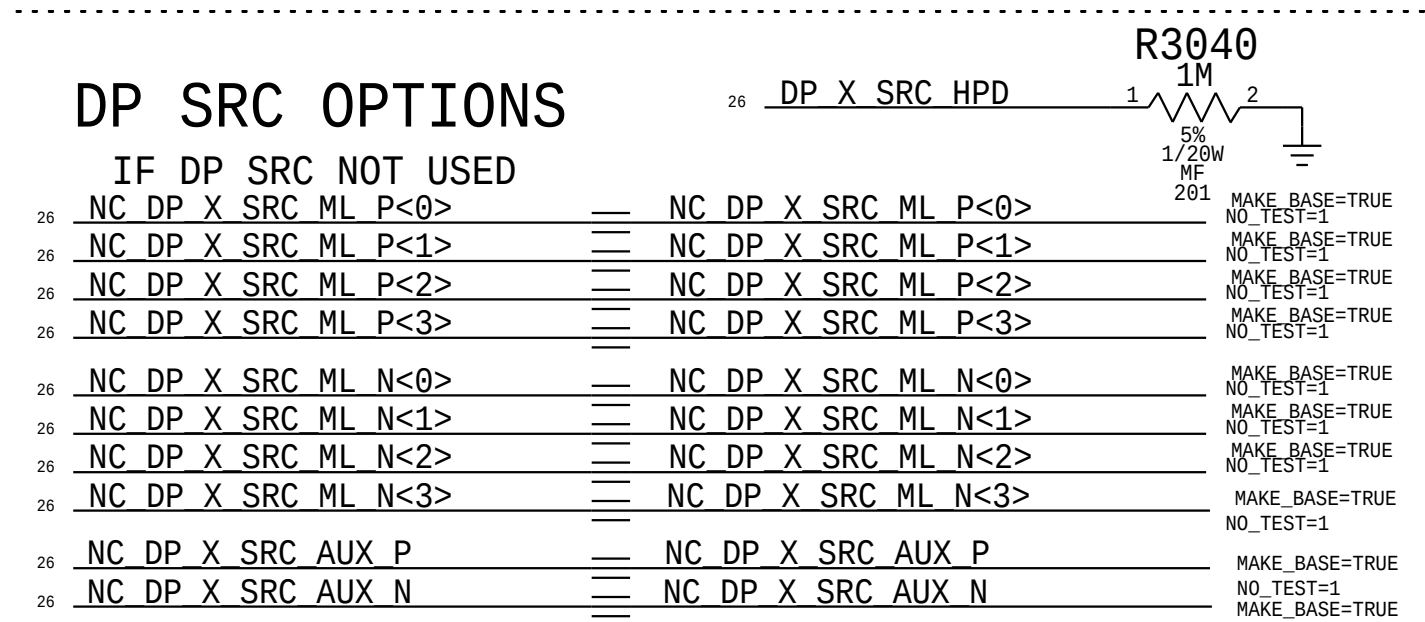
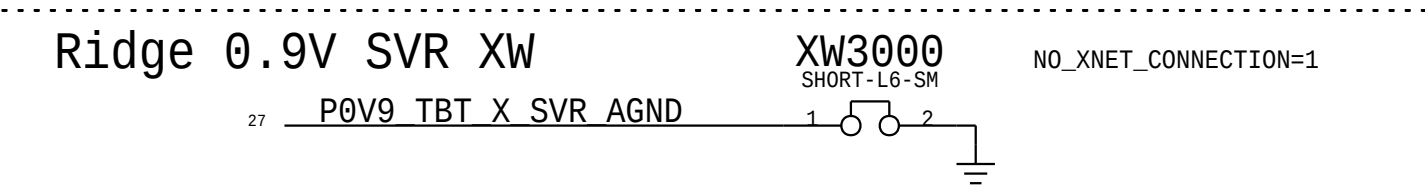
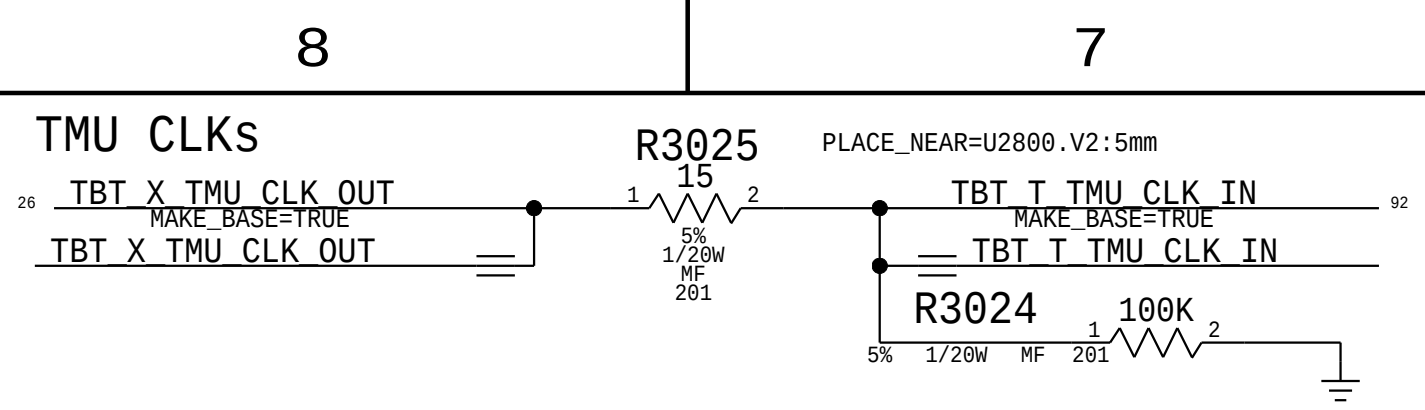
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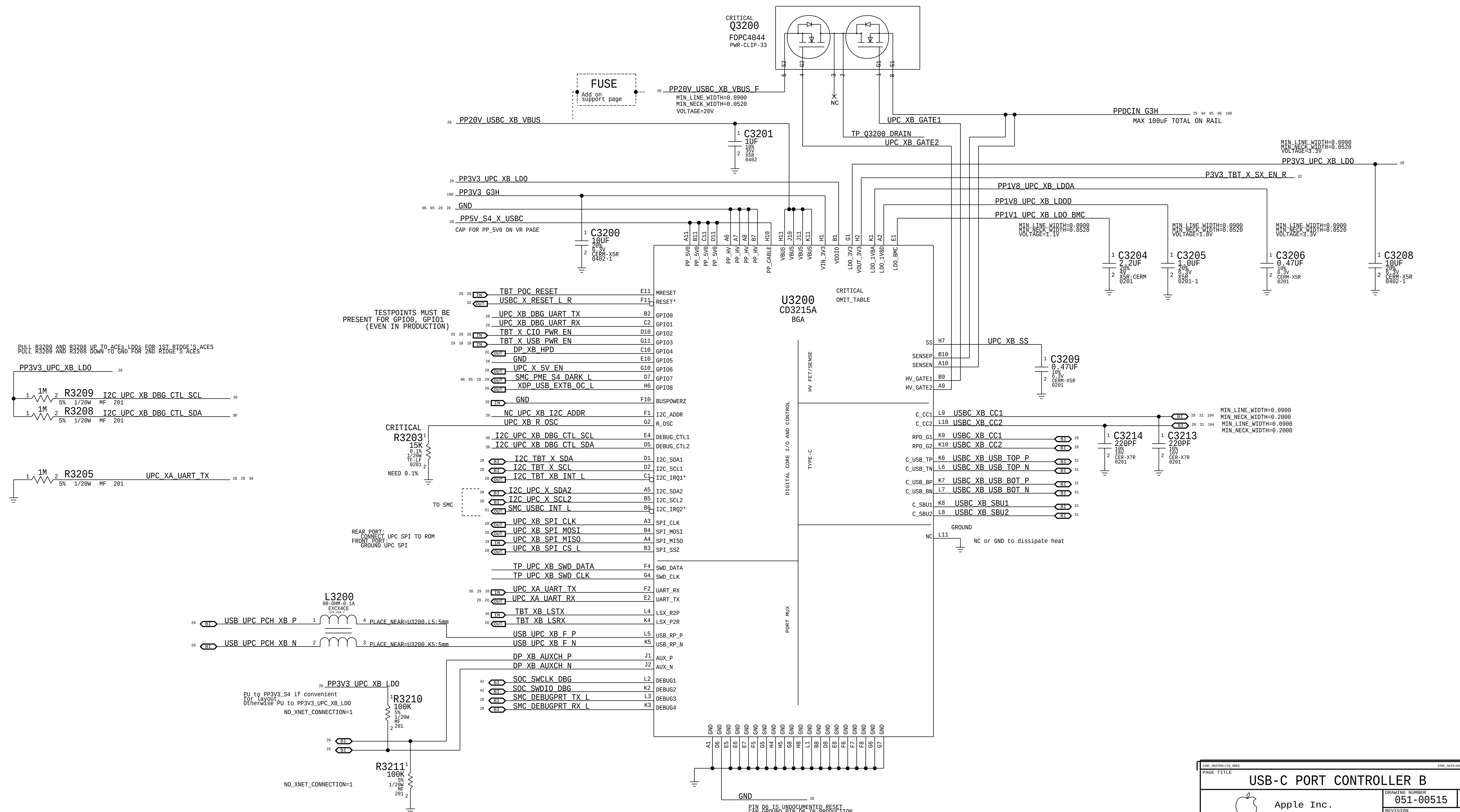


D



A

SECONDARY ACE USB-C PORT CONTROLLER (UPC)



DRAWING NUMBER 051-00515		SIZE D
REVISION 9.0.0		BRANCH dvt-fab09-0
PAGE 32 OF 145		SHEET 30 OF 119

BOM_COST_GROUP=USB-C

D

C

B

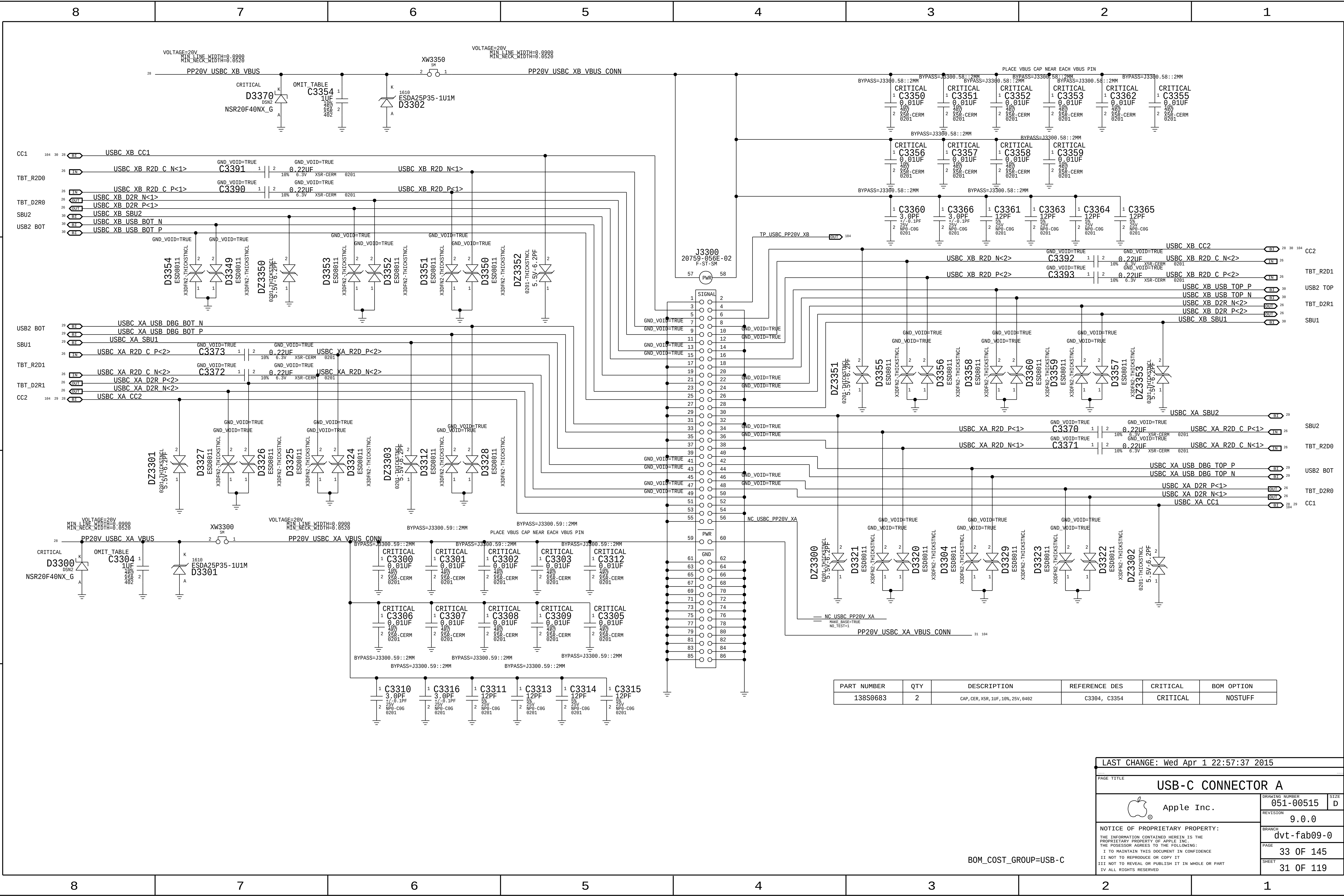
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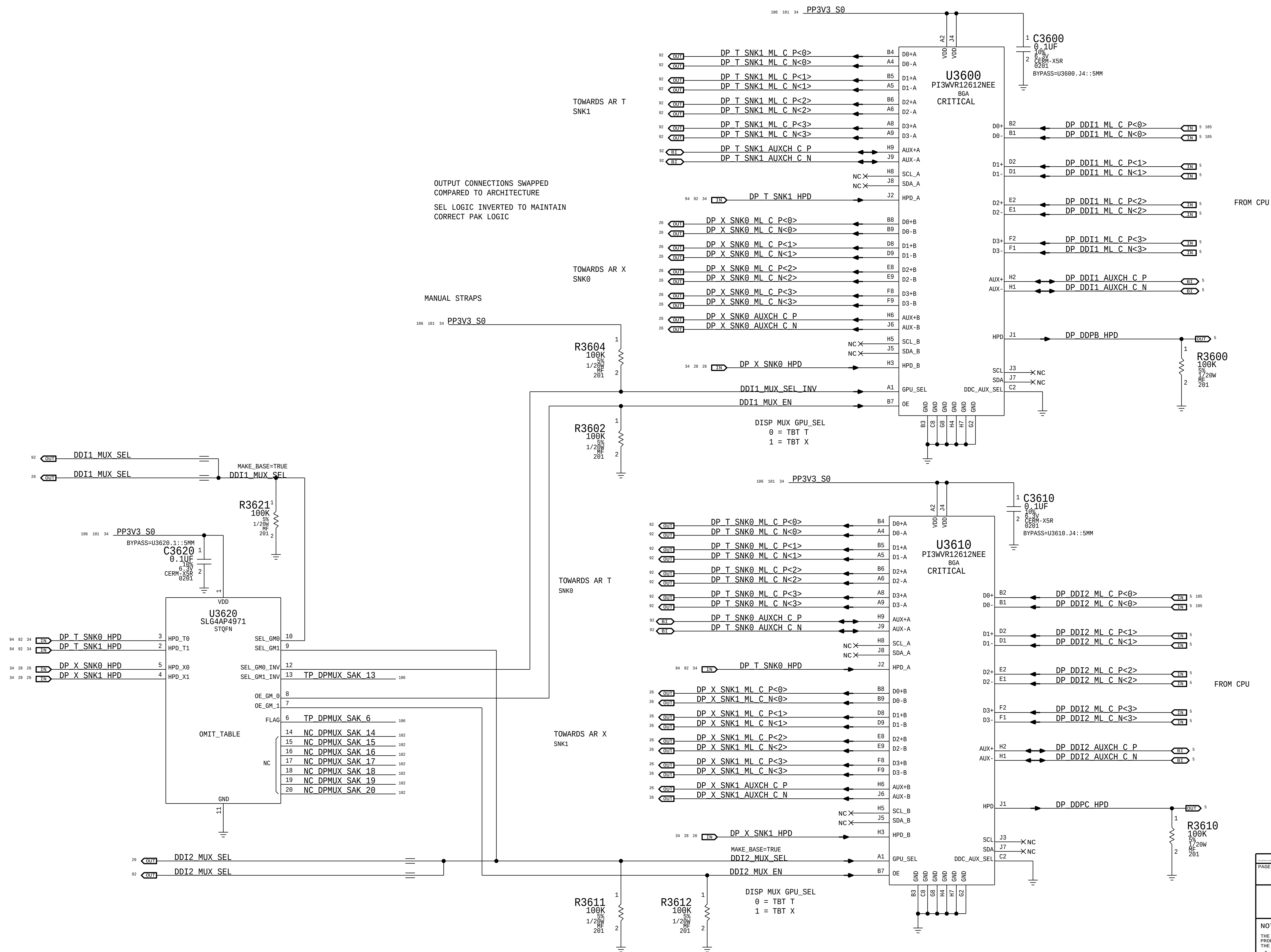
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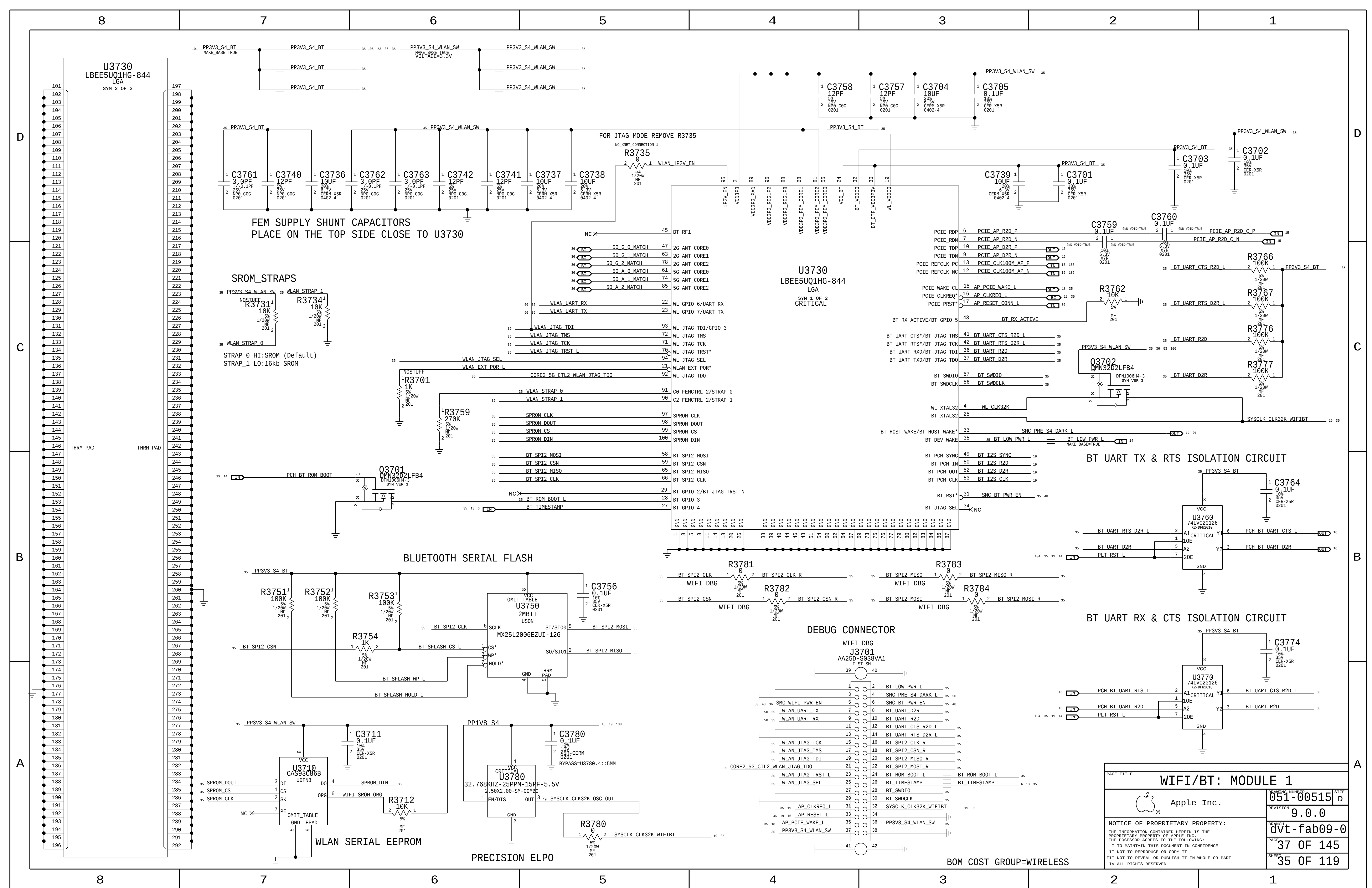
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138S0683	2	CAP, CER, X5R, 1UF, 10%, 25V, 0402	C3304, C3354	CRITICAL	NOSTUFF

LAST CHANGE: Wed Apr 1 22:57:37 2015			
PAGE TITLE			
USB-C CONNECTOR A			
 Apple Inc.		DRAWING NUMBER	051-00515
		REVISION	9.0.0
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		PAGE	33 OF 145
		SHEET	31 OF 119

BOM_COST_GROUP=USB-C



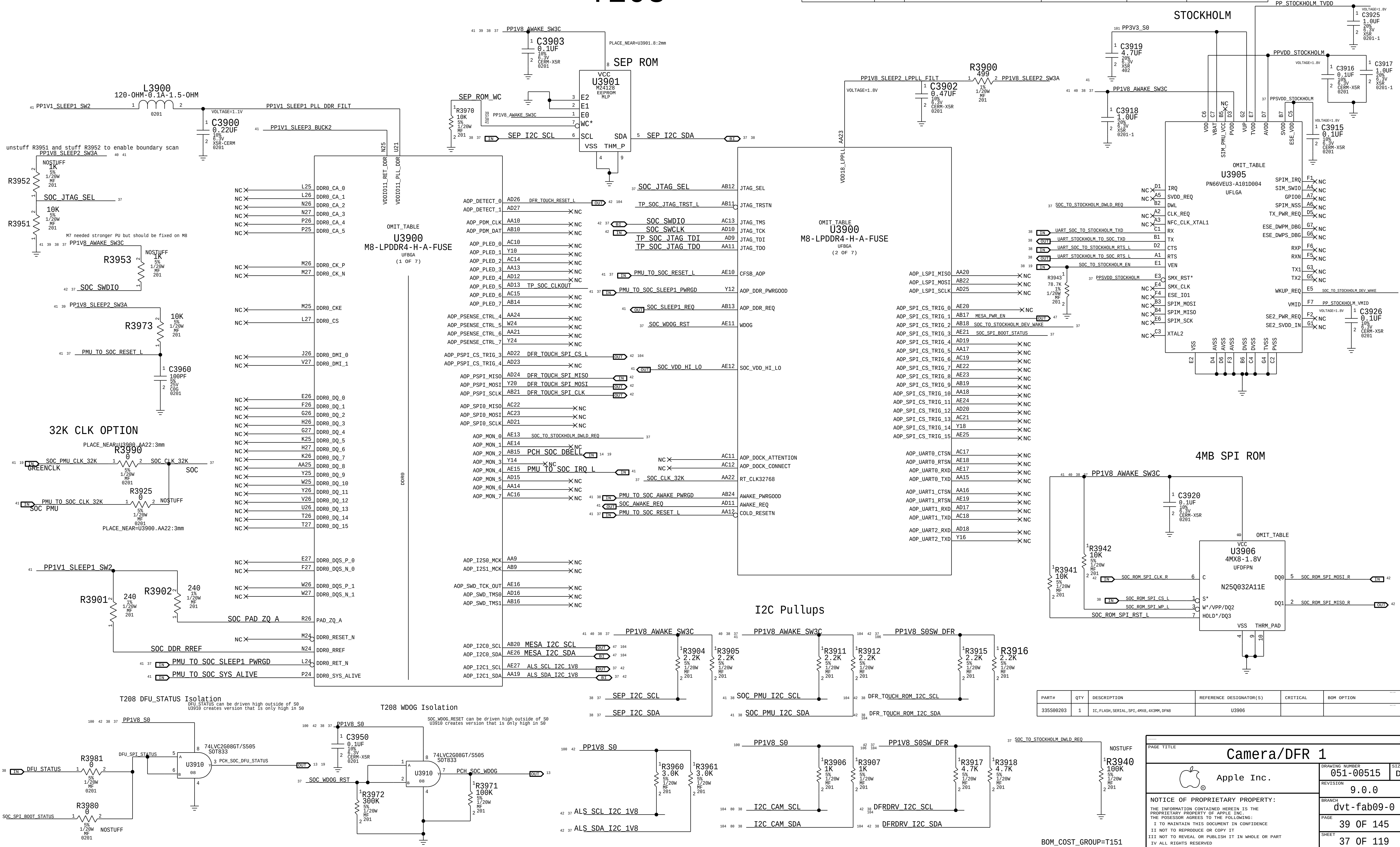
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		SHEET		34 OF 119	

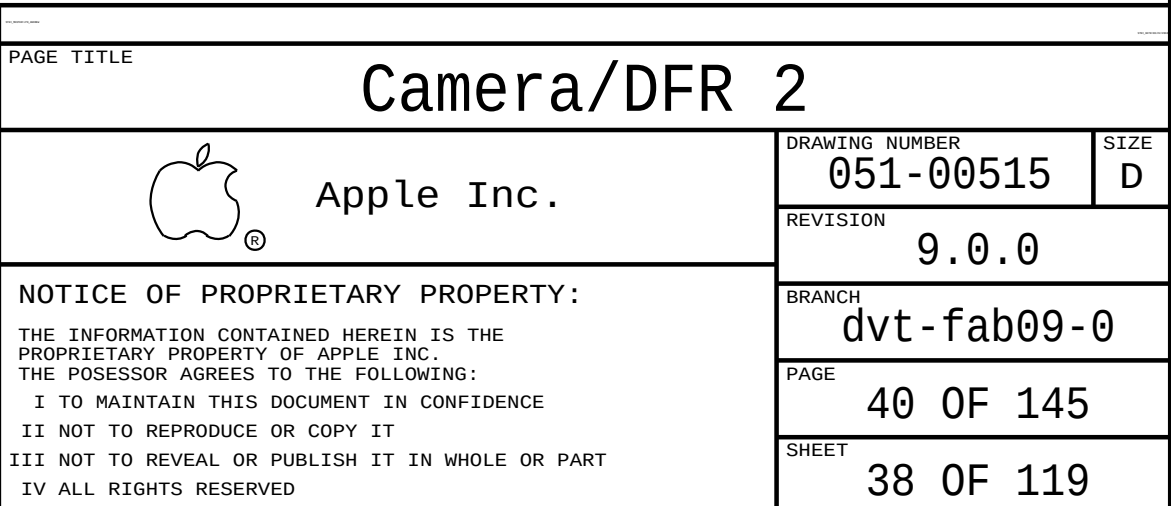


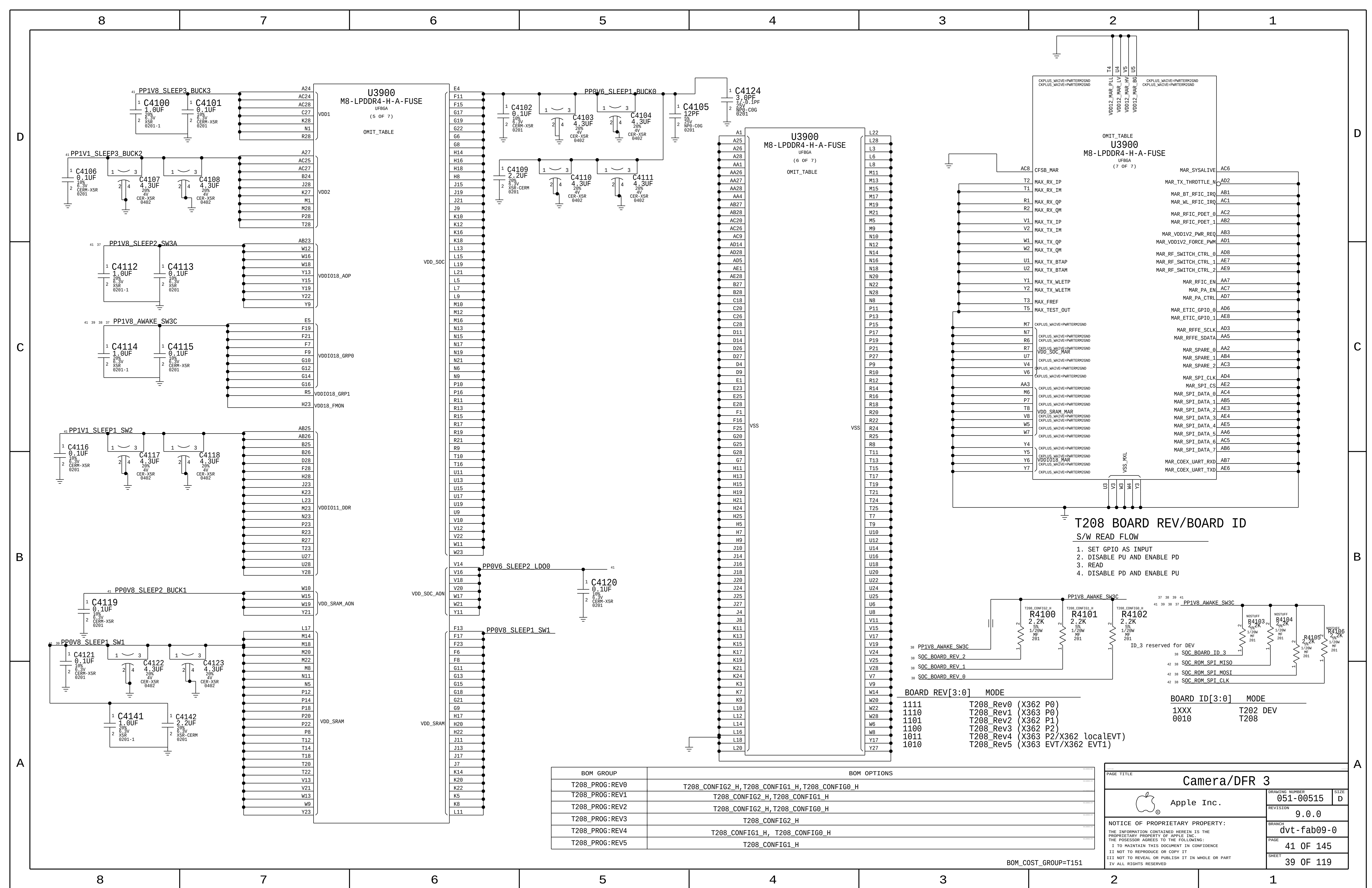
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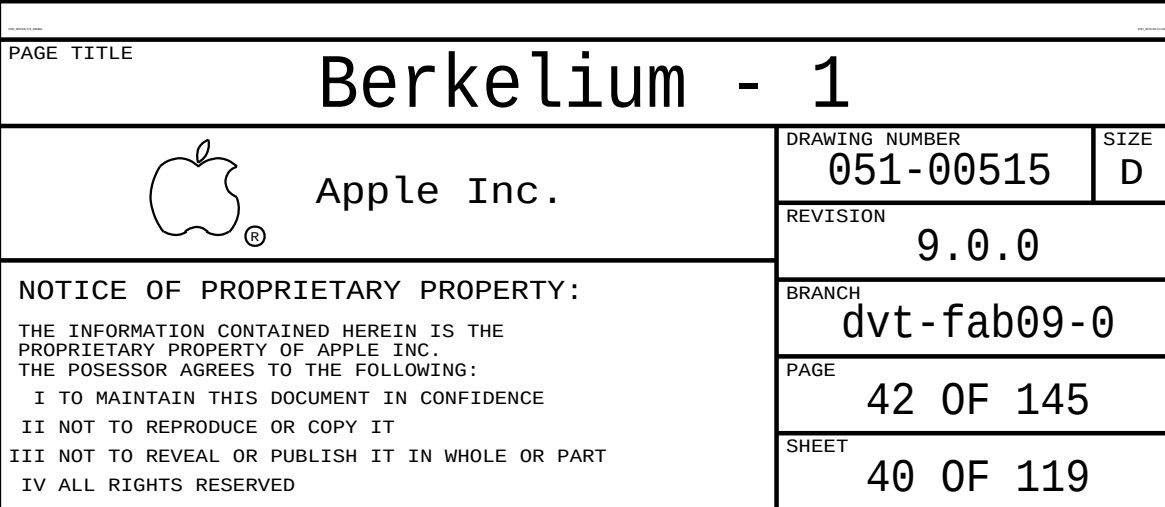
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338S00097	1	IC, RTM2, MP, PN549A1, P61D0	U3905	CRITICAL	SE:PROD

T208

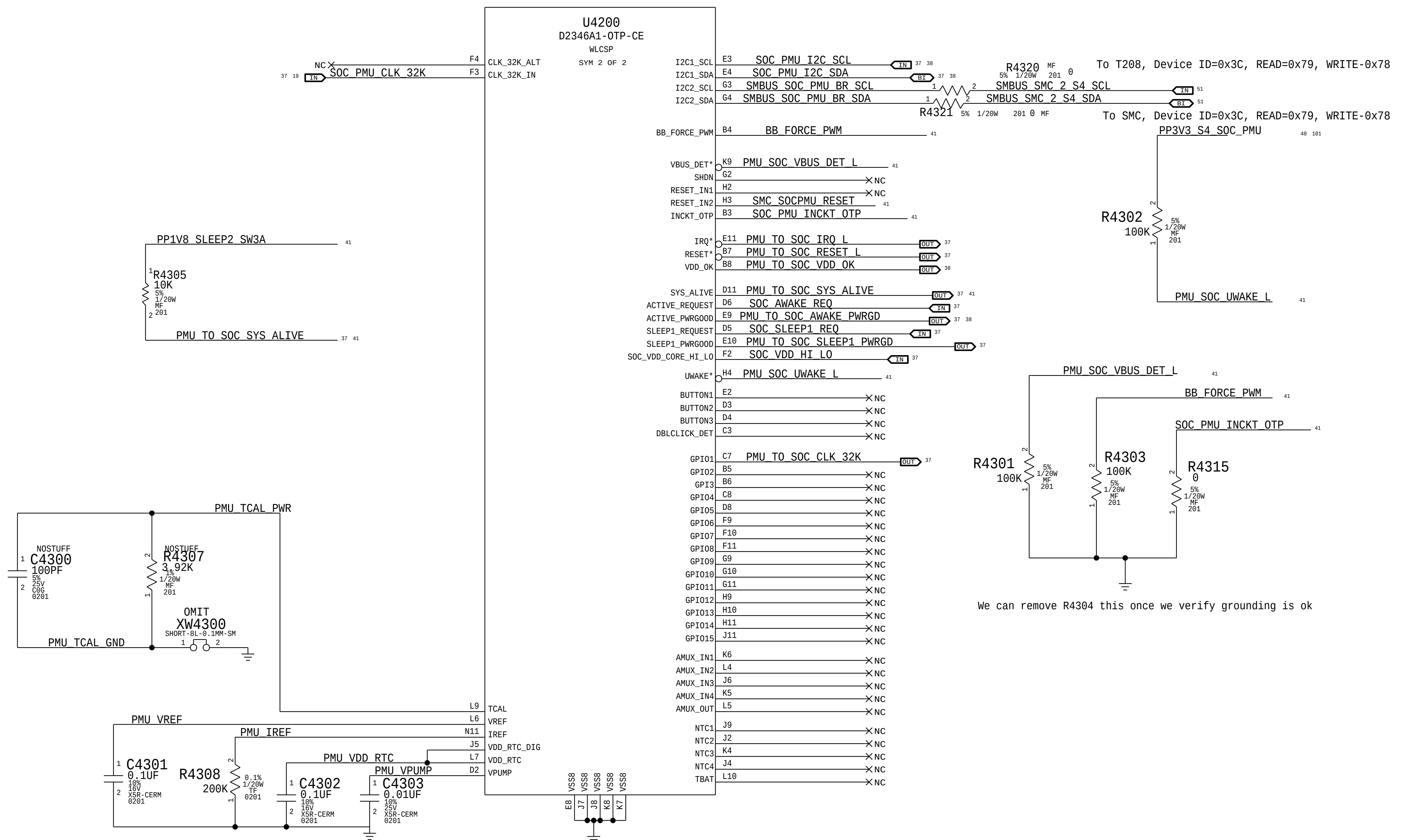






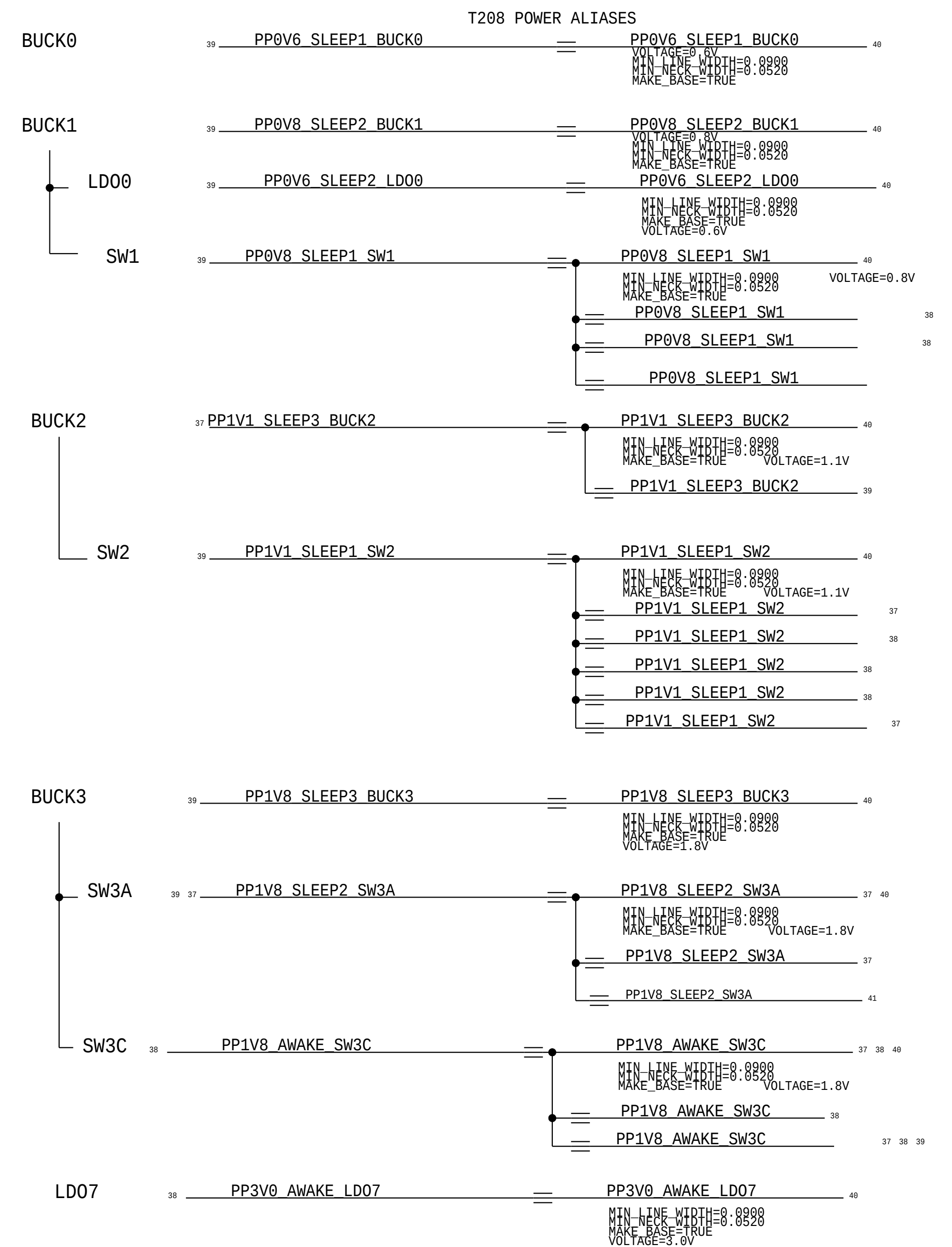
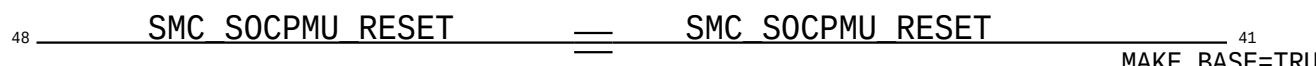


Berkelium - 2



We can remove R4304 this once we verify grounding is ok

Signal Aliase



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T208 Support

DFR Connectors

3.3V DFR Switch

DFR Disp Conn

MIPID FILTERING

SWD DEBUG MUX

T208 I2C Mapping

SPI TERM

T208 LEVEL SHIFTING

ALS I2C

ALS/DEBUG UART

ACE SWD DBG

T208 Support



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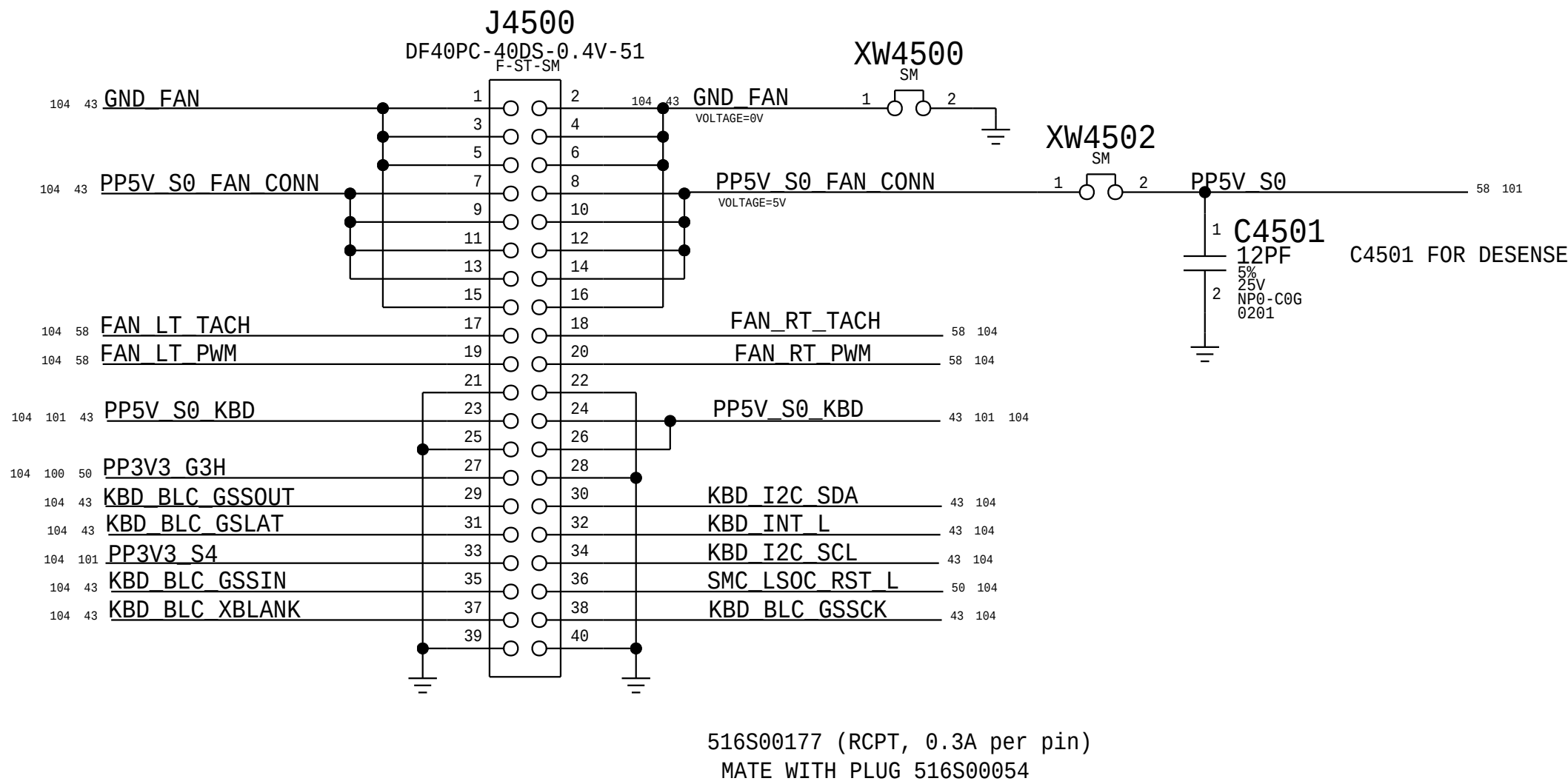
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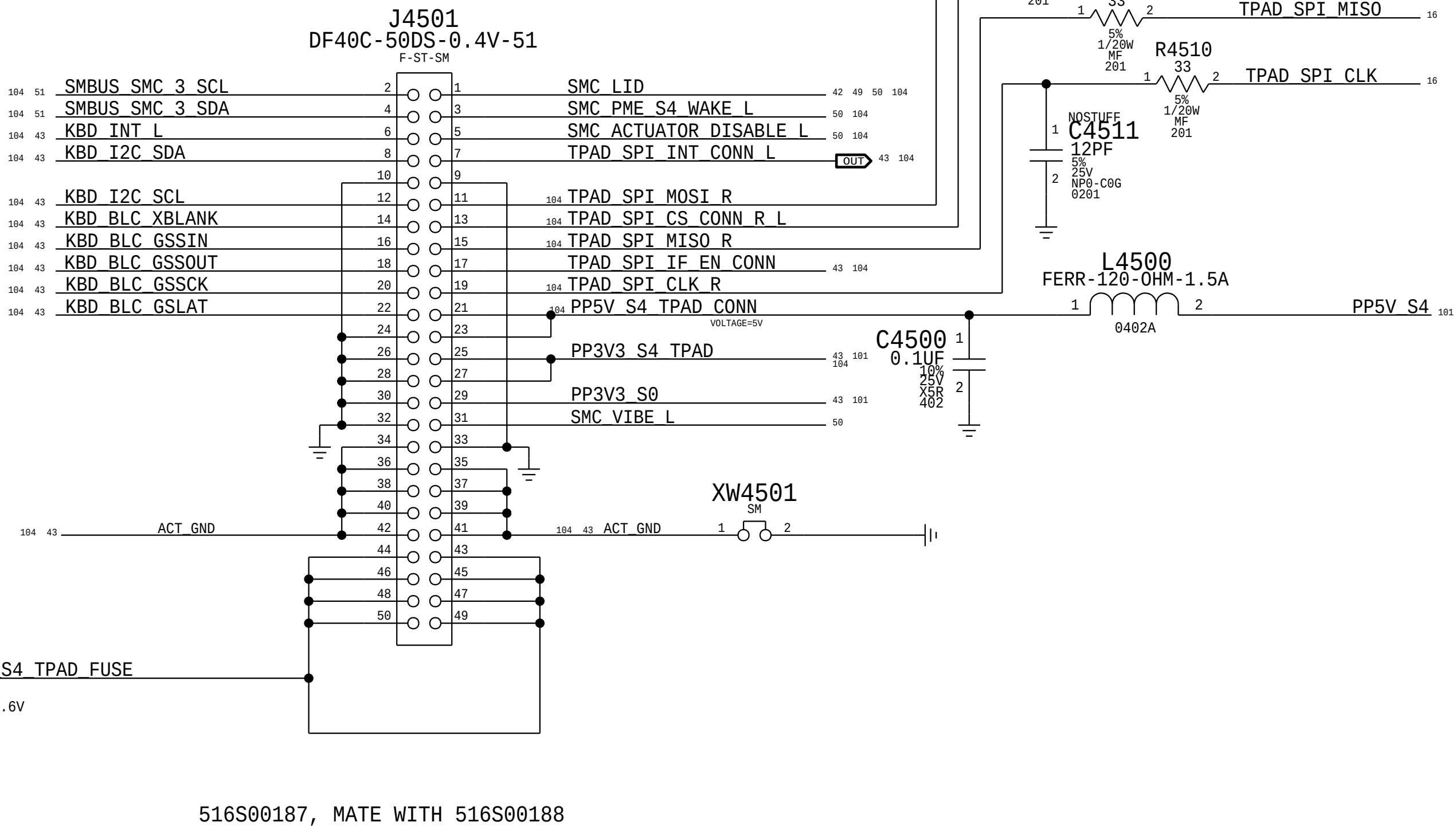
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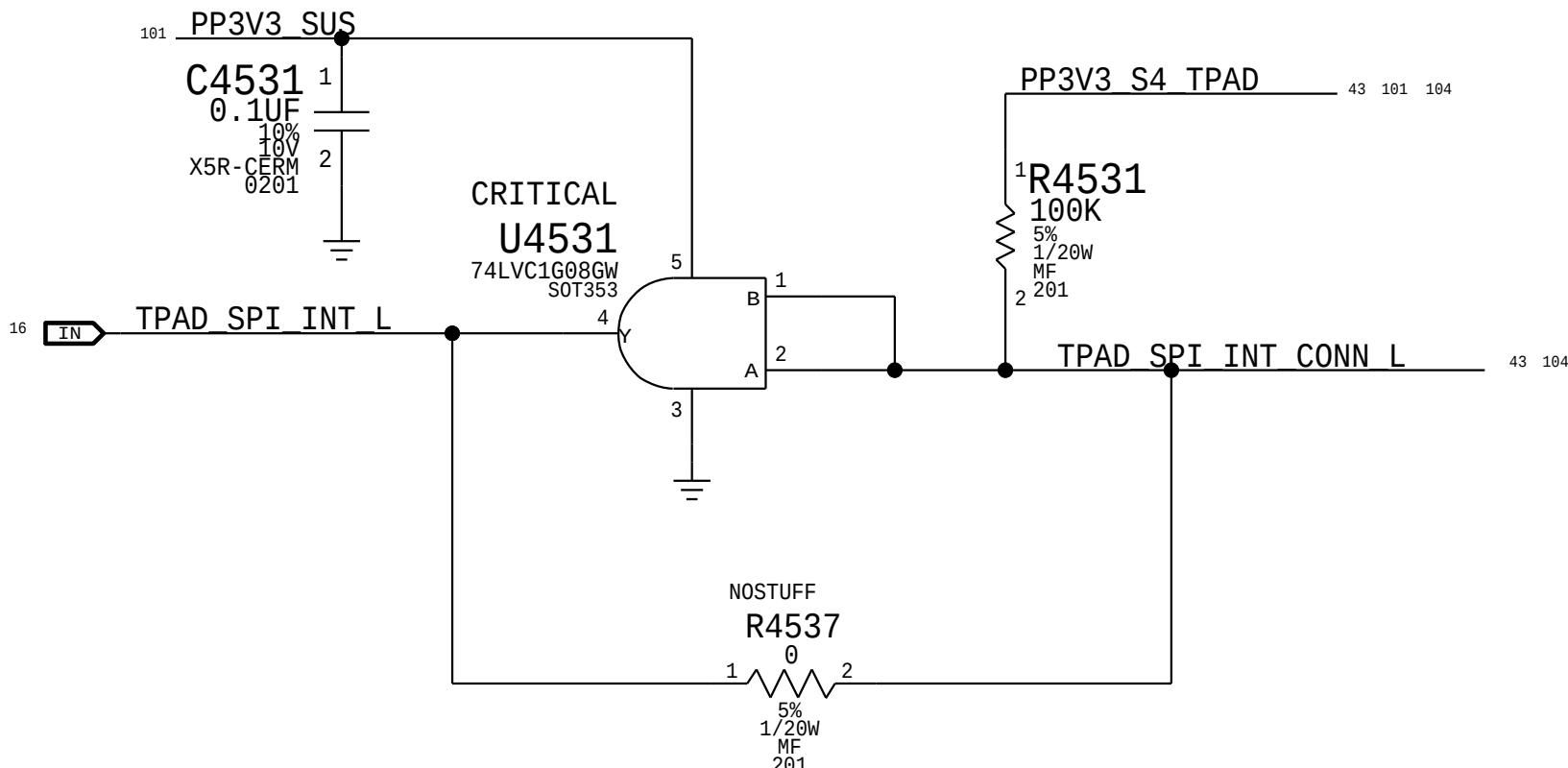
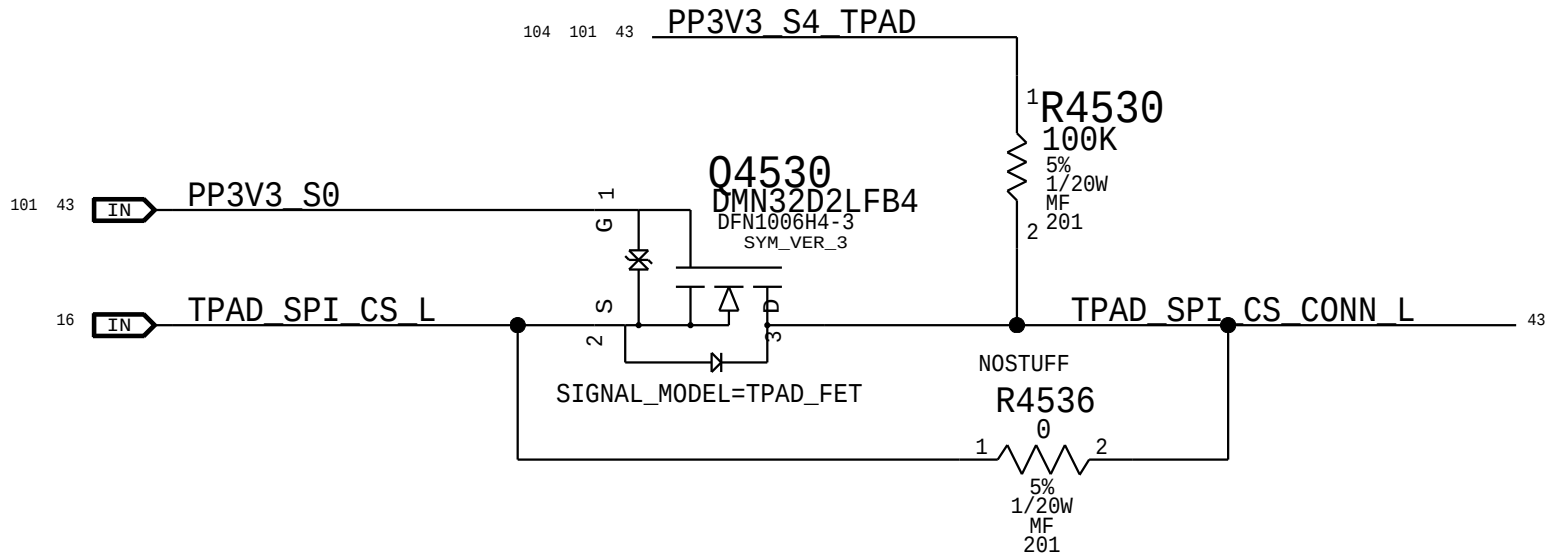
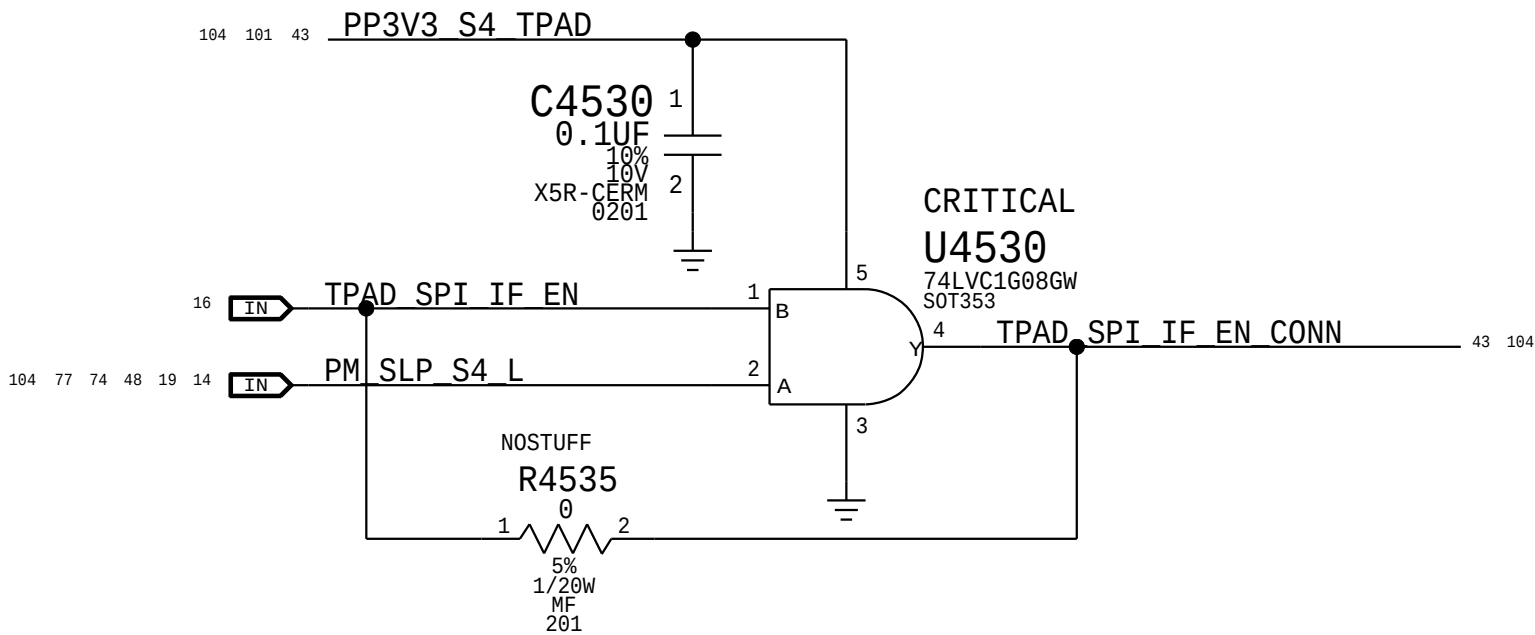
KBD CONNECTOR



TPAD CONNECTOR



TRACKPAD ISOLATION



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		43 OF 119	

BOM_COST_GROUP=KEYBOARD

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		SHEET	44 OF 119

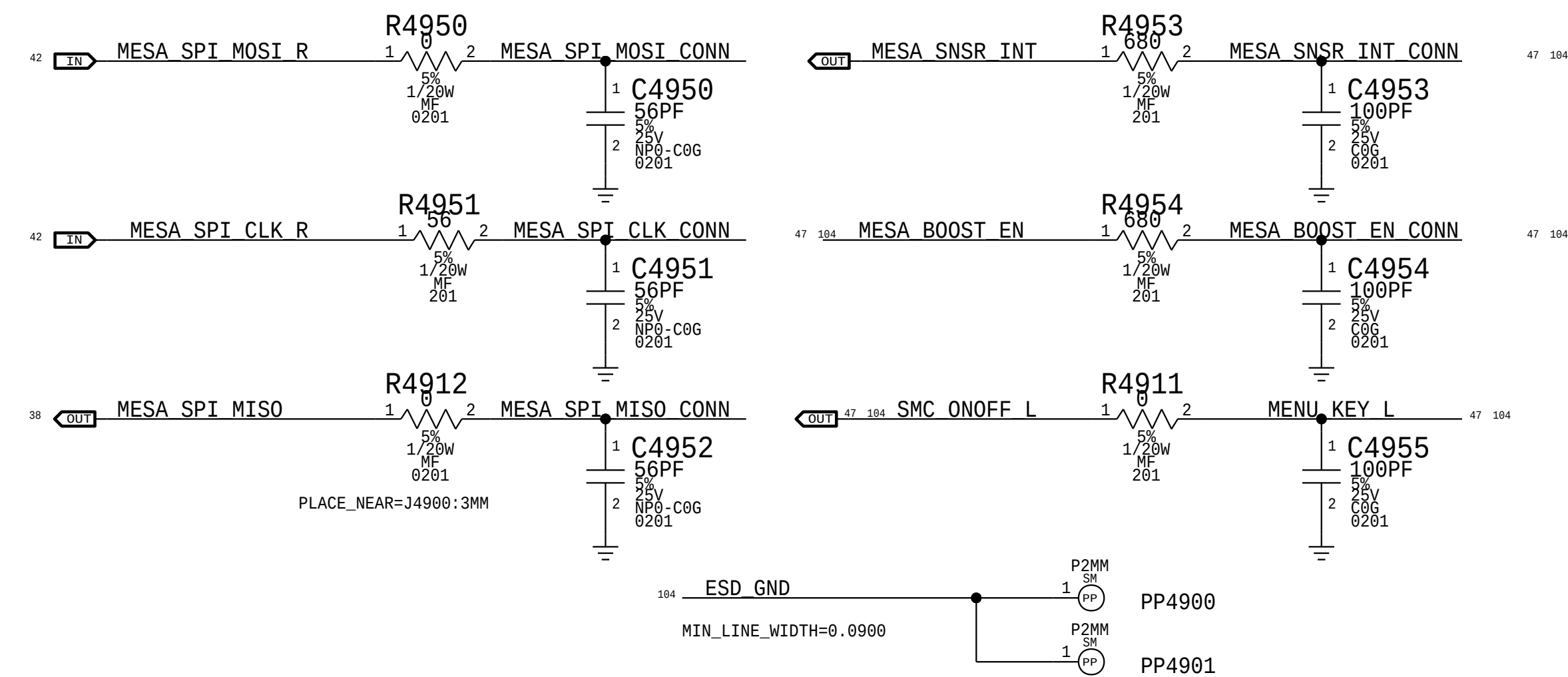
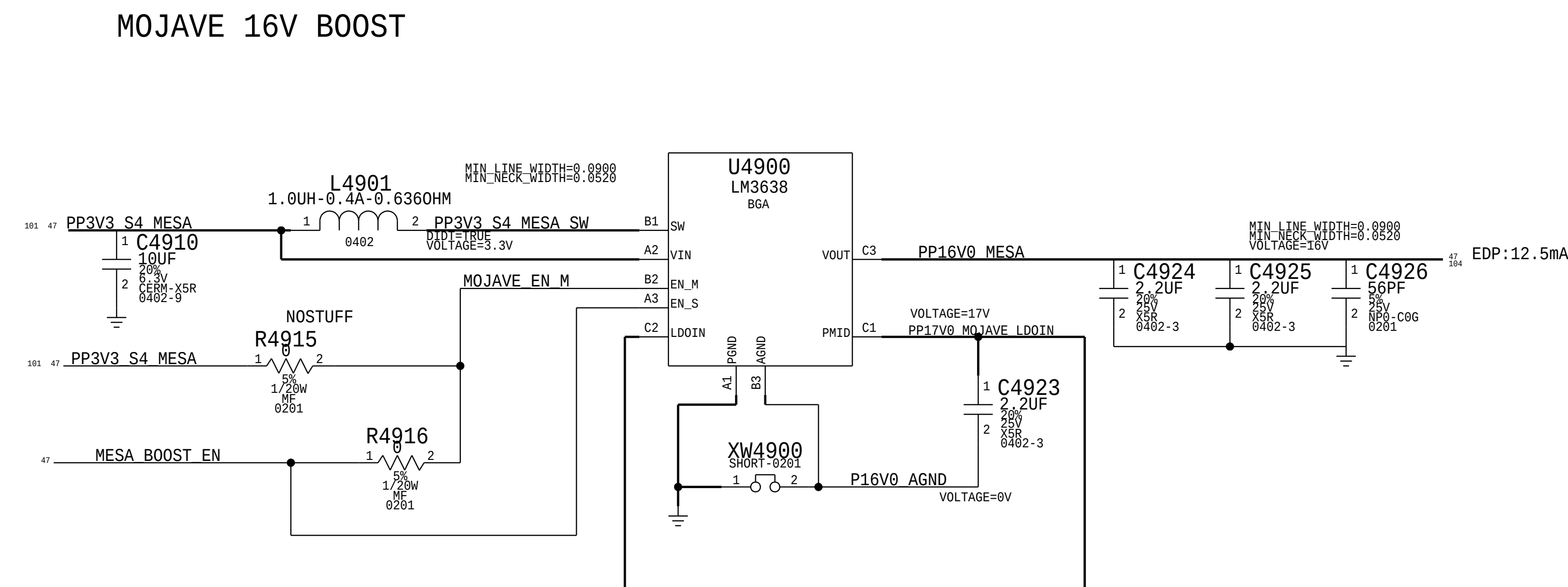
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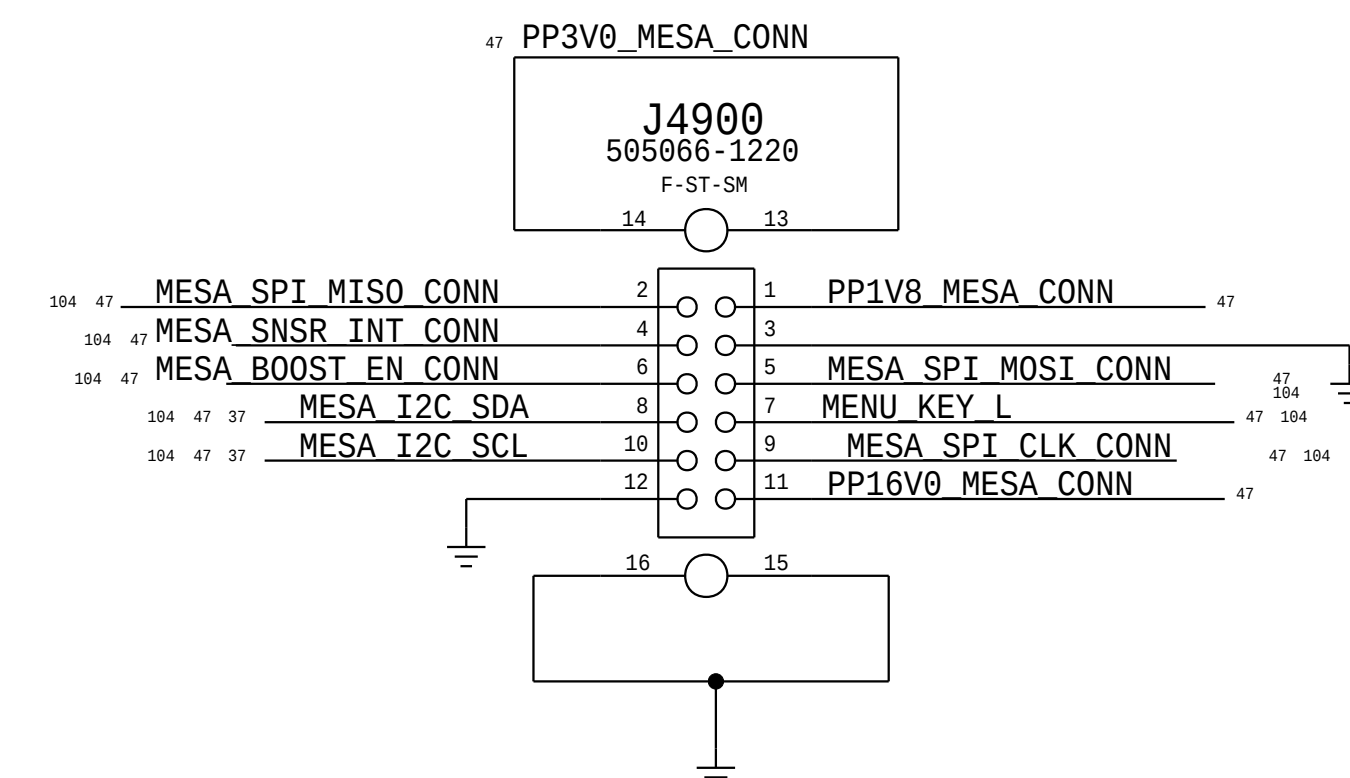
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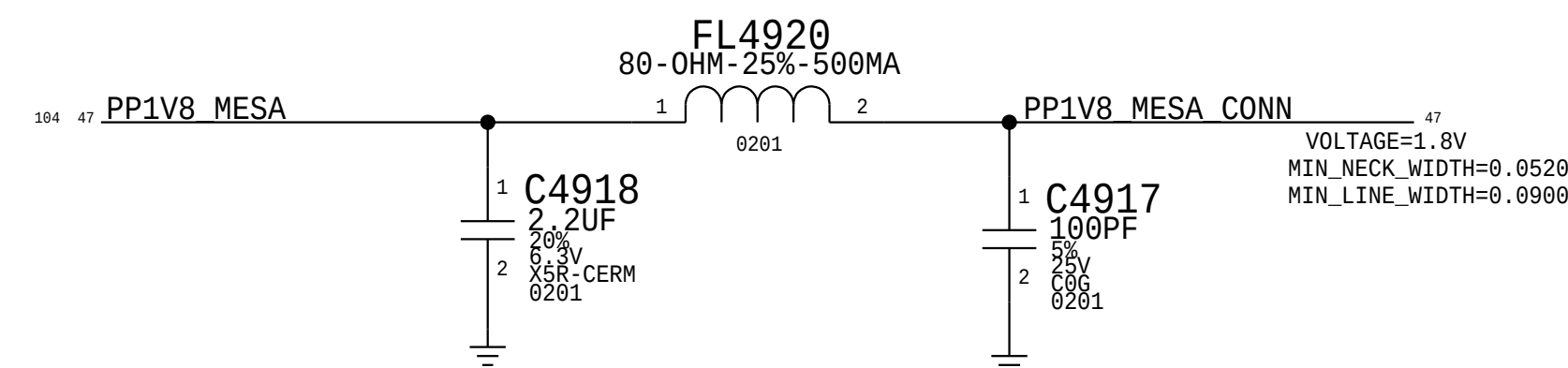
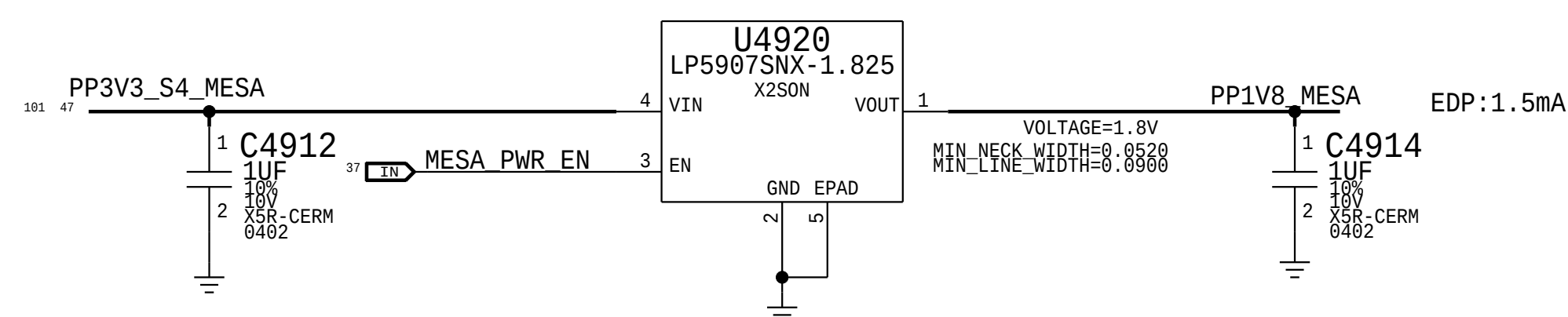
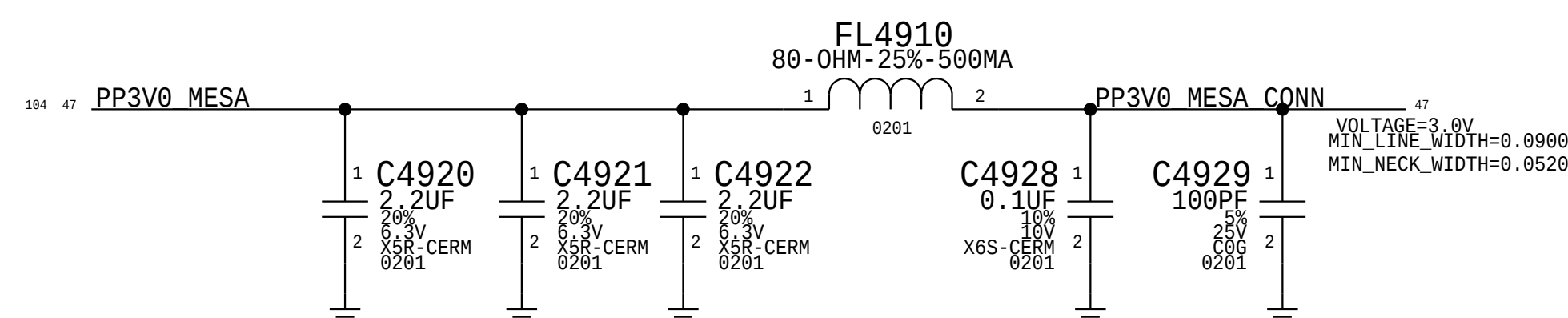
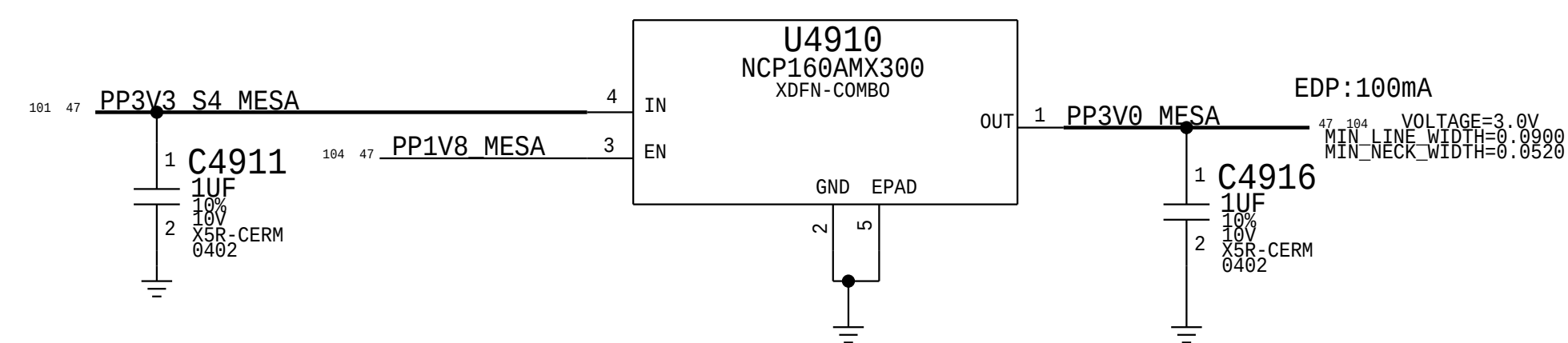
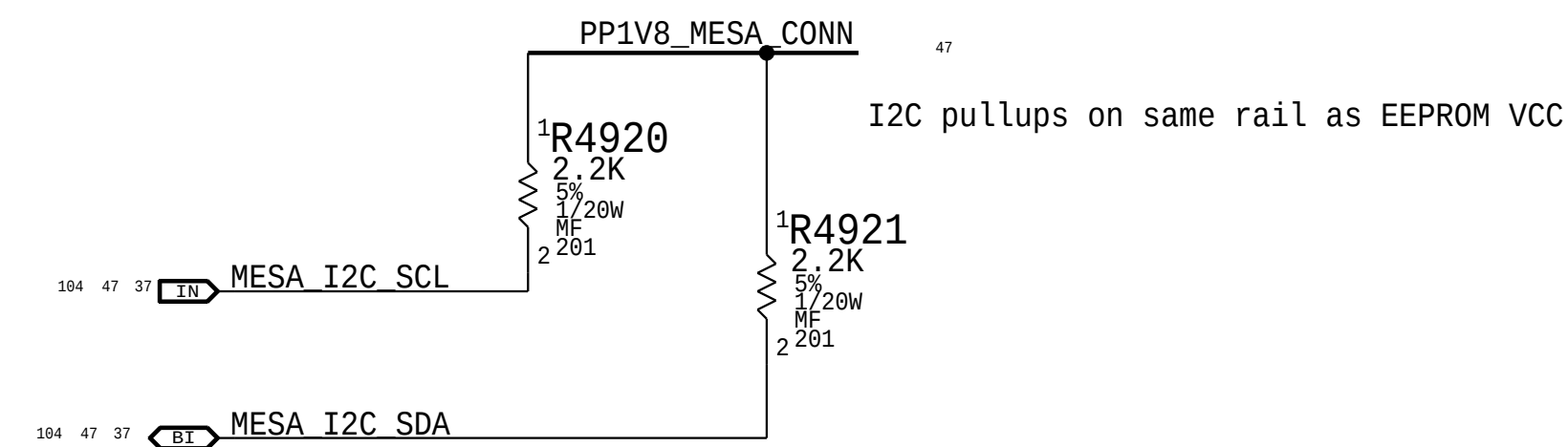


Proto1 Connector for X434/X435 Support
PLUG (516S00115) - X434/ X435 Jumper
Recptacle (516S00203) - X362/X363 MLB



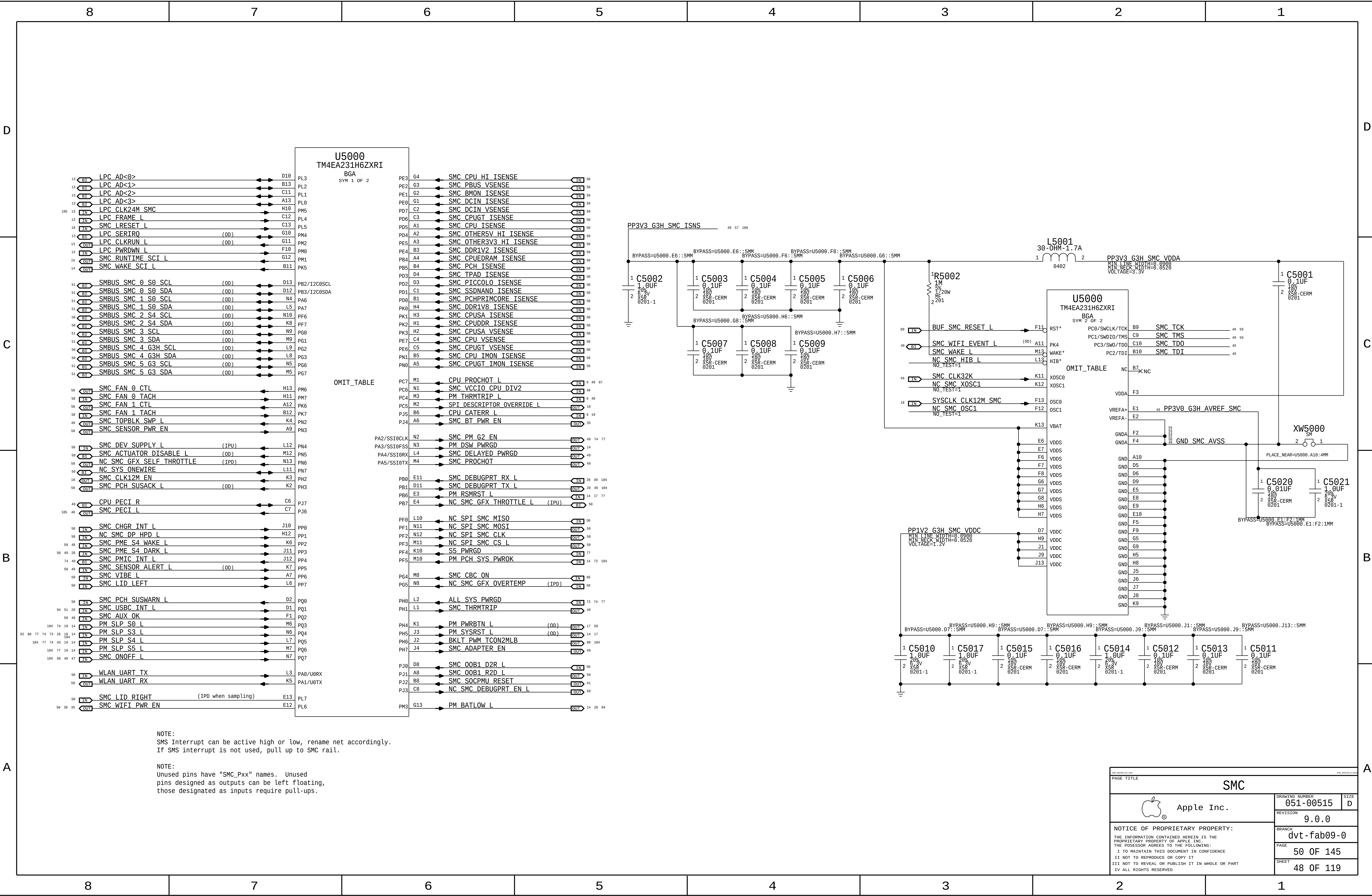
Mesa Power Sequencing Requirements

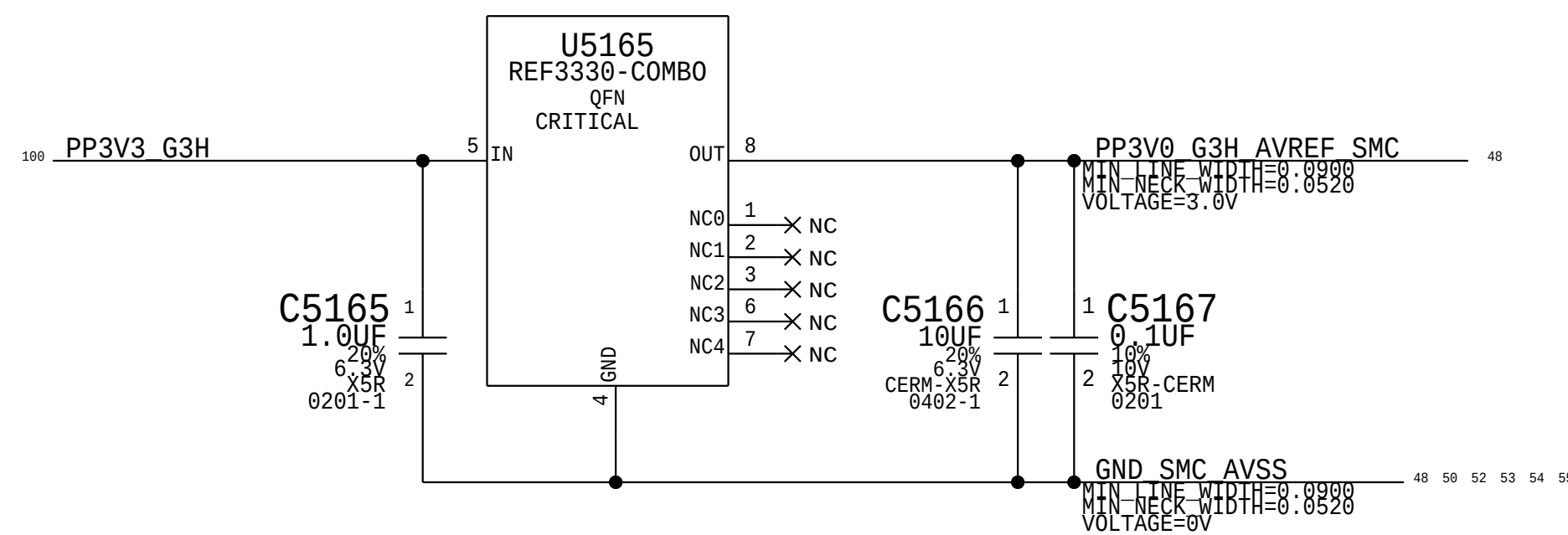
Power On: 1V8 -> 3V3 -> 16V0



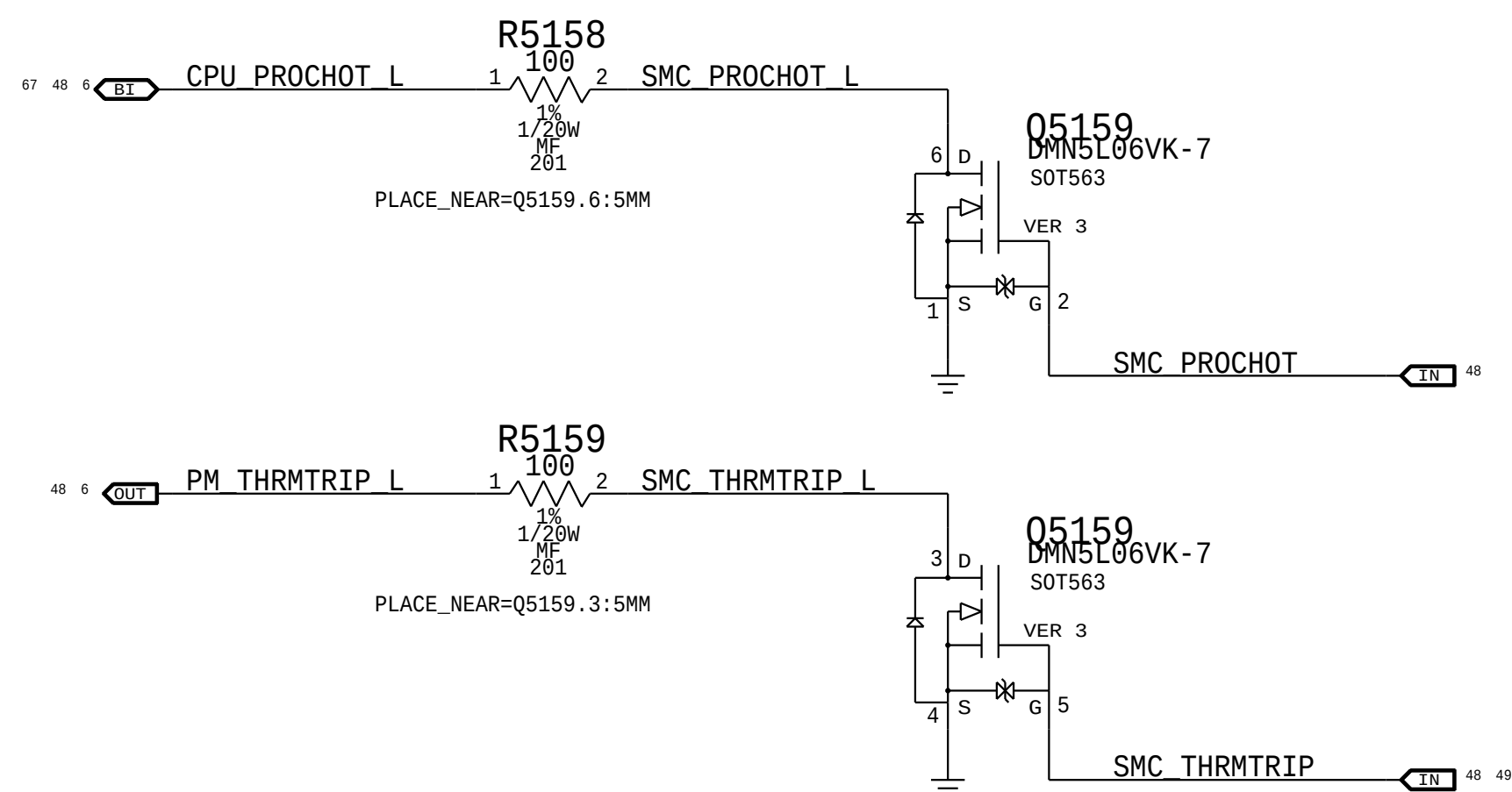
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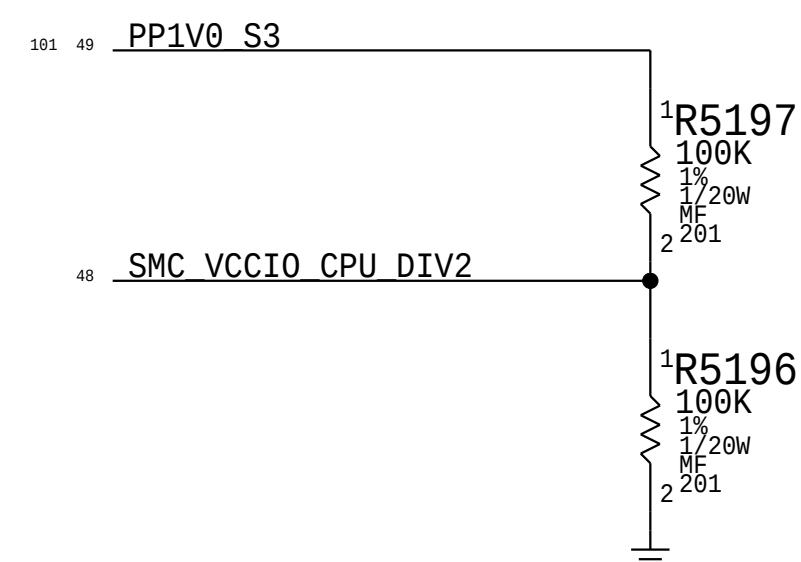
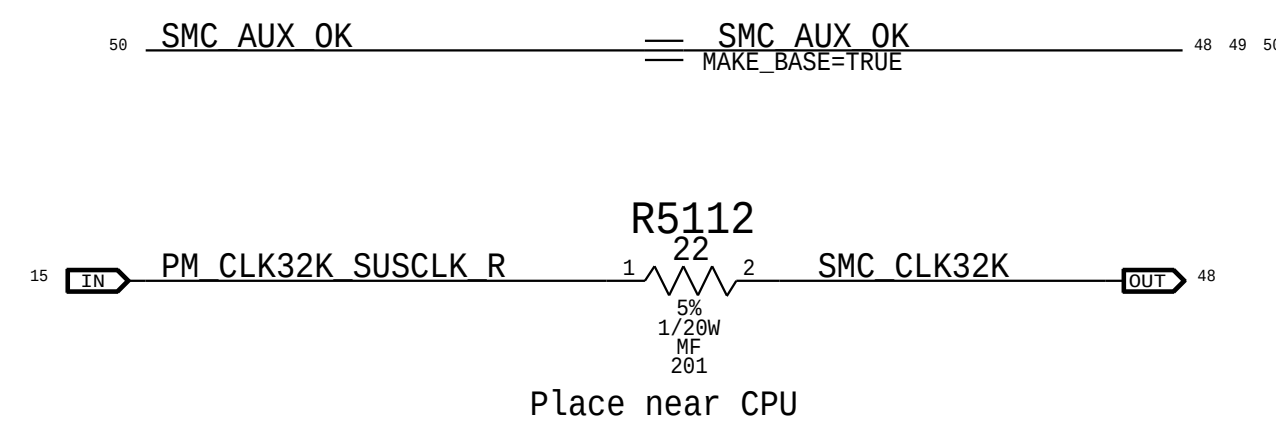
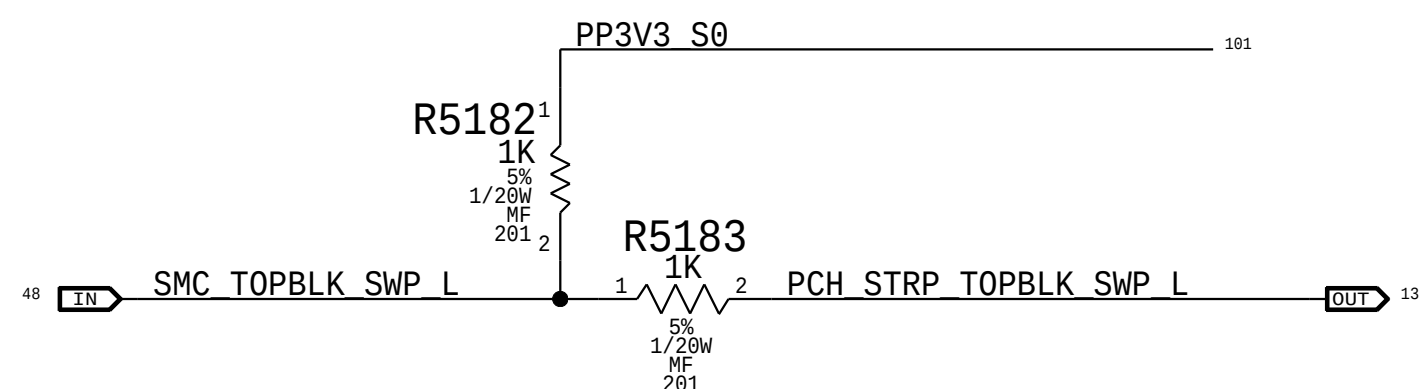
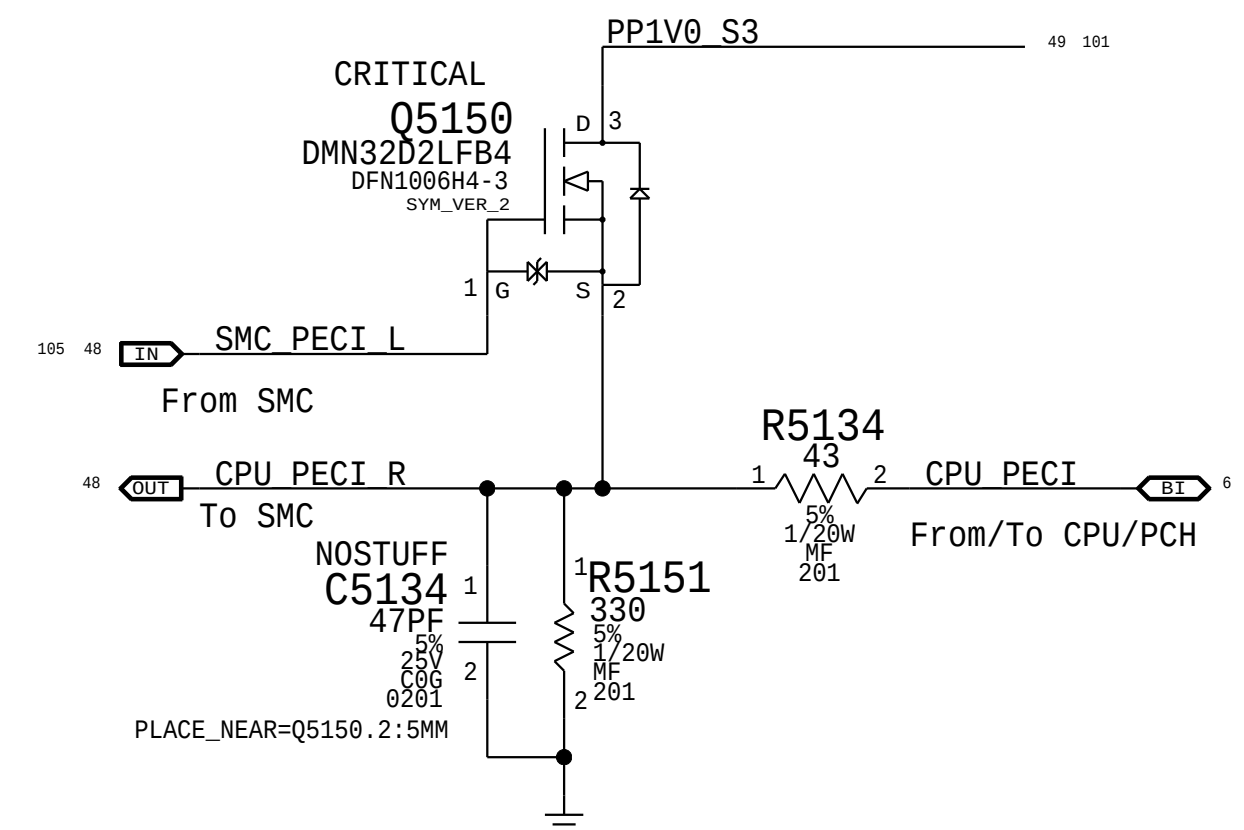




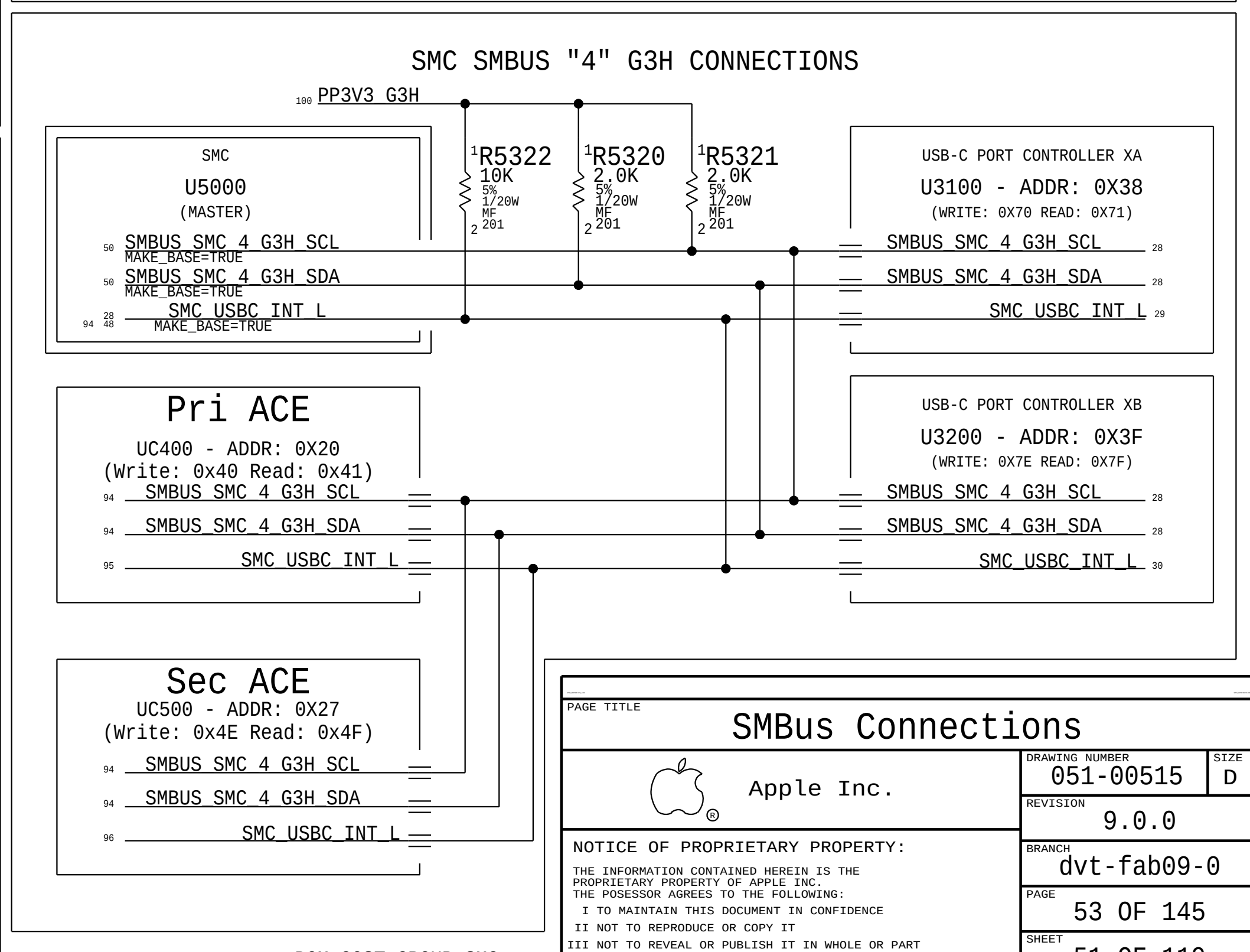
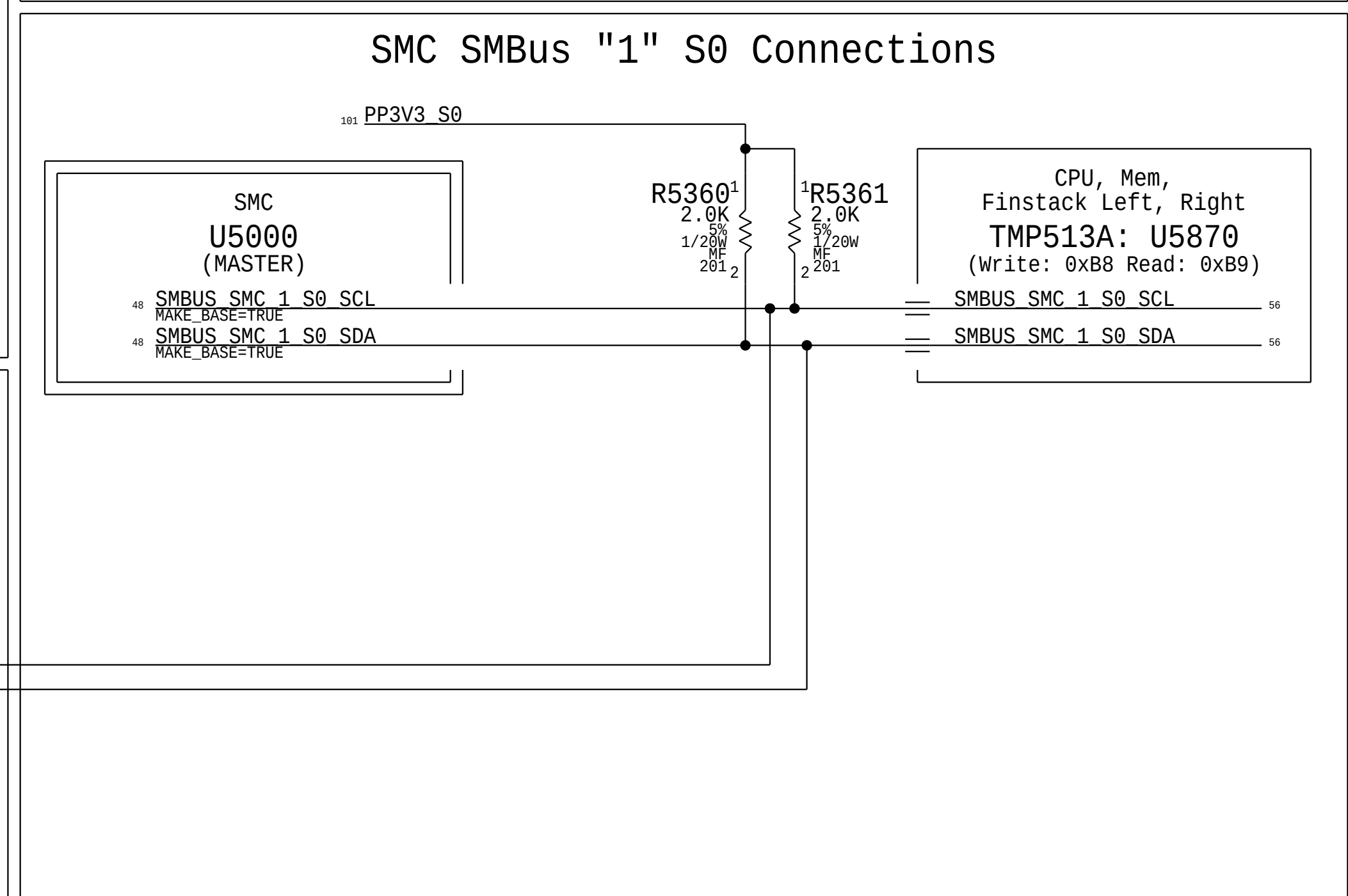
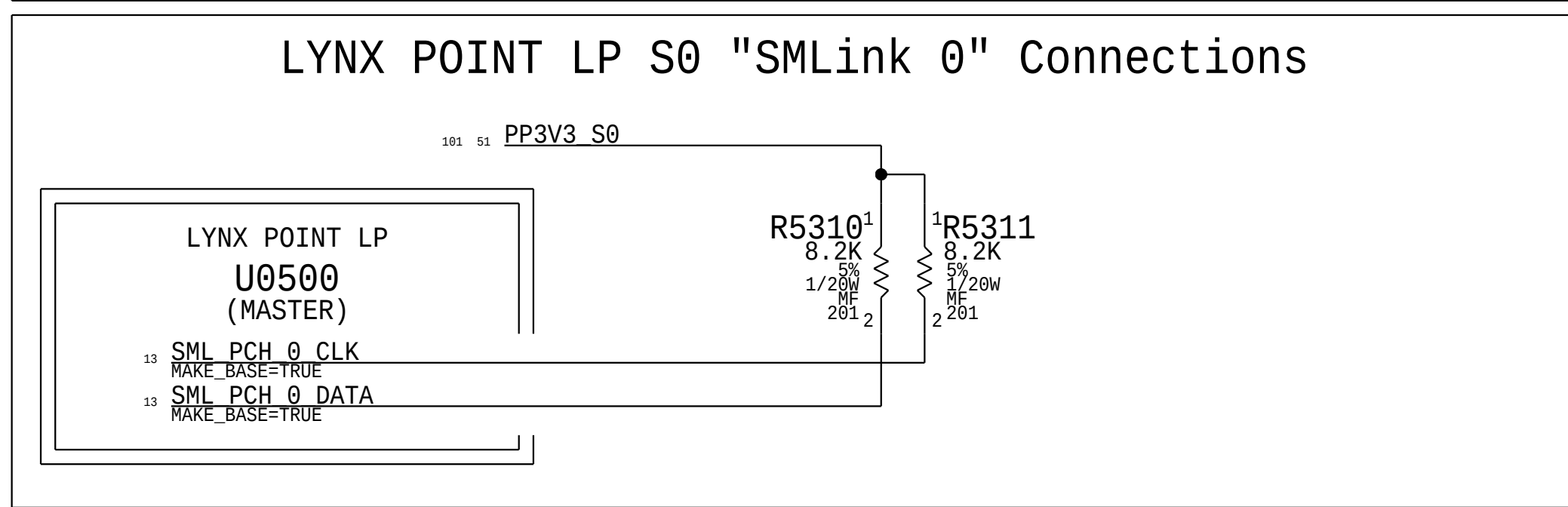
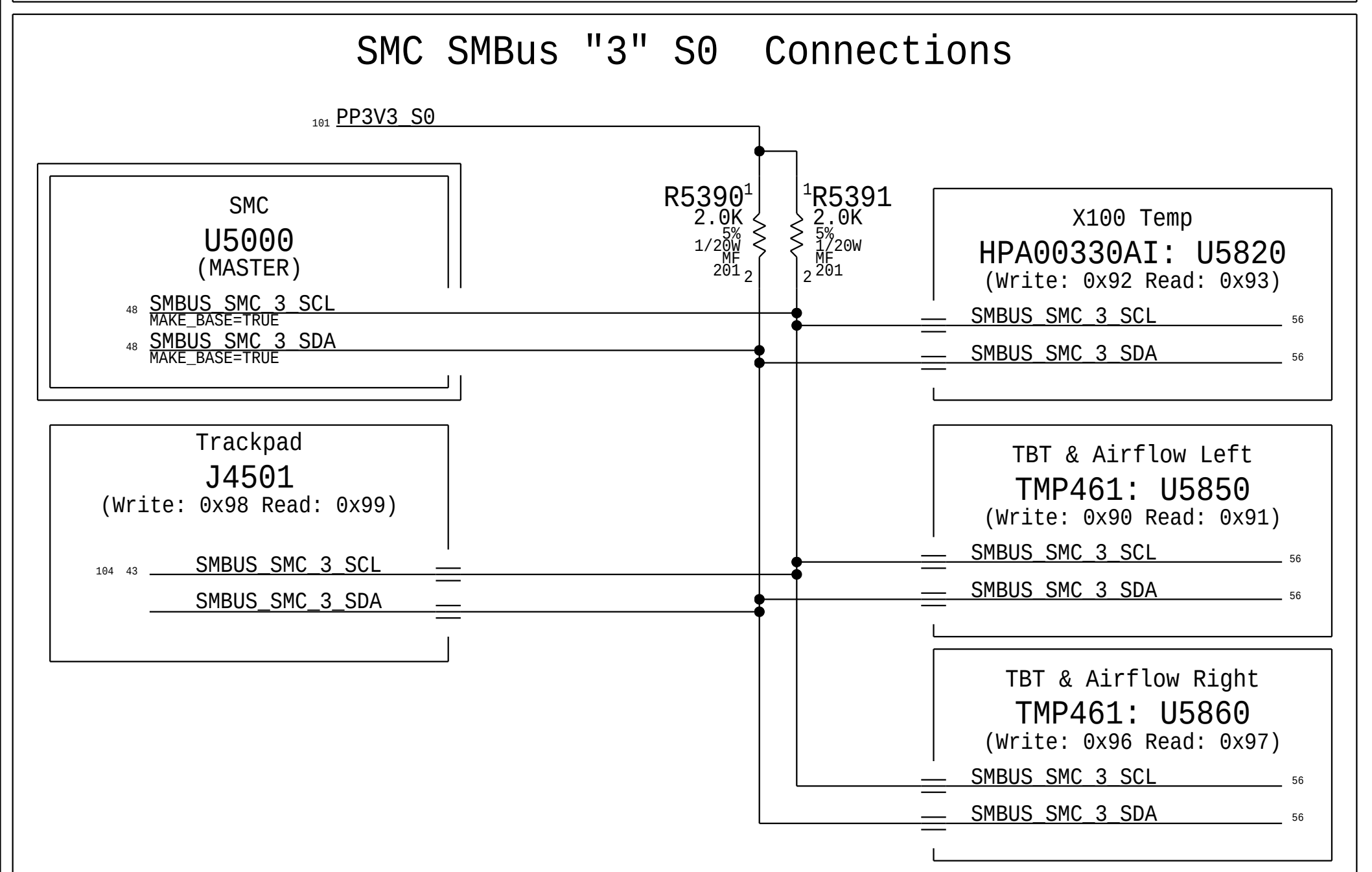
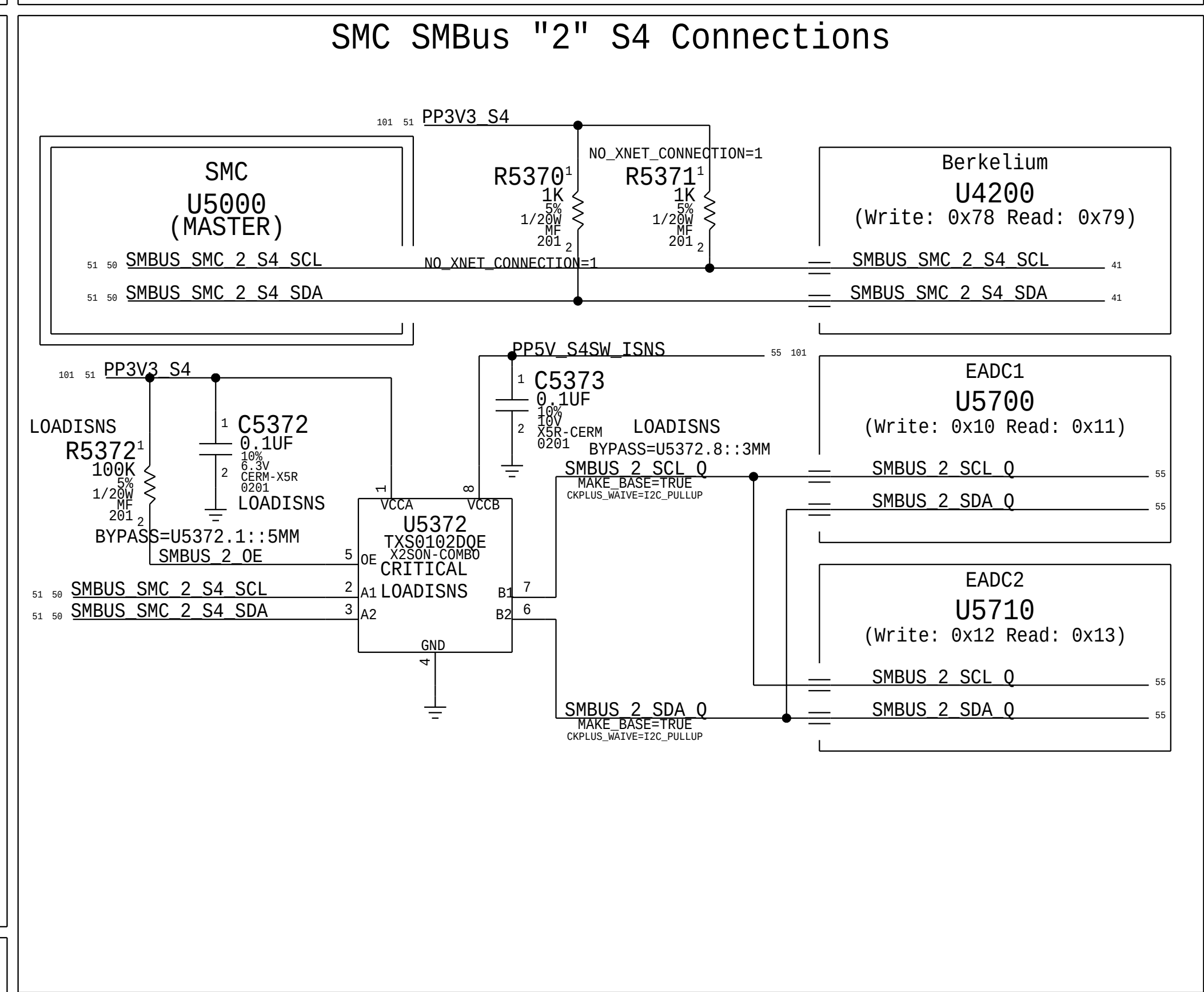
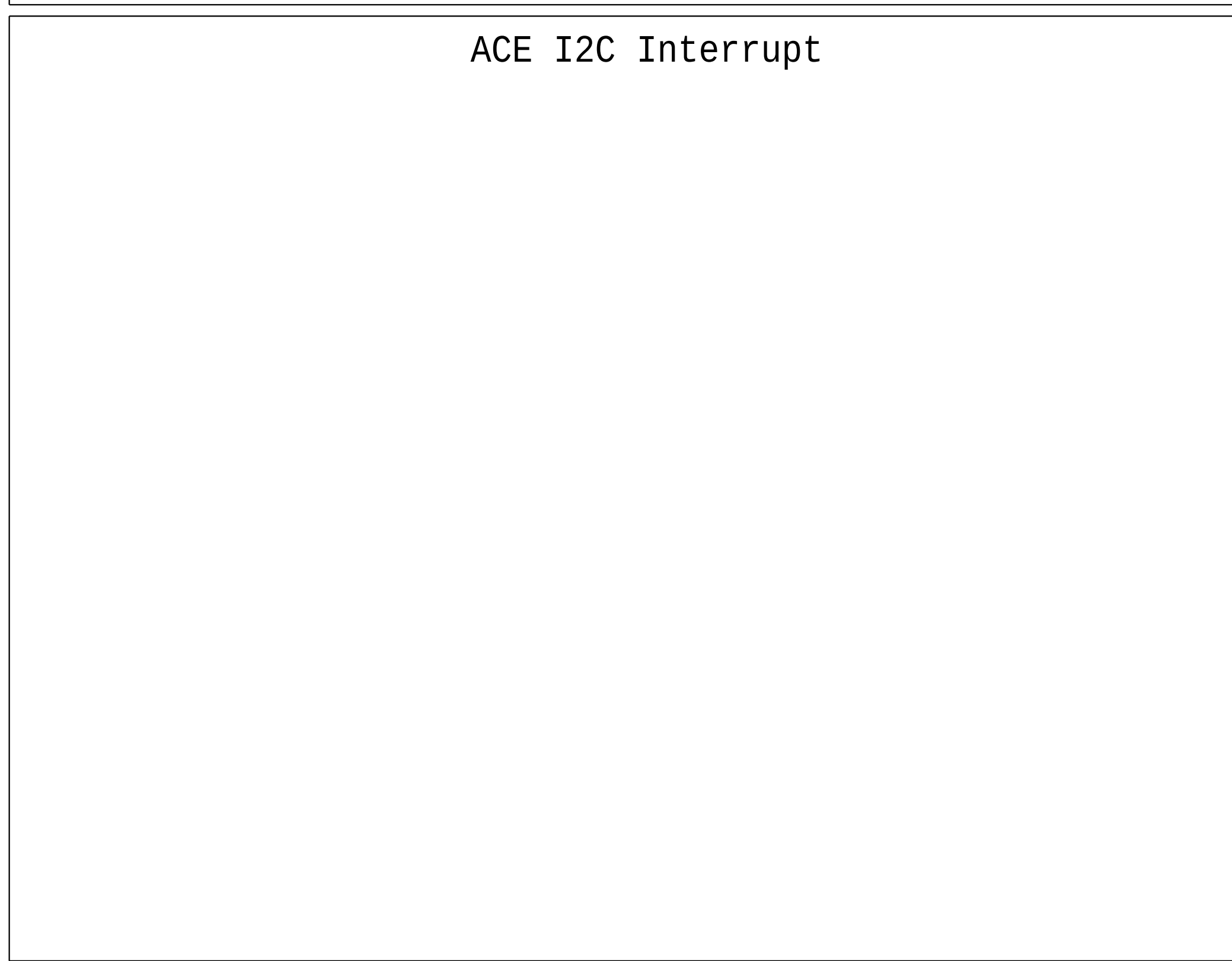
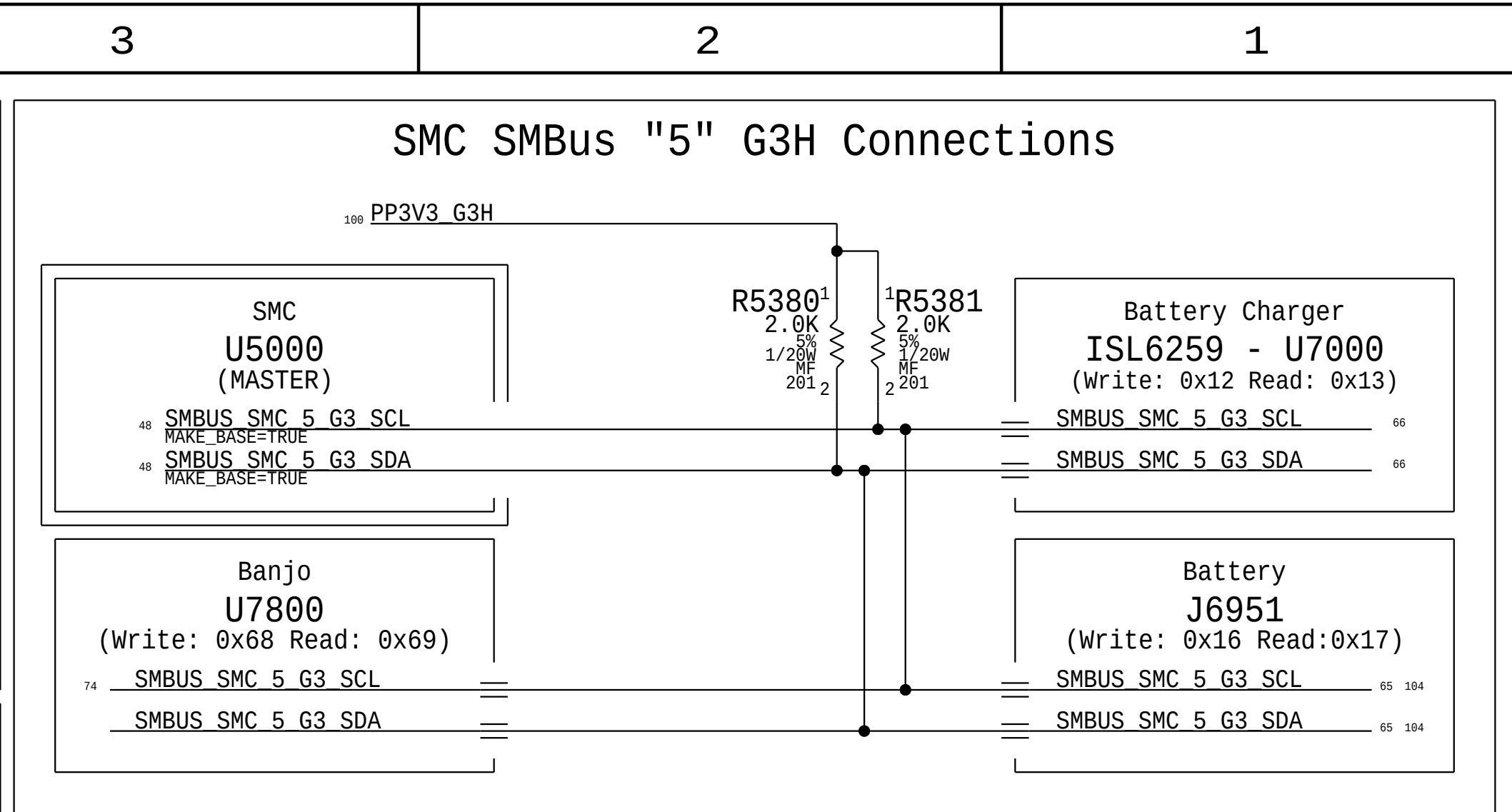
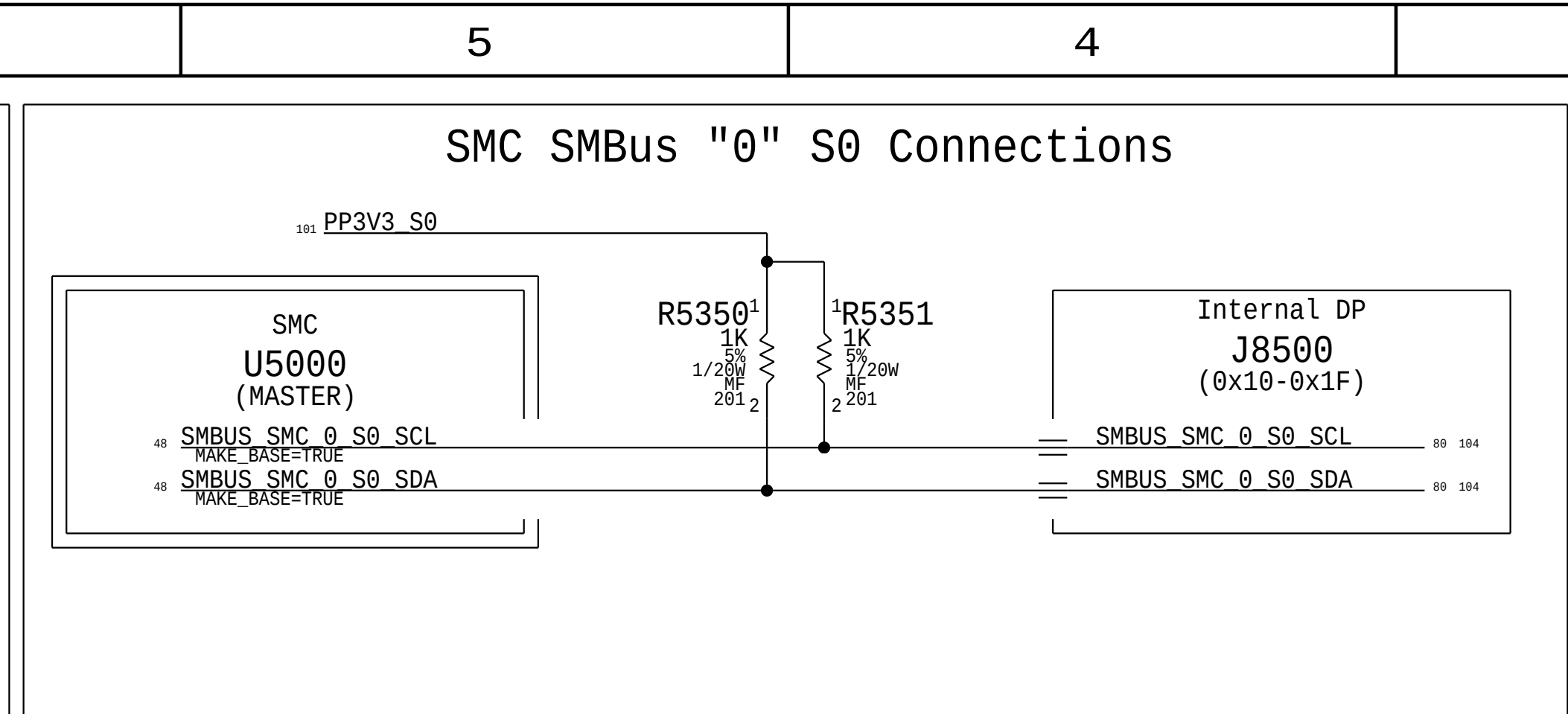
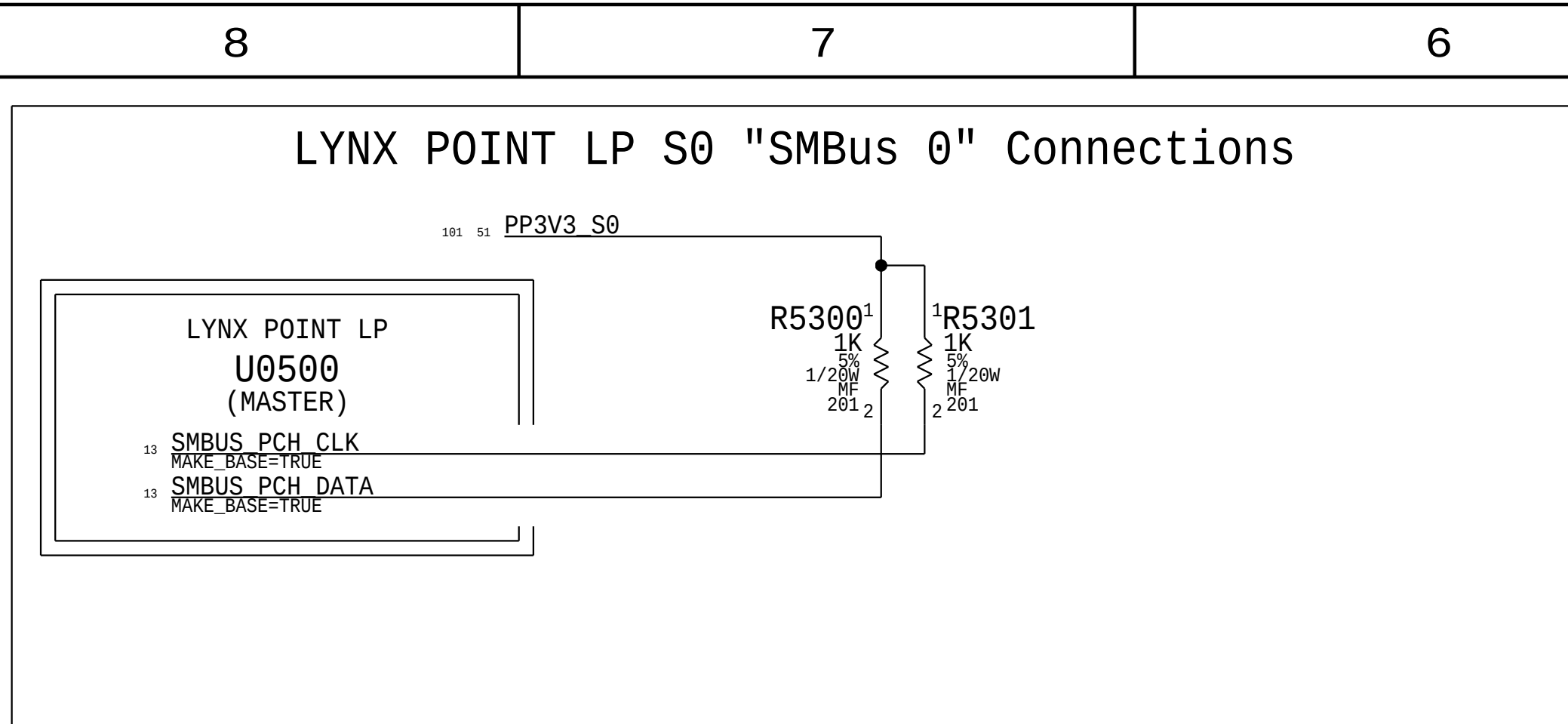
PROCHOT/THRMTRIP Support



PECI Support

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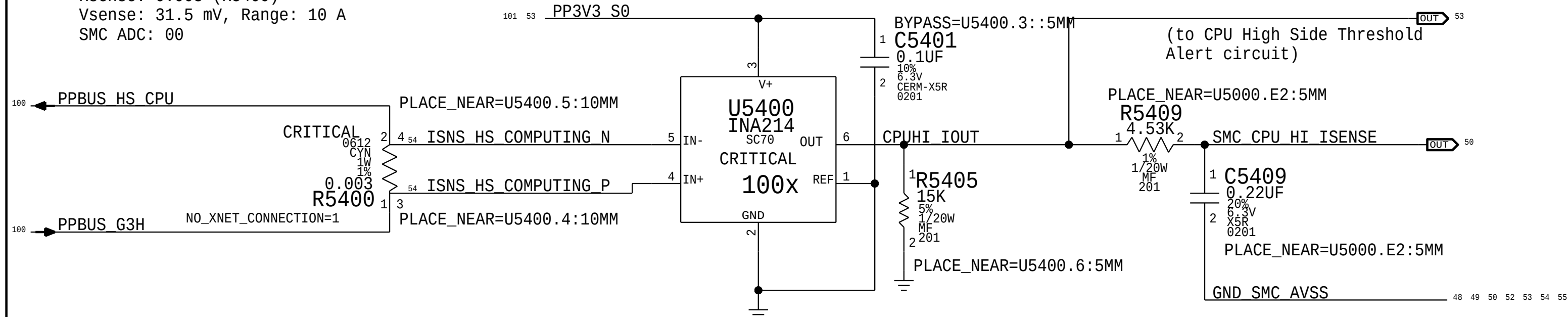
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		SHEET	51 OF 119	

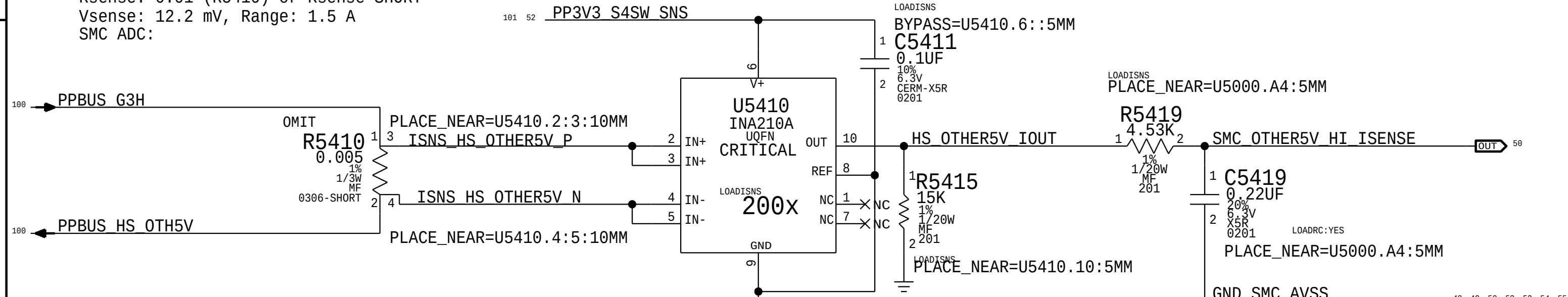
CPU High Side Current Sense (IC0R)

Gain: 100x, EDP: 10.5 A
Rsense: 0.003 (R5400)
Vsense: 31.5 mV, Range: 10 A
SMC ADC: 00



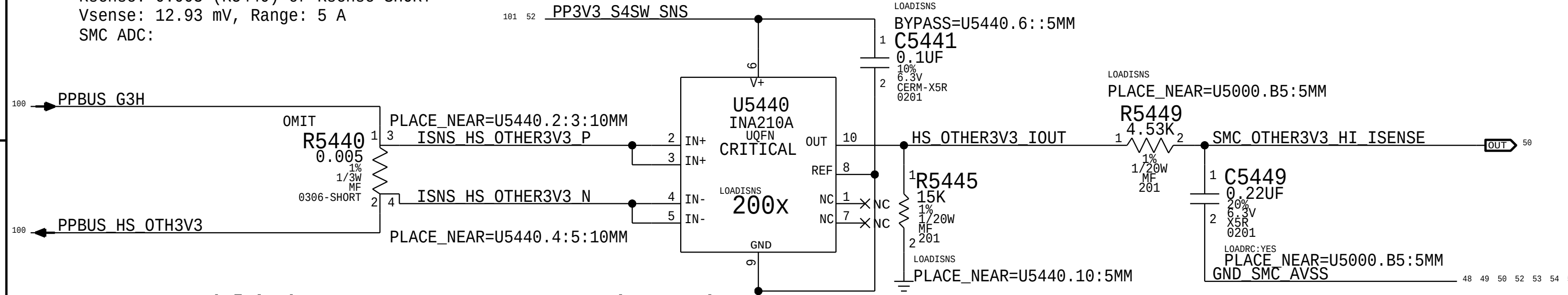
OTHER 5V High Side Current Sense (IO5R)

Gain: 200x, EDP: 1.22 A
Rsense: 0.01 (R5410) or Rsense SHORT
Vsense: 12.2 mV, Range: 1.5 A
SMC ADC:



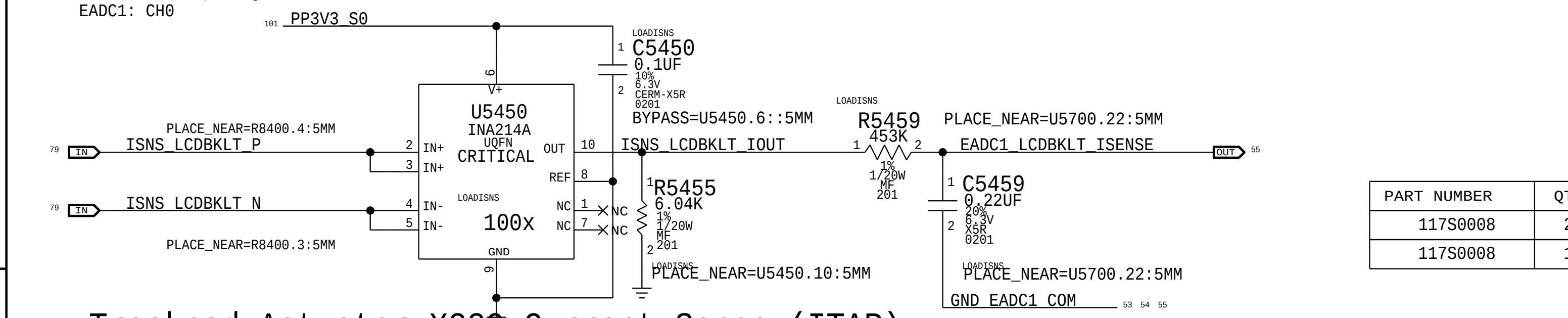
OTHER 3.3V High Side Current Sense (IO3R)

Gain: 200x, EDP: 4.31 A
Rsense: 0.003 (R5440) or Rsense SHORT
Vsense: 12.93 mV, Range: 5 A
SMC ADC:



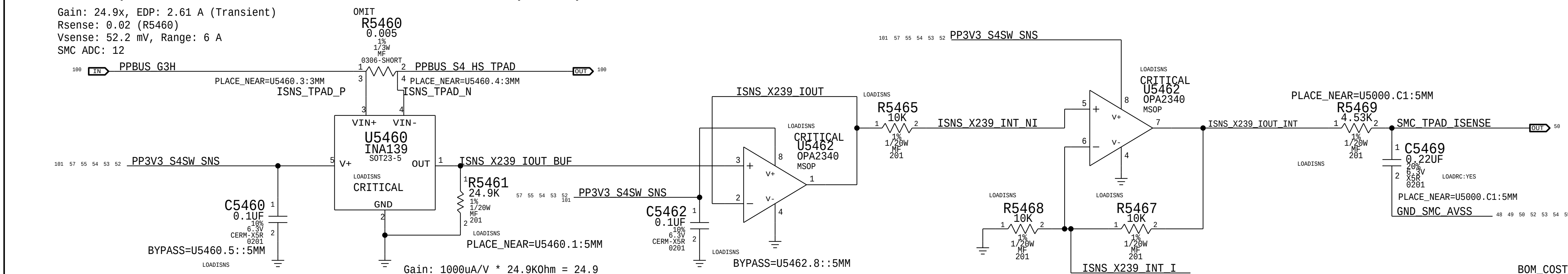
LCD Backlight Current Sense (IBLR)

Gain: 100x, EDP: 1 A
Rsense: 0.025 (R8400)
Vsense: 25 mV, Range: 2.4 A
EADC1: CH0



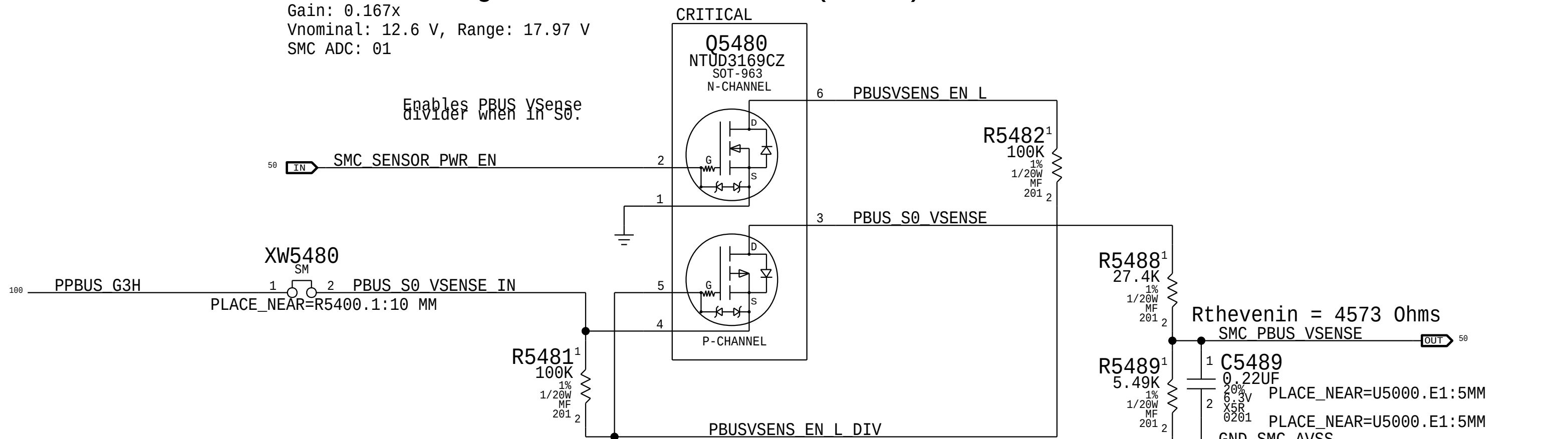
Trackpad Actuator X239 Current Sense (ITAR)

Gain: 24.9x, EDP: 2.61 A (Transient)
Rsense: 0.02 (R5460)
Vsense: 52.2 mV, Range: 6 A
SMC ADC: 12



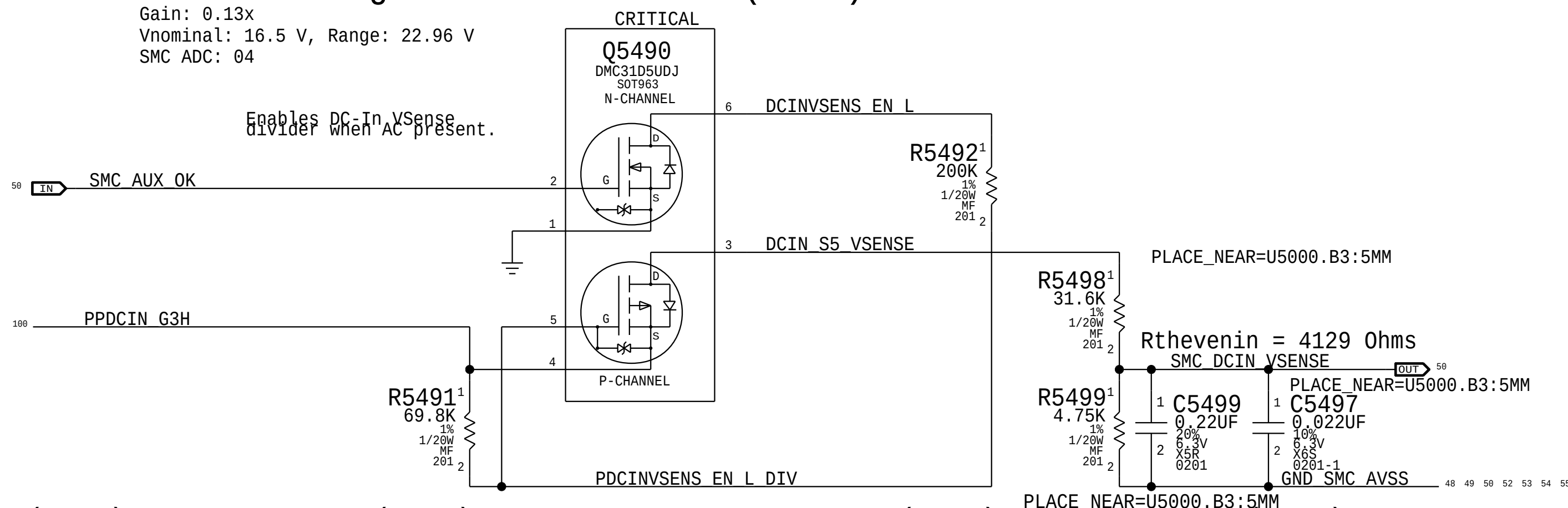
PBUS Voltage Sense & Enable (VP0R)

Gain: 0.167x
Vnominal: 12.6 V, Range: 17.97 V
SMC ADC: 01



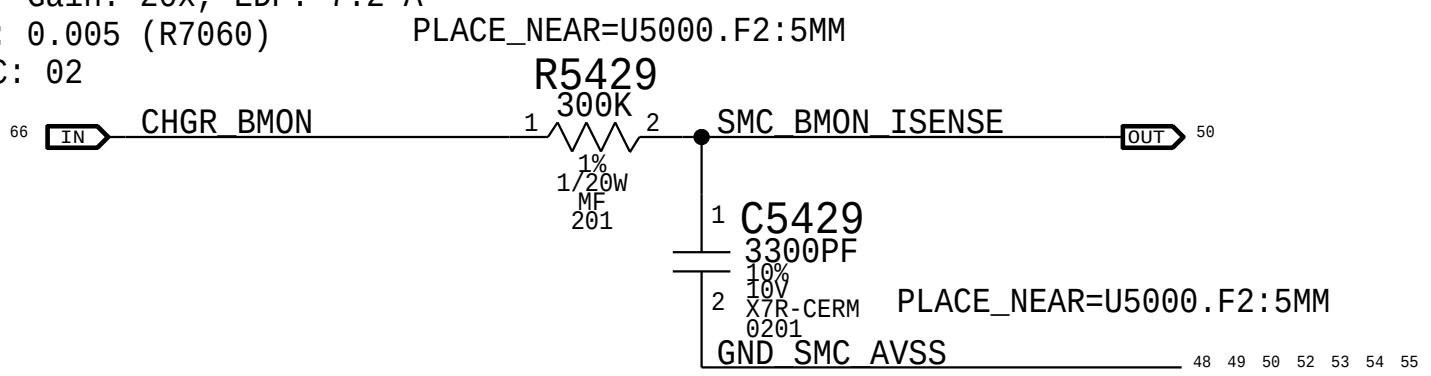
DC In Voltage Sense & Enable (VD0R)

Gain: 0.13x
Vnominal: 16.5 V, Range: 22.96 V
SMC ADC: 04



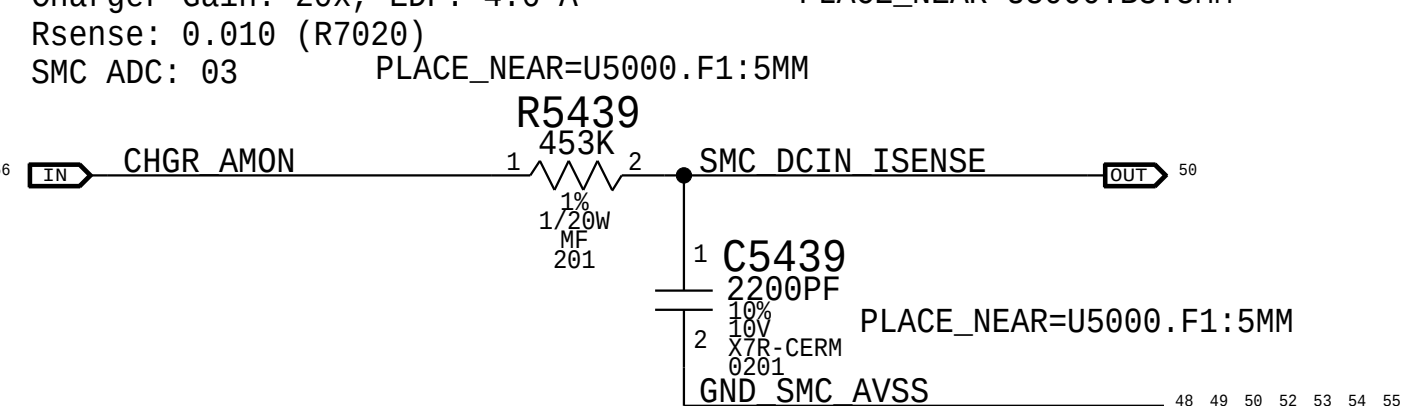
Charger (BMON) Current Sense (IPBR)

Charger Gain: 20x, EDP: 7.2 A
Rsense: 0.005 (R7060)
SMC ADC: 02



DC-IN (AMON) Current Sense (ID0R)

Charger Gain: 20x, EDP: 4.6 A
Rsense: 0.010 (R7020)
SMC ADC: 03

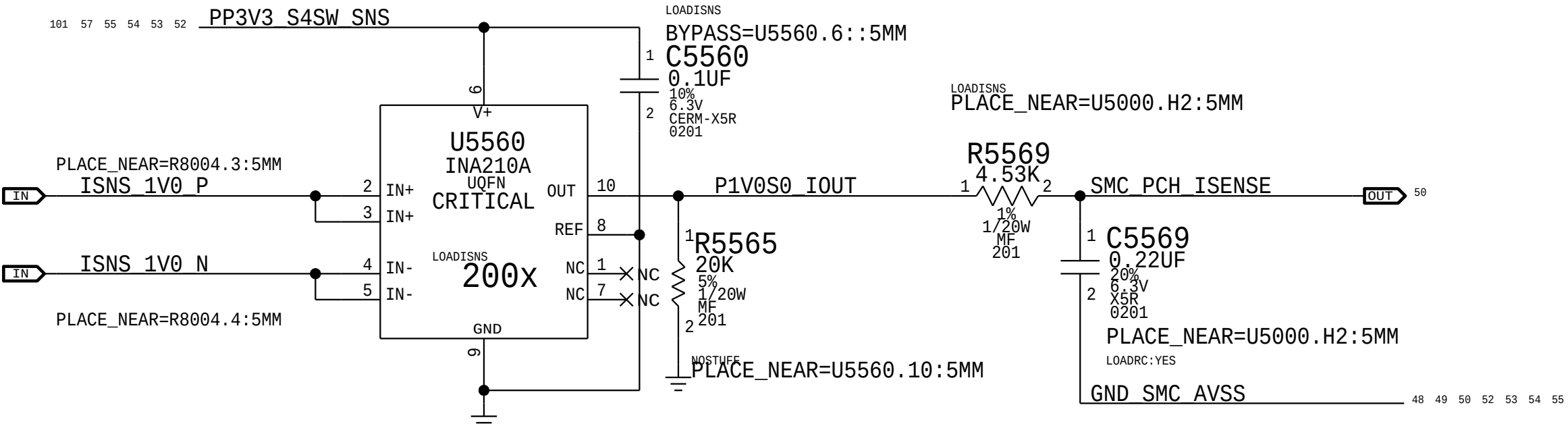


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
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117S0008	1	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	C5469		LOADRC:NO

Power Sensors: High Side			
Apple Inc.		DRAWING NUMBER	051-00515
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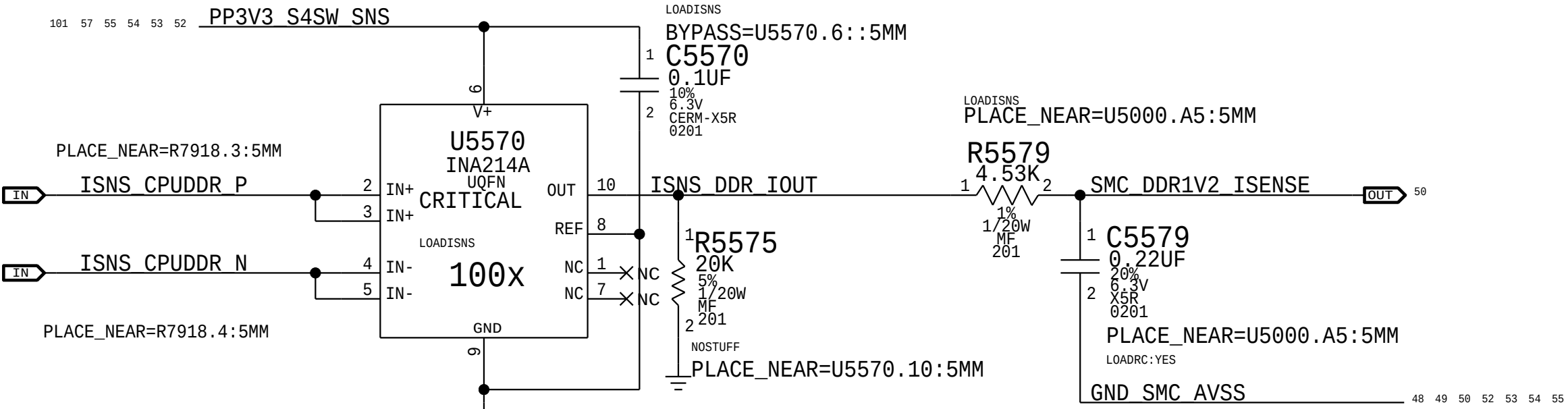
PCH 1.0V Current Sense (IS1C)

Gain: 200x, EDP: 3.29 A
Rsense: 0.003 (R8004) or Rsense SHORT
Vsense: 9.87 mV, Range: 5 A
SMC ADC: 11



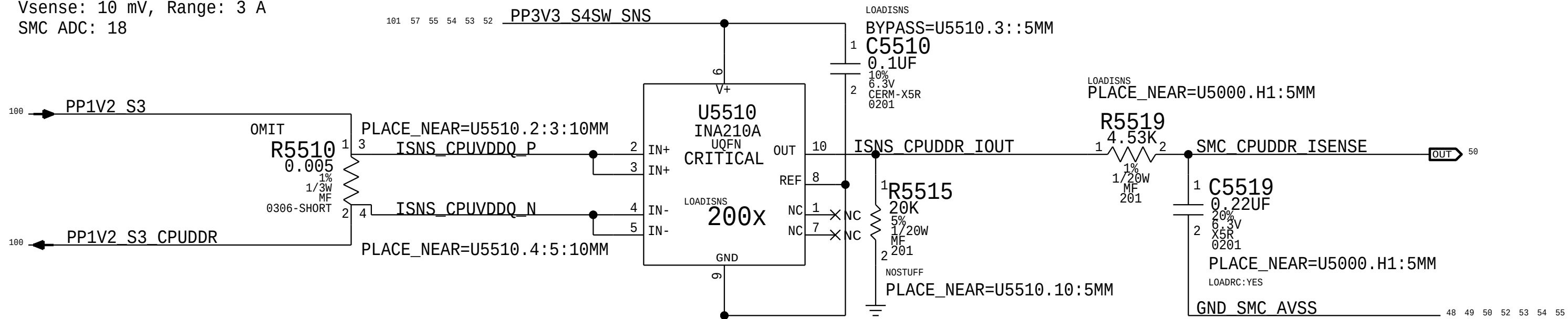
DDR 1.2V S3 (CPU & Memory) Current Sense (IM0C)

Gain: 100x, EDP: 8.21 A
Rsense: 0.003 (R7918) or XWTBD
Vsense: 24.63 mV, Range: 10 A
SMC ADC: 09



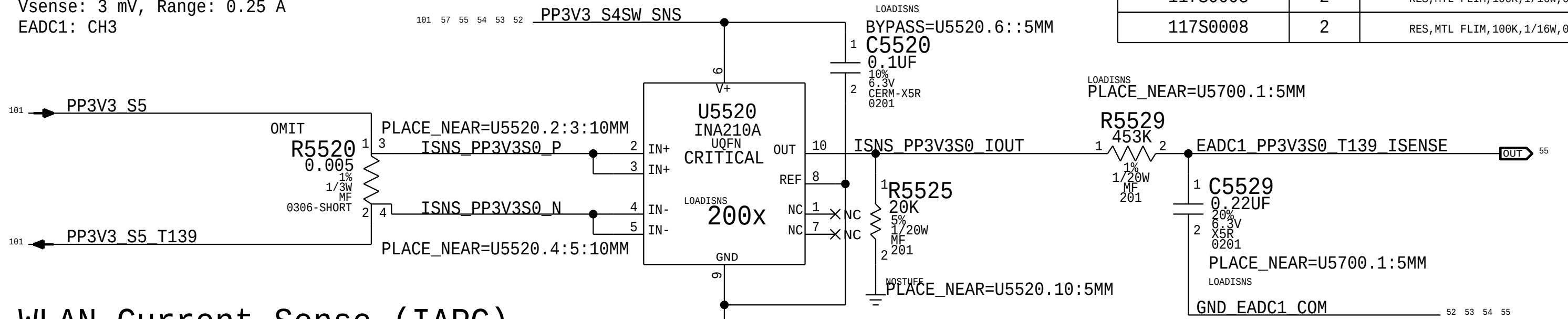
CPU DDR 1.2V S3 (CPU Only) Current Sense (IMCC)

Gain: 200x, EDP: 2 A
Rsense: 0.005 (R5510) or Rsense SHORT
Vsense: 10 mV, Range: 3 A
SMC ADC: 18



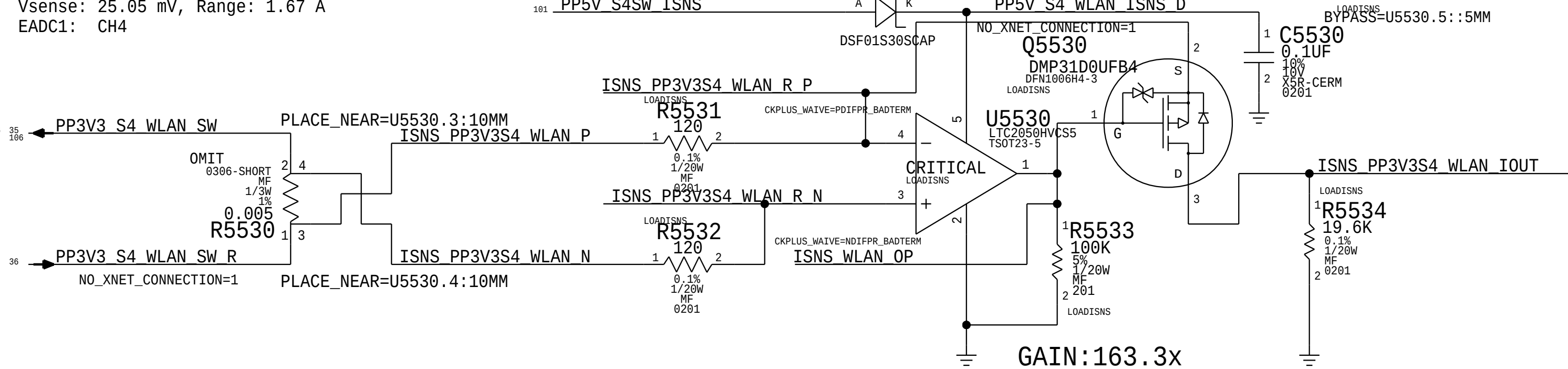
T139 Current Sense (IF3C)

Gain: 200x, EDP: 0.06 A
Rsense: 0.05 (R5520) or Rsense SHORT
Vsense: 3 mV, Range: 0.25 A
EADC1: CH3



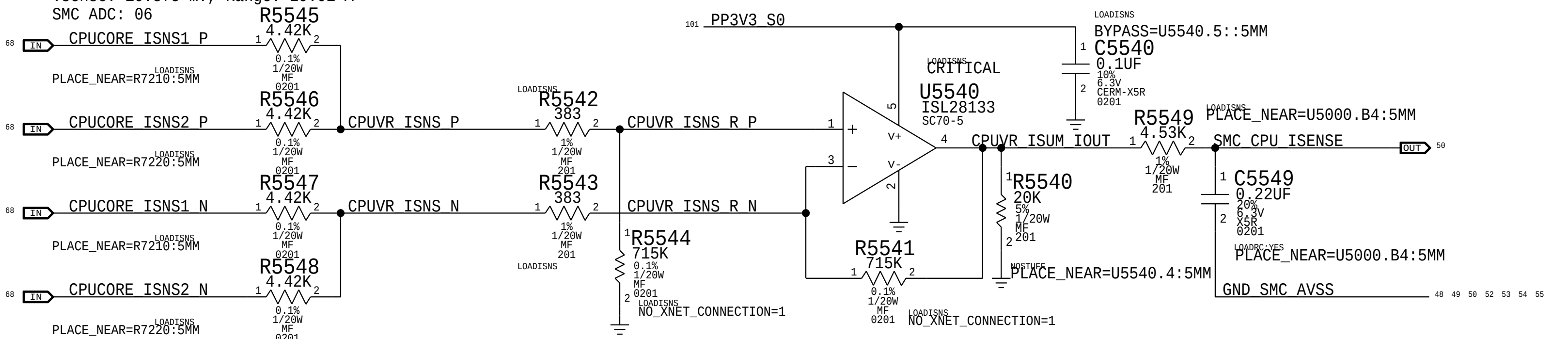
WLAN Current Sense (IAPC)

Gain: 163.3x, EDP: 1.67 A
Rsense: 0.015 (R5530) or Rsense SHORT
Vsense: 25.05 mV, Range: 1.67 A
EADC1: CH4



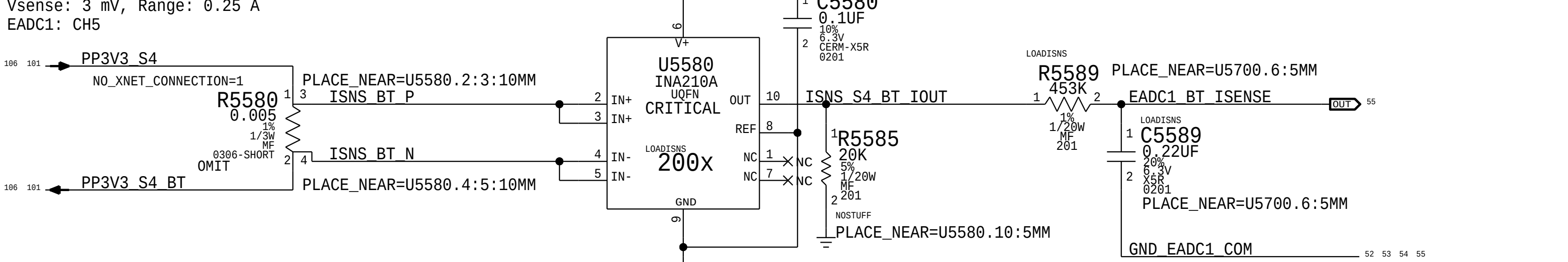
CPU Fixed Current Sense (ICAC)

Gain: 275.74x, EDP: 29 A
Rsense: 2x of 0.00075 (R7310, R7320), Rsum: 0.000375
Vsense: 10.875 mV, Range: 29.01 A
SMC ADC: 06



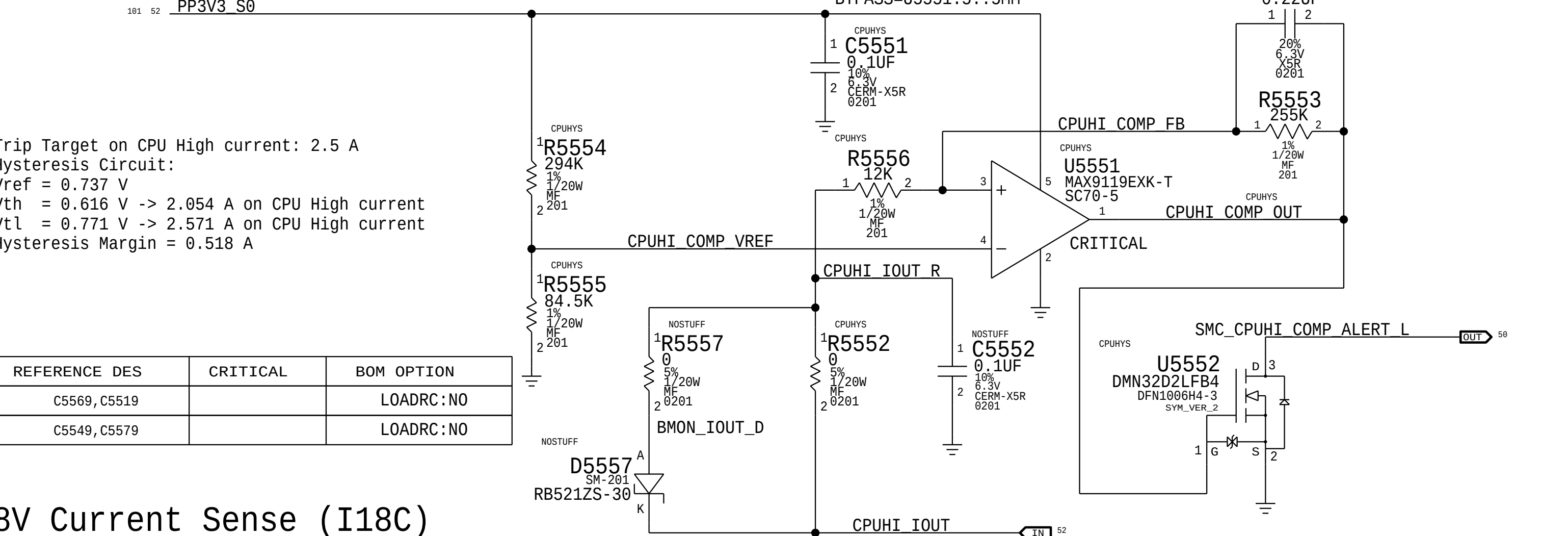
BT Current Sense (IBTC)

Gain: 200x, EDP: 0.06 A
Rsense: 0.05 (R5580)
Vsense: 3 mV, Range: 0.25 A
EADC1: CH5



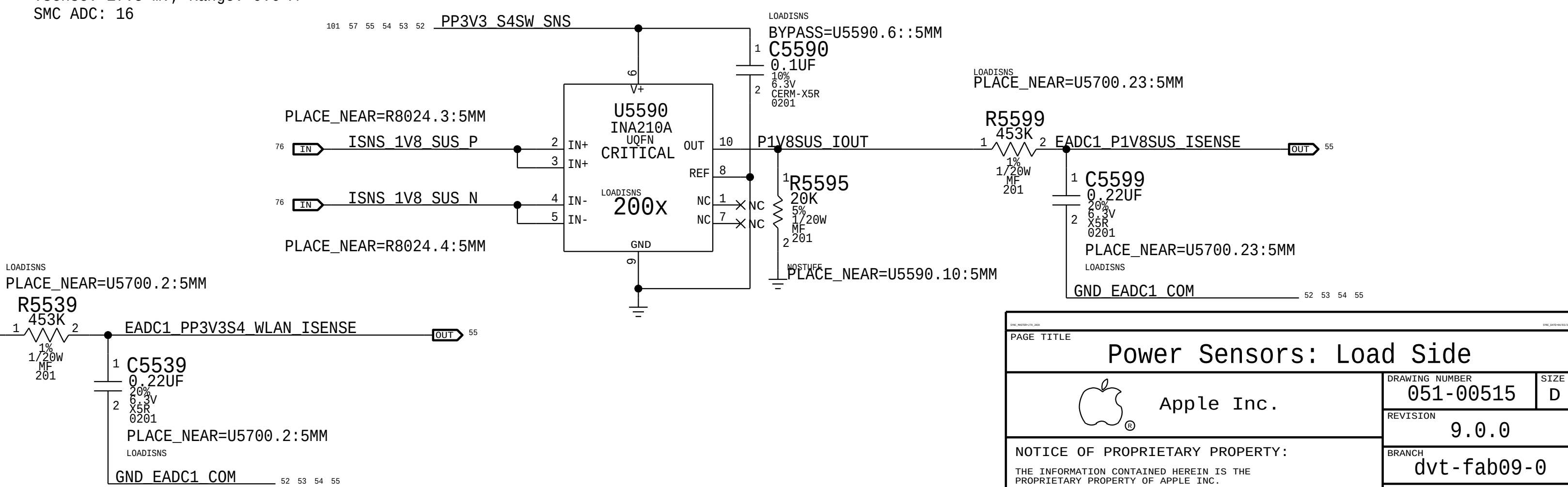
CPU High Side Current (IC0R) Threshold Alert

Gain: 100x
Rsense: 0.003 (R5400)



1.8V Current Sense (I18C)

Gain: 200x, EDP: 0.7 A
Rsense: 0.025 (R8024) or Rsense SHORT
Vsense: 17.5 mV, Range: 0.6 A
SMC ADC: 16

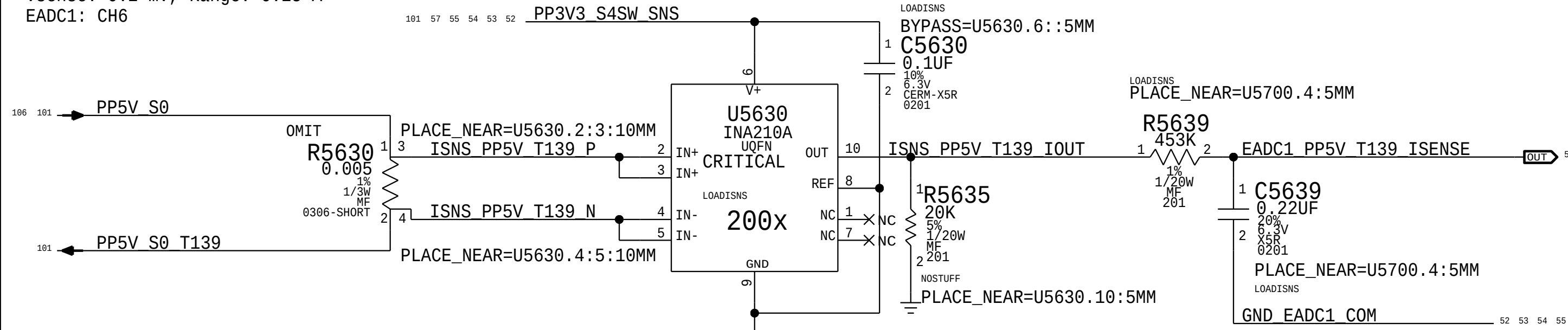


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5569,C5519		LOADRC:NO
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5549,C5579		LOADRC:NO

PAGE TITLE		DRAWING NUMBER		SIZE
Power Sensors: Load Side		051-00515		D
Apple Inc.		REVISTION		9.0.0
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		PAGE		55 OF 145
		SHEET		53 OF 119

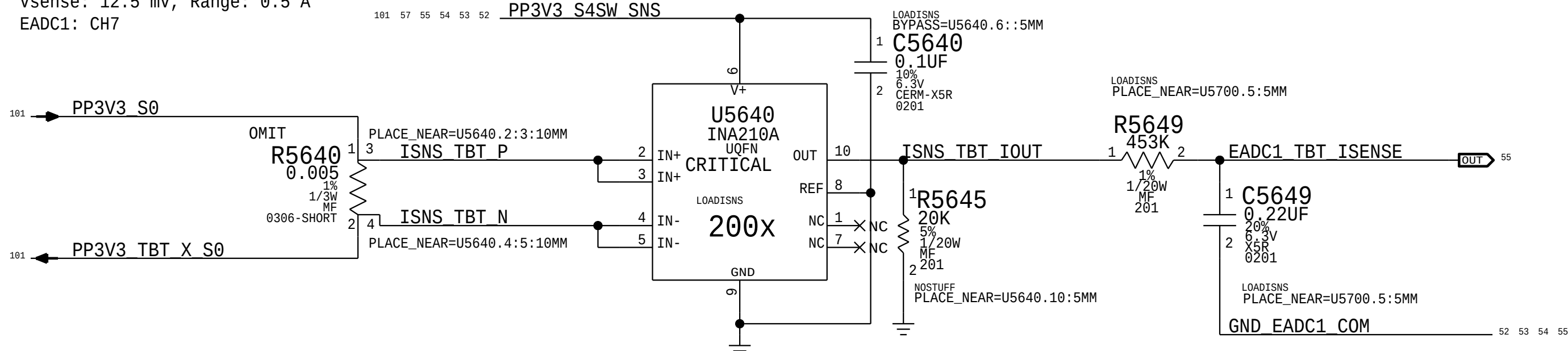
T139 5V Current Sense (IF5C)

Gain: 200x, EDP: 0.004 A
Rsense: 0.05 (R5630) or Rsense SHORT
Vsense: 0.2 mV, Range: 0.25 A
EADC1: CH6



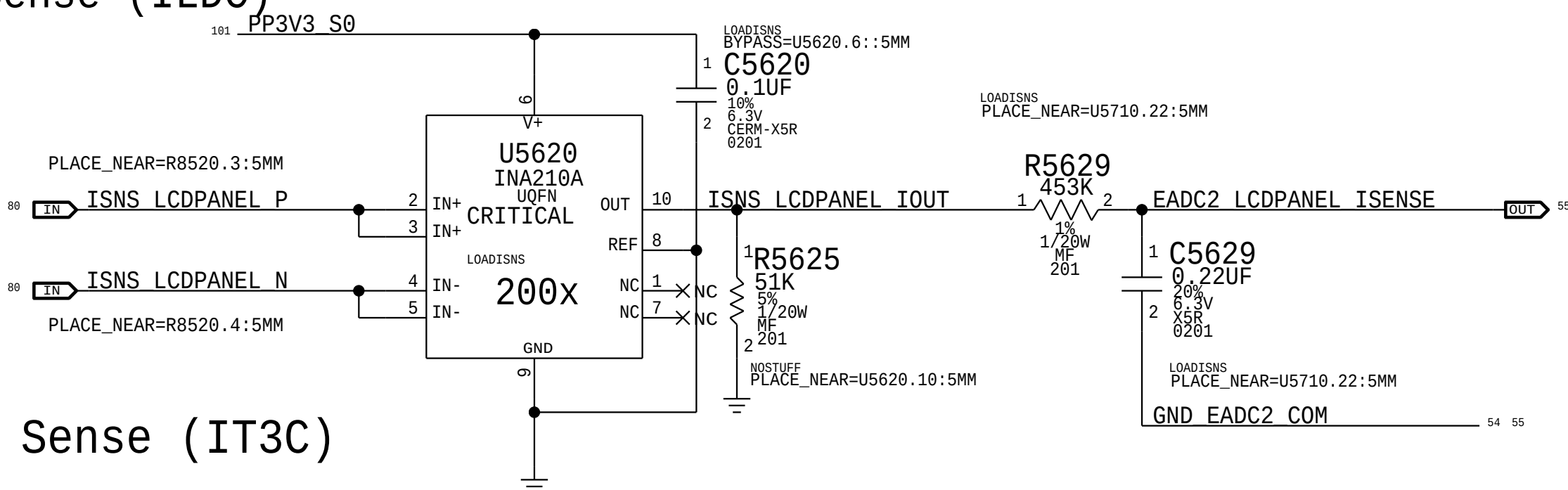
Thunderbolt TBT Current Left (IULC)

Gain: 200x, EDP: 0.5 A
Rsense: 0.025 (R5640) or Rsense SHORT
Vsense: 12.5 mV, Range: 0.5 A
EADC1: CH7



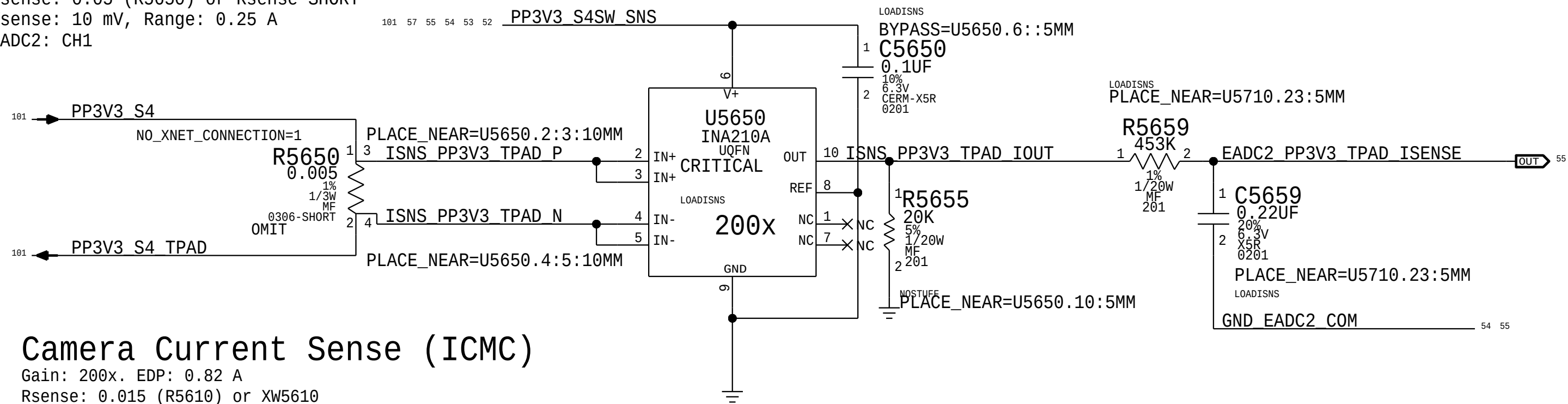
LCD Panel Current Sense (ILDC)

Gain: 200x, EDP: 1 A
RSense: 0.01 (R8520) or Rsense SHORT
Vsense: 5 mV, Range: 1.25 A
EADC2: CH0



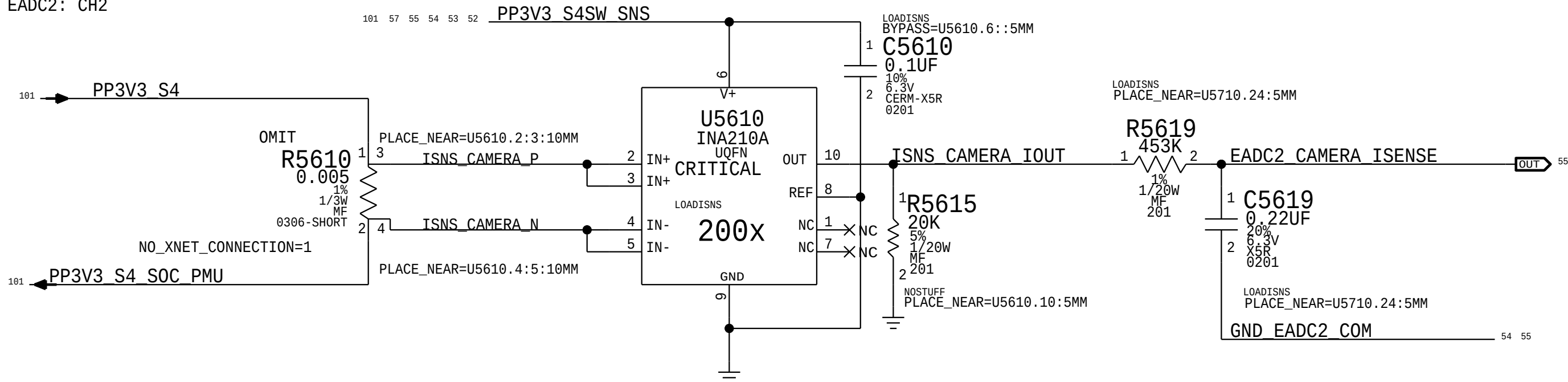
Trackpad 3V Current Sense (IT3C)

Gain: 200x, EDP: 0.2 A
Rsense: 0.05 (R5650) or Rsense SHORT
Vsense: 10 mV, Range: 0.25 A
EADC2: CH1

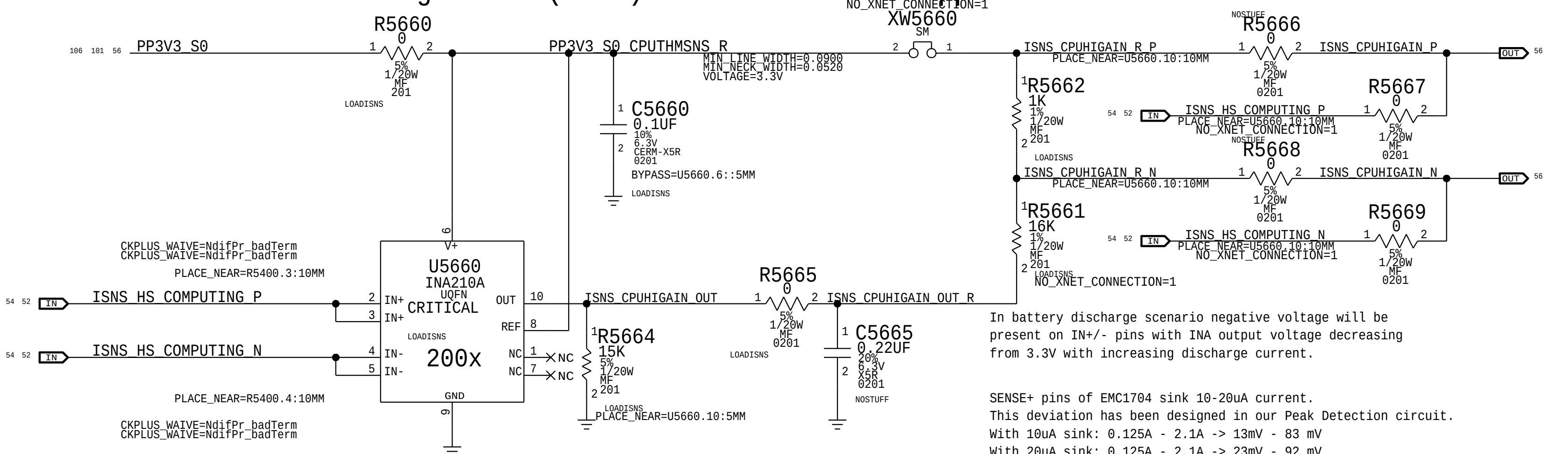


Camera Current Sense (ICMC)

Gain: 200x, EDP: 0.82 A
Rsense: 0.015 (R5610) or XW5610
Vsense: 12.3 mV, Range: 0.83 A
EADC2: CH2

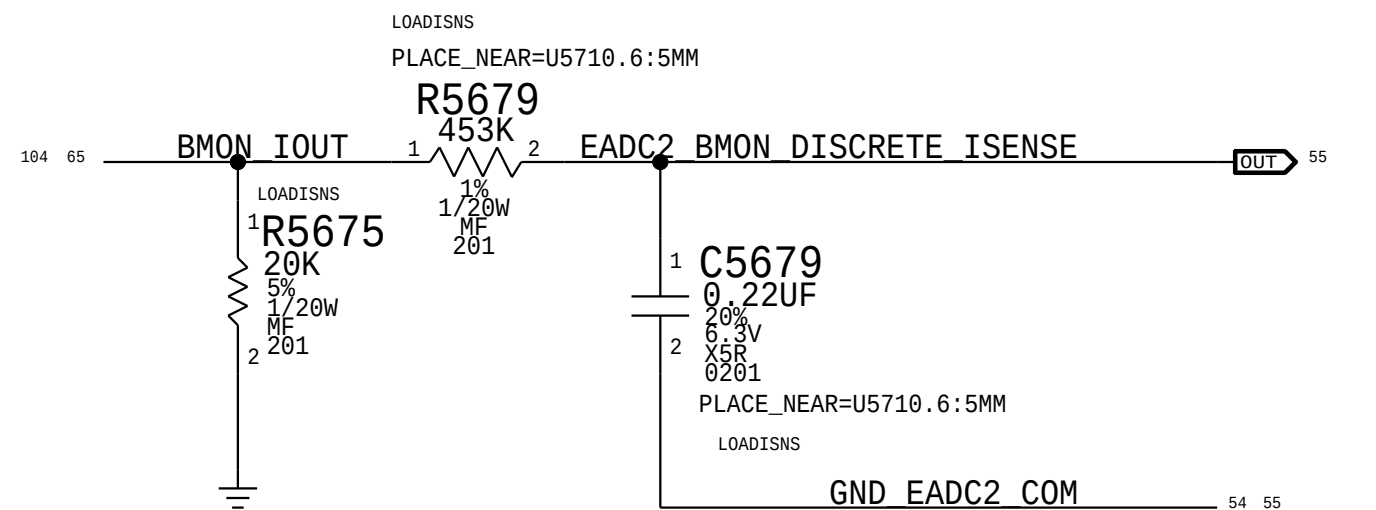


CPU High Side (IC0R) Peak Detection Support



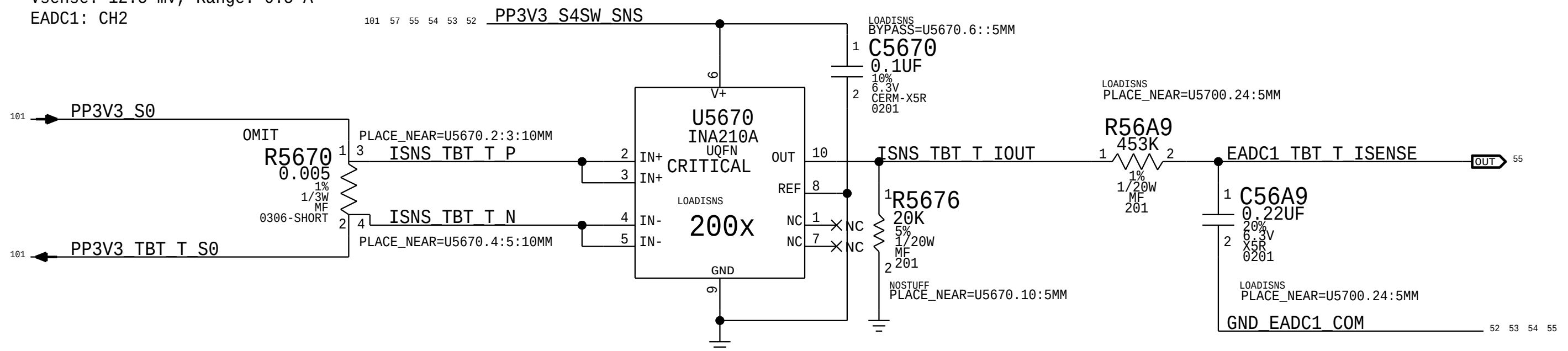
Battery Discrete Current Sense (IB0L)

Gain: 2940x, EDP: 8 A
Rsense: 0.003 (R5670) or Rsense SHORT
Vsense: 24 mV, Range: 0.28 A
EADC2: CH5



Thunderbolt TBT Current Right (IURC)

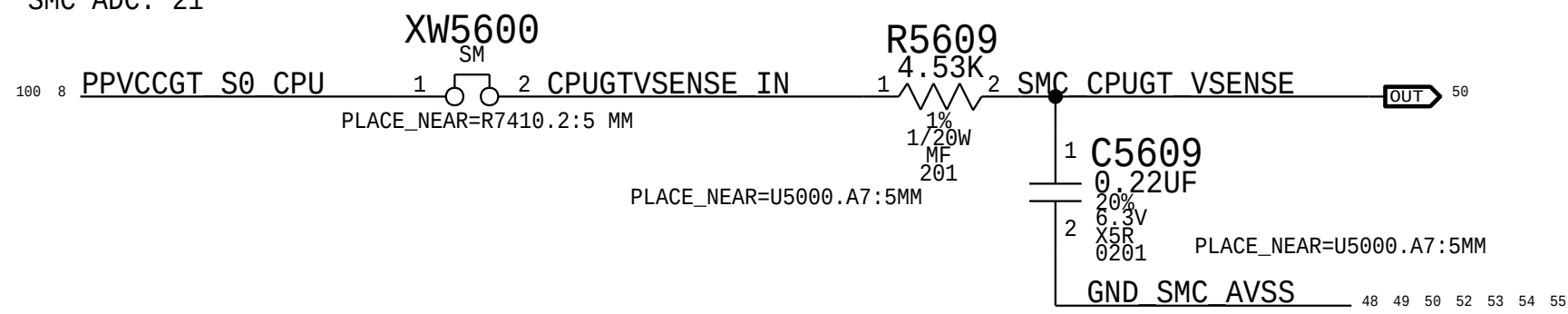
Gain: 200x, EDP: 0.5 A
Rsense: 0.025 (R5670) or Rsense SHORT
Vsense: 12.5 mV, Range: 0.5 A
EADC1: CH2



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
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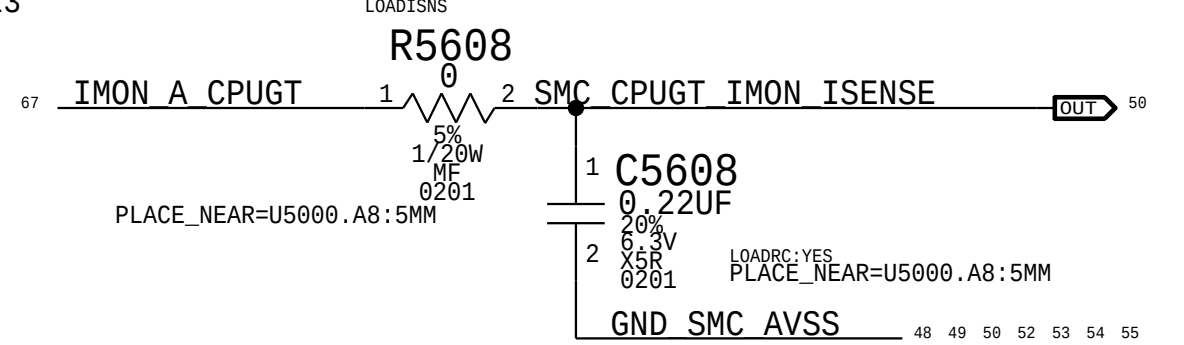
CPU GT Voltage Sense (VCGC)

SMC ADC: 21



CPU GT IMON Current Sense (ICGM)

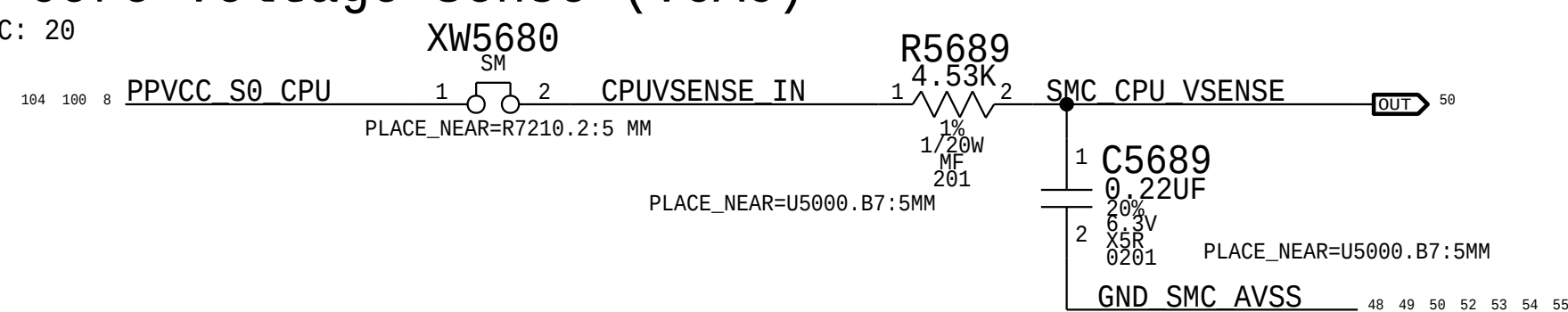
Gain: 1 A / 17.963 mV, Range: 64 A.
SMC ADC: 23



With R7154 (Ri) set to 294 Ohm,
R7410 (Rsen) set to 0.75 mOhm,
R7194 set to 84.5 kOhm,
Num Phases (N) is 3, and Io (ICMax) is 64A,
then 1A of Io gives 17.963mV at the Vimon.

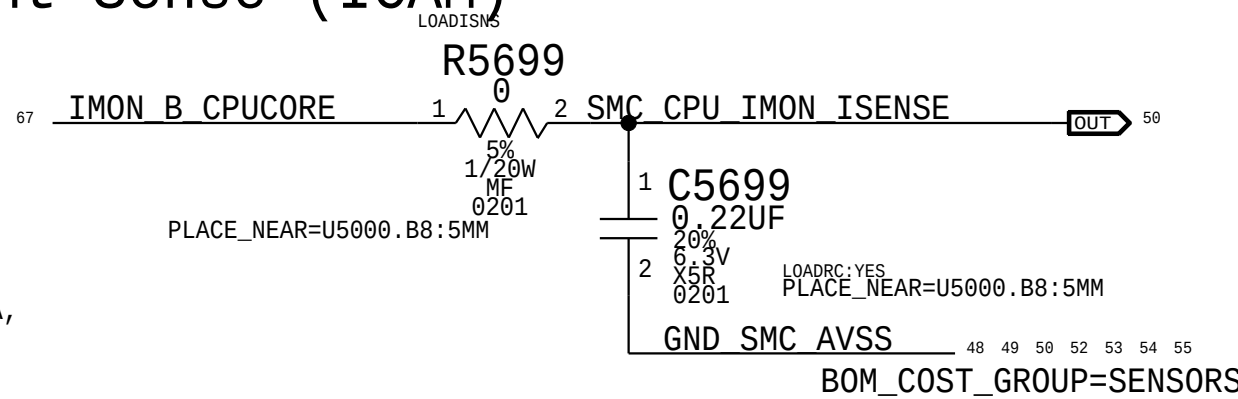
CPU Core Voltage Sense (VCAC)

SMC ADC: 20



CPU Core IMON Current Sense (ICAM)

Gain: 1 A / 34.223 mV, Range: 29 A.
SMC ADC: 22



With R7150 (Ri) set to 226 Ohm,
R7210 (Rsen) set to 0.75 mOhm,
R7160 set to 82.5 kOhm,
Num Phases (N) is 2, and Io (ICMax) is 29A,
then 1A of Io gives 34.223mV at the Vimon.

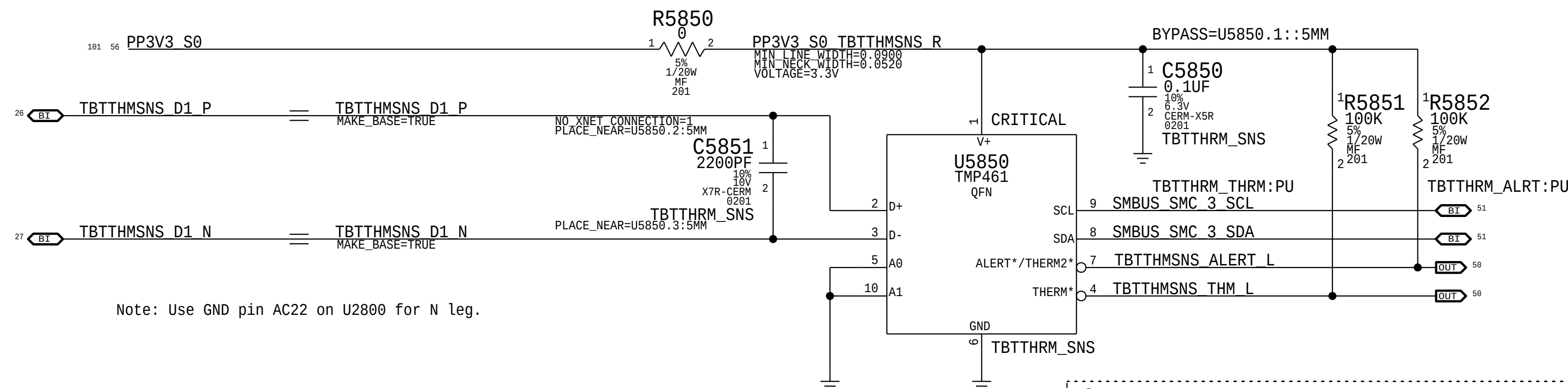
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Power Sensors: Extended			
 Apple Inc.	DRAWING NUMBER	SIZE	
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	BRANCH	dvt-fab09-0	
	PAGE	56 OF 145	
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Thermal Sensor A:
Thunderbolt Die, Airflow Left

I2C Write: 0xD8, I2C Read: 0xD9

Thermal Diode: TBT Die (TBT1)

Placement Note:
The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AC22.



U5850 I2C Address: TMP461 is 0x90/0x91.

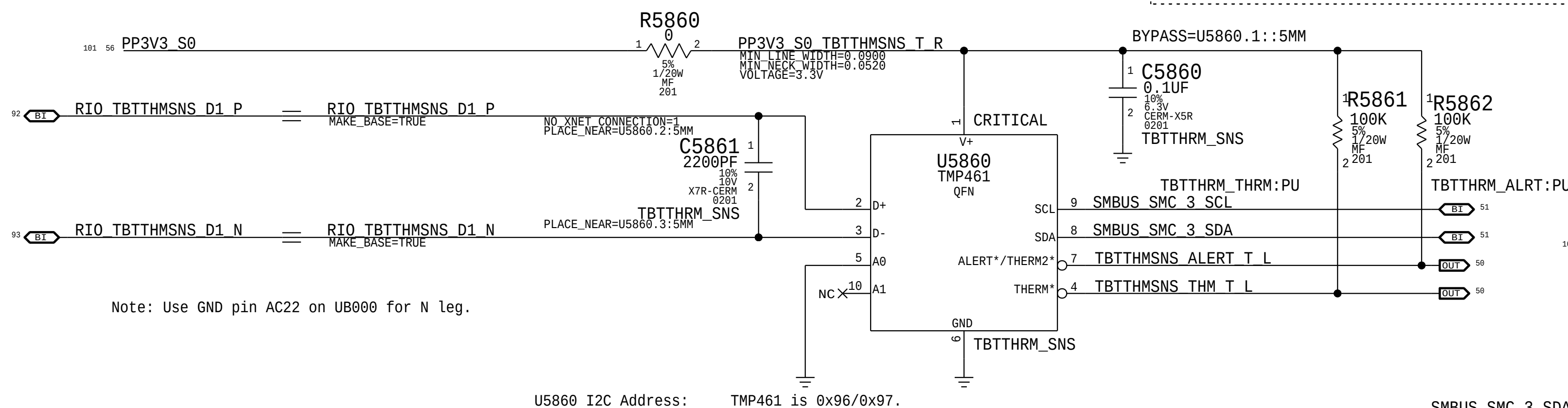
Note: Use GND pin AC22 on U2800 for N leg.

Thermal Sensor C:
Thunderbolt Die, Air Flow Right

I2C Write: 0xB8, I2C Read: 0xB9

Thermal Diode: TBT Die (TBT2)

Placement Note:
The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AC22.



U5860 I2C Address: TMP461 is 0x96/0x97

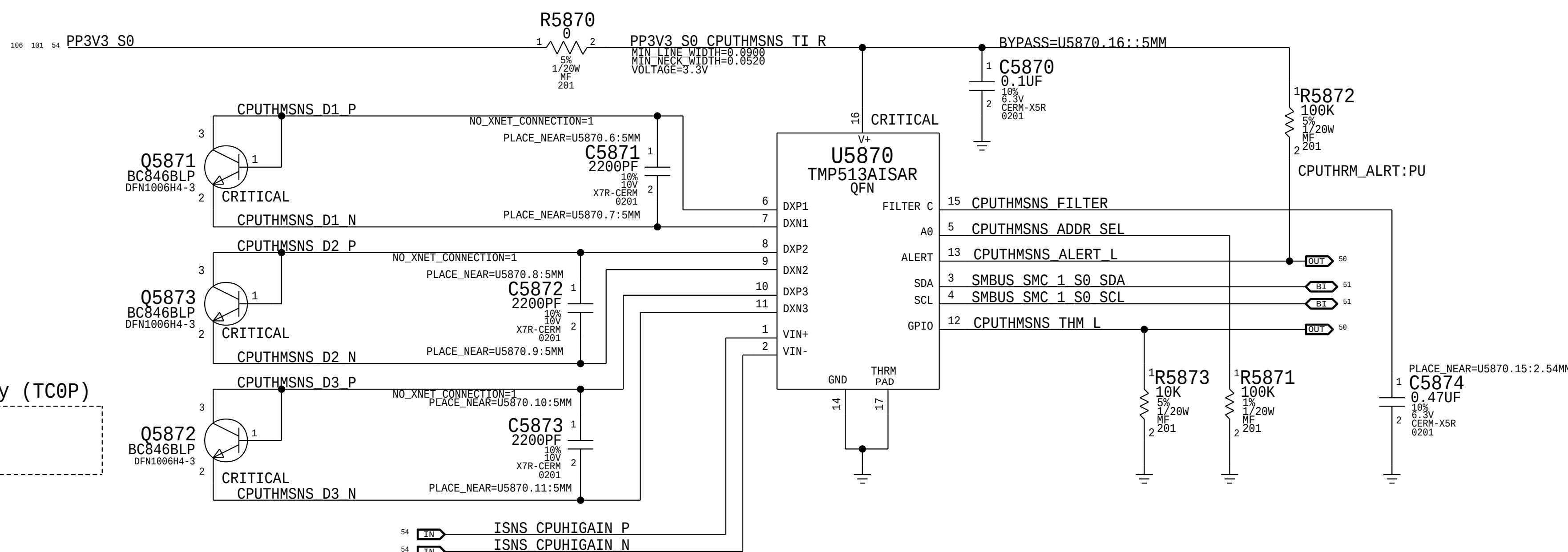
Placement Note:
Place U5860 on the BOTTOM side, on the right portion
of the board, 1" to the left of USB connector.

Thermal Sensor B & CPU High Peak Detection:
CPU Proximity, Memory Proximity, Fin Stack Left, Fin Stack Right

I2C Write: 0xB8, I2C Read: 0xB9

Thermal Diode: Fin Stack Left (Th2H)

Placement Note:
Place Q5871, Airflow thermal indicator, above
the X100, on the TOP side.



Placement note:
PLACE U5820 ON BOTTOM NEAR X100

Thermal Diode: CPU Proximity (TCOP)

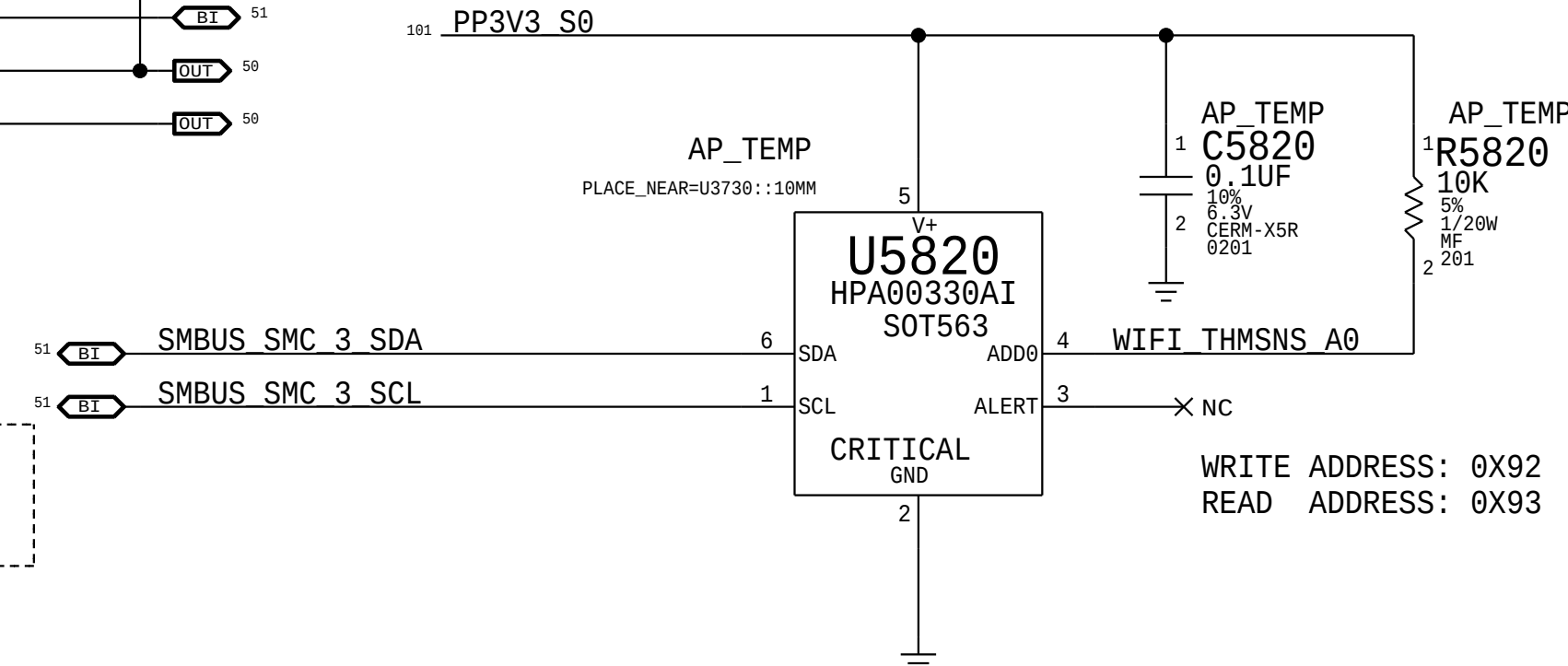
Placement Note:
Place Q5873 under the CPU,
on the BOTTOM side.

Thermal Diode: Memory Proximity (TM0P)

Placement Note:
Place Q5872 between two rows of Memory devices,
between channel A and B, on the BOTTOM side.

Thermal Sensor: Fin Stack Right (Th1H)

Placement Note:
Place U5870 at corner near right Fan,
on the TOP side.



```
WRITE ADDRESS: 0X92
READ  ADDRESS: 0X93
```

PAGE TITLE		DRAWING NUMBER		SIZE
Thermal Sensors		051-00515		D
 Apple Inc.		REVISION		
		9.0.0		
NOTICE OF PROPRIETARY PROPERTY:		BRANCH		
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		PAGE		
		58 OF 145		
		SHEET		
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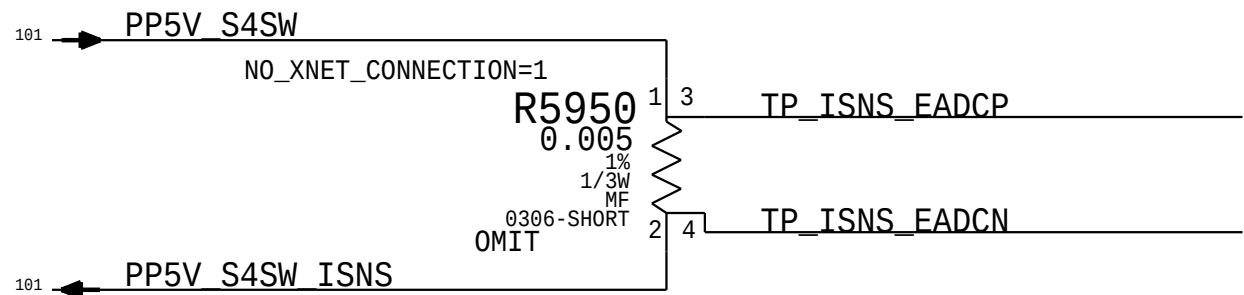
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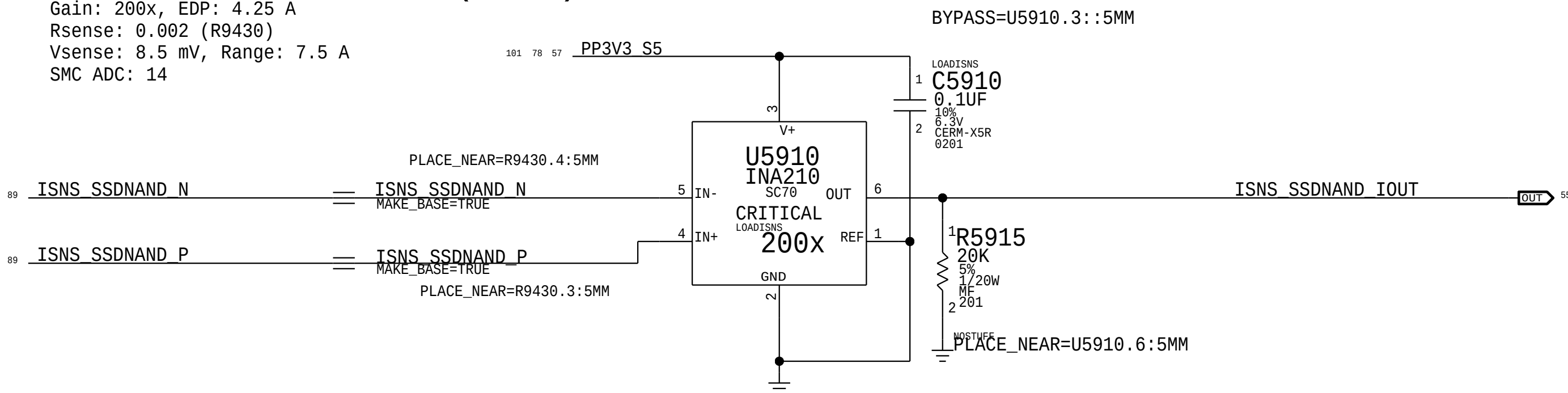
EADC Current Sense (IADC)

Gain: 200x, EDP: 200m A
Rsense: 0.050 (R5950) or Rsense SHORT
Vsense: 10 mV, Range: 0.25 A
EADC2: CH7



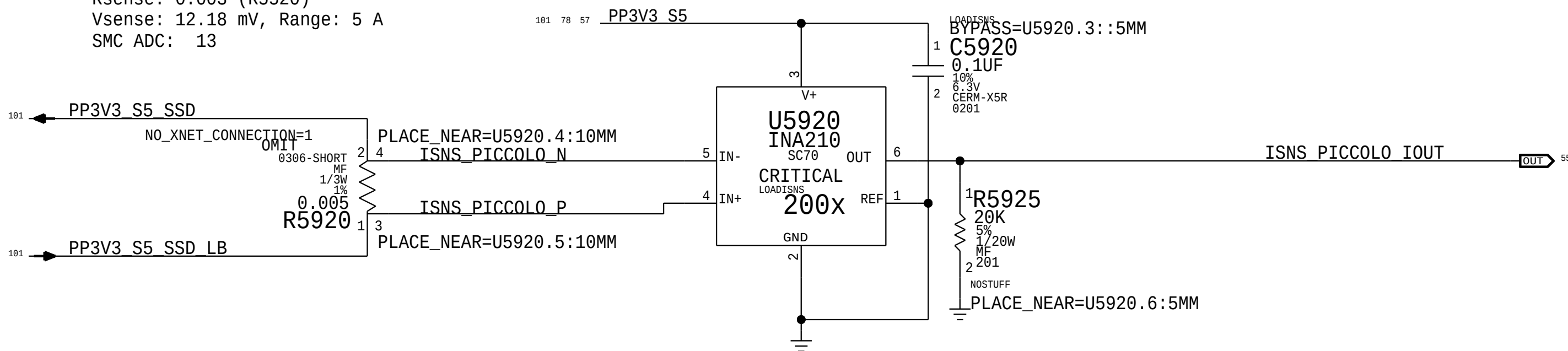
NAND Current Sense (IHNC)

Gain: 200x, EDP: 4.25 A
Rsense: 0.002 (R9430)
Vsense: 8.5 mV, Range: 7.5 A
SMC ADC: 14



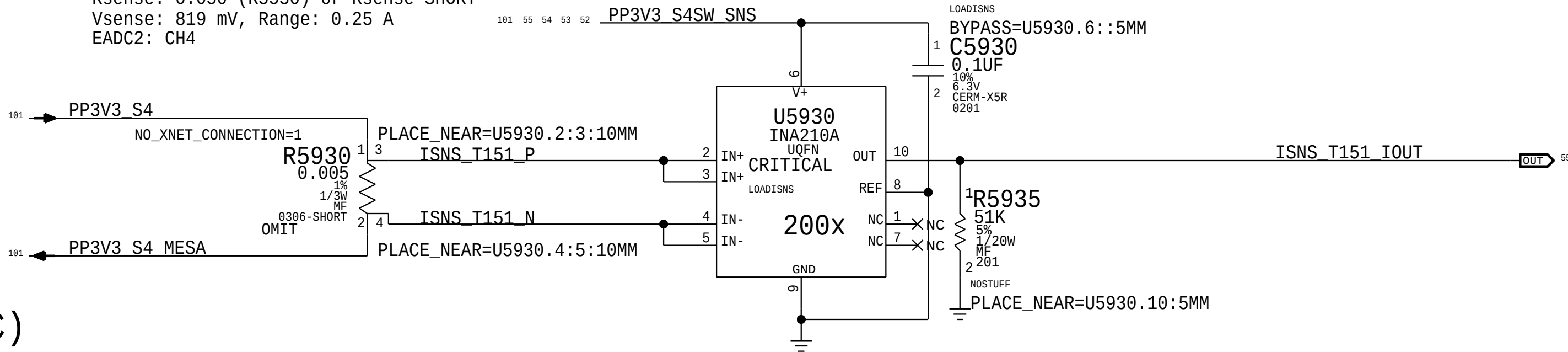
Piccolo Current Sense (IHCC)

Gain: 200x, EDP: 4.06 A
Rsense: 0.003 (R5520)
Vsense: 12.18 mV, Range: 5 A
SMC ADC: 13



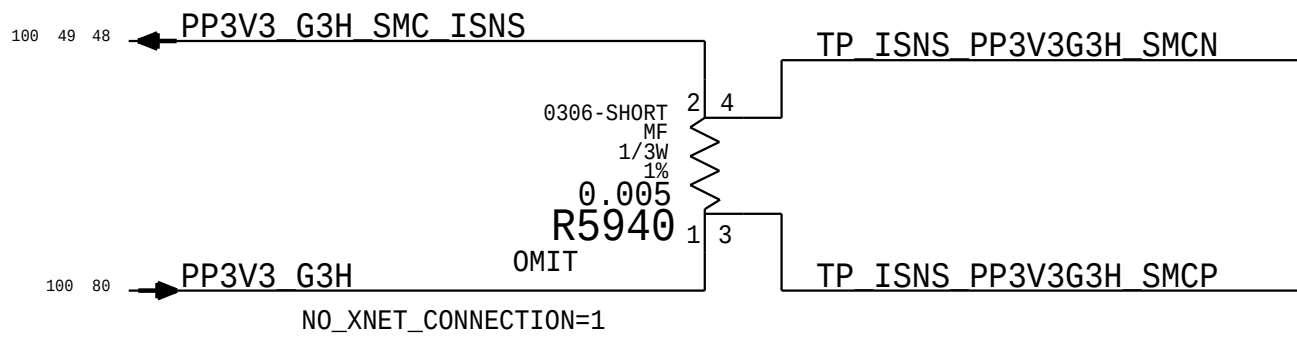
T151 Current Sense (IIDC)

Gain: 200x, EDP: 163.8m A
Rsense: 0.050 (R5530) or Rsense SHORT
Vsense: 819 mV, Range: 0.25 A
EADC2: CH4



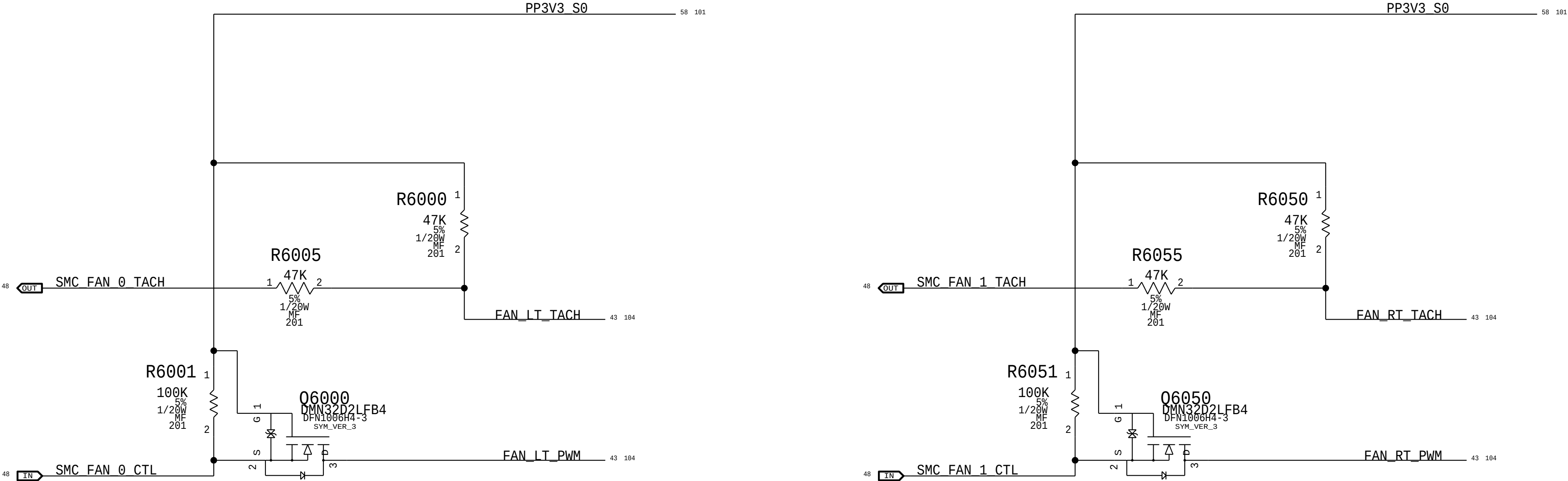
SMC Current Sense (ISMC)

Gain: 391.67x, EDP: 0.2 A
Rsense: 0.05 (R5940) or Rsense SHORT
Vsense: 10 mV, Range: 0.21A
EADC2: CH6

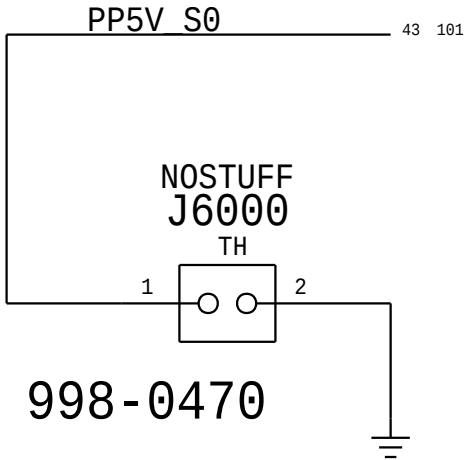


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FAN CONTROL



FOR DEBUG FAN POWER



SYNC_MASTER=J79_JACK		SYNC_DATE=88/21/2015	
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 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
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	9.0.0		
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BOM_COST_GROUP=FAN

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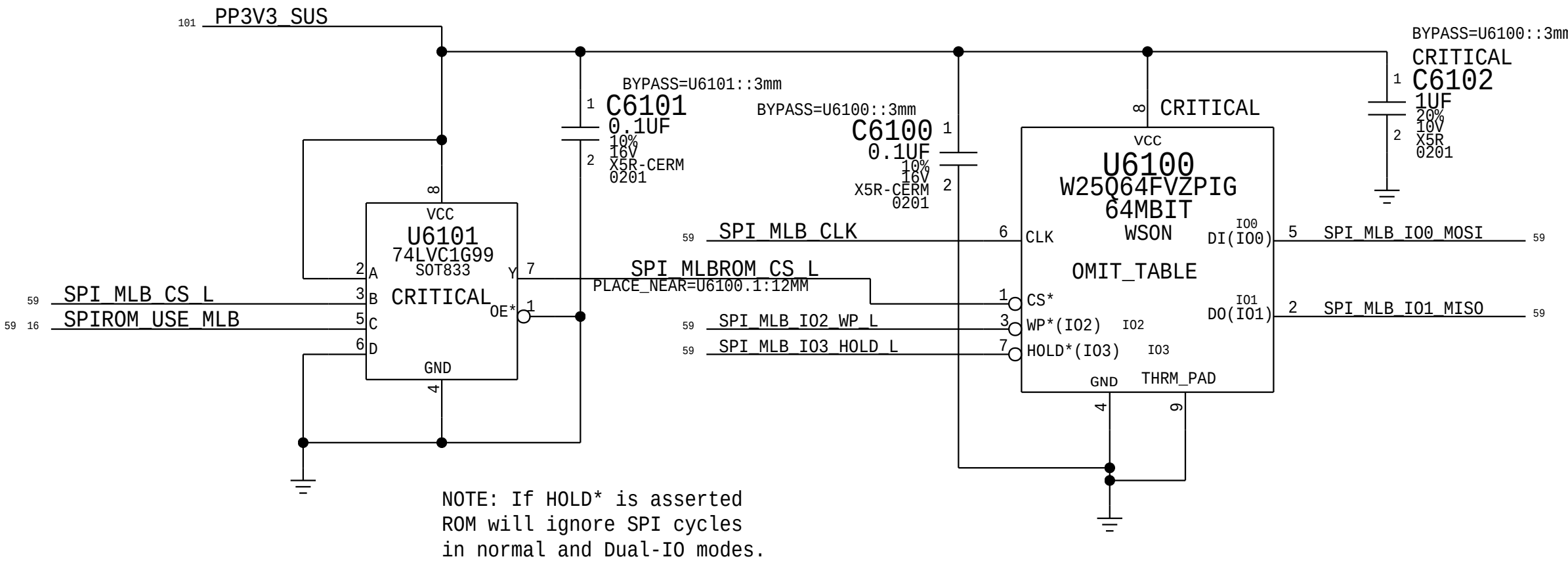
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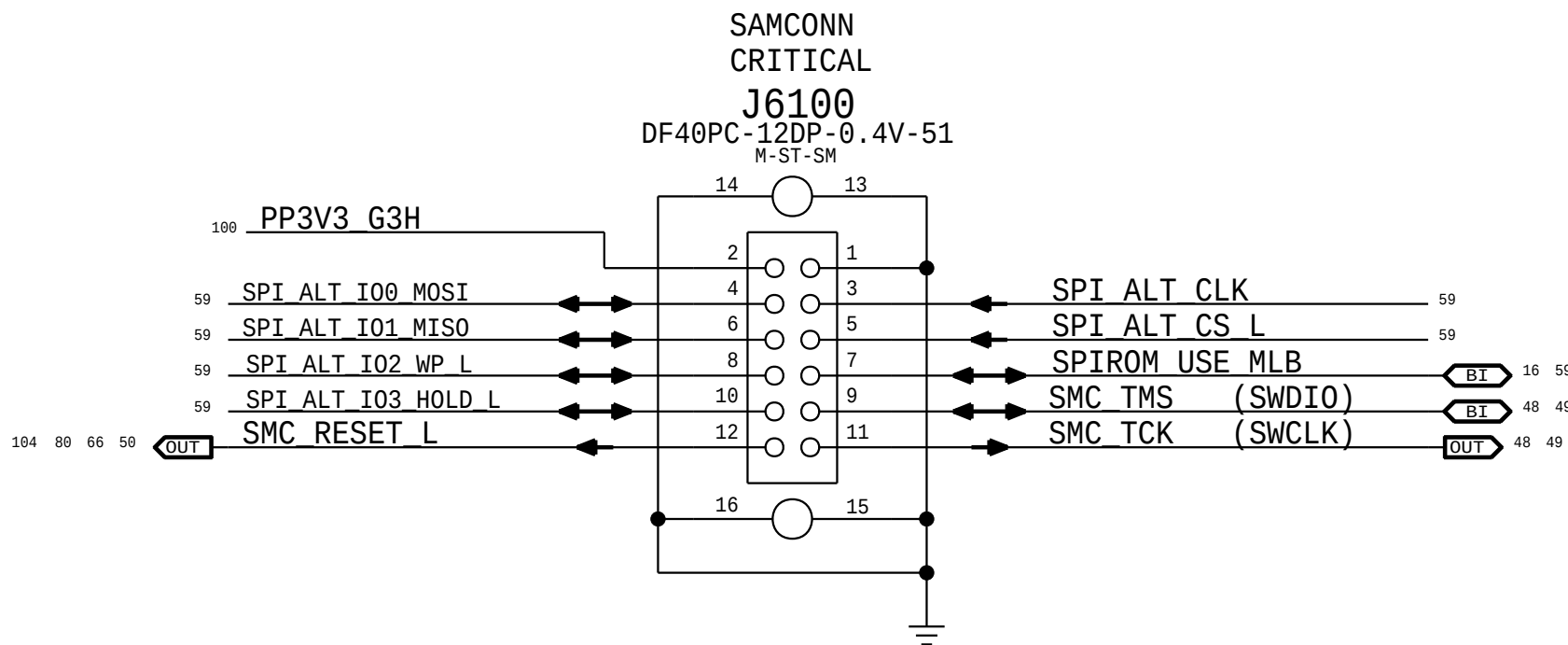
A

SPI ROM

Quad-I/O Mode (Mode 0 & 3) supported.
SPI Frequency: 50MHz for CPU, 20MHz for SMC.



SPI+SWD SAM Connector



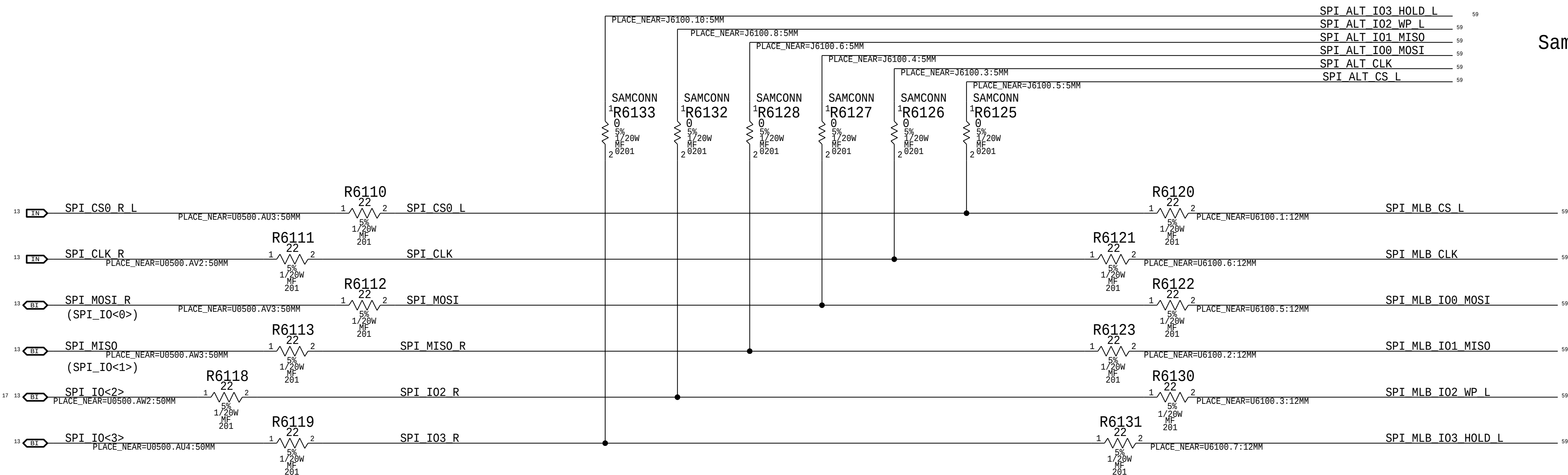
Quad SPI and QPI instructions require the non-volatile Quad Enable bit (QE) in Status Register-2 to be set. When QE=1, the /WP pin becomes I/O2 and /HOLD pin becomes I/O3.

SPI Bus Series Termination (Modified per PDG)

CPU Master

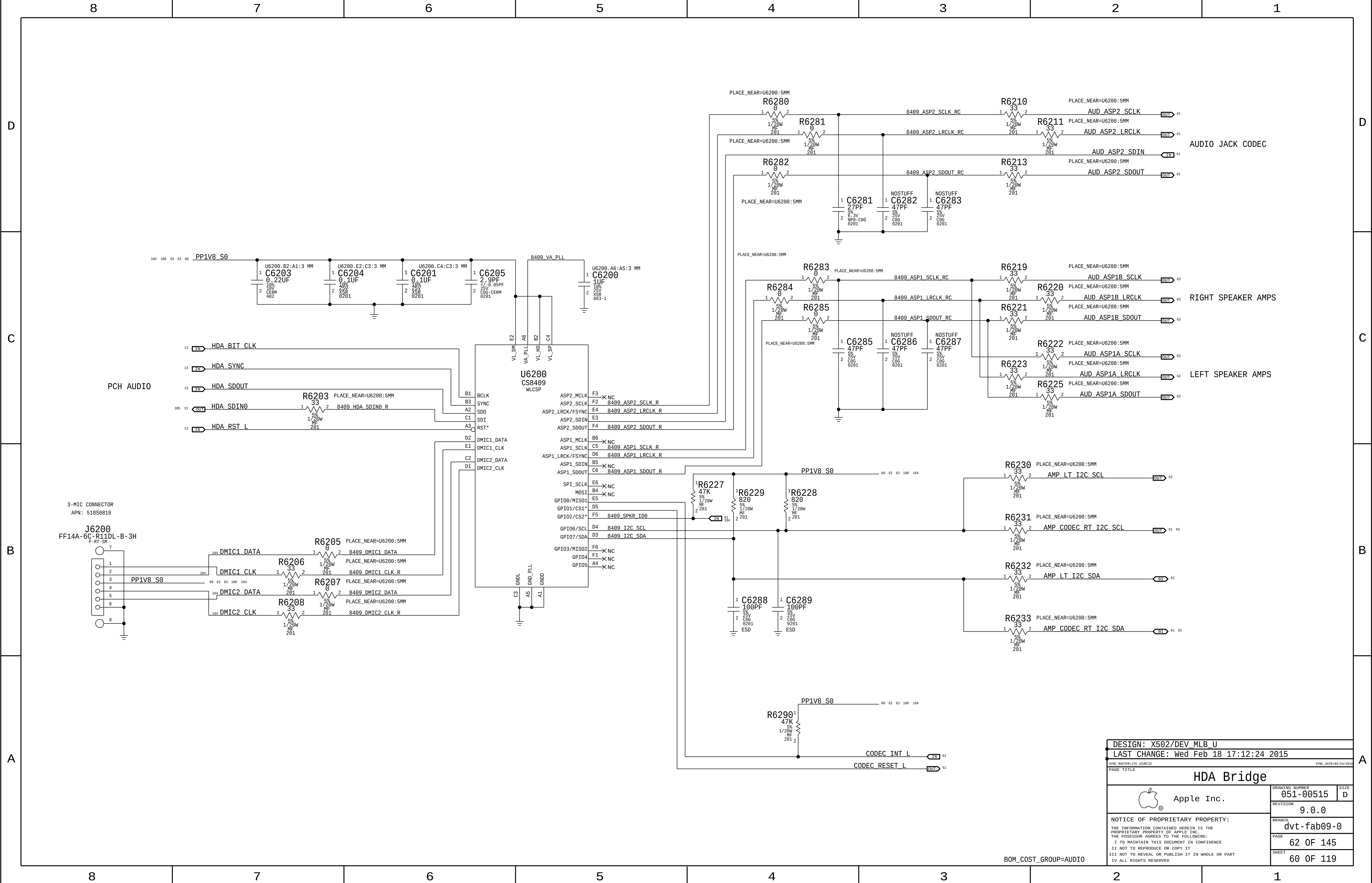
Sam Card ROM Slave

SPI ROM Slave



SYNC_MASTER=J52_MLB		SYNC_DATE=05/12/2015	
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 Apple Inc.		DRAWING NUMBER	051-00515
		SIZE	D
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		BRANCH	dvt-fab09-0
		PAGE	61 OF 145
		SHEET	59 OF 119

BOM_COST_GROUP=CPU & CHIPSET



DESIGN: X502/DEV_MLB_U		
LAST CHANGE: Wed Feb 18 17:12:24 2015		
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PAGE TITLE		
HDA Bridge		
 Apple Inc.	DRAWING NUMBER	051-00515
	REVISION	9.0.0
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	PAGE	62 OF 145
	SHEET	60 OF 119
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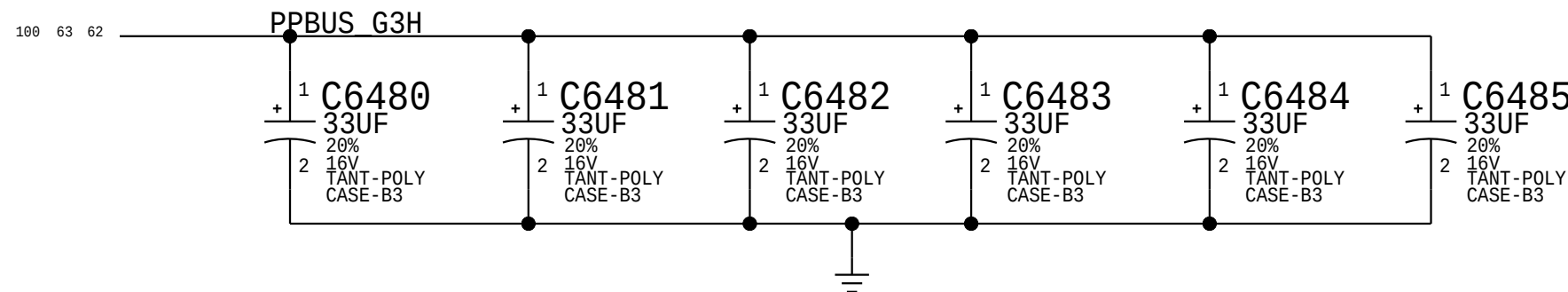
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APN: 353S4074
GAIN = TBD

TDM SLOT 1

TDM SLOT 2

LEFT TWEETER SPEAKER CONNECTOR

LEFT WOOFER SPEAKER CONNECTOR



BOM_COST_GROUP=AUDIO

DESIGN: X502/DEV_MLB_U		
LAST CHANGE: Wed Feb 18 17:12:24 2015		
PAGE TITLE		
Left Speaker Amps & Conn		
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	PAGE	64 OF 145
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2X MONO SPEAKER AMPLIFIERS
APN: 35354073
GAIN = TBD

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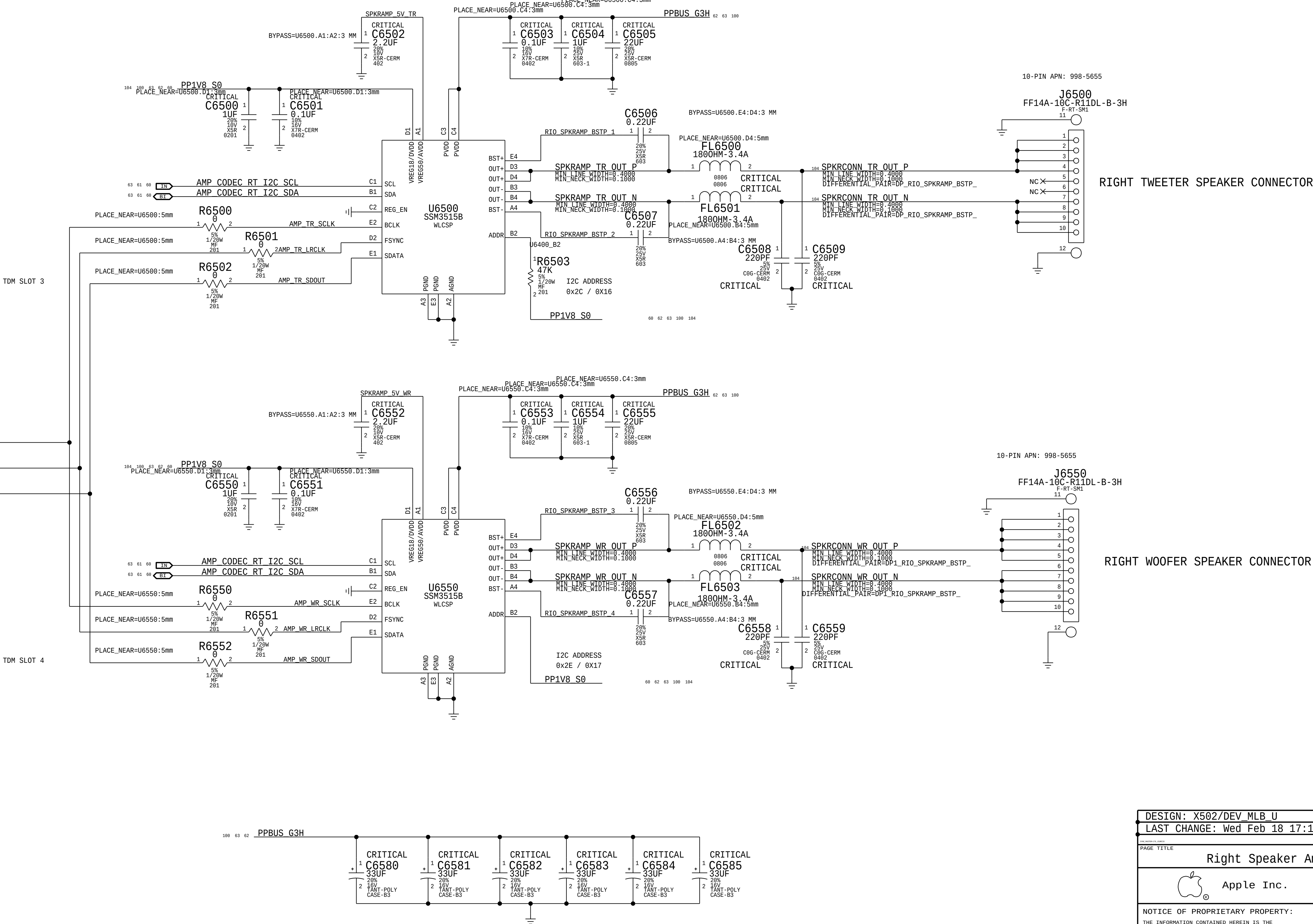
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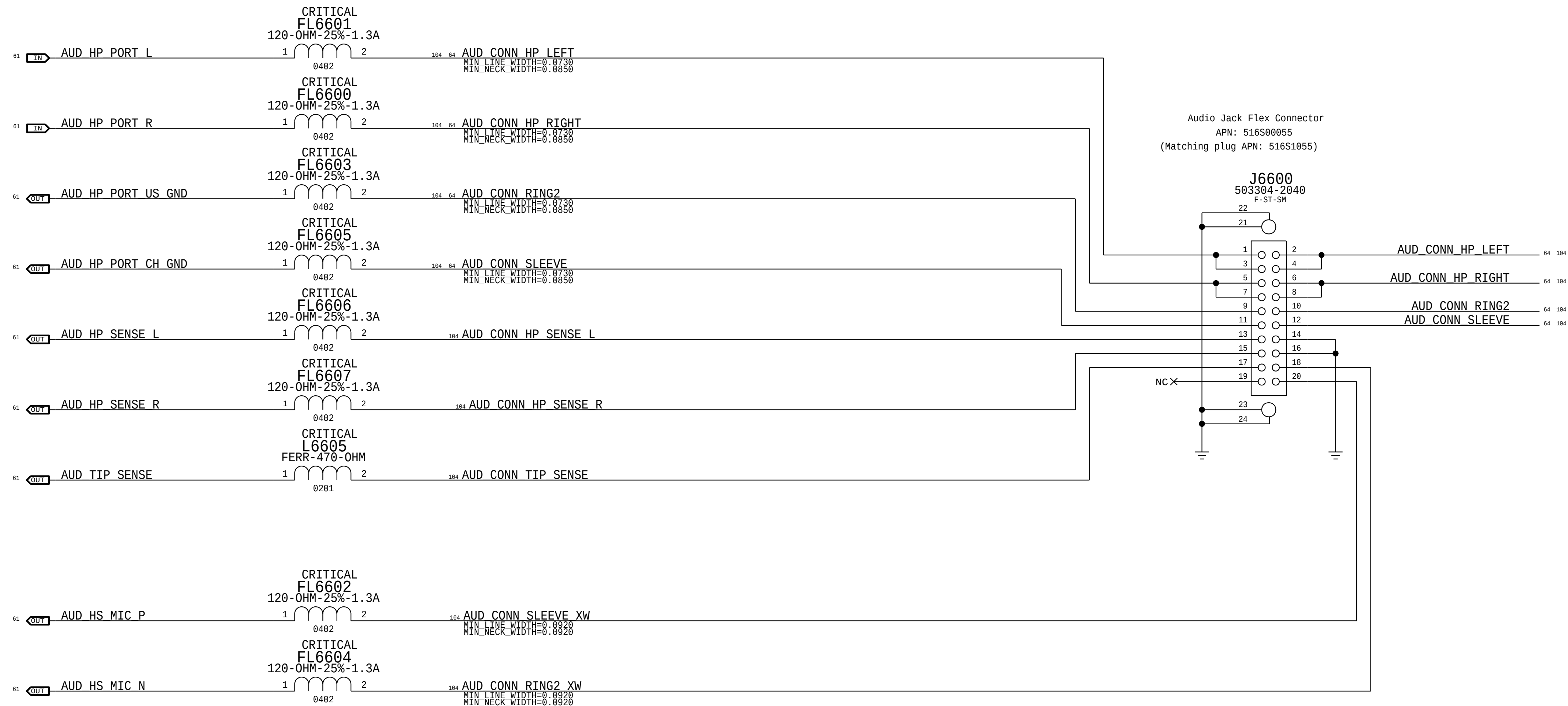
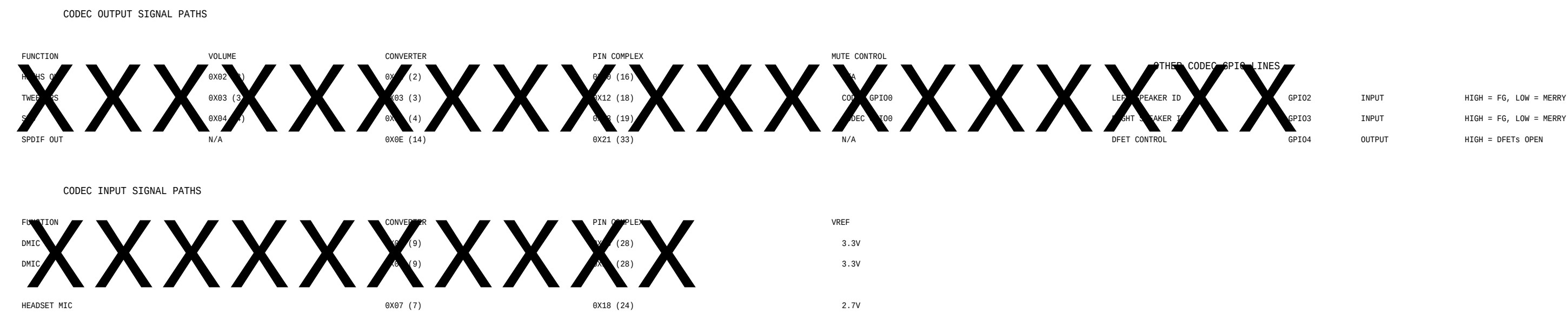
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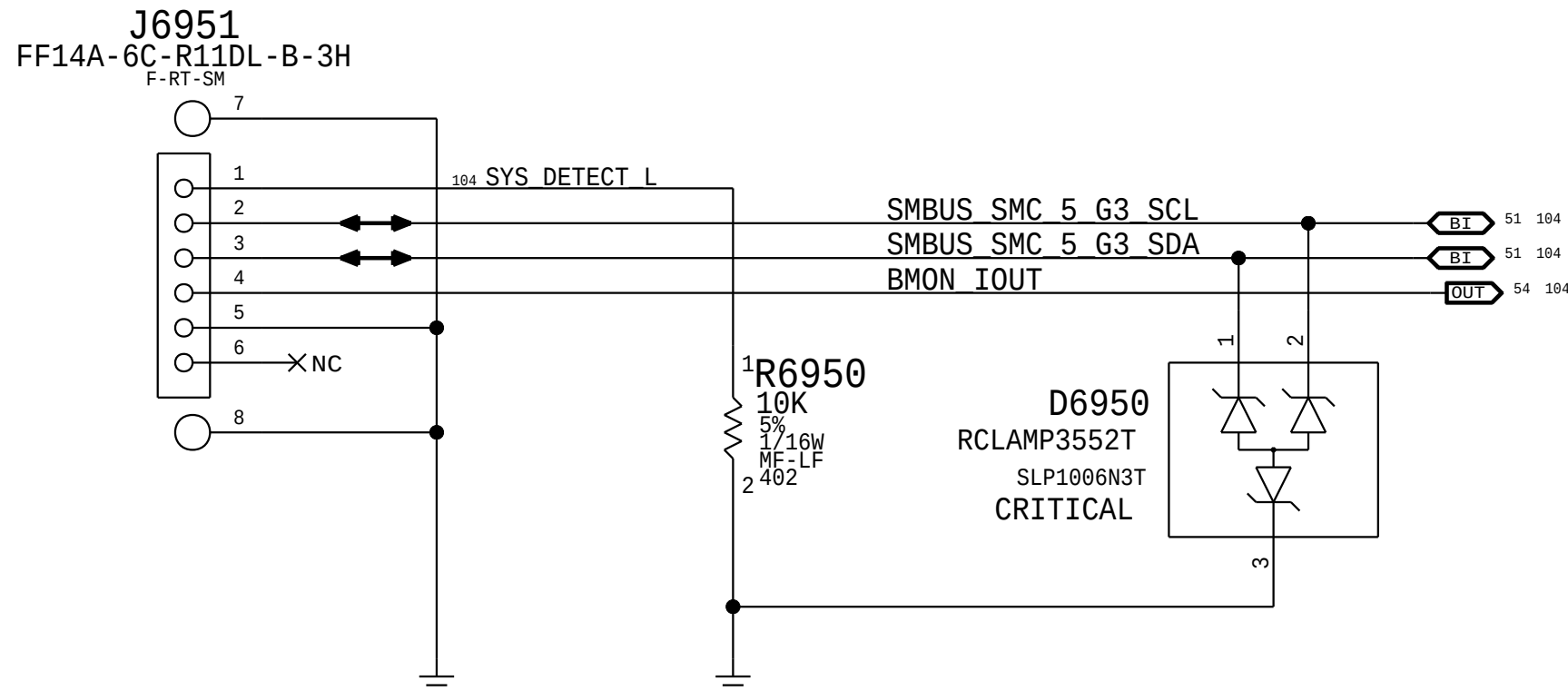
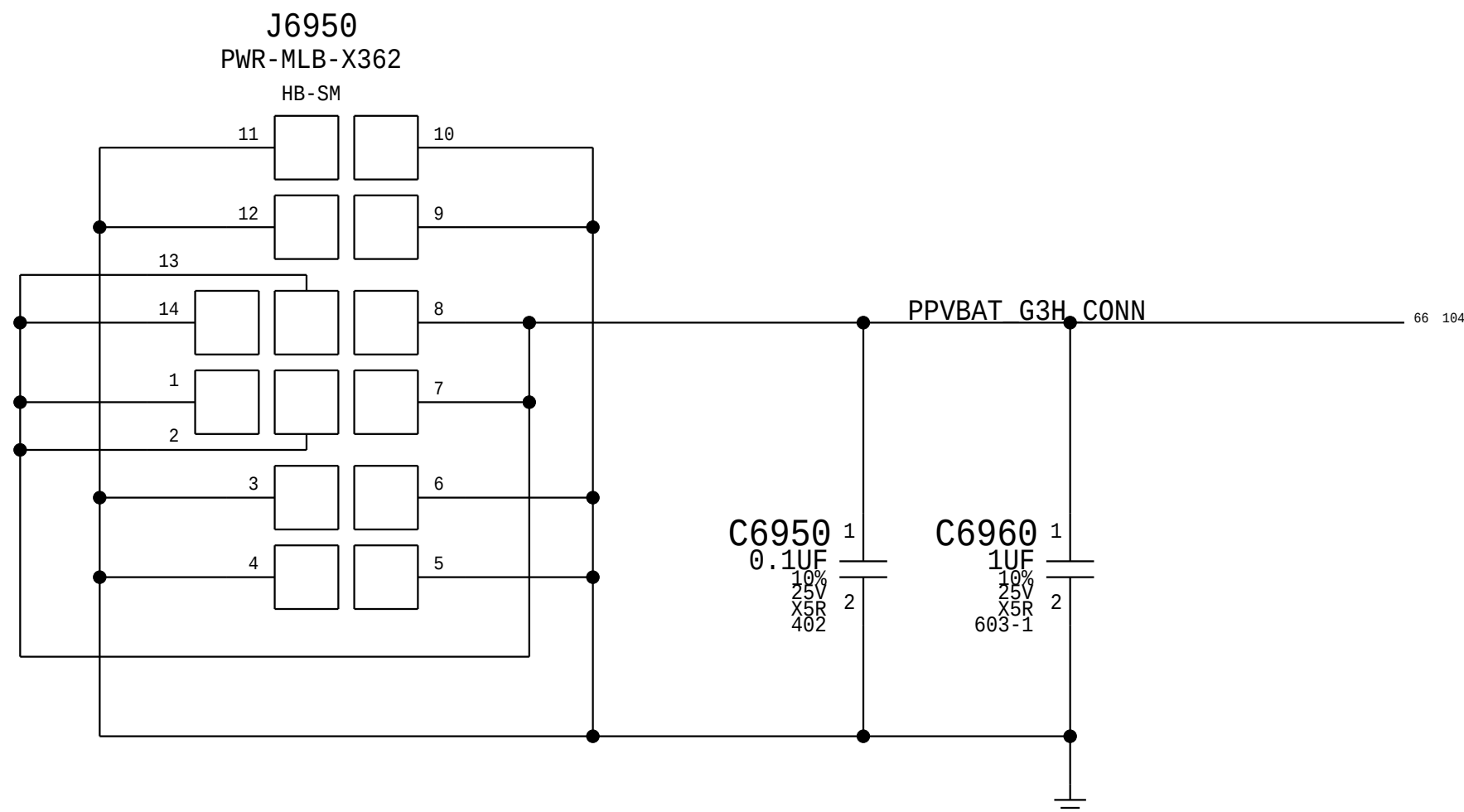
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DESIGN: X502/DEV MLB U		
LAST CHANGE: Wed Feb 18 17:12:24 2015		
DNC MASTER-178_200810		DNC MASTER-112-18-2015
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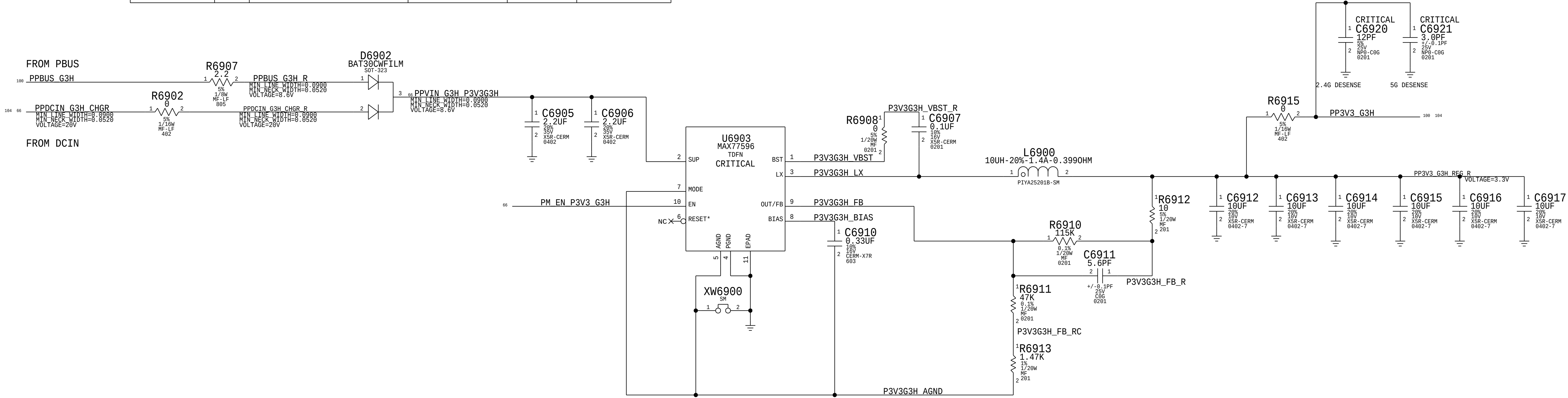
BOM_COST_GROUP=AUDIO

J79 Battery Hotbar Flex Pads



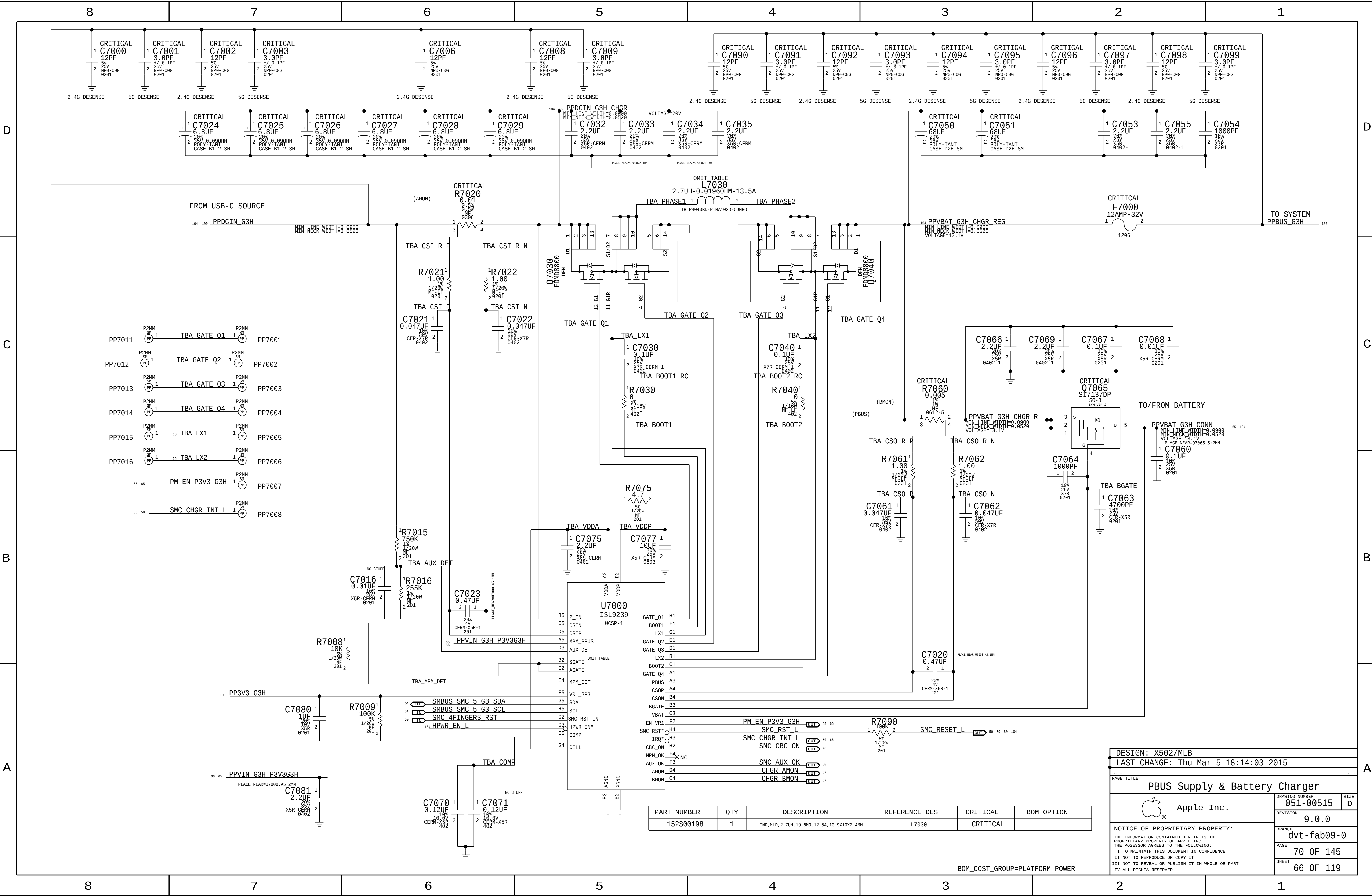
BMU POWER FLEX HOTBAR'd TO THE MLB:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
632-00566	1	PCBA, FLEX, BMU PWR, X362	J6950	CRITICAL	



PAGE TITLE		
DC-In & Battery Connectors		
 Apple Inc.	DRAWING NUMBER	051-00515
	REVISION	9.0.0
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	PAGE	69 OF 145
	SHEET	65 OF 119

BOM_COST_GROUP=PLATFORM POWER



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
152S00198	1	IND, MLD, 2.7UH, 19.6MO, 12.5A, 10.9X18X2.4MM	L7030	CRITICAL	

DESIGN: X502/MLB
LAST CHANGE: Thu Mar 5 18:14:03 2015

PAGE TITLE
PBUS Supply & Battery Charger

Apple Inc.

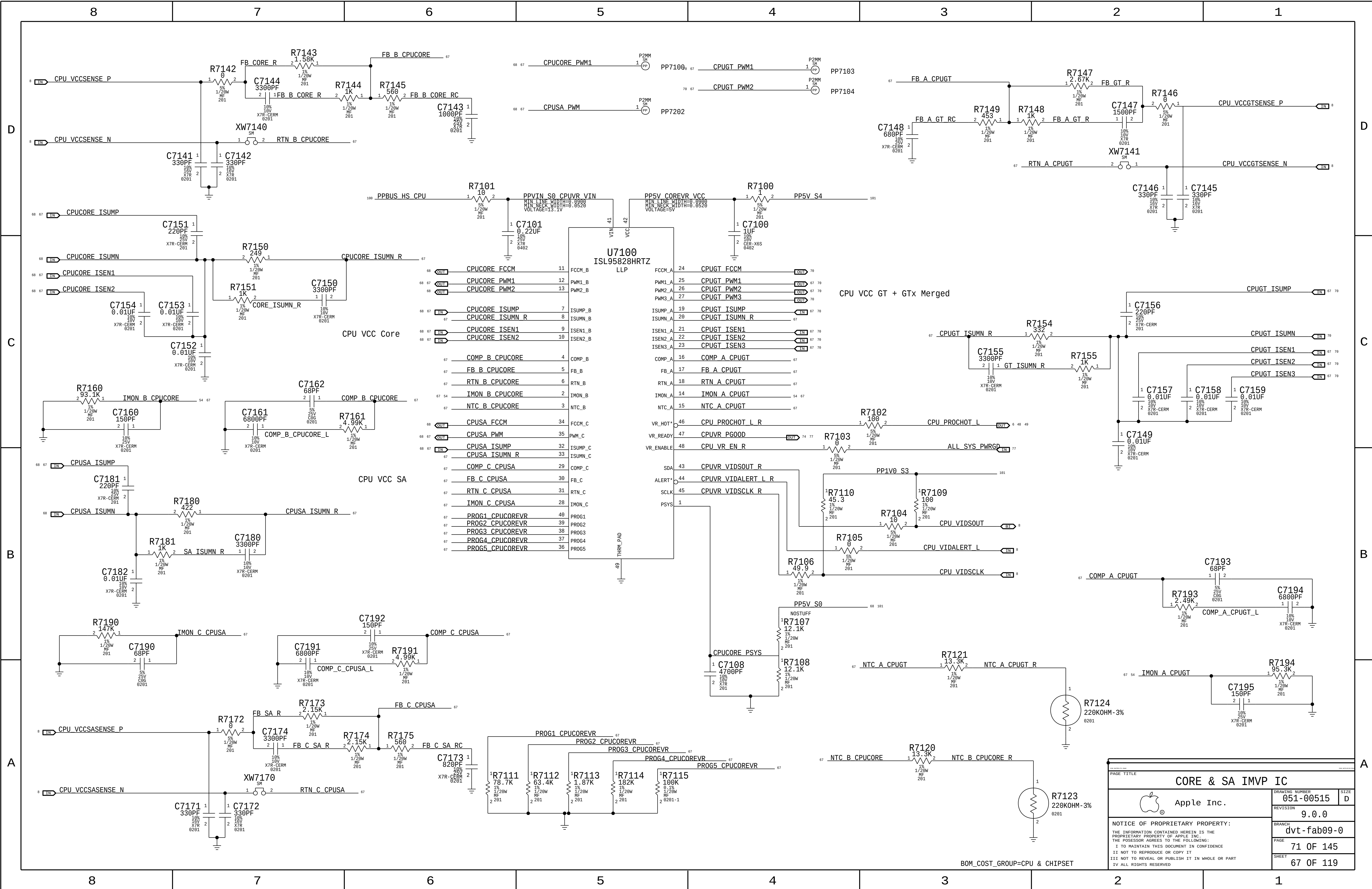
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dvt-fab09-0

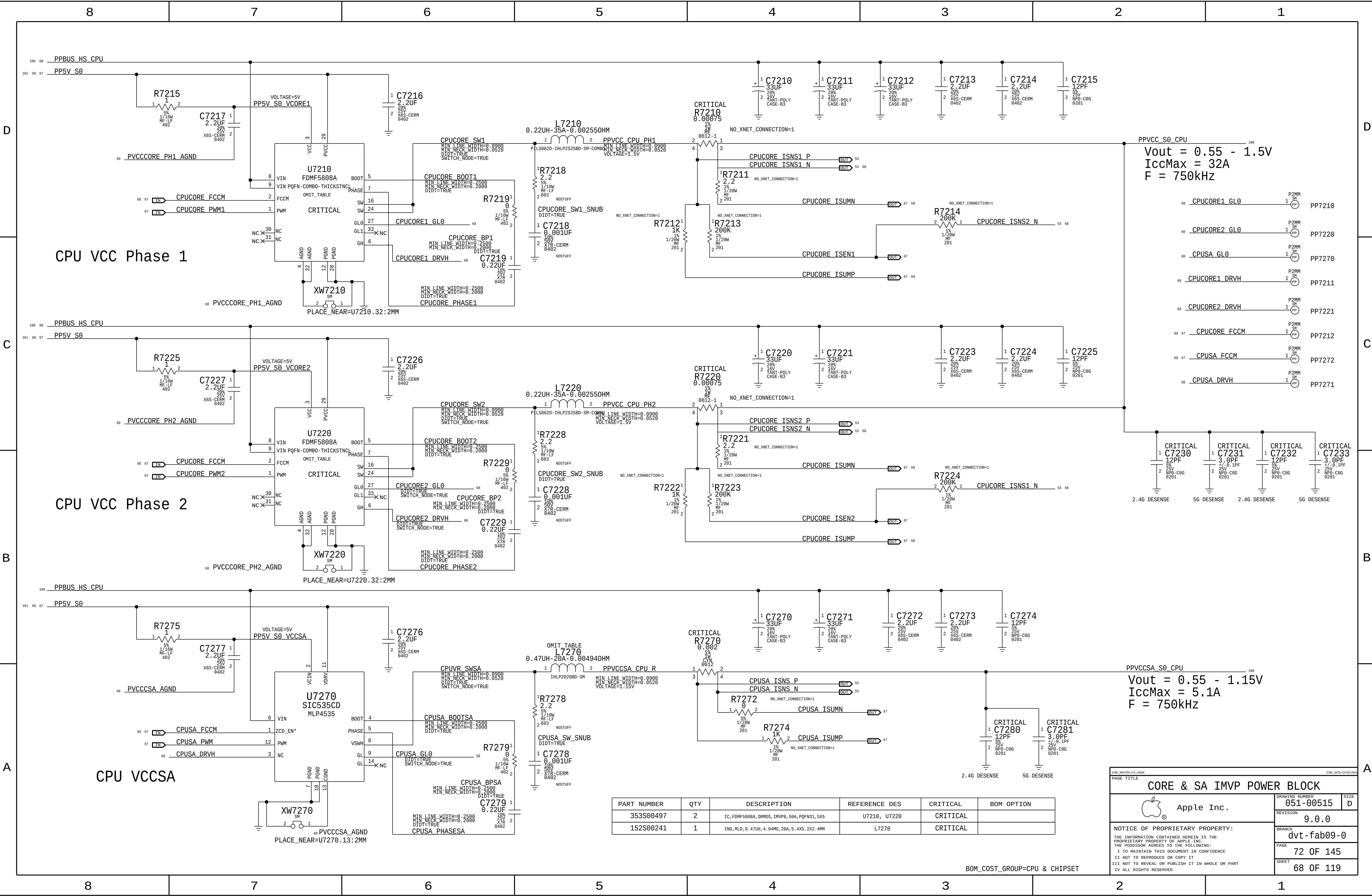
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70 OF 145

SHEET
66 OF 119

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BOM_COST_GROUP=PLATFORM POWER





PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	2	IC, FDMF5808A, DRMOS, IMVP8, 50A, PQFN31, 5X5	U7210, U7220	CRITICAL	
152S00241	1	IND, MLD, 0.47UH, 4.94mOhm, 20A, 5.4X5.2X2.4MM	L7270	CRITICAL	

Apple Inc.

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DRAWING NUMBER
051-00515

REVISTION
9.0.0

BRANCH
dvt-fab09-0

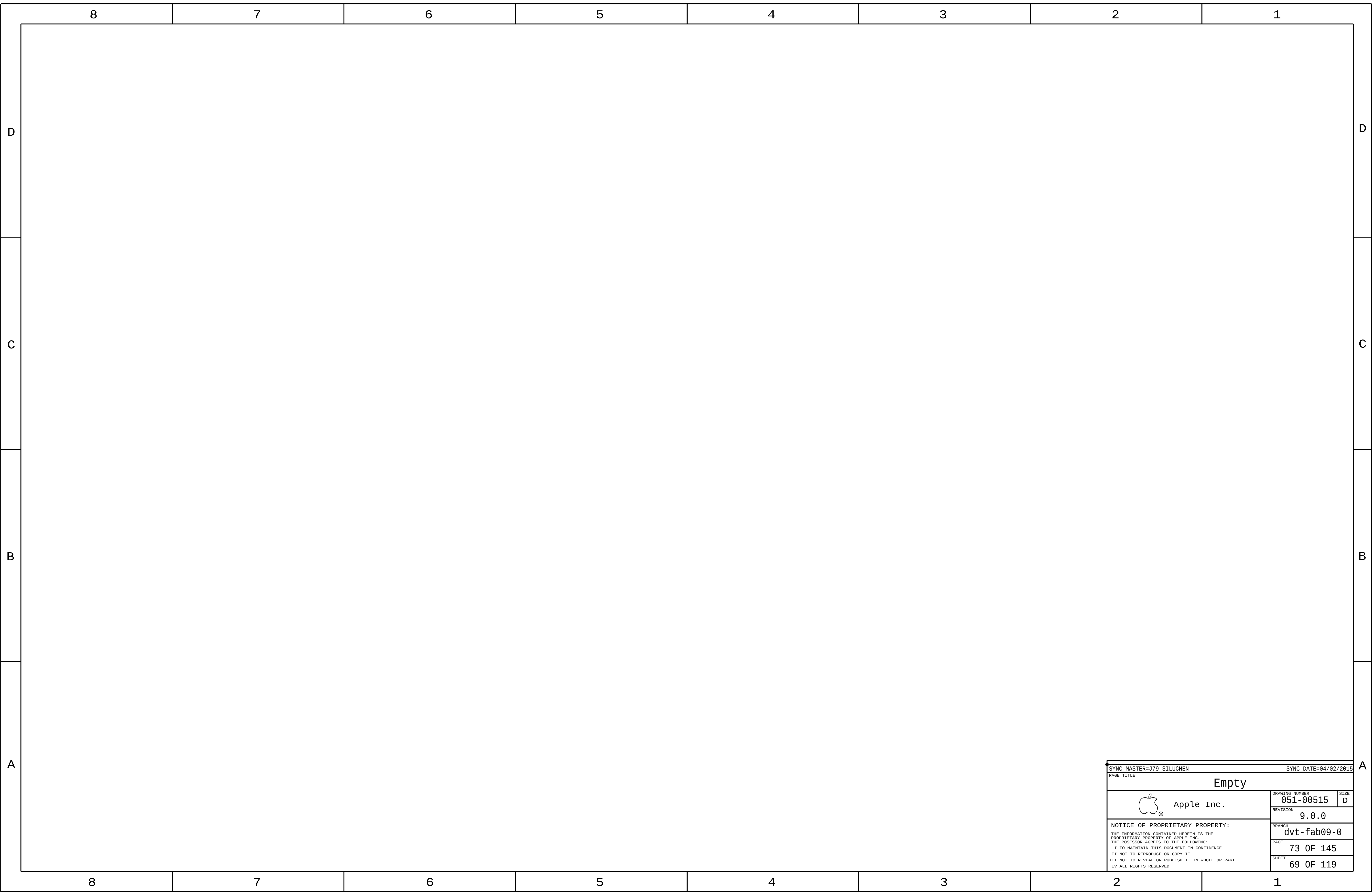
PAGE
72 OF 145

SHEET
68 OF 119

051-00515

D

BOM_COST_GROUP=CPU & CHIPSET



8

7

6

5

4

3

2

1

D

CPU VCCGT/GTx Phase 1

PLACE_NEAR=U7410.32:2MM
70 PVCCCGT_PH1_AGNDVout = 0.55 - 1.5V
IccMax = 64A
F = 750KHz

D

C

CPU VCCGT/GTx Phase 2

PLACE_NEAR=U7420.32:2MM
70 PVCCCGT_PH2_AGND

C

B

CPU VCCGT/GTx Phase 3

PLACE_NEAR=U7430.32:2MM
70 PVCCCGT_PH3_AGND

B

A

A

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	3	IC, FDMF5808A, DRMOS, 1MVP8, 50A, PQFN31, 5X5	U7410, U7420, U7430	CRITICAL	

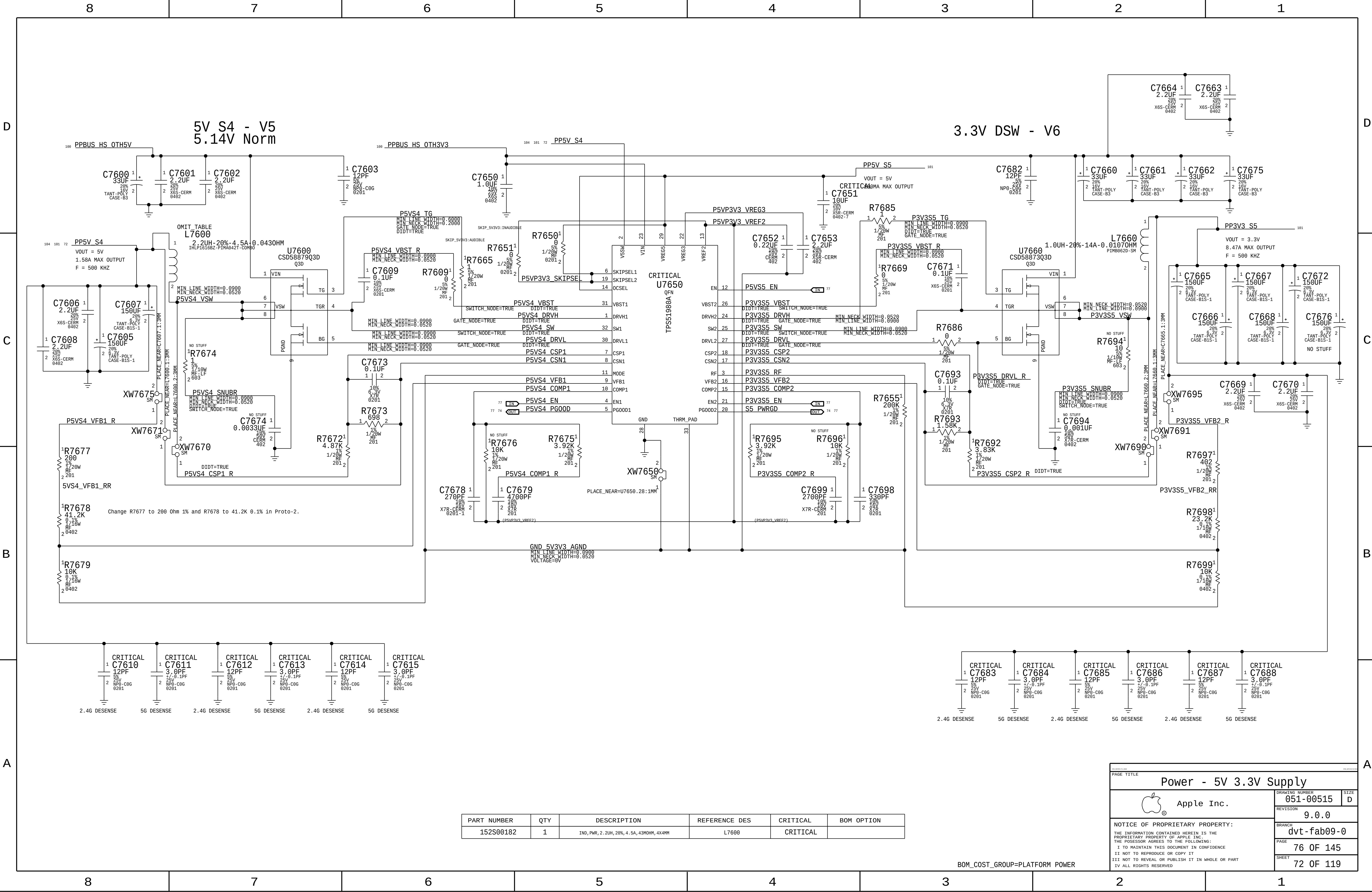
BOM_COST_GROUP=CPU & CHIPSET

GT & GTX IMVP POWER BLOCK		DRAWING NUMBER	051-00515	SIZE	D
Apple Inc.		REVISION	9.0.0		
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		SHEET	70 OF 119		

8		7		6		5		4		3		2		1		
																D
																C
																B
																A
8		7		6		5		4		3		2		1		

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										REVISION					9.0.0					D				
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										PAGE					75 OF 145									
										SHEET					71 OF 119									

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										BRANCH		dvt-fab09-0			
										PAGE		75 OF 145			
										SHEET		71 OF 119			

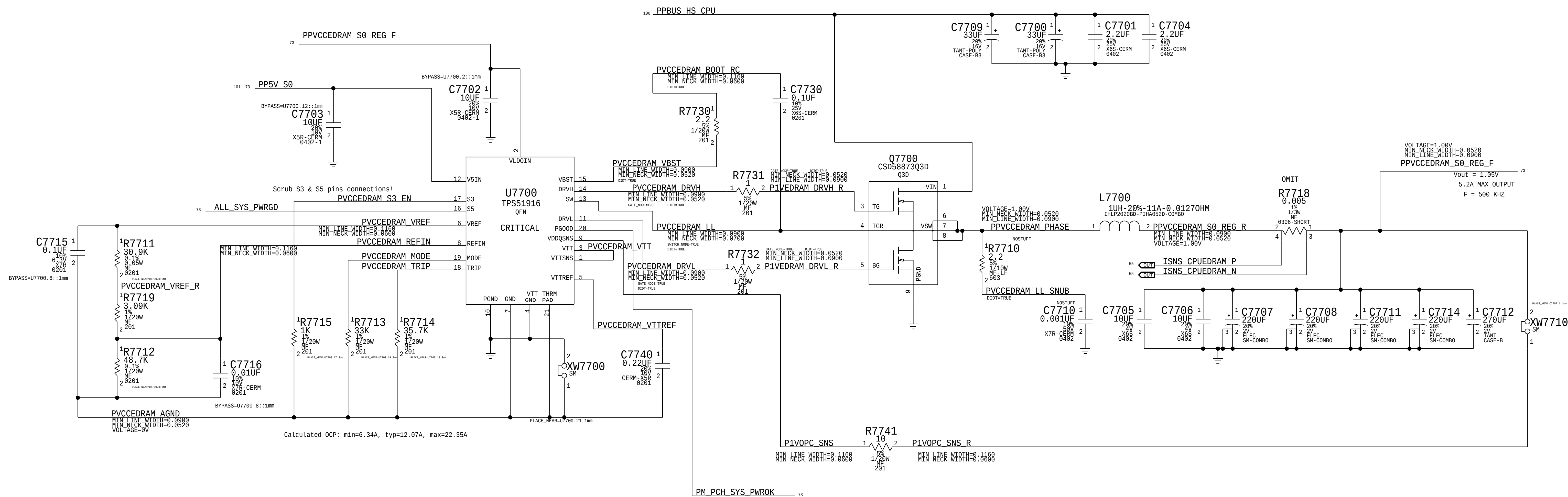


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
152S00182	1	IND, PWR, 2.2UH, 20%, 4.5A, 43MOHM, 4X4MM	L7600	CRITICAL	

PAGE TITLE		
Power - 5V 3.3V Supply		
 Apple Inc.	DRAWING NUMBER	051-00515
	REVISION	9.0.0
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	PAGE	76 OF 145
	SHEET	72 OF 119

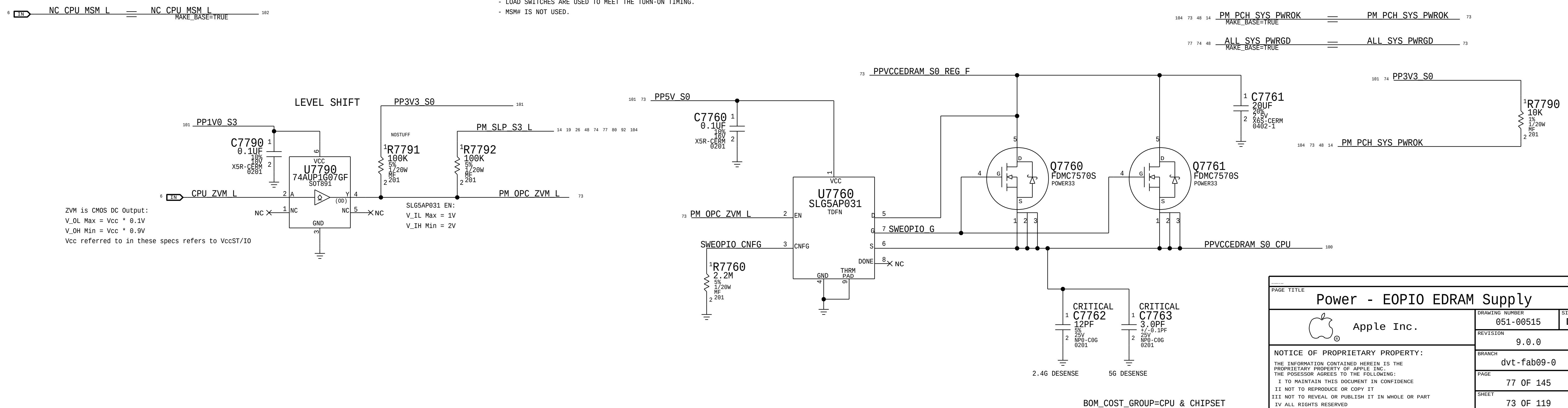
BOM_COST_GROUP=PLATFORM POWER

1V EDRAM & EOPIO



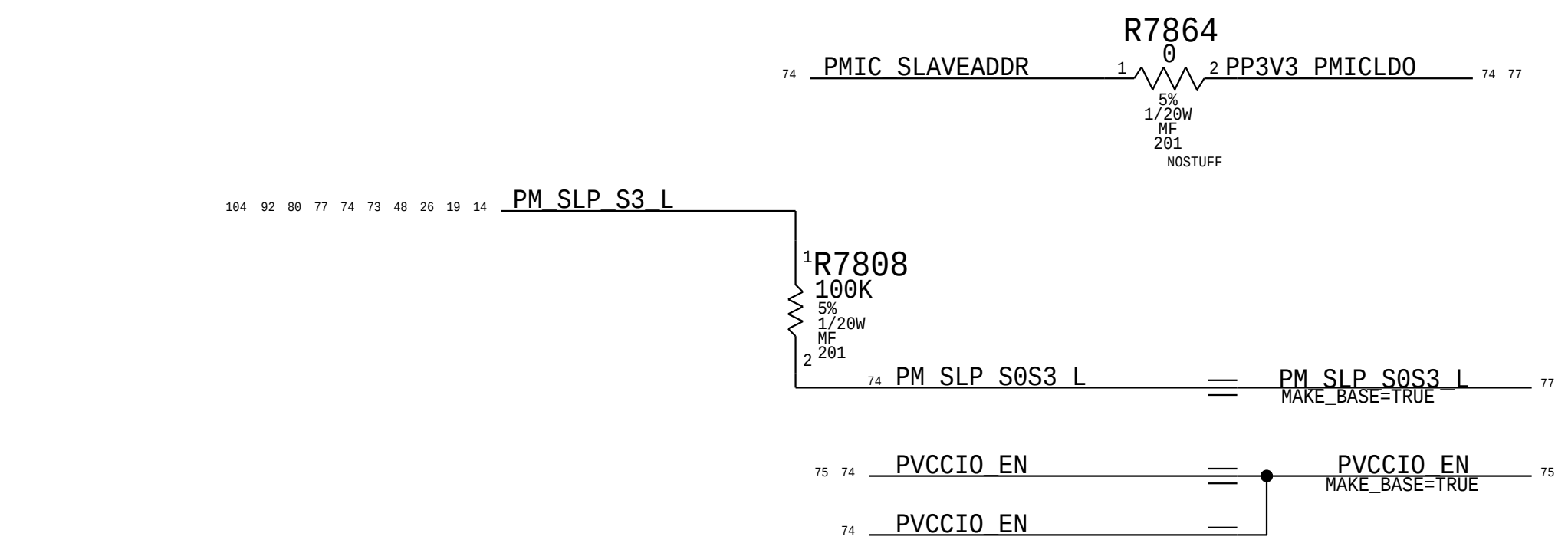
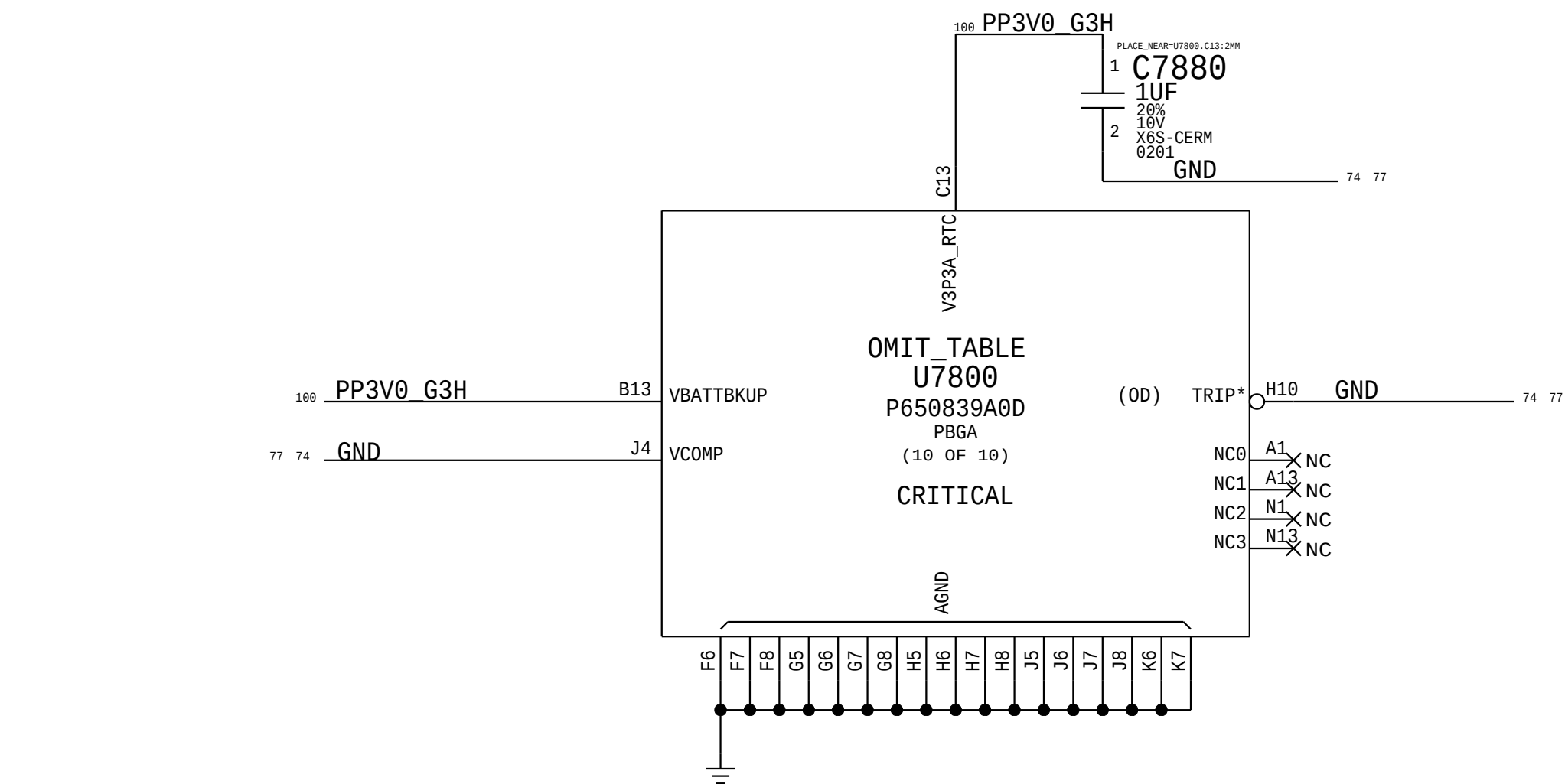
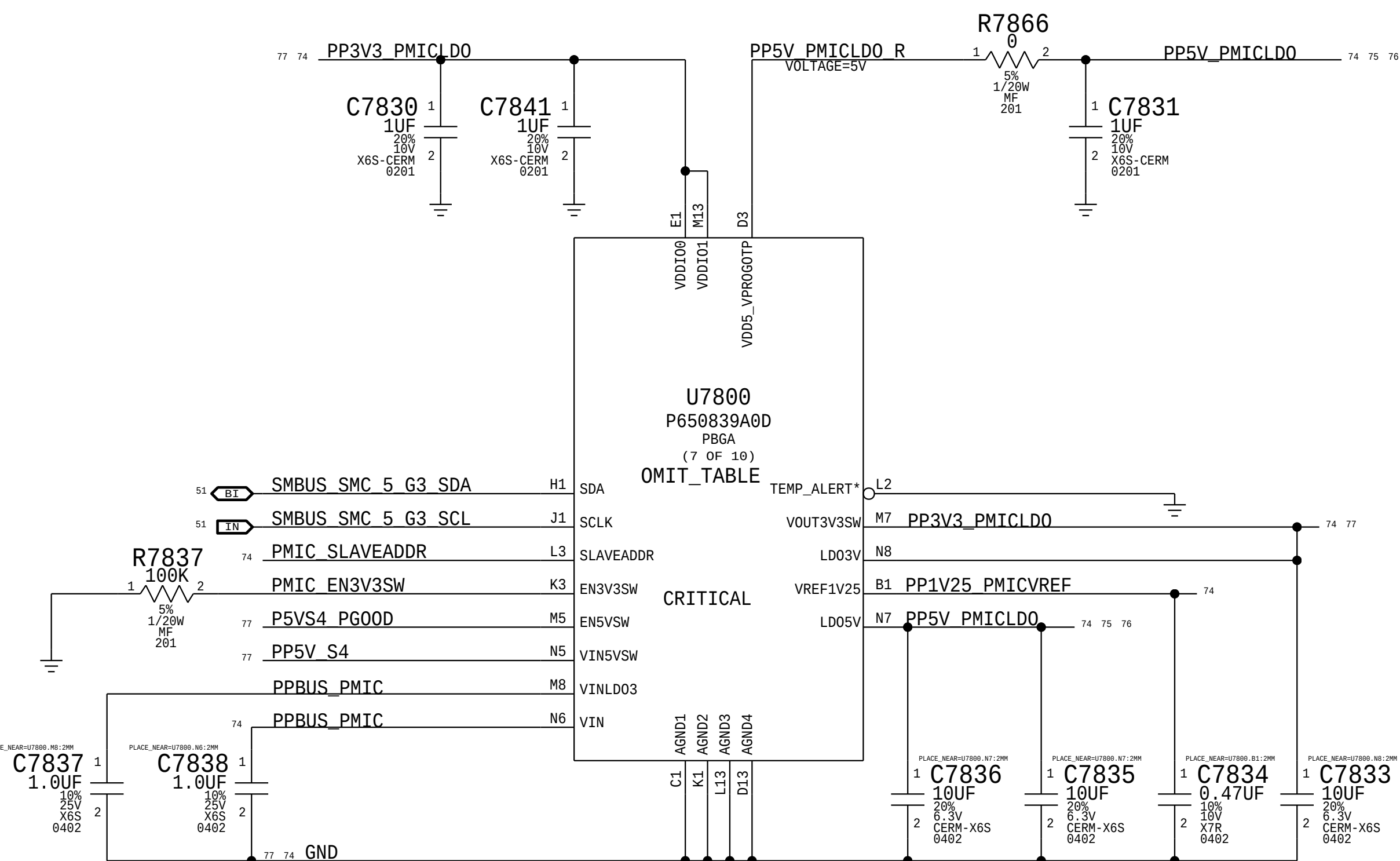
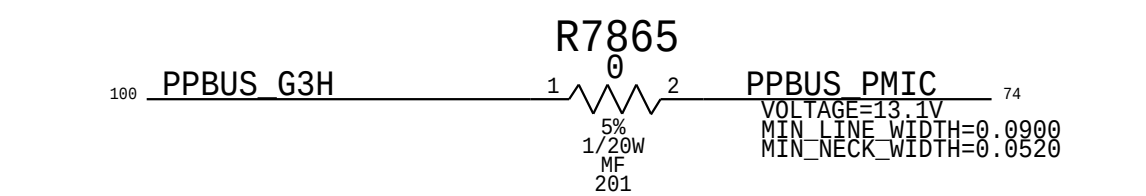
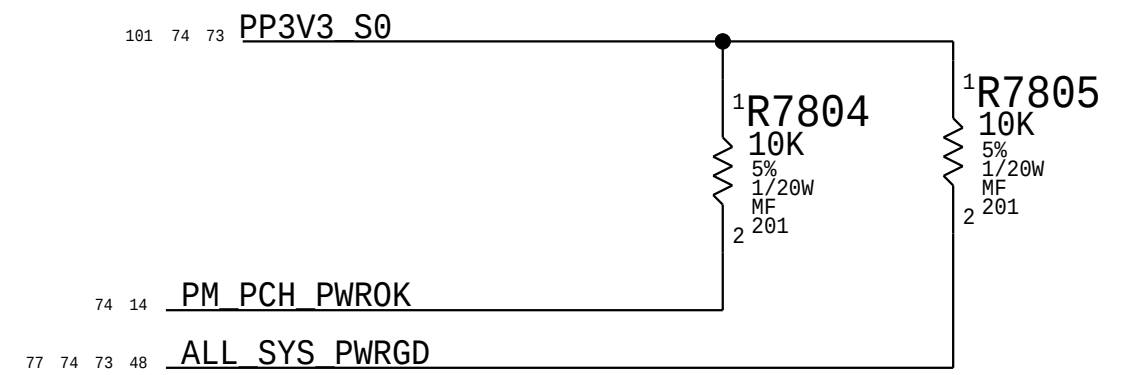
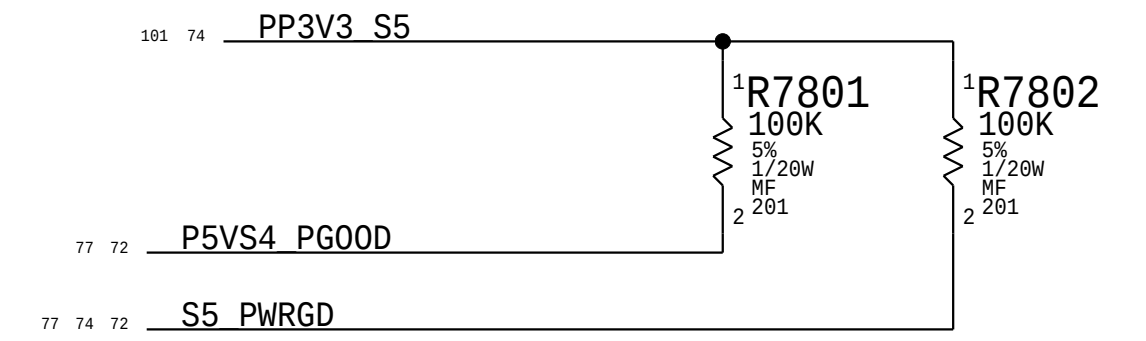
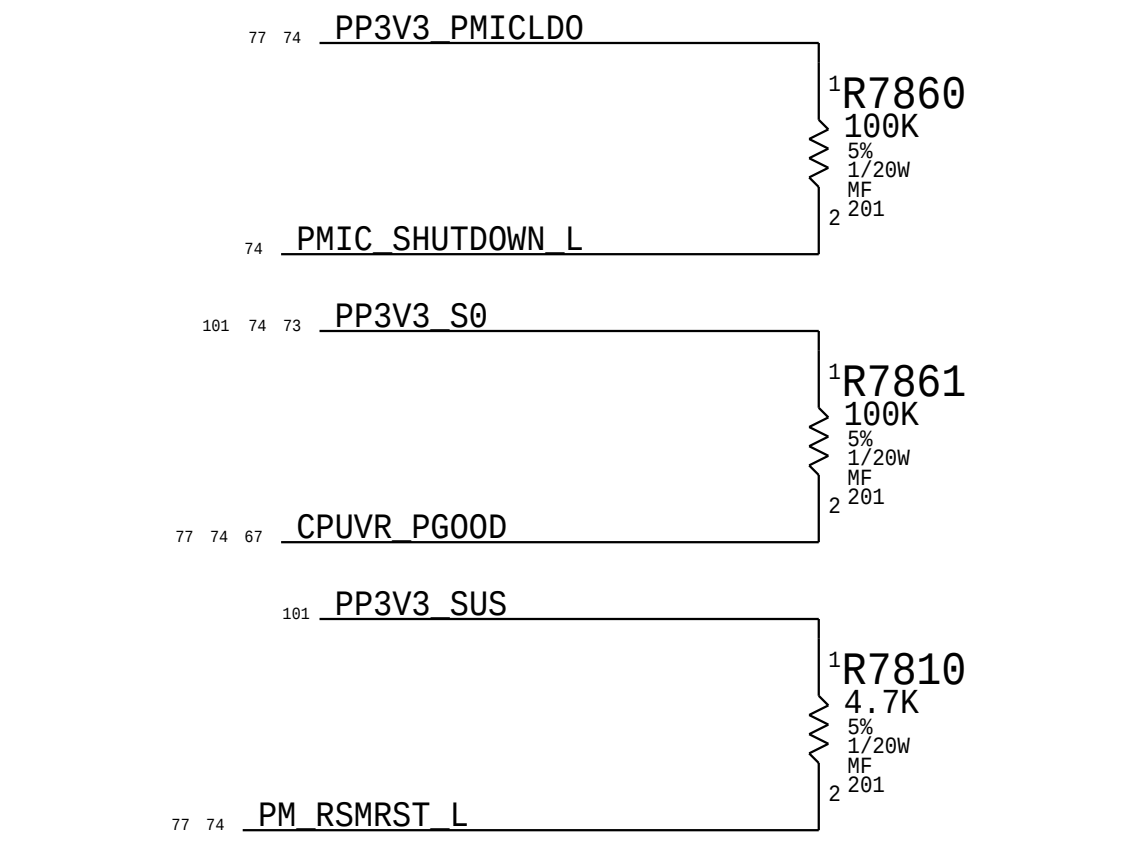
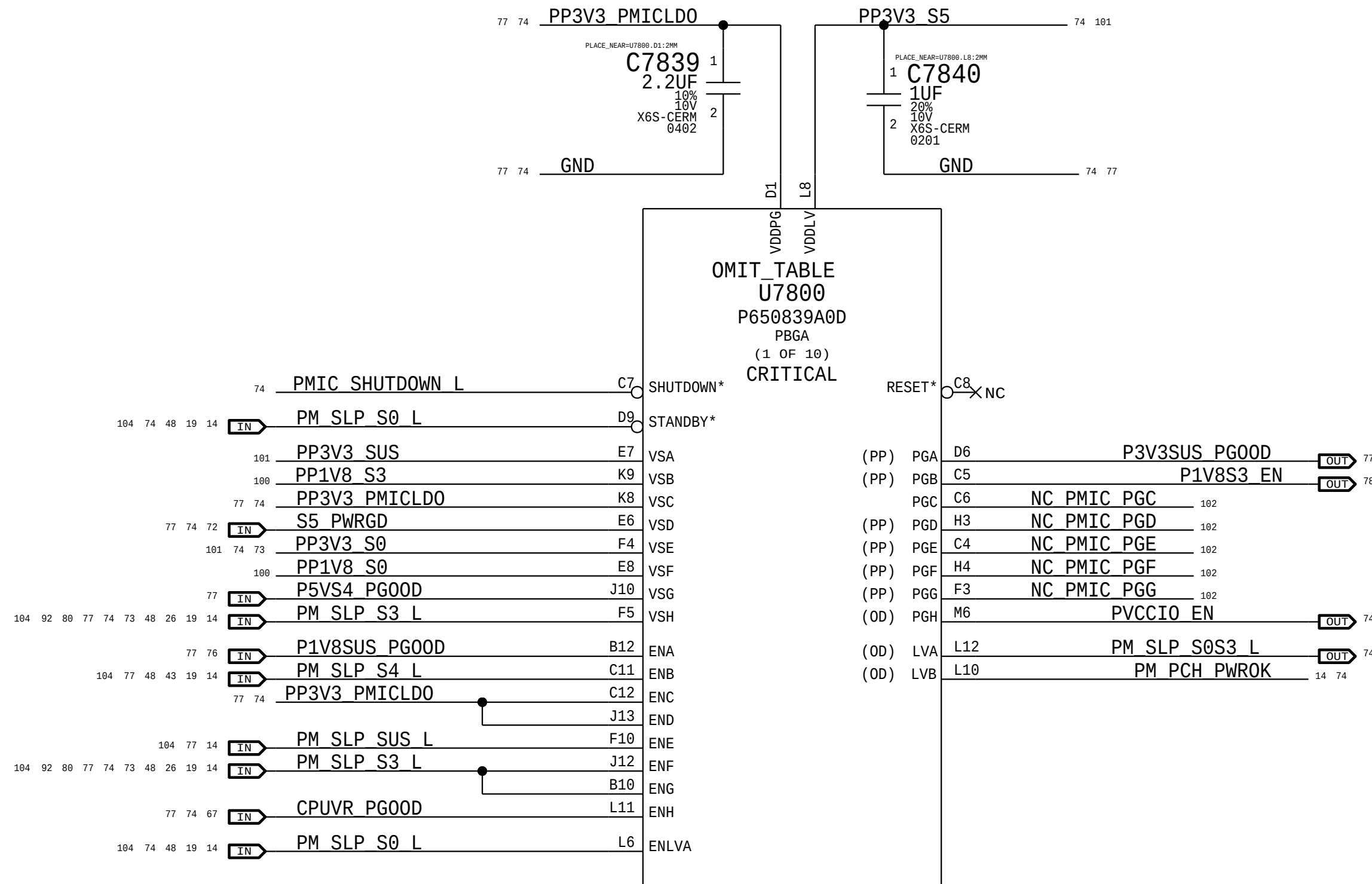
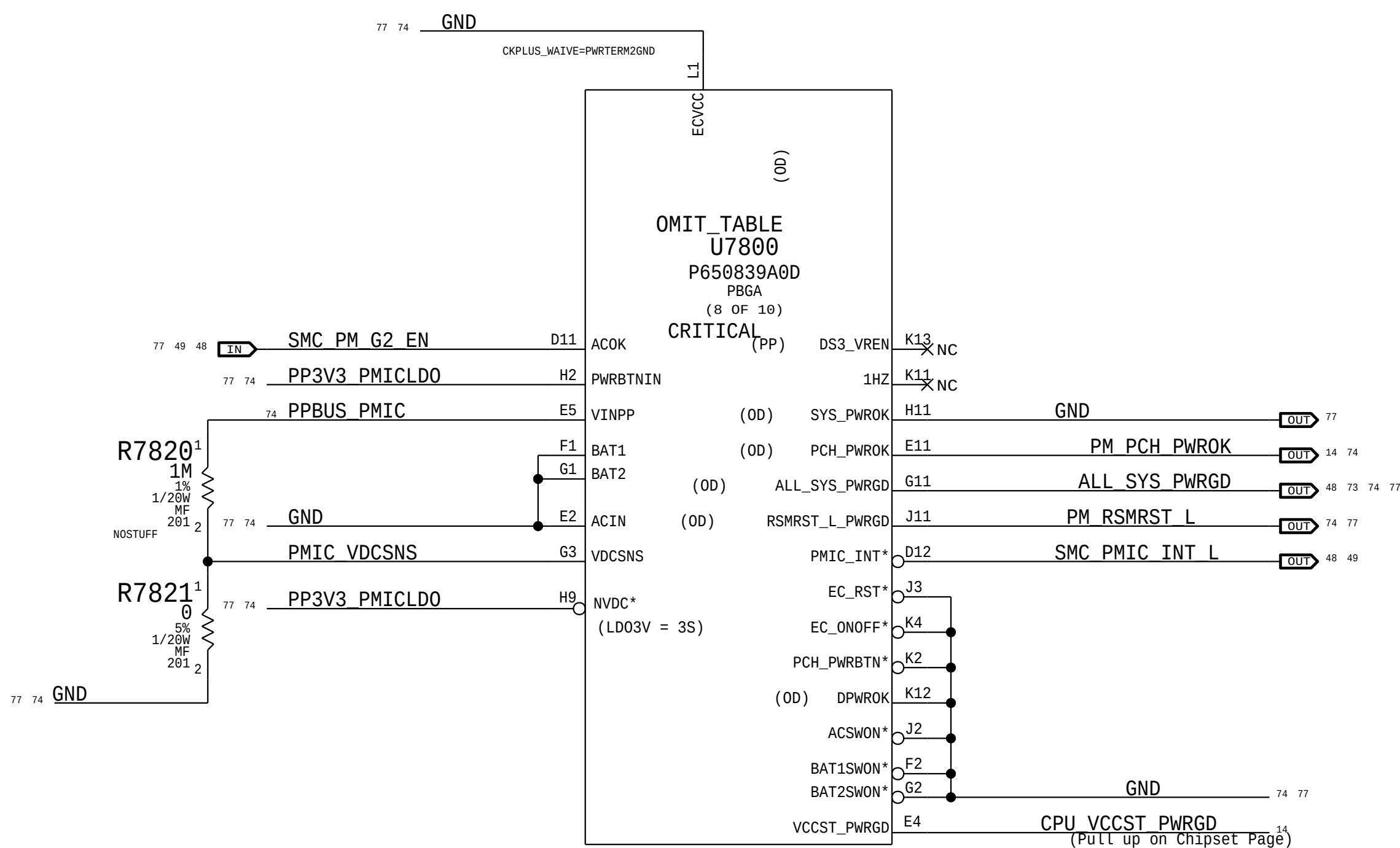
THE FOLLOWING SHORTCUTS ARE USED:

- OPC (EDRAM) IS POWERED FROM ONE VR.
- LOAD SWITCHES ARE USED TO MEET THE TURN-ON TIMING
- MSM# IS NOT USED.



PAGE TITLE			
Power - EOPiO EDRAM Supply			
	Apple Inc.		DRAWING NUMBER 051-00515
			SIZE D
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		PAGE 77 OF 145	
		SHEET 73 OF 119	

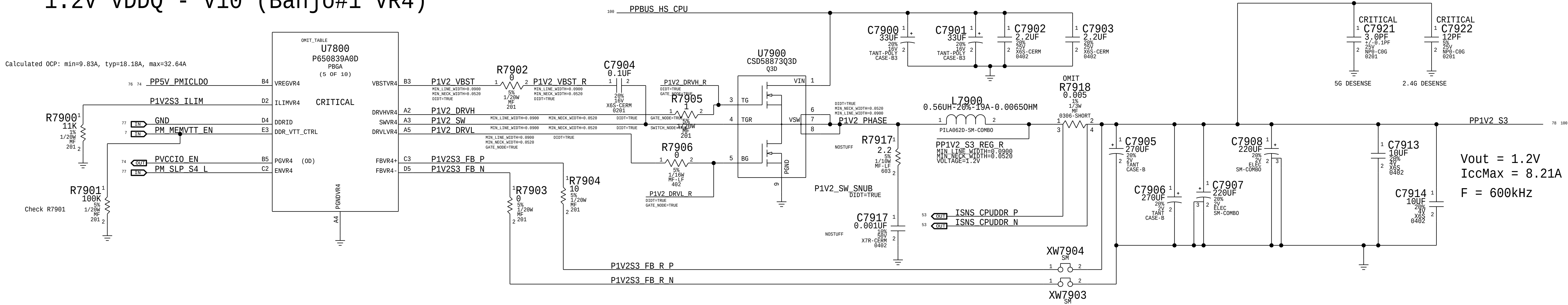
BANJO - PMIC Control



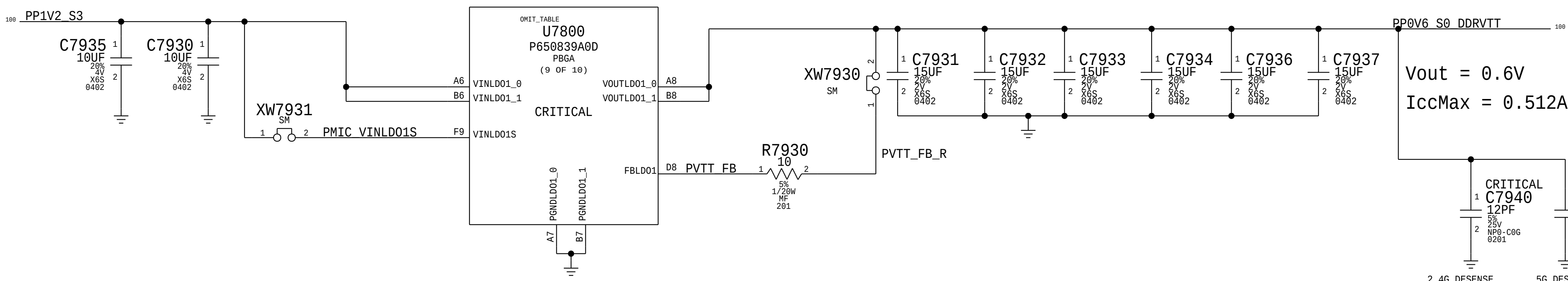
DWC PROTECTOR LTD 22082		DWC PROTECTOR LTD 22082	
PAGE TITLE			
PMIC-1 & Power Control			
	Apple Inc.		DRAWING NUMBER 051-00515
			SIZE D
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		BRANCH dvt-fab09-0	
		PAGE 78 OF 145	
		SHEET 74 OF 119	

BOM_COST_GROUP=PLATFORM POWER

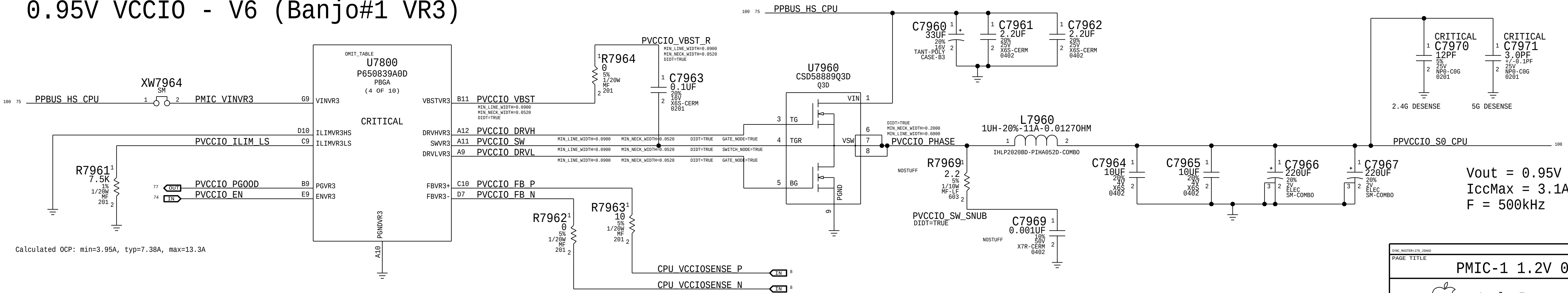
1.2V VDDQ - V10 (Banjo#1 VR4)



0.6V VTT - V13 (Banjo#1 LD01)



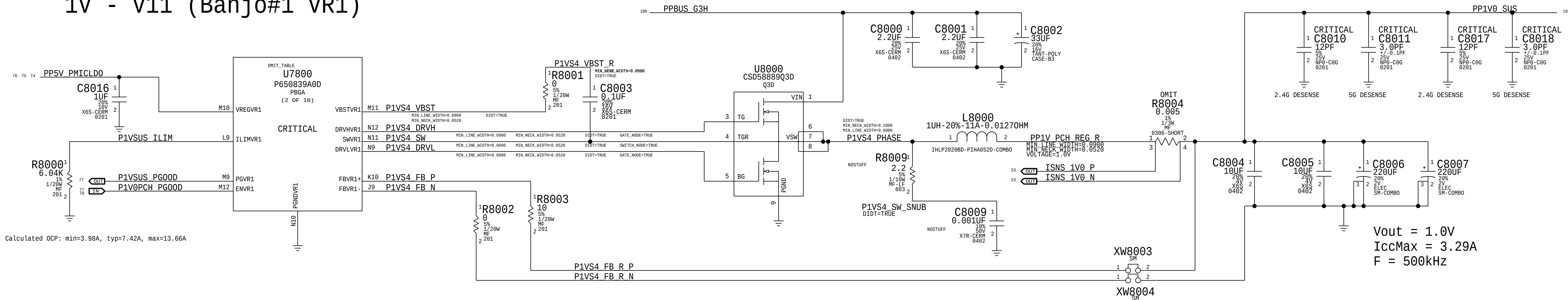
0.95V VCCIO - V6 (Banjo#1 VR3)



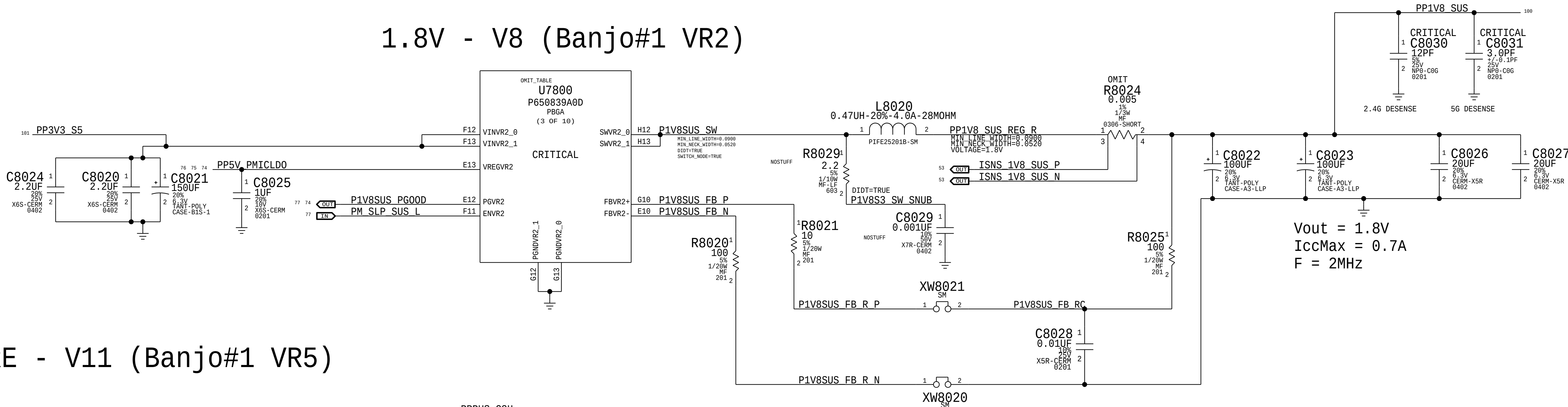
DRAWING NUMBER 051-00515		SIZE D
REVISION 9.0.0		BRANCH dvt-fab09-0
PAGE 79 OF 145		SHEET 75 OF 119

BOM_COST_GROUP=PLATFORM POWER

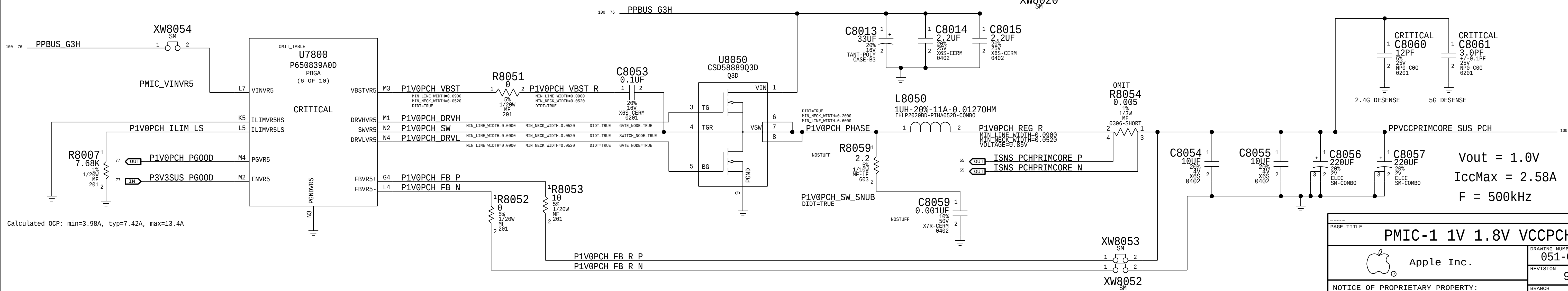
1V - V11 (Banjo#1 VR1)



1.8V - V8 (Banjo#1 VR2)

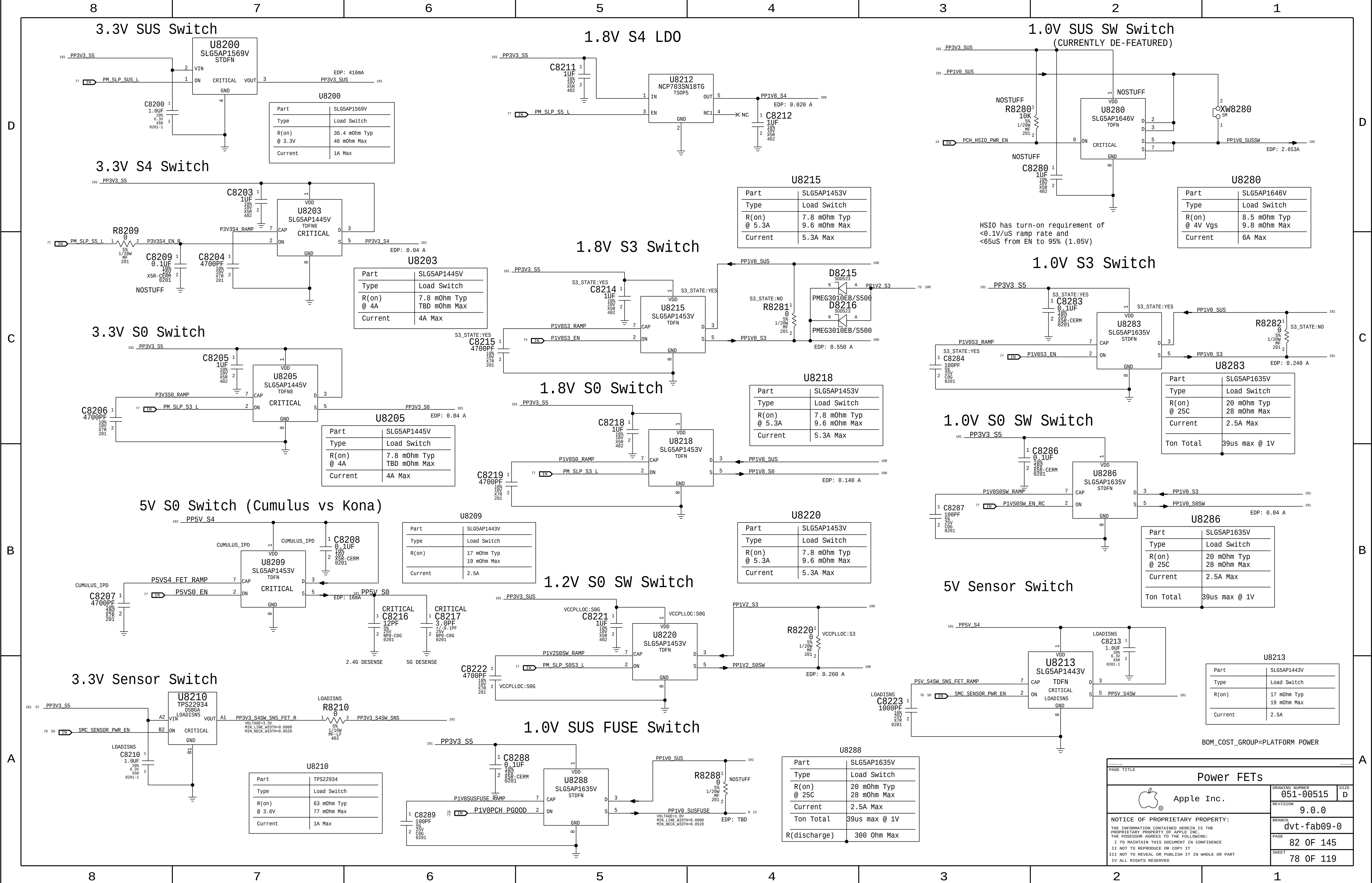


1.0V PCH CORE - V11 (Banjo#1 VR5) 0.7V LPM



PAGE TITLE		DRAWING NUMBER		SIZE
PMIC-1 1V 1.8V VCCPCH		051-00515		D
Apple Inc.		REVISTION		9.0.0
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		PAGE		80 OF 145
		SHEET		76 OF 119

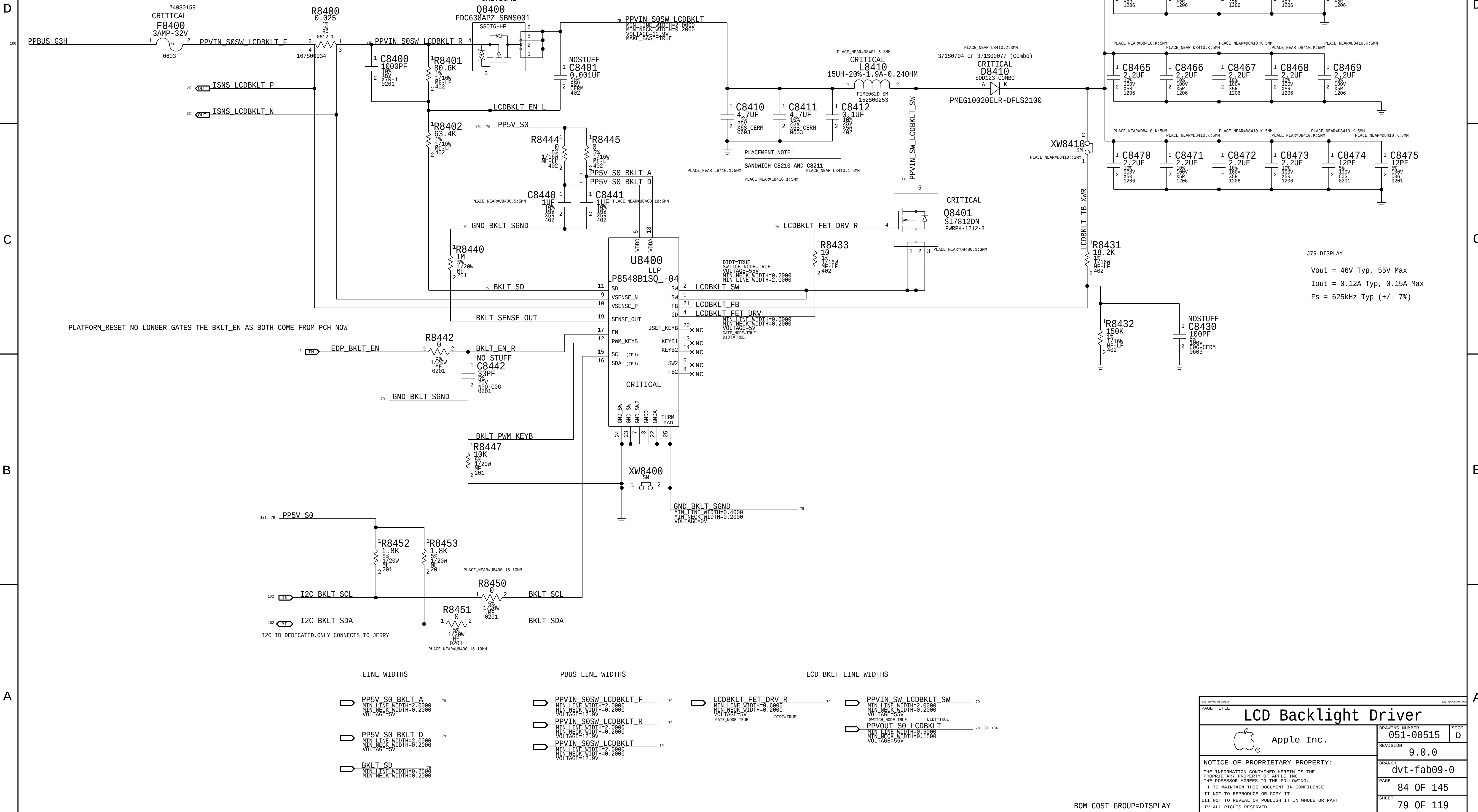
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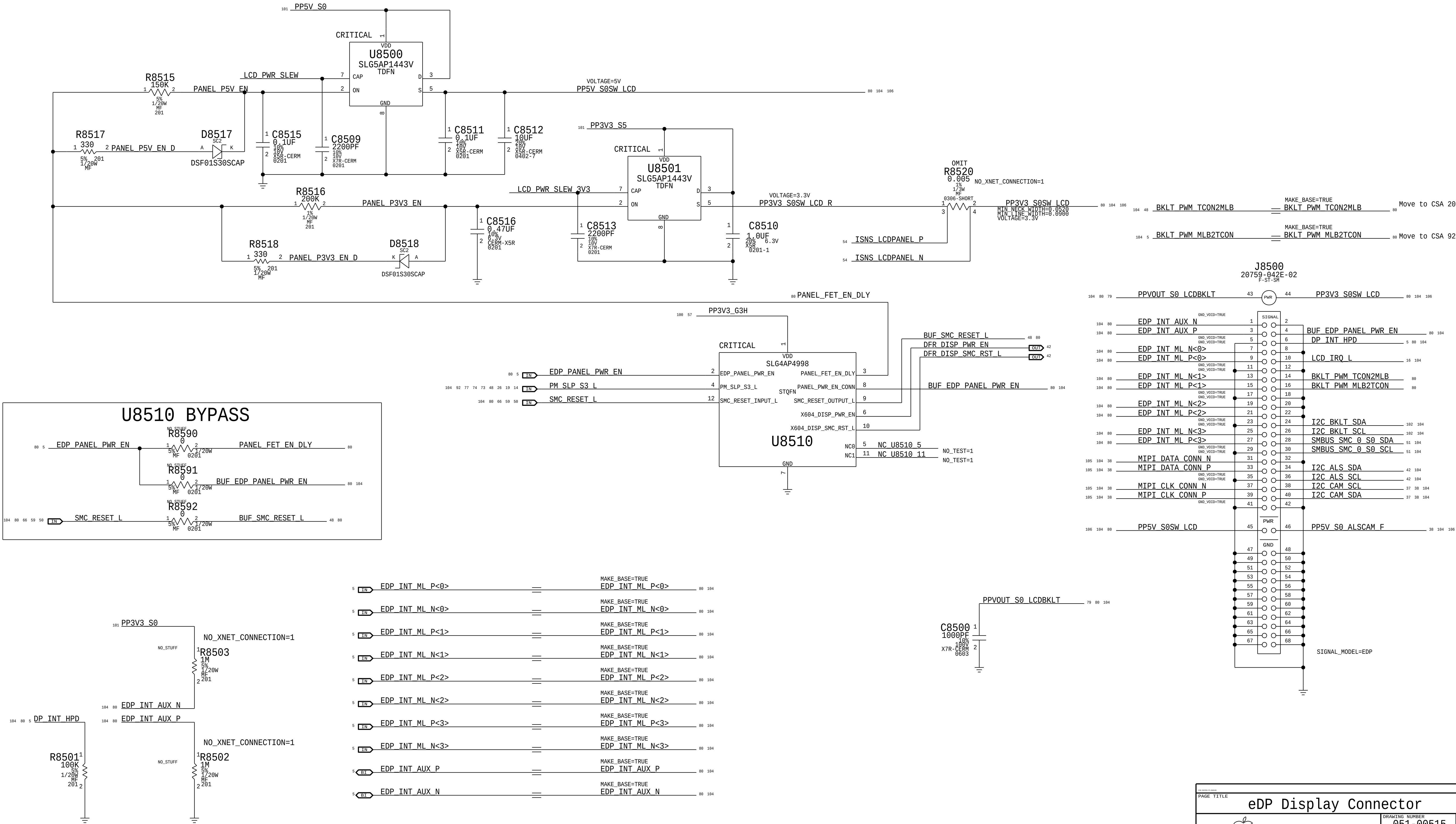
Page Notes

Power aliases required by this page:

- PPVIN_S0SW_LCDBKLTFT (9-12.6V LCD BACKLIGHT INPUT)
- PP5V_S0_BKLT (5V BACKLIGHT DRIVER INPUT)



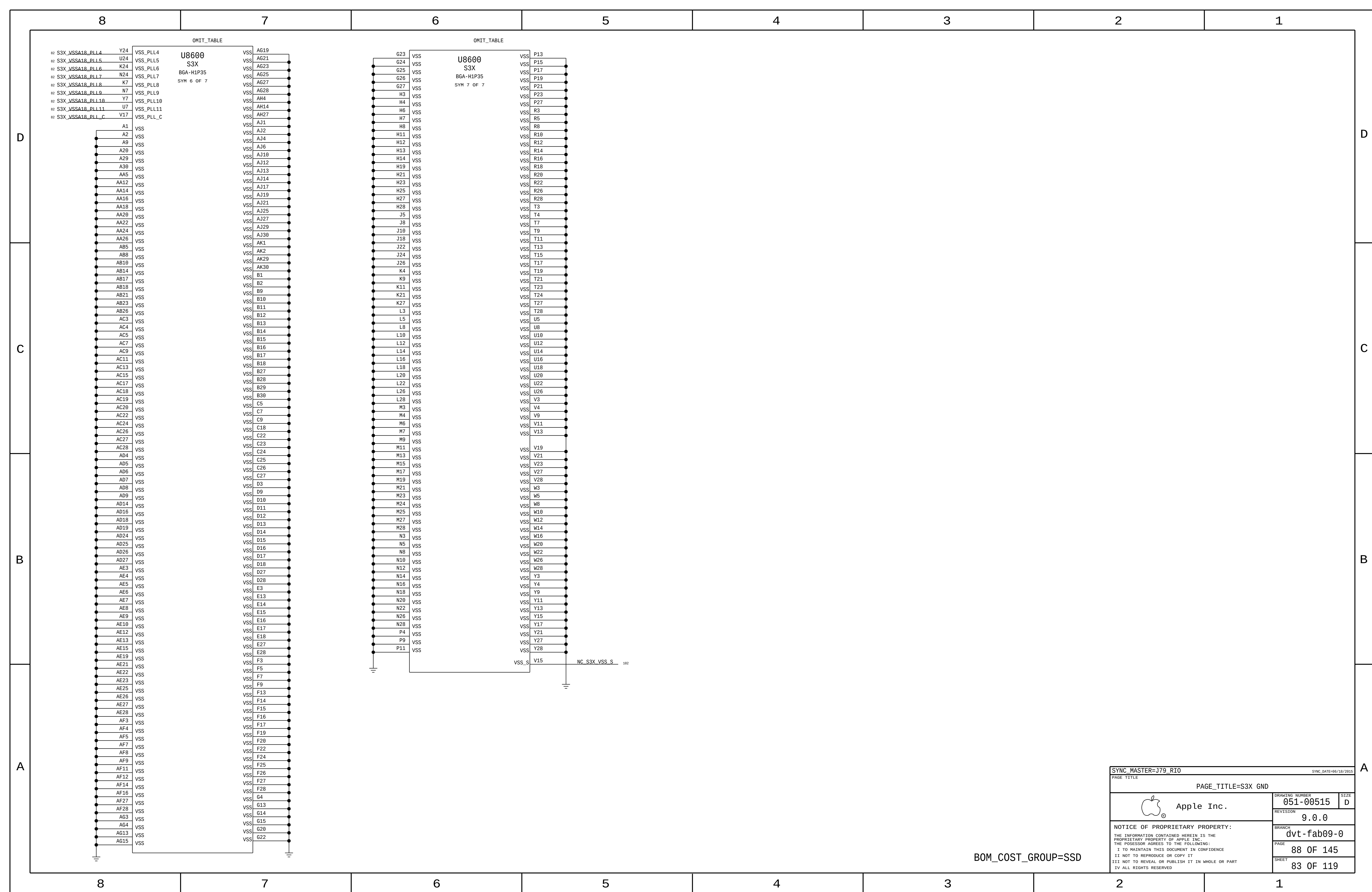
LCD PANEL INTERFACE (eDP) + Camera (MIPI)

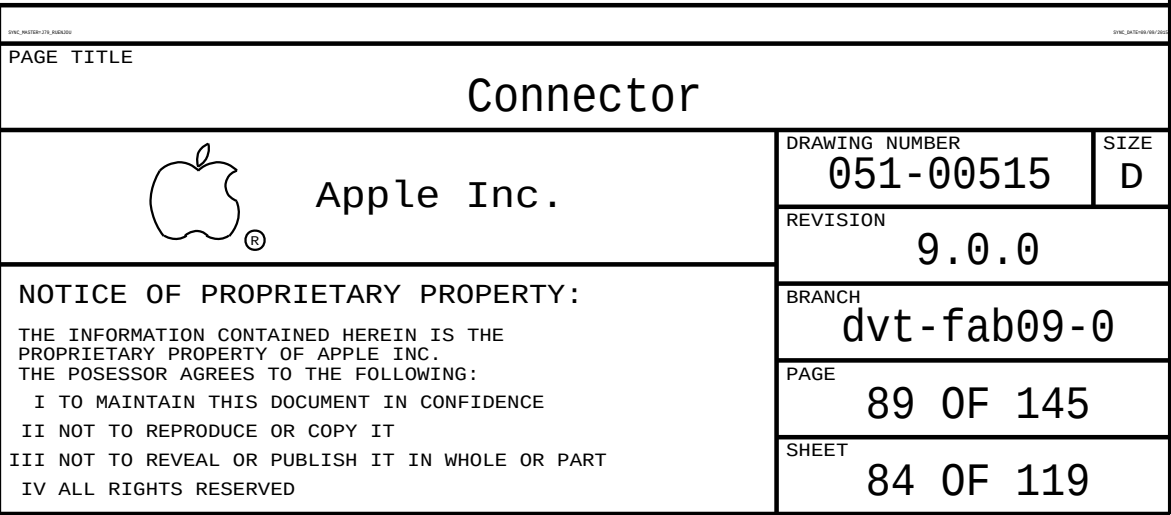


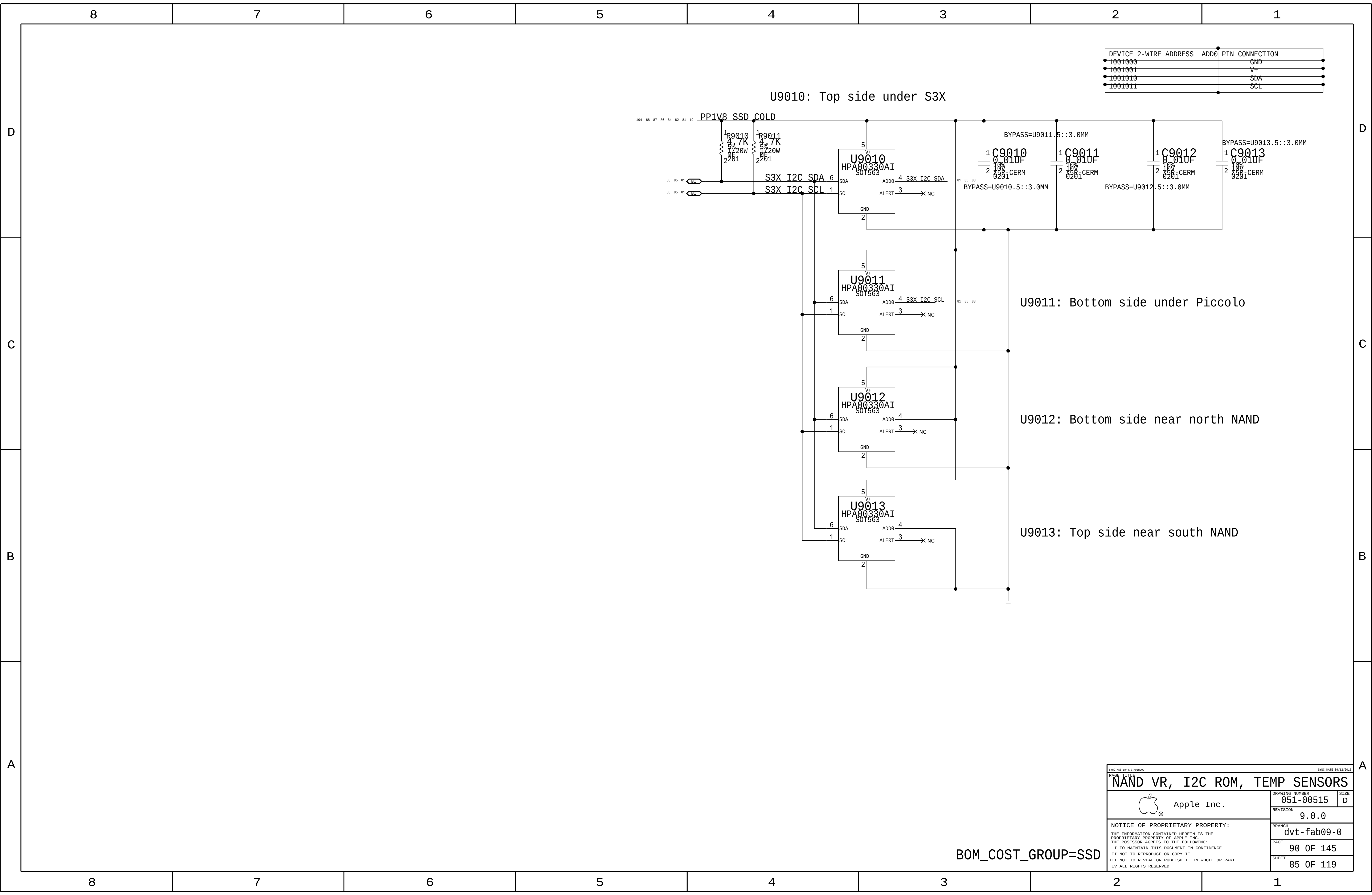
LCD Panel HPD & AUX strapping

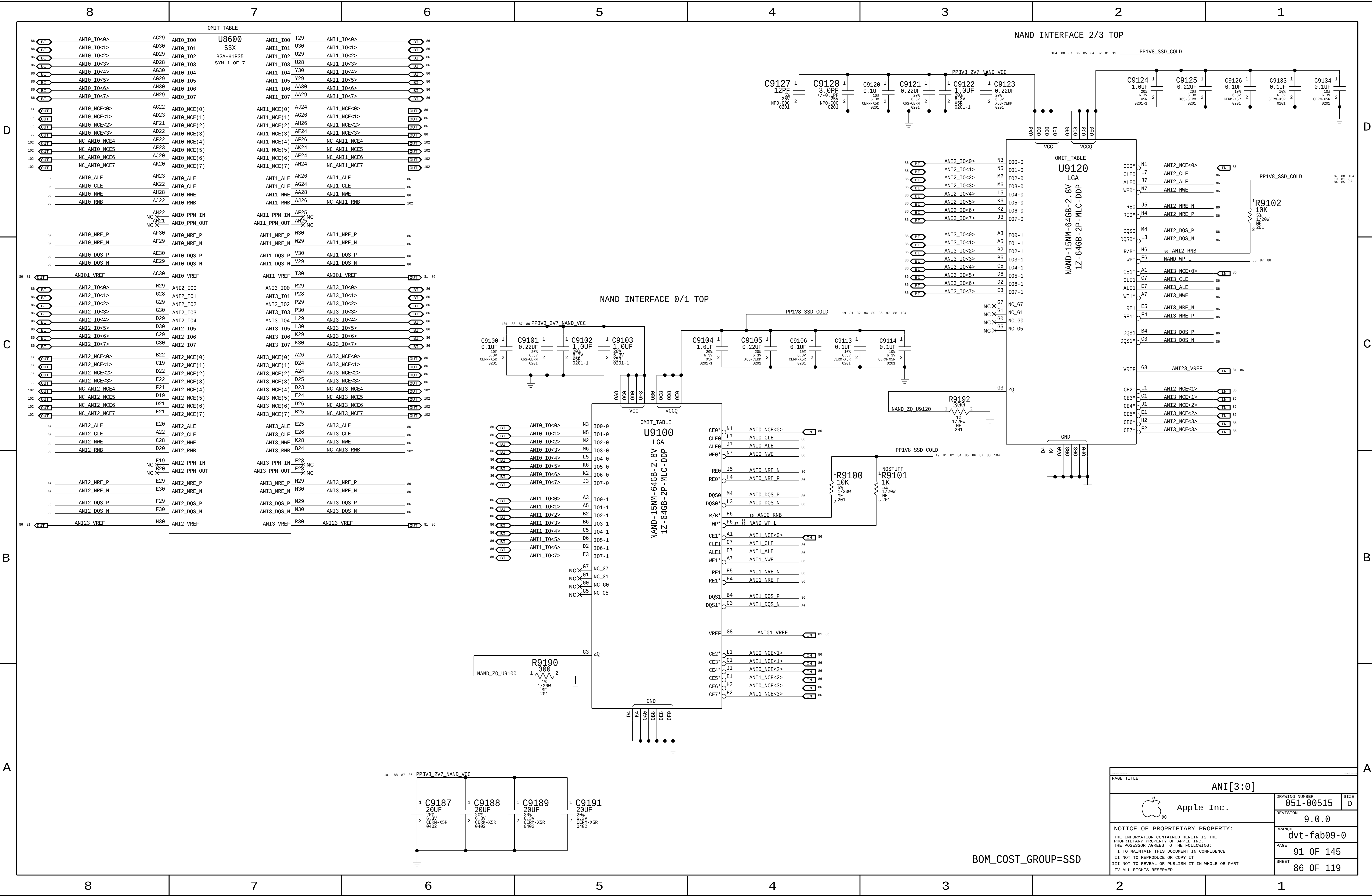
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eDP Display Connector		
 Apple Inc.	DRAWING NUMBER	051-00515
	REVISION	9.0.0
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	PAGE	85 OF 145
	SHEET	80 OF 119

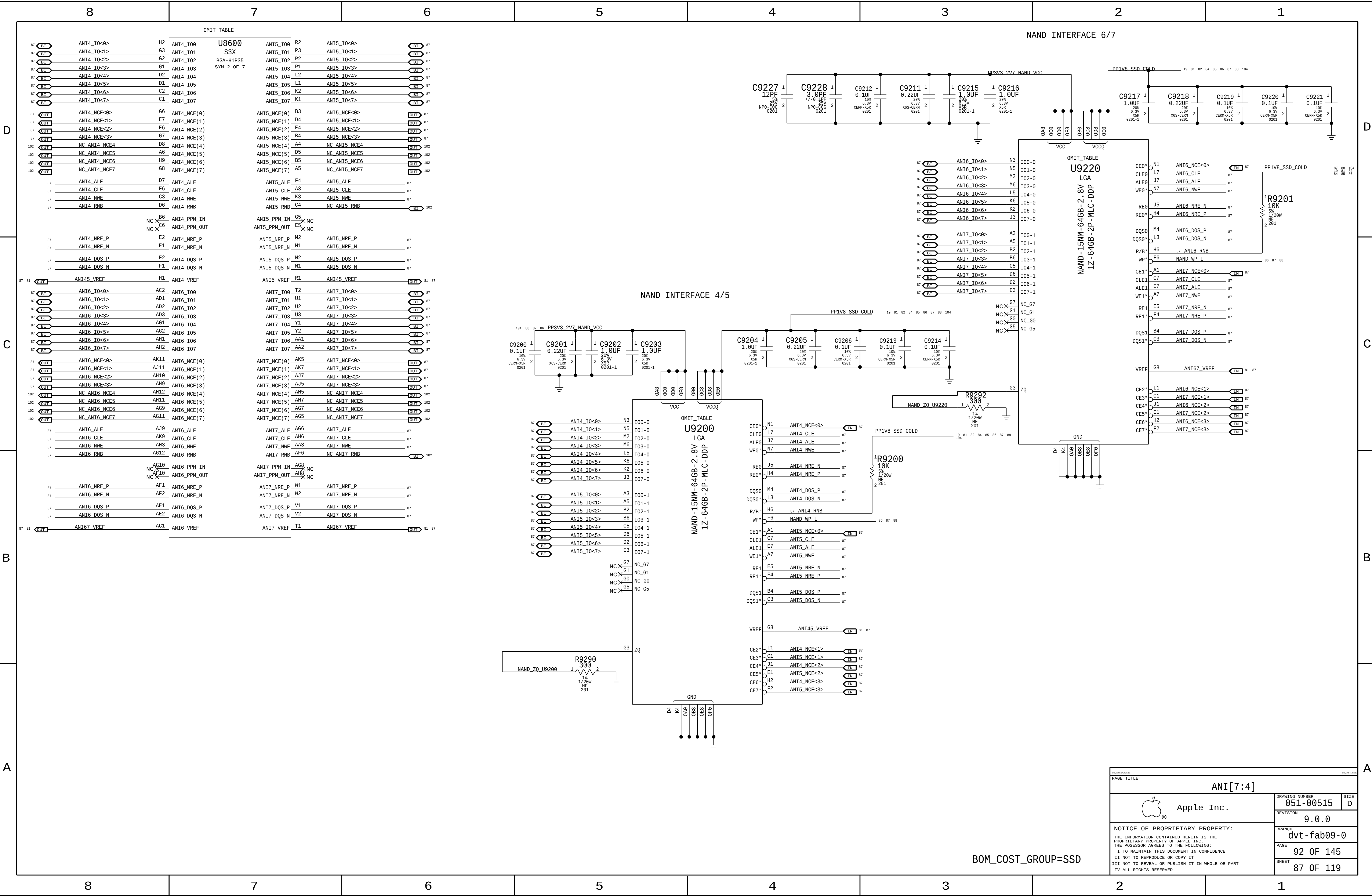
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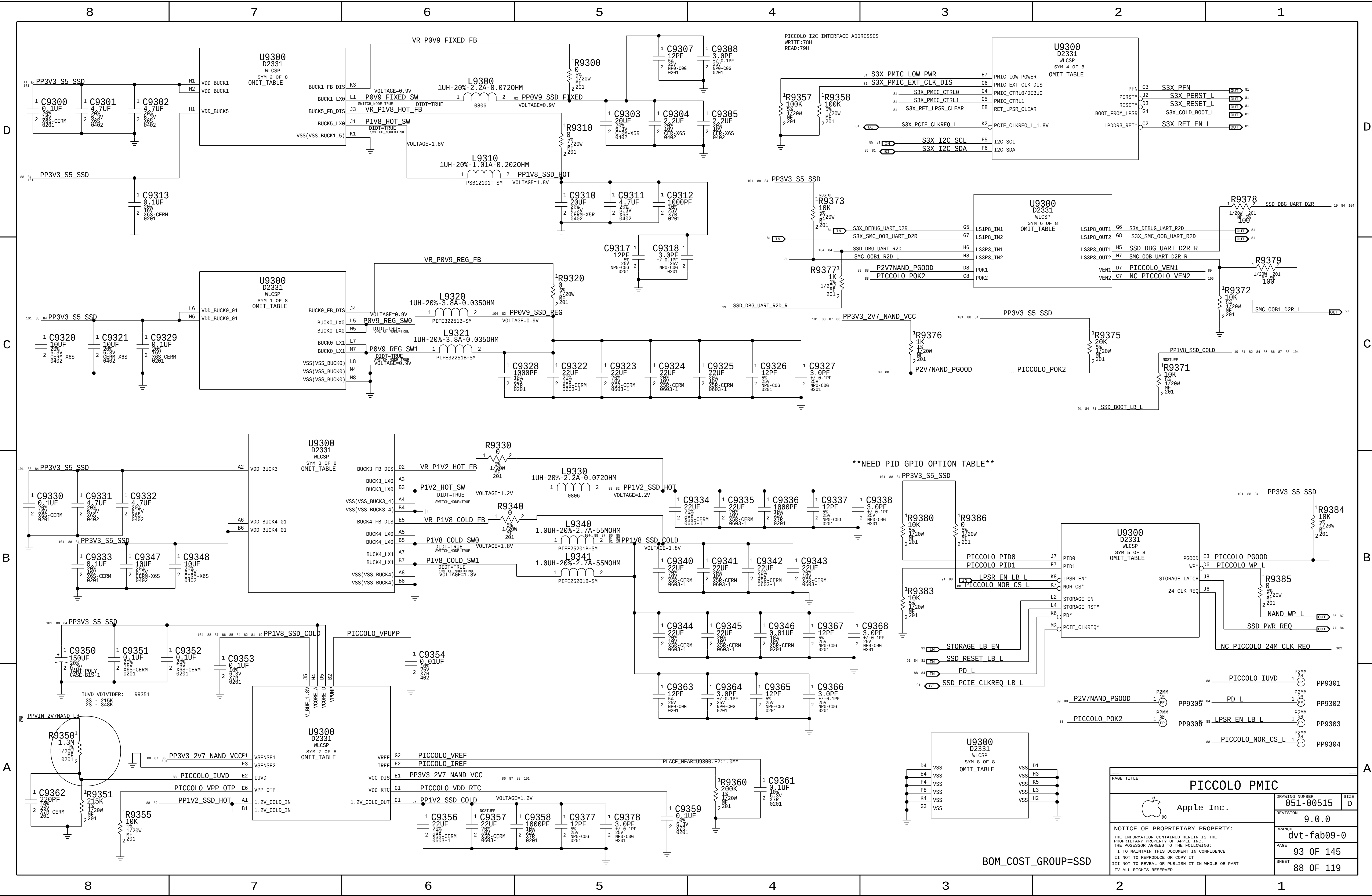






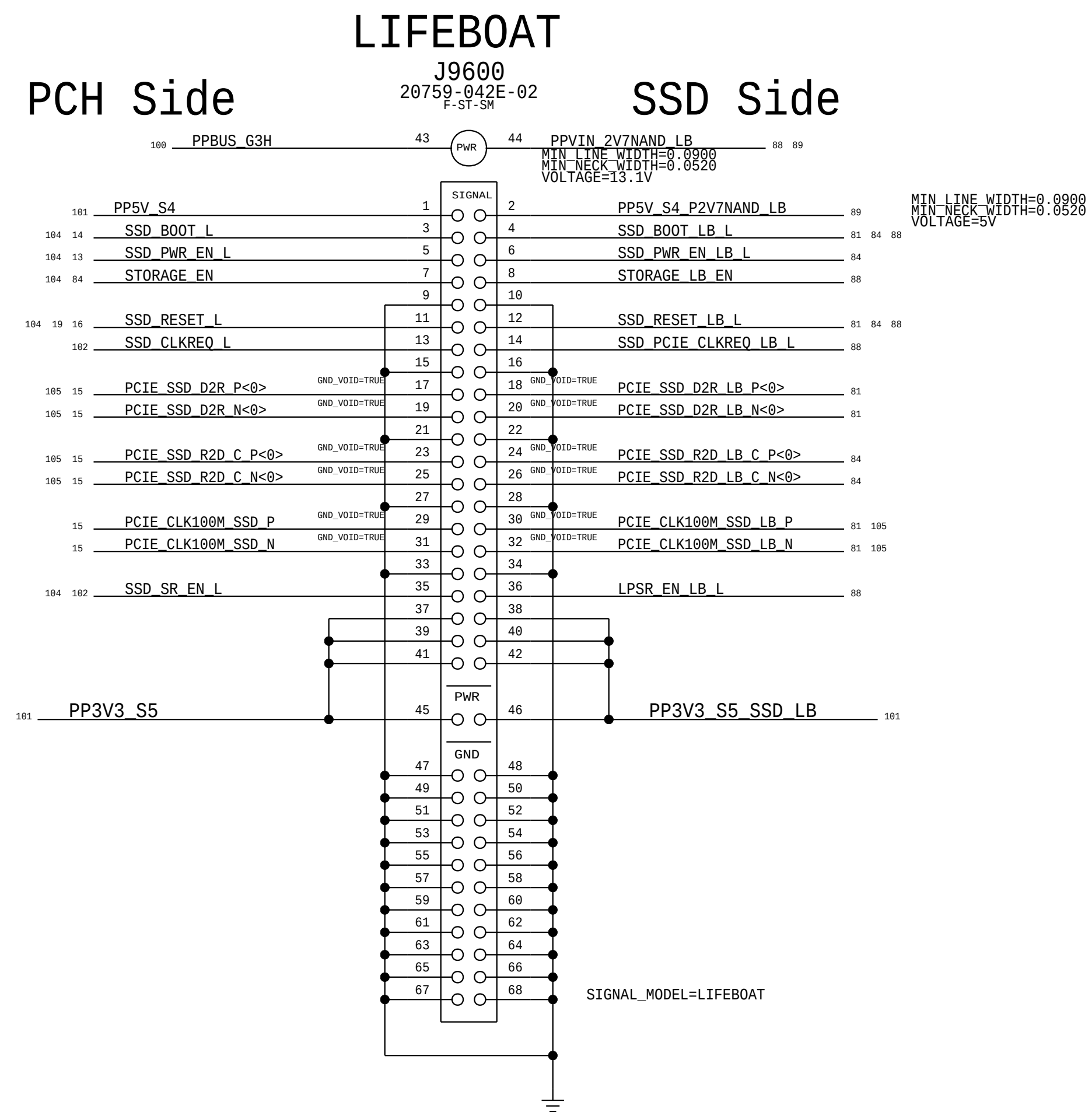




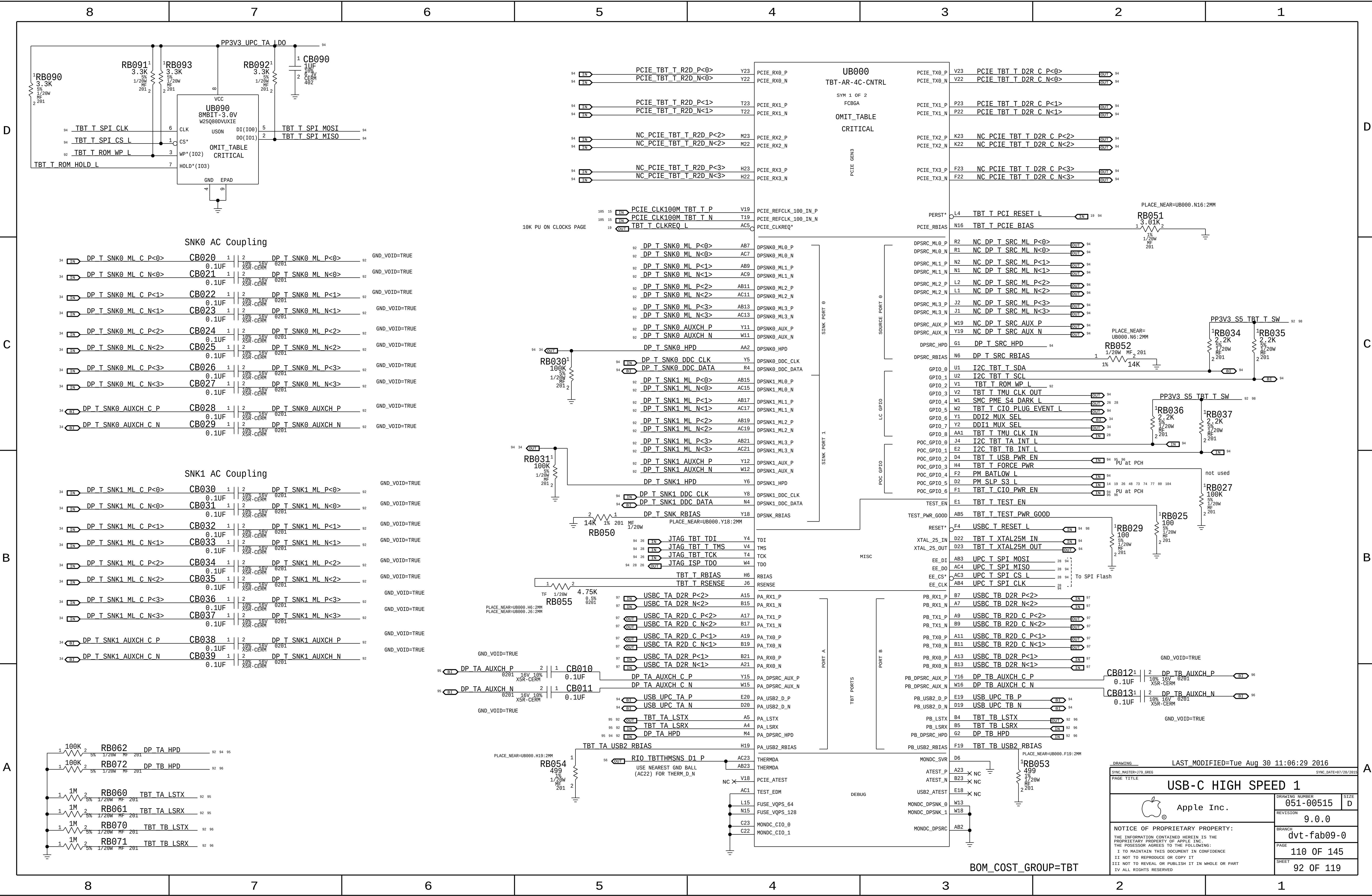


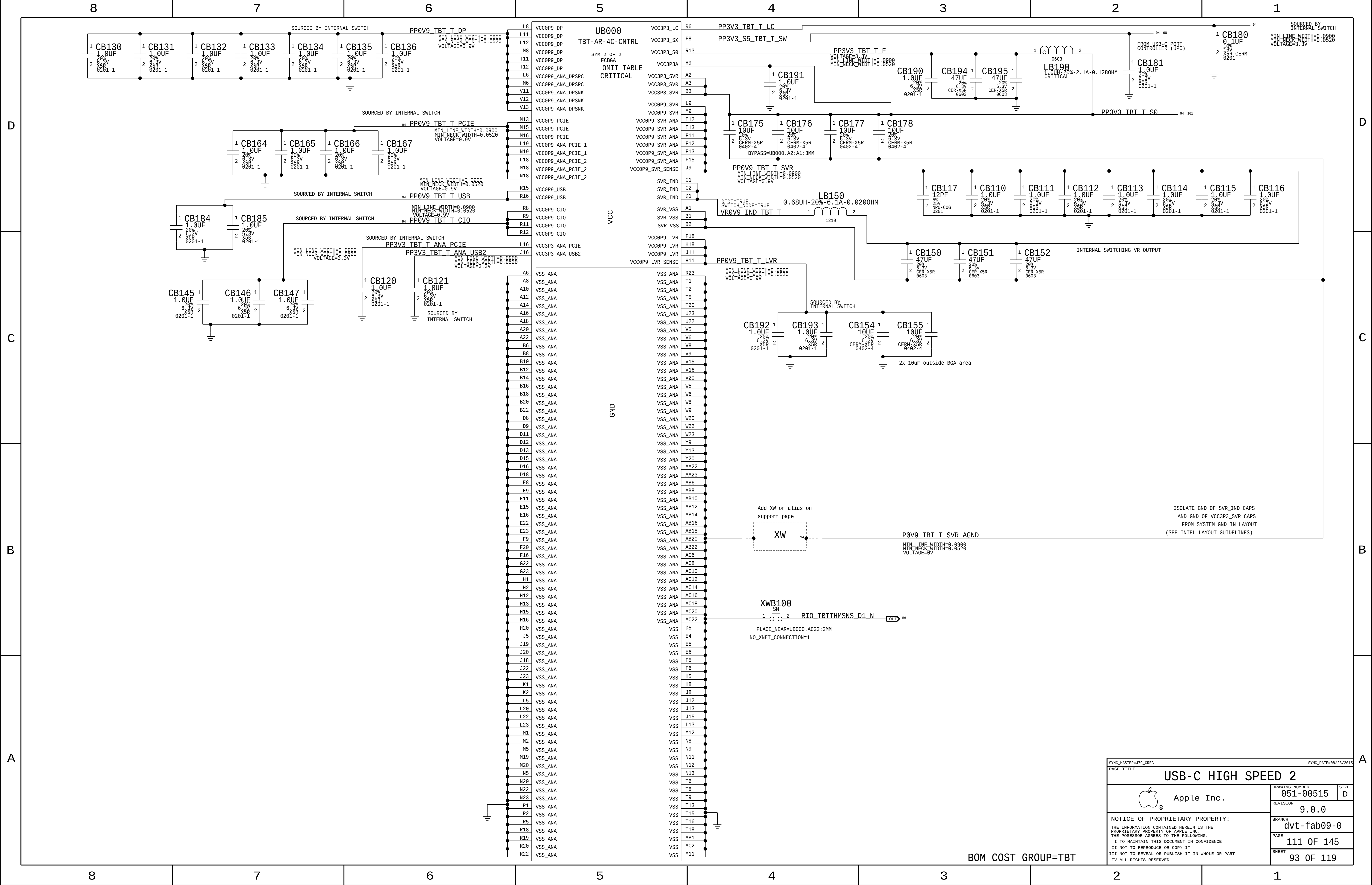
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C							C
B							B
A							A
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		PAGE	95 OF 145
		SHEET	90 OF 119



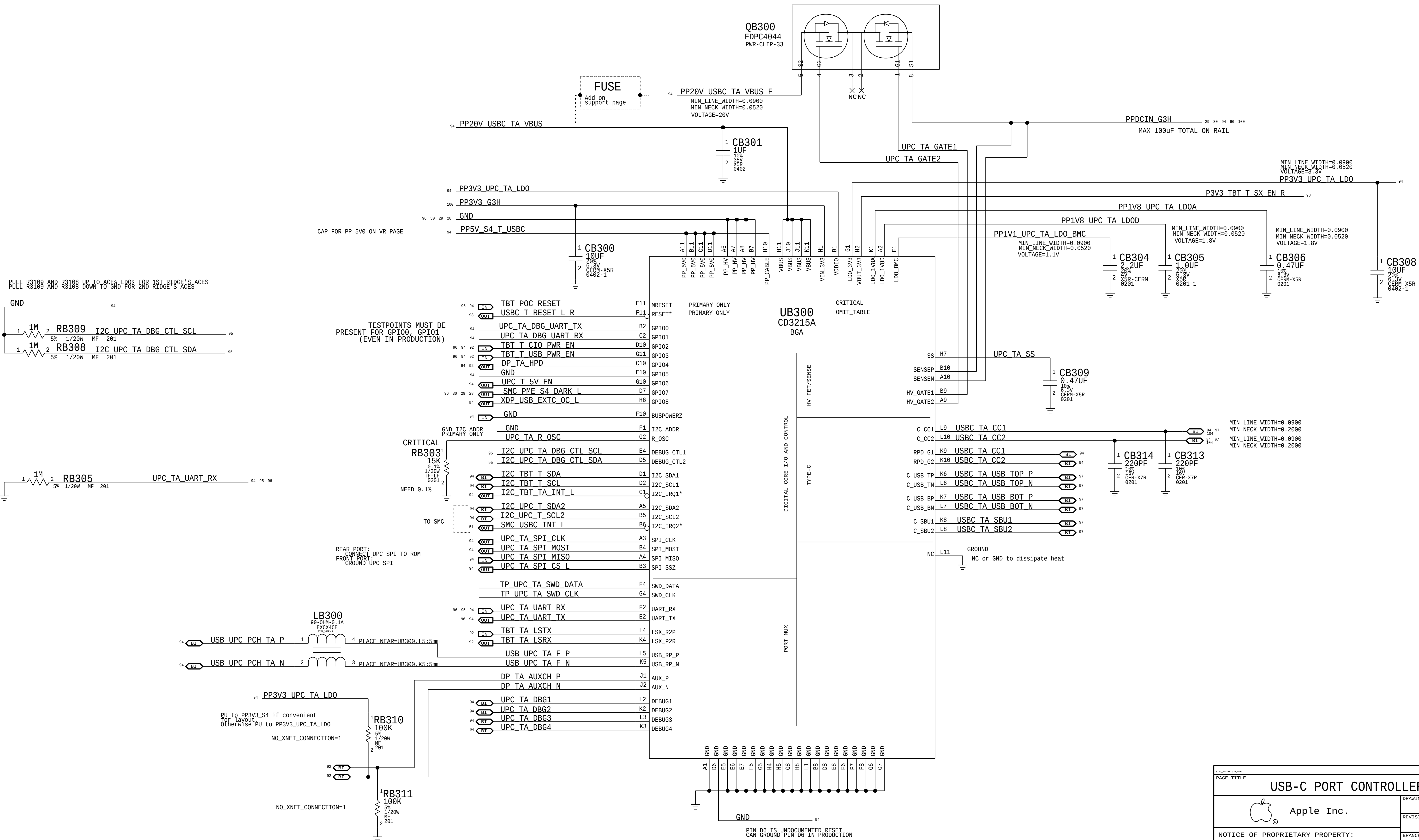
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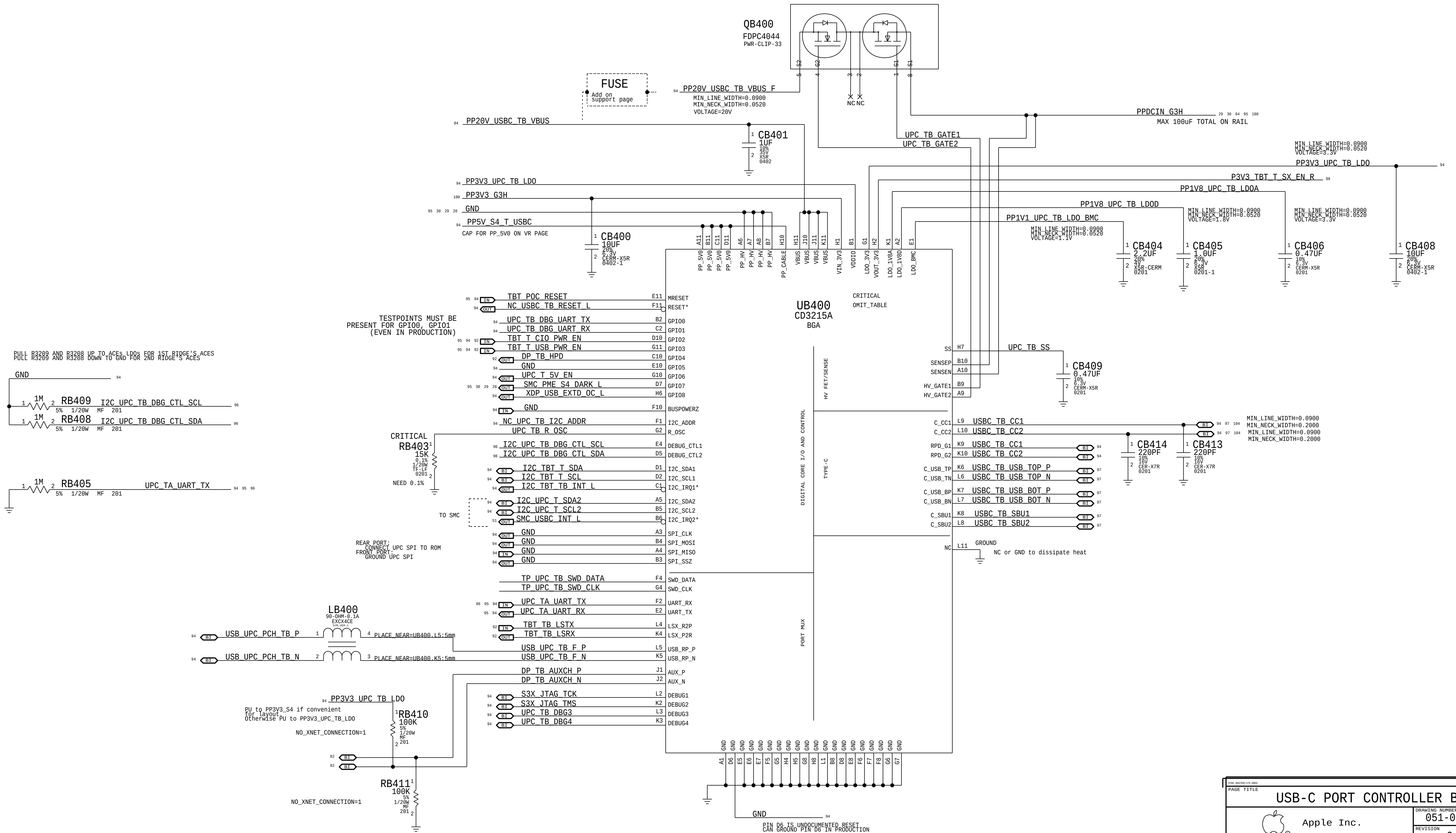


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USB-C HIGH SPEED 2			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
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	9.0.0		
	BRANCH		
	dvt-fab09-0		
	PAGE		
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PRIMARY ACE USB-C PORT CONTROLLER (UPC)



SECONDARY ACE USB-C PORT CONTROLLER (UPC)



BOM_COST_GROUP=USB-C

DWG. PART-0718-010_002 PAGE TITLE		DWG. DATE: 05/12/2019	
USB-C PORT CONTROLLER B			
 Apple Inc.	DRAWING NUMBER 051-000515		SIZE D
	REVISION 9.0.0		
	NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		
BRANCH dvt-fab09-0		PAGE 114 OF 145	
SHEET 96 OF 119			

D

C

B

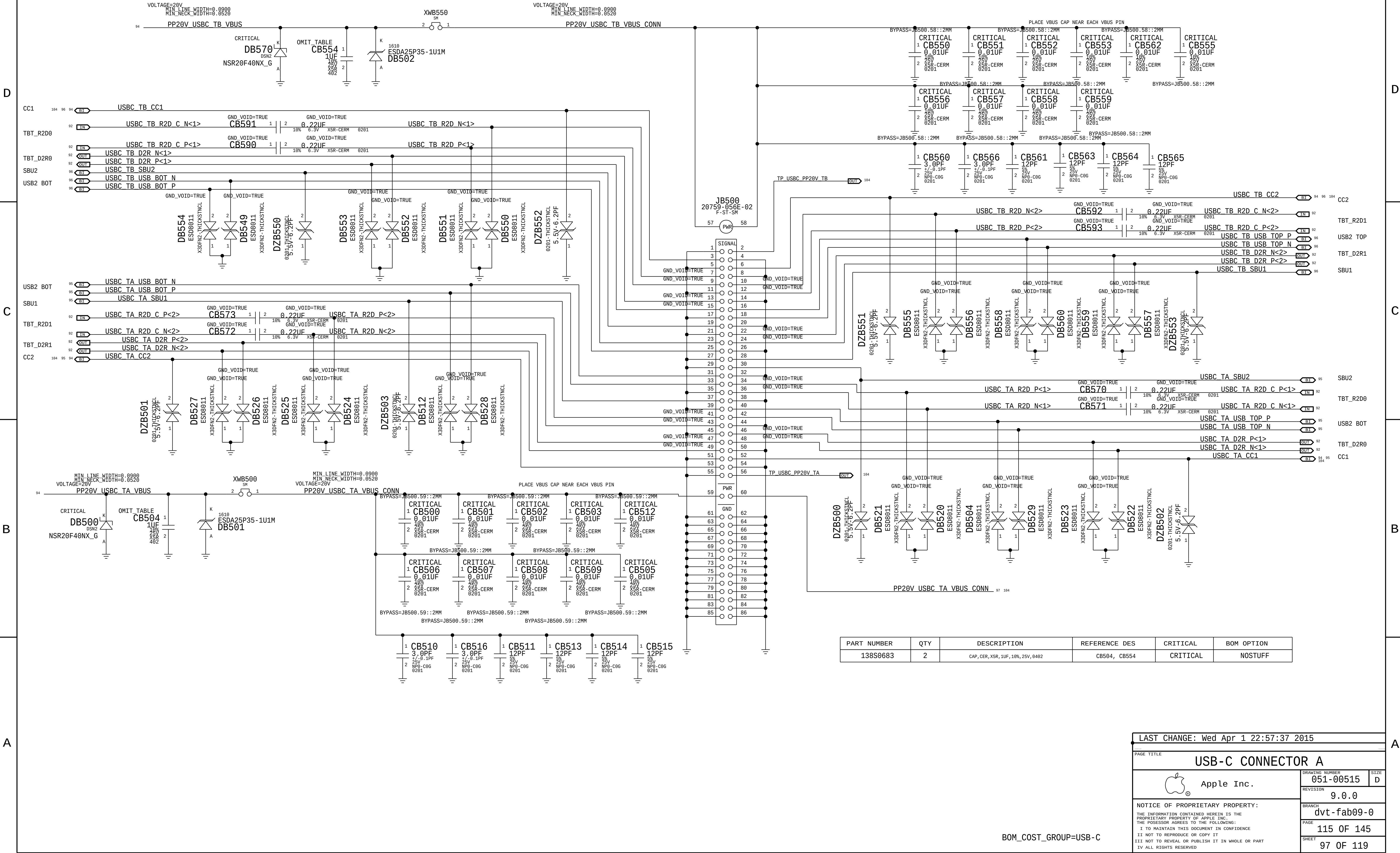
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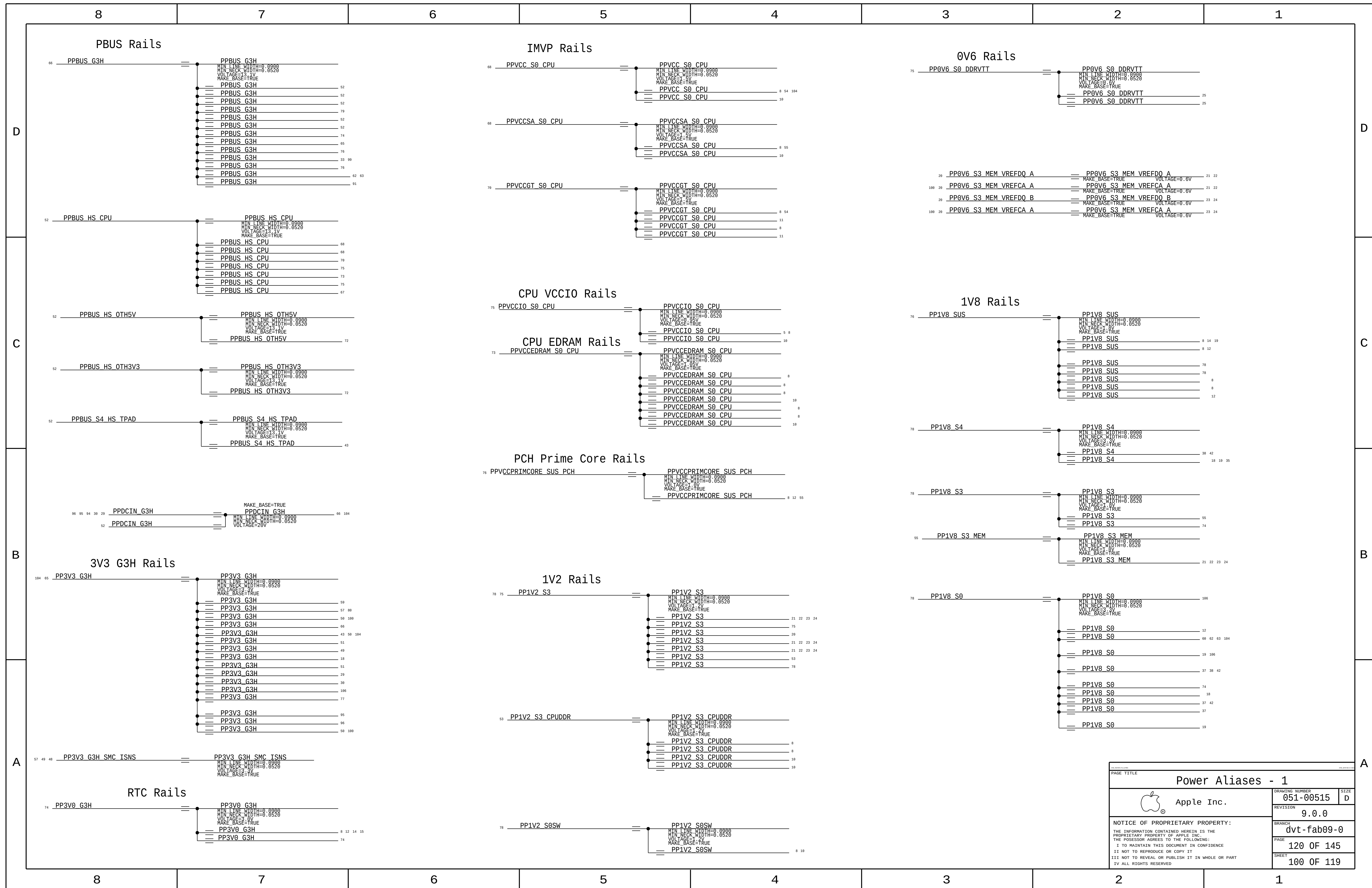
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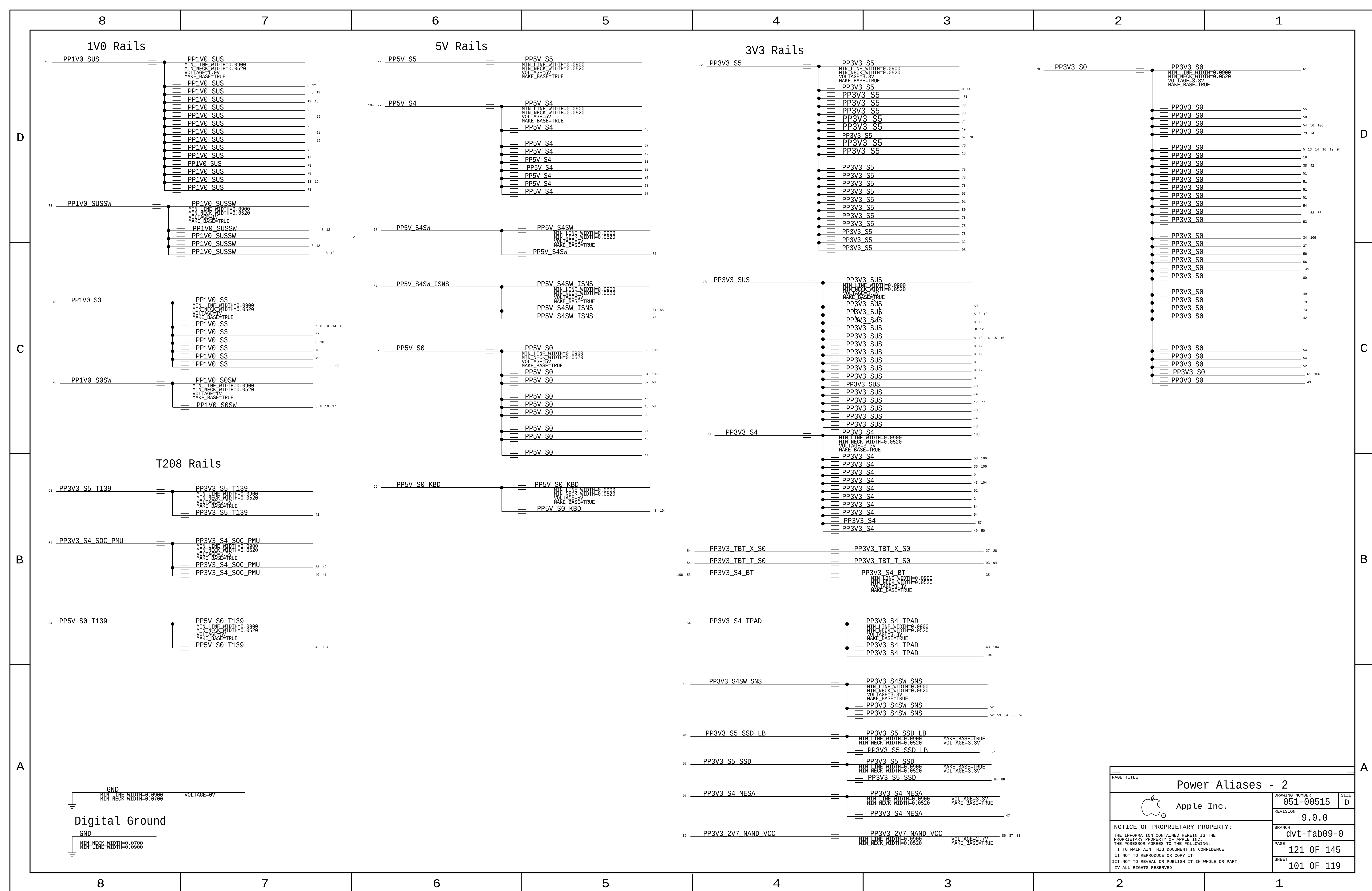
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LAST CHANGE: Wed Apr 1 22:57:37 2015					
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 Apple Inc.				DRAWING NUMBER	051-00515
				REVISTION	9.0.0
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				PAGE	115 OF 145
				SHEET	97 OF 119

BOM_COST_GROUP=USB-C





UNUSED TP ALIASES

UNUSED TP ALIASES

UNUSED SIGNAL ALIAS

83	NC S3X VSS S	==	NC S3X VSS S
82	NC S3X VDD S	==	NC S3X VDD S
81	NC S3X PMIC CTRL 2	==	NC S3X PMIC CTRL 2
81	NC S3X PCIE ATB1	==	NC S3X PCIE ATB1
81	NC S3X PCIE ATB0	==	NC S3X PCIE ATB0
81	NC S3X DT1	==	NC S3X DT1
81	NC S3X DT0	==	NC S3X DT0
81	NC S3X DDR ATO	==	NC S3X DDR ATO
74	NC PMIC PGG	==	NC PMIC PGG
74	NC PMIC PGF	==	NC PMIC PGF
74	NC PMIC PGE	==	NC PMIC PGE
74	NC PMIC PGD	==	NC PMIC PGD
74	NC PMIC PGC	==	NC PMIC PGC
88	NC PICCOLO 24M CLK REQ	==	NC PICCOLO 24M CLK REQ
14	NC PCH SLP A L	==	NC PCH SLP A L
14	NC PCH PME L	==	NC PCH PME L
14	NC PCH LANPHYPC	==	NC PCH LANPHYPC
14	NC PCH GPP F9	==	NC PCH GPP F9
14	NC PCH GPP F8	==	NC PCH GPP F8
14	NC PCH GPP F10	==	NC PCH GPP F10
16	NC PCH GPP D4	==	NC PCH GPP D4
16	NC PCH GPP D3	==	NC PCH GPP D3
16	NC PCH GPP D1	==	NC PCH GPP D1
14	NC PCH GPP D0	==	NC PCH GPP D0
14	NC PCH GPD7	==	NC PCH GPD7
99	NC P5VUSBC T PG00D	==	NC P5VUSBC T PG00D
34	NC DPMUX SAK 20	==	NC DPMUX SAK 20
34	NC DPMUX SAK 19	==	NC DPMUX SAK 19
34	NC DPMUX SAK 18	==	NC DPMUX SAK 18
34	NC DPMUX SAK 17	==	NC DPMUX SAK 17
34	NC DPMUX SAK 16	==	NC DPMUX SAK 16
34	NC DPMUX SAK 15	==	NC DPMUX SAK 15
34	NC DPMUX SAK 14	==	NC DPMUX SAK 14

6	NC CPU RSVD BB69	==	NC CPU RSVD BB69
6	NC CPU RSVD BB68	==	NC CPU RSVD BB68
6	NC CPU RSVD BA70	==	NC CPU RSVD BA70
6	NC CPU RSVD BA68	==	NC CPU RSVD BA68
6	NC CPU RSVD AW71	==	NC CPU RSVD AW71
6	NC CPU RSVD AW70	==	NC CPU RSVD AW70
6	NC CPU RSVD AK13	==	NC CPU RSVD AK13
6	NC CPU RSVD AK12	==	NC CPU RSVD AK12
9	NC CPU NCTFVSS C1	==	NC CPU NCTFVSS C1
9	NC CPU NCTFVSS BB70	==	NC CPU NCTFVSS BB70
9	NC CPU NCTFVSS BA71	==	NC CPU NCTFVSS BA71
9	NC CPU NCTFVSS BA1	==	NC CPU NCTFVSS BA1
9	NC CPU NCTFVSS B71	==	NC CPU NCTFVSS B71
9	NC CPU NCTFVSS AV1	==	NC CPU NCTFVSS AV1
9	NC CPU NCTFVSS A70	==	NC CPU NCTFVSS A70
9	NC CPU NCTFVSS A5	==	NC CPU NCTFVSS A5
6	NC CPU BB5	==	NC CPU BB5
6	NC CPU BB3	==	NC CPU BB3
6	NC CPU AY4	==	NC CPU AY4
6	NC CPU AU5	==	NC CPU AU5
6	NC CPU AT5	==	NC CPU AT5
73	NC CPU MSM L	==	NC CPU MSM L
28	NC USBC XA RESET L	==	NC USBC XA RESET L
34	NC USBC TB RESET L	==	NC USBC TB RESET L

UNUSED SIGNALS

14	NC_PCH_SLP_WLAN_L	==	TRUE	NC_PCH_SLP_WLAN_L	105
102	NC_SPI_CS1_L	==	TRUE	NC_SPI_CS1_L	102 105
102	NC_SPI_CS2_L	==	TRUE	NC_SPI_CS2_L	102 105
102	NC_SPI_CS1_L	==	TRUE	NC_SPI_CS1_L	102 105
102	NC_SPI_CS2_L	==	TRUE	NC_SPI_CS2_L	102 105
15	NC_PCIE_CAMERA_D2R_N	==	NC_PCIE_CAMERA_D2R_N	TRUE	
15	NC_PCIE_CAMERA_D2R_P	==	NC_PCIE_CAMERA_D2R_P	TRUE	
15	NC_PCIE_CAMERA_R2D_C_N	==	NC_PCIE_CAMERA_R2D_C_N	TRUE	105
15	NC_PCIE_CAMERA_R2D_C_P	==	NC_PCIE_CAMERA_R2D_C_P	TRUE	105
15	NC_PCIE_CLK100M_CAMERA_N	==	NC_PCIE_CLK100M_CAMERA_N	105	TRUE
15	NC_PCIE_CLK100M_CAMERA_P	==	NC_PCIE_CLK100M_CAMERA_P	105	TRUE
15	NC_USB_EXT_A_N	==	NC_USB_EXT_A_N	TRUE	
15	NC_USB_EXT_A_P	==	NC_USB_EXT_A_P	NO_TEST=1	TRUE
15	NC_USB_EXT_B_N	==	NC_USB_EXT_B_N	TRUE	
15	NC_USB_EXT_B_P	==	NC_USB_EXT_B_P	NO_TEST=1	TRUE
6	NC_XDP_BPM_L<1>	==	NC_XDP_BPM_L<1>	TRUE	
6	NC_XDP_BPM_L<2>	==	NC_XDP_BPM_L<2>	NO_TEST=1	TRUE
6	NC_XDP_BPM_L<3>	==	NC_XDP_BPM_L<3>	NO_TEST=1	TRUE
15	NC_USB3_EXTB_D2R_N	==	NC_USB3_EXTB_D2R_N	TRUE	
15	NC_USB3_EXTB_D2R_P	==	NC_USB3_EXTB_D2R_P	NO_TEST=1	TRUE
15	NC_USB3_EXTB_R2D_C_N	==	NC_USB3_EXTB_R2D_C_N	NO_TEST=1	TRUE
15	NC_USB3_EXTB_R2D_C_P	==	NC_USB3_EXTB_R2D_C_P	NO_TEST=1	TRUE

86	NC_ANI0_NCE4	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI0_NCE4
86	NC_ANI0_NCE5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI0_NCE5
86	NC_ANI0_NCE6	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI0_NCE6
86	NC_ANI0_NCE7	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI0_NCE7
86	NC_ANI1_NCE4	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI1_NCE4
86	NC_ANI1_NCE5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI1_NCE5
86	NC_ANI1_NCE6	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI1_NCE6
86	NC_ANI1_NCE7	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI1_NCE7
86	NC_ANI2_NCE4	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI2_NCE4
86	NC_ANI2_NCE5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI2_NCE5
86	NC_ANI2_NCE6	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI2_NCE6
86	NC_ANI2_NCE7	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI2_NCE7
86	NC_ANI3_NCE4	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI3_NCE4
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86	NC_ANI3_NCE7	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI3_NCE7
87	NC_ANI4_NCE4	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI4_NCE4
87	NC_ANI4_NCE5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI4_NCE5
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87	NC_ANI4_NCE7	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI4_NCE7
87	NC_ANI5_NCE4	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI5_NCE4
87	NC_ANI5_NCE5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI5_NCE5
87	NC_ANI5_NCE6	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI5_NCE6
87	NC_ANI5_NCE7	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI5_NCE7
87	NC_ANI6_NCE4	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI6_NCE4
87	NC_ANI6_NCE5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI6_NCE5
87	NC_ANI6_NCE6	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI6_NCE6
87	NC_ANI6_NCE7	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI6_NCE7
87	NC_ANI7_NCE4	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI7_NCE4
87	NC_ANI7_NCE5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI7_NCE5
87	NC_ANI7_NCE6	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI7_NCE6
87	NC_ANI7_NCE7	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI7_NCE7
86	NC_ANI1_RNB	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI1_RNB
86	NC_ANI3_RNB	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI3_RNB
87	NC_ANI5_RNB	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI5_RNB
87	NC_ANI7_RNB	==	MAKE_BASE=TRUE	NO_TEST=1	NC_ANI7_RNB

81	NC_S3X_GPI01	==	MAKE_BASE=TRUE	NO_TEST=1	NC_S3X_GPI01
81	NC_S3X_BOOTSTRAP6	==	MAKE_BASE=TRUE	NO_TEST=1	NC_S3X_BOOTSTRAP6
81	NC_S3X_BOOTSTRAP1	==	MAKE_BASE=TRUE	NO_TEST=1	NC_S3X_BOOTSTRAP1

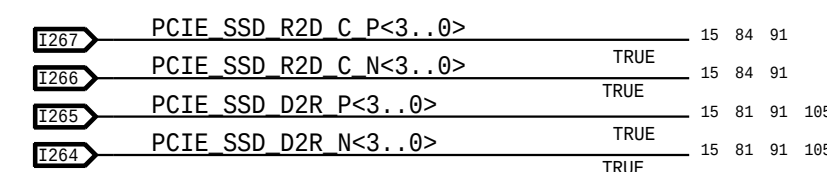
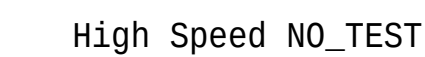
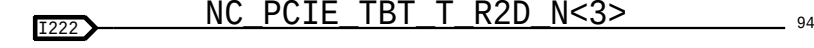
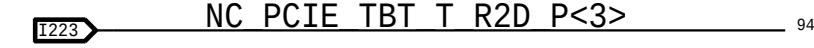
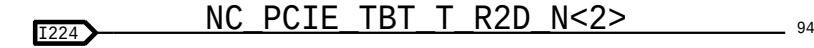
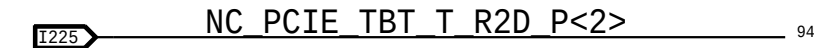
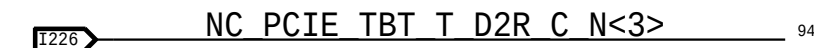
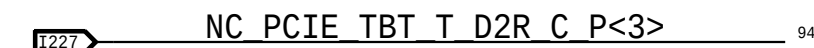
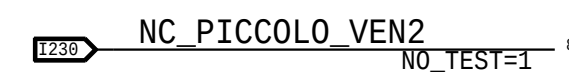
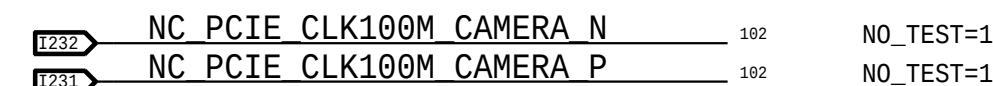
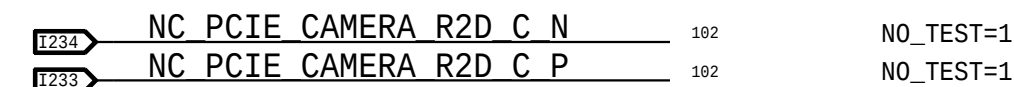
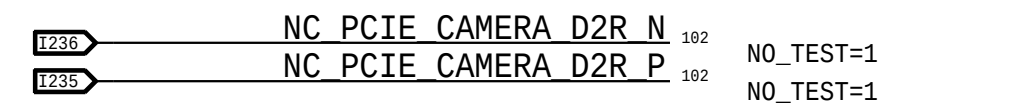
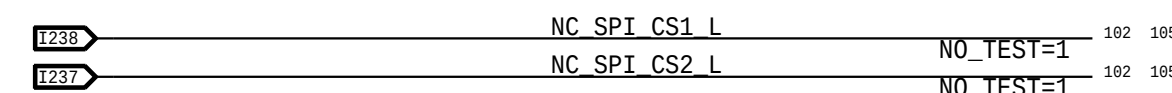
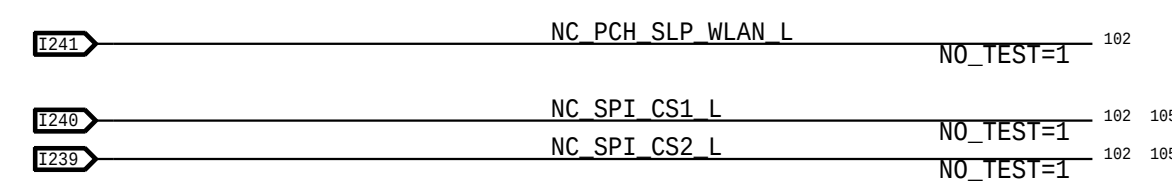
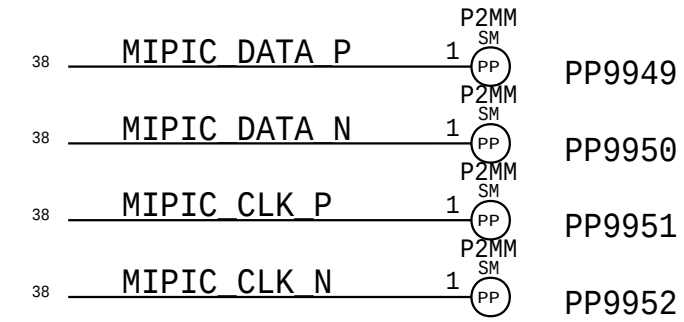
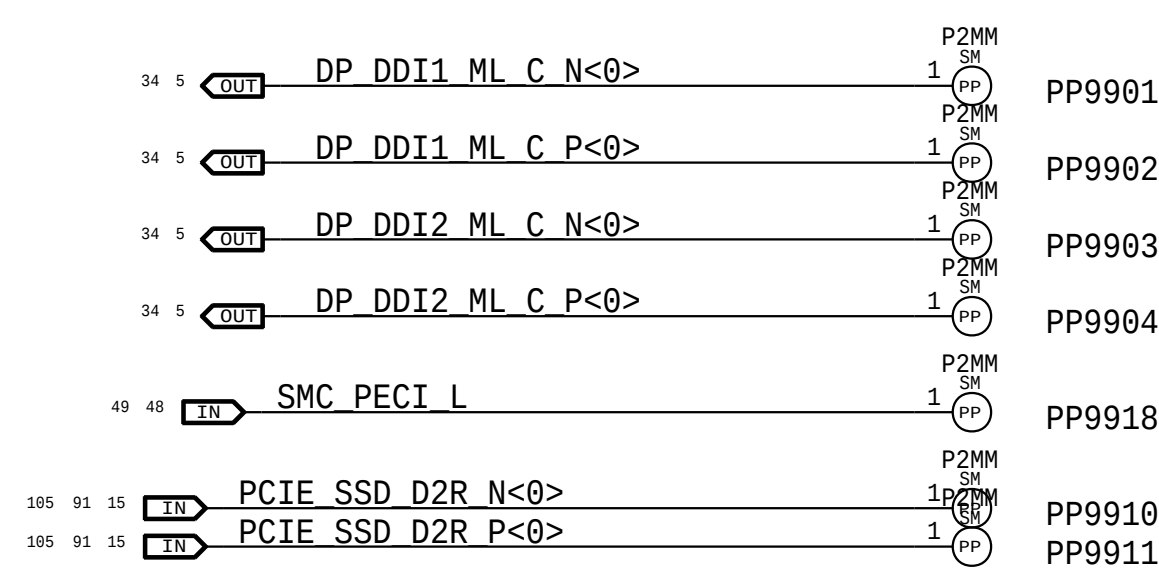
S3X Aliasing

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14	SSD_SR_EN_L	==	SSD_SR_EN_L	91 104

EPD PANEL

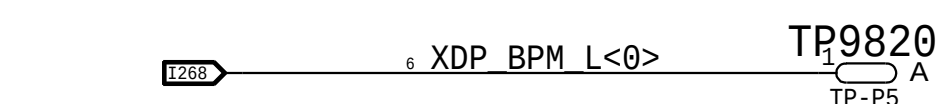
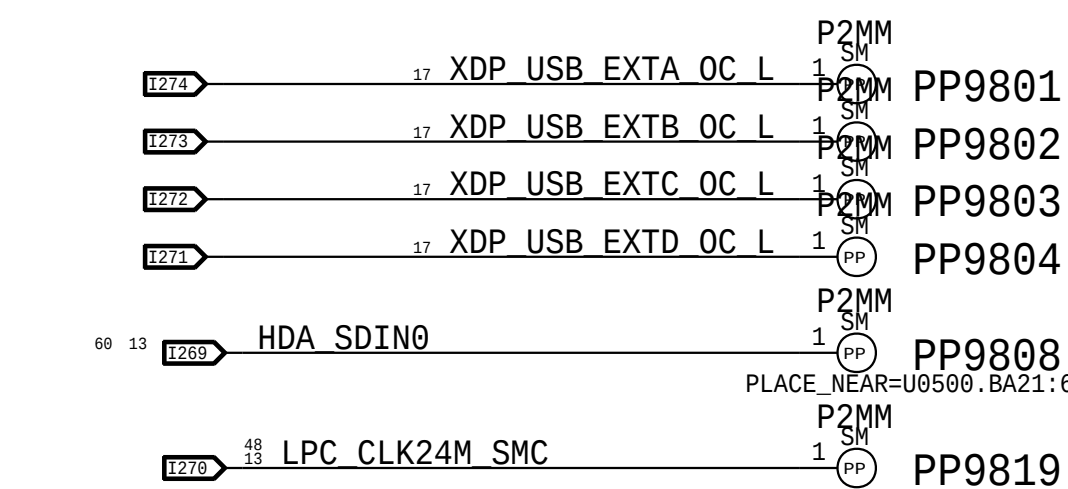
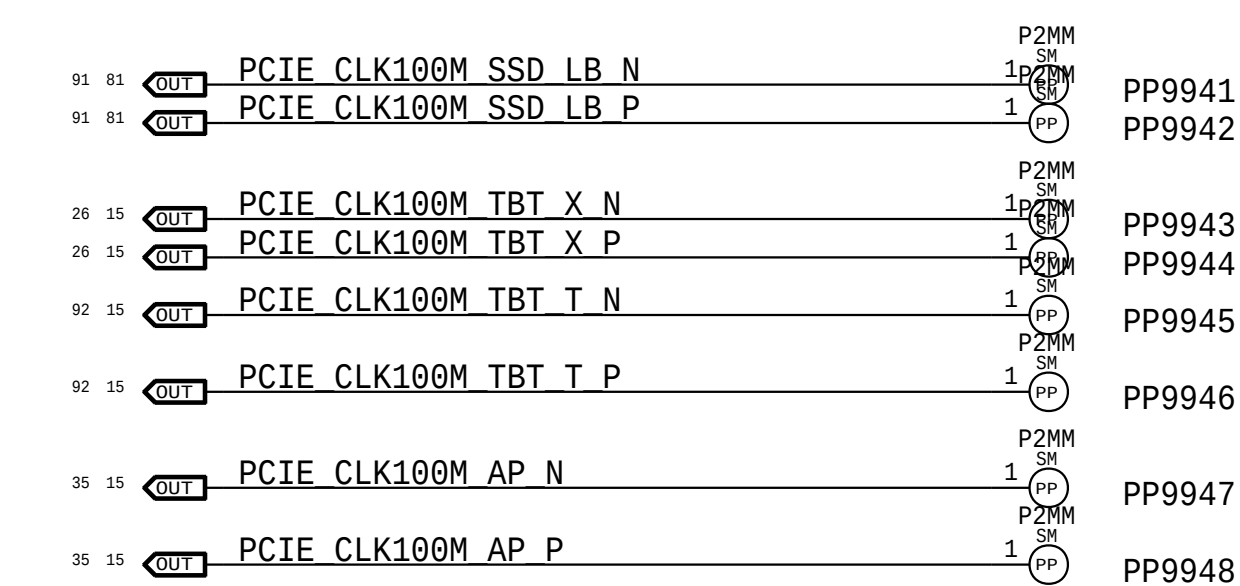
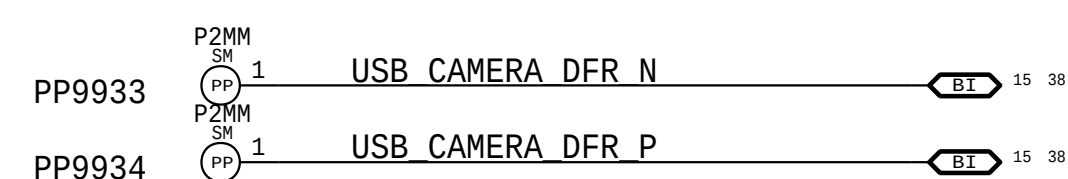
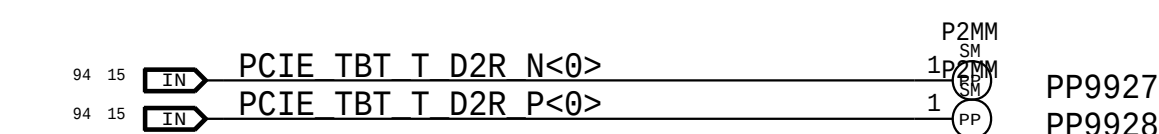
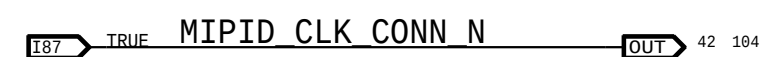
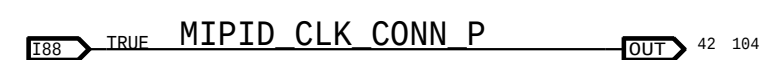
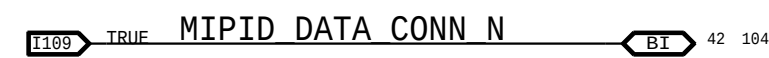
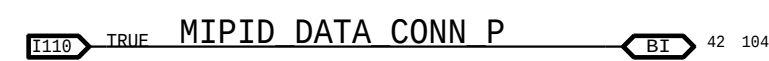
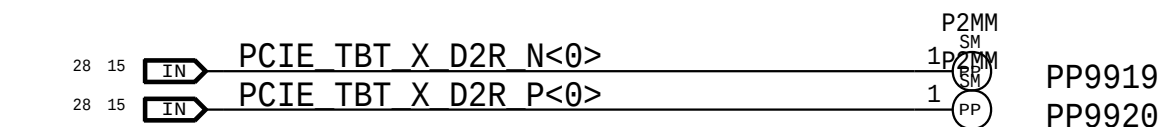
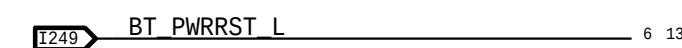
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79	I2C_BKLT_SDA	==	TRUE	I2C_BKLT_SDA	89 104

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		SHEET	102 OF 119

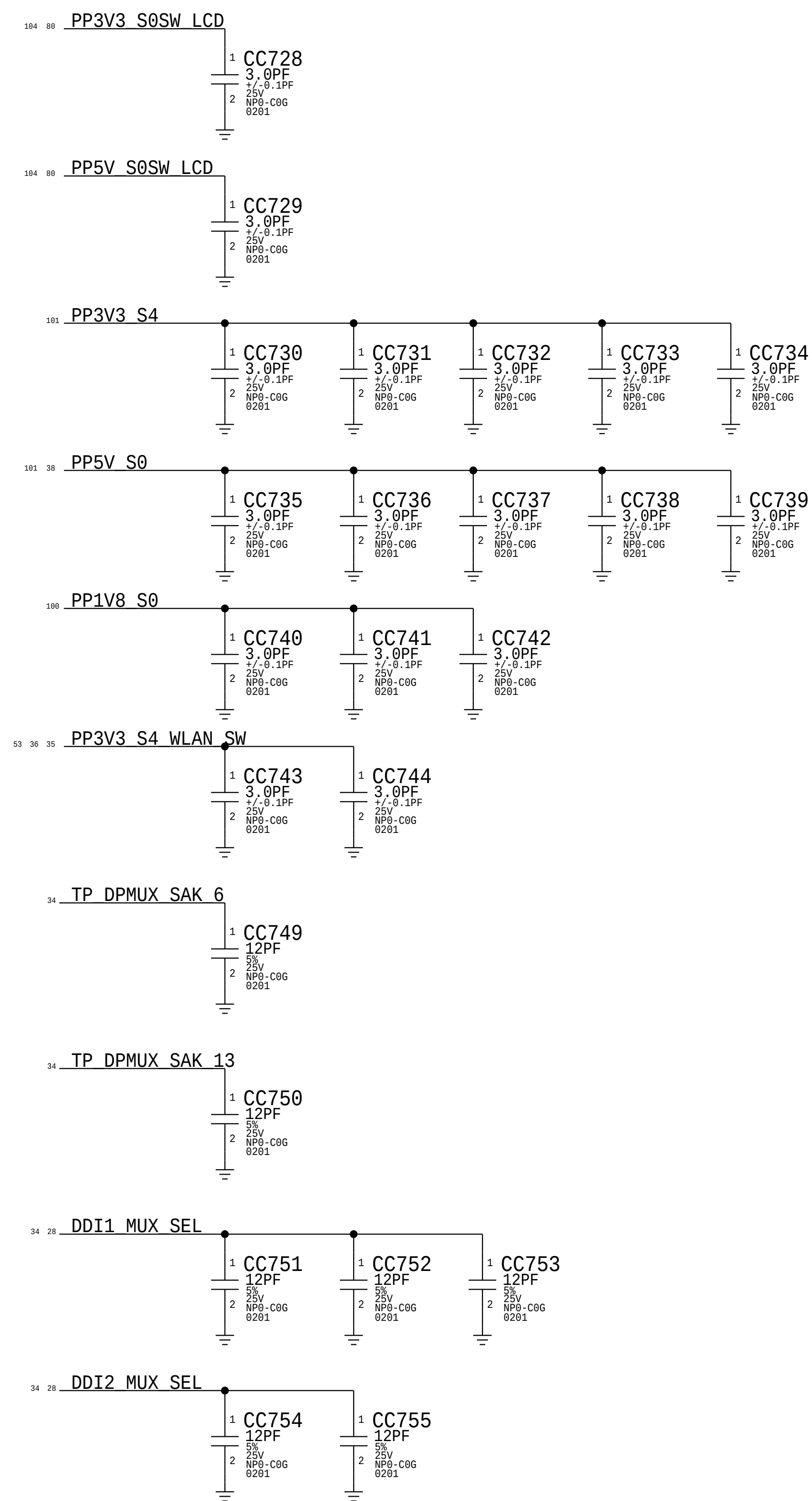
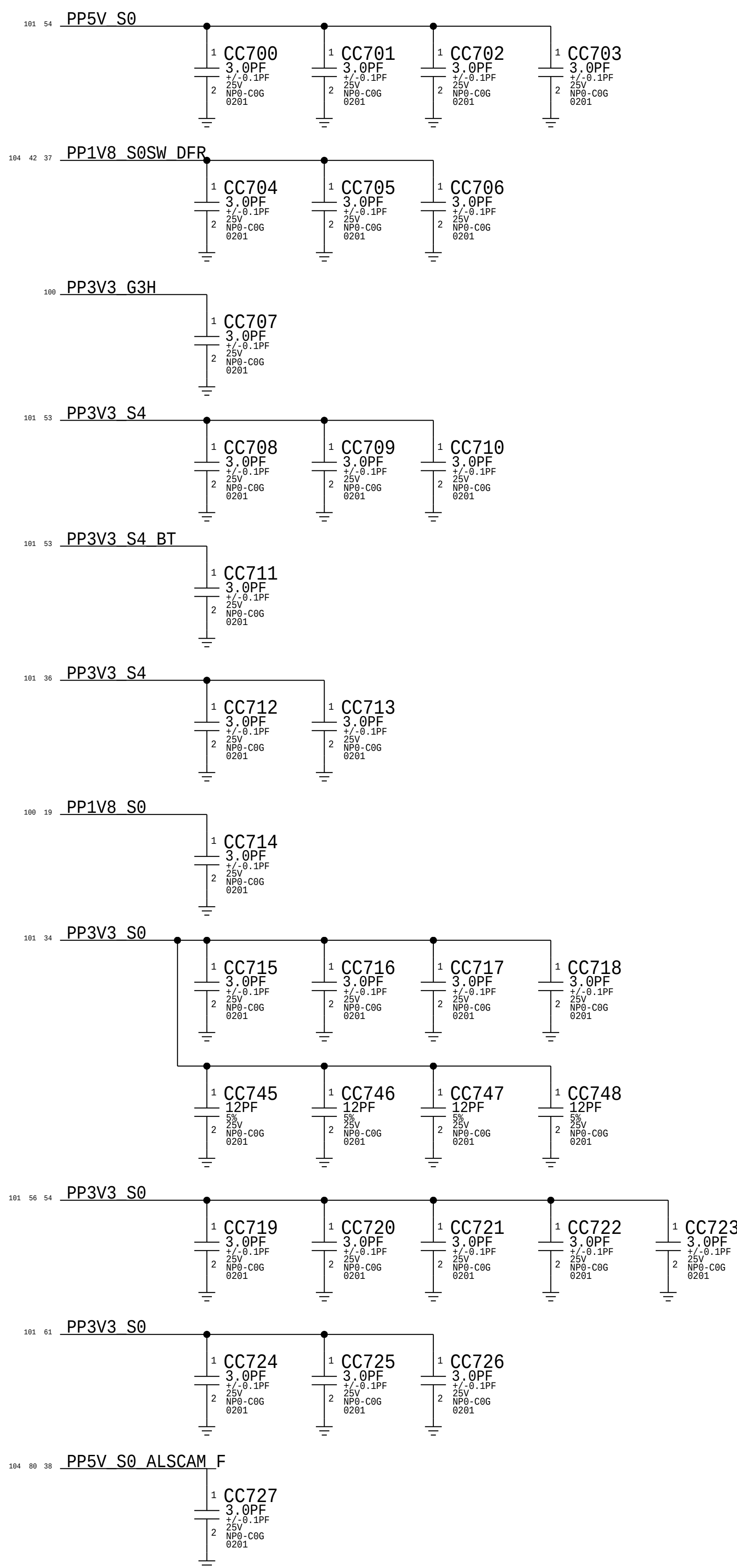


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(Nets with offpages not used on this project)



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Memory Bus Constraints															Memory Net Properties															Memory to Power Spacing																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
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MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP	MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	MEM_75D	*	=75_OHM_DIFF	=75_OHM_DIFF	=75_OHM_DIFF	=75_OHM_DIFF	=75_OHM_DIFF	=75_OHM_DIFF	MEM_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	<table><tr><th colspan="2">ELECTRICAL CONST SET</th><th colspan="2">NET TYPE</th></tr><tr><th colspan="2"></th><th>PHYSICAL</th><th>SPACING</th></tr><tr><td colspan="2"></td><td></td><td></td></tr><tr><td colspan="2"></td><td></td><td></td></tr><tr><td colspan="2"></td><td></td><td></td></tr><tr><td colspan="2"></td><td></td><td></td></tr><tr><td colspan="2"></td><td></td><td></td></tr><tr><td colspan="2"></td><td></td><td></td></tr><tr><td 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SMC SMBus & Charger Net Properties

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MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_20THER	*	=4X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?
MIPICKL_20THER	*	=7X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_20THER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICKL_20THER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	*	=2X_DIELECTRIC	?	S2_DATA2SELF	TOP, BOTTOM	=4X_DIELECTRIC	?
S2_DQS20WINDATA	*	=2X_DIELECTRIC	?	S2_DQS20WINDATA	TOP, BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CMD	*	=2X_DIELECTRIC	?	S2_CMD2CMD	TOP, BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CTRL	*	=2X_DIELECTRIC	?	S2_CMD2CTRL	TOP, BOTTOM	=4X_DIELECTRIC	?
S2_CTRL2CTRL	*	=2X_DIELECTRIC	?	S2_CTRL2CTRL	TOP, BOTTOM	=4X_DIELECTRIC	?
S2_20THERMEM	*	=4X_DIELECTRIC	?	S2_20THERMEM	TOP, BOTTOM	=6X_DIELECTRIC	?
S2MEM_2PWR	*	=4X_DIELECTRIC	?	S2MEM_2PWR	TOP, BOTTOM	=4X_DIELECTRIC	?
S2MEM_2GND	*	=2X_DIELECTRIC	?	S2MEM_2GND	TOP, BOTTOM	=4X_DIELECTRIC	?
S2MEM_20THER	*	=6X_DIELECTRIC	?	S2MEM_20THER	TOP, BOTTOM	=10X_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_20THER
S2_MEM_DQS*	*	*	S2MEM_20THER
S2_MEM_CMD	*	*	S2MEM_20THER
S2_MEM_CTRL	*	*	S2MEM_20THER
S2_MEM_CLK	*	*	S2MEM_20THER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS20WNDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS20WNDATA

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

[illegible]

SYNC_MASTER=YHARTANTO_J44		SYNC_DATE=01/09/2013	
PAGE TITLE			
 Apple Inc.		DRAWING NUMBER 051-00515	
		SIZE D	
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		BRANCH dvt-fab09-0	
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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_45S	*	=1T01_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.1 MM	0.1 MM
THERM_45S	*	=1T01_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.1 MM	0.1 MM
DIG_AUDIO	*	=1T01_DIFFPAIR	=1T01_DIFFPAIR	=1T01_DIFFPAIR	=1T01_DIFFPAIR	0.1 MM	0.1 MM
ANL_AUDIO	*	=1T01_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
ANL_AUDIO_WIDE	*	=1T01_DIFFPAIR	0.3 MM	0.3 MM	10 MM	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2X_DIELECTRIC	?
THERM	*	=2X_DIELECTRIC	?
AUDIO	*	=2X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
GND	PCIE *	*	GND_P2MM
GND	SATA *	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SB_POWER	SATA *	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_45S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.070 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_40S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D	TOP			0.100 MM	500 MIL		
CPU_27P4S	BOTTOM			0.230 MM	100 MIL		
USB3_85D	TOP			0.100 MM	500 MIL		
USB3_85D	ISL10			0.075 MM			0.090 MM
DP_85D	ISL9			0.075 MM			0.090 MM
PCIE_85D	ISL10			0.075 MM			0.090 MM

DP, SATA, HDMI, PCIE CONSTRAINT RELAXATIONS

Alternate diffpair width/gap through BGA fanout areas (95-ohm diff)

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
DP_85D	BGA	P65_BGA
PCIE_85D	BGA	P65_BGA
CLK_PCIE_85D	BGA	P65_BGA
HDMI_85D	BGA	P65_BGA

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SENSE_45S	*	SENSE_45S
THERM_45S	*	THERM_45S
DIG_AUDIO	*	DIG_AUDIO
ANL_AUDIO	*	ANL_AUDIO

J79 Specific Net Properties

ELECTRICAL CONST SET	NET TYPE	
	PHYSICAL	SPACING

J79 Specific Net Properties

[illegible]

SYNC_MASTER=YHARTANTO_344		SYNC_DATE=01/04/2013	
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Sensors & Audio Constraints			
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Change List:

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<RDAR://COMPONENT/XXXXXX> J79 HW EE SCHEMATIC | PROTO 0
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Kismet:

AFP://KISMET.APPLE.COM/KISMET-PROJECTS/J79

Useful Wiki Links:

Schematic Conventions - <https://hmts.ecs.apple.com/wiki/index.php/User:Wferry/SchConventions>

Schematic Design Wiki - https://hmts.ecs.apple.com/wiki/index.php/Schematic_Design

MobileMac HW Radar:

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<rdar://component/XXXXXX> MobileMac HW | Task
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<rdar://component/XXXXXX> MobileMac HW | Schematic
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<rdar://component/XXXXXX> MobileMac HW | New Bugs
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<rdar://component/XXXXXX>	MobileMac HW	Layout
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<rdar://component/XXXXXX> MobileMac HW	Investigation
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<rdar://component/XXXXXX>	MobileMac	HW	Architecture
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Other Info:

Page Allocations - <rdar://problem/19817053> 2015 Schematic Page Allocations

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